Afe Babalola University
Ado-Ekiti

B.Eng. Degree Programme in

Electrical/Electronics Engineering

SELF-STUDY REPORT (SSR)

FOR

ACCREDITATION OF ENGINEERING PROGRAMMES IN NIGERIAN UNIVERSITIES

by

COUNCIL FOR THE REGULATION OF ENGINEERING IN NIGERIA

Established by Decrees 55/70 and 27/92 (amendment) now the Engineers (Registration, etc.) Act, CAP E11, 2004

May, 2026.

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SECTION ONE: GENERAL

PRELIMINARY INFORMATION ON THE INSTITUTION

1. PRELIMINARY INFORMATION ON THE INSTITUTION

1.1. Name and Address of the Institution

Afe Babalola University Ado-Ekiti (ABUAD),
KM 8.5, Afe Babalola Way,
P.M.B 5454, Ado-Ekiti.
Ado-Ekiti, Nigeria.
Telephone: +2348050506152
+2348127772121
+2348038308348
Website: www.abuad.edu.ng

1.2. Name and Address of the Proprietor of the University

Aare Afe Babalola *OFR, CON, SAN, LL.D, D.LL, D.Litt. FCI.Arb, FNIALS.*
KM 8.5, Afe Babalola Way,
P.M.B 5454, Ado-Ekiti.
Ado-Ekiti, Nigeria

1.3. Name of the Vice Chancellor

Prof. Elisabeta S. OLARINDE
FCArb, FCN

1.4. Vision, Mission, and Core Values of the University

1.4.1. Vision of the University

"to be a world class educational centre of excellence in academics, character, sports and vocational development"

1.4.2. Mission of the University

Afe Babalola University is a result-oriented institution for producing highly skilled and socially relevant graduates capable of applying scientific knowledge for the resolution of social problems. The University is equally committed to transforming students into expert thinkers, innovative managers and resourceful technocrats in all fields of learning.

1.4.3. Core goals of the University

As a unique University, it seeks to achieve the following goals:

- Encourage a balance between academic excellence, professional ethics, and practical competence among students;
- Encourage daily consistency and personal integrity among the students. To pursue academic excellence through outstanding teaching and the promotion of research and community service on the national and international level. To create and promote interdisciplinary and international scholarship and education, while sustaining foundational disciplines;
- Define and lead emerging disciplines and intellectual paradigms that create knowledge and to promote the translation of that knowledge to the benefit of society;
- Develop new learning opportunities that build on these paradigms and create student engagement in the context of a community of scholars; and
- Foster effective concern for environmental well-being, justice, peace and compassion and commitment to their promotion in the contemporary world.

1.5. Establishment of the University

Date Established: 2009

Afe Babalola University is the brainchild of Chief Afe Babalola, B.Sc. (Econs) Lond; LL.B (Hons) Lond; BL; FFPA; FNIALS, FCI Arb; LL.D; Senior Advocate of Nigeria (SAN); Doctor of Letters (LL.D); Officer of the Order of the Federal Republic (OFR), and Commander of the Order of the Niger (CON), the immediate past Pro-Chancellor and Chairman of the Governing Council of the University of Lagos, Nigeria, and the immediate past Chairman of the Committee of Pro-Chancellors of Nigerian Universities.

It is a product of a mind disturbed by the incurable problems afflicting government universities and the quality of graduates being produced each year. As a Pro-chancellor and Chairman of Pro-chancellors, he found that these Universities problems were due to lack of commitment on behalf of the staff and want of emotional attachment to their various institutions as in the case of a private or personal establishment. It is common knowledge that workers in public institutions often display lackadaisical attitude towards government work with the justification that it is not their father's business.

The Promoter believes that a time has come when private individuals should stand up to the challenge of providing affordable and qualitative tertiary institutions as it is done in other nations like the ivy league of Universities in the United States and other leading institutions in Europe. These are classic examples of how and why private universities are more successful than government universities evidenced by the quality of graduates they produce and their performances in the competitive world field. For example, Japan which has the largest number of private Universities is indisputably a leader in the field of technology which goes to show that private

institutions offer brighter prospects of qualitative tertiary education. This is because the proponents of these great private universities have dreams, ideas, and objectives about the type of university they want. They established code of conduct and disciplinary measures which are pursued religiously and rigorously.

Location

The University is located on a 130 hectares land at an altitude of over 1500 ft above sea level along Ado-Ekiti - Ijan Road, directly opposite the Federal Polytechnic, Ado-Ekiti in Ekiti State, which ipso facto provides cool and ideal climate for learning and sport activities.

Objectives

In order to achieve these goals, we pursue the following objectives:

(a) Establish in Ado Ekiti a world class institution of higher learning which combines the acquisition of knowledge with its practical application in training graduates of high integrity and of international recognition;

(b) Provide an environment of international standards for learning, research, acquisition of specialized skills and comprehensive training of students empowering them to apply knowledge and skill to the benefit of society;

(c) Complement the efforts of Government in bridging the yawning unemployment gap by creating employment opportunities for Nigerians.

(d) Develop and continuously up-date a modern library including a digital library to encourage research, publication and to promote a "community of learning" atmosphere, global interaction as well as promote scholarship and conduct research in all fields of human endeavor to advance learning and provide solutions to developmental problems for the benefit of society and mankind;

(e) Develop and offer specialized, flexible and innovative programmes for specialized and non-specialized undergraduate and post graduate degrees and non-degree programmes, thus creating a balance between academic excellence, professional ethics and practical competence among a critical mass of human capital who are imbued with a high sense of responsibility, individual initiative as well as with capacity for analytical, creative and practical thinking;

(f) Promote effective communication among the academic community, students and the public in order to produce specialists competent to bring about research aimed at creative inventions and practical contribution towards the advancement of knowledge, technological development and for the creation of wealth for Nigeria.

(g) Establish a centre for entrepreneurial studies to stimulate job creation abilities in students from the onset of their studies;

(h) Establish a Creative and Lifelong Education Centre for the purpose of facilitating non-degree and vocational skills programmes;

(i) Encourage and promote the advancement of learning and hold out to all persons without distinction of race, creed, sex, religious or political conviction the opportunity to acquire university education; and

(j) Undertake other activities appropriate for teaching, research and community service as expected of a university of high standard.

Strategies for Achieving Stated Objectives

The major University-wide strategies to be adopted in achieving the objectives of Afe Babalola University are premised on a platform of these basic characteristics:

Assistance by way of scholarships and job opportunities on campus to brilliant but indigent students;

* Excellence in educational programmes that is affordable to students;

* Commitment to attracting and retaining world-class teaching and non-teaching staff, along with competitive employee compensation and support for success;

* Institutional agility and resource flexibility in order to capture emerging opportunities, and to be readily and creatively responsible to the needs of constituent stakeholders at the state, national, and international levels.

* Undergraduate and post-graduate programmes in various disciplines that meet the students' need for learning and the societal need for solutions to arts, science and technological challenges. The University is accessible to diverse makers, and offers curricula designed to serve their needs.

* Courses are broad-ranged and integrated to facilitate the required creativity and productivity of students.

* The University emphasizes student-centered learning environments engaging academic variety and student diversity, with the goals of providing a holistic and collaborative approach to learning; learning through discovery, experience, and innovation; and effective use of information.

* The University has developed innovative means of providing financial assistance to

deserving indigent students, with due regard to competitive recruitment of qualified students, cost of education and student need, enrolment characteristics, scholastic achievement, and educational enrichment.

* The University has developed information technology and widespread access as a catalyst in the execution of other strategies, especially in the creation of new mechanisms for effective engagement of service.

* The University has created and expanded academic facilities lecture rooms, e-learning platforms, ICT center, library, laboratories, and studios – to support innovative teaching and learning.

* The University provides learning experiences for students through hands-on community service opportunities and internship opportunities that complement their career objectives.

* The University publicizes and promotes institutional achievements and staff and student accomplishments among the constituent stakeholders and enhance national and international visibility.

* There will be formal assessment of student outcomes at the University level at least every other year and, more frequently, there will be assessment of the outcomes and effectiveness of academic programmes, courses, and targeted initiatives in the interest of continuous improvement.

* All students will take compulsory courses that cut across all programmes being undertaken. They include courses such as Phonetics and Use of English, Communication Skills, Logic, Creative Thinking, African Culture and Ethics, Entrepreneurship as well as basic education in practical agriculture.

* The University operates a non-discriminatory students' admissions policy on the basis of gender, tribe, religion, political leanings or physical disability. Only candidates that are highly qualified and able to benefit from good university education will however stand a good chance of being admitted.

* Admission of students is through the UTME including a selection process that will from time to time be determined by the University.

* The method of instruction is face-to-face interaction – a normal 17 weeks semester for roper educational impartation of knowledge. The use of modern communication facilities available in imparting knowledge to students will be employed. Practical and tutorials are also designed and students are given opportunities to think out projects and work under supervision. On-line approach to student-teacher communication is in place to facilitate student-teacher interactions. To this end every staff and student will own a computer for their use.

* There will be continuous assessment as part of the grading system to ensure that students are up to date in their studies.

* The University undertakes a periodic curriculum review that takes into account feedback from organizations and employs of labour and NUC Minimum Academic Standards.

* All students will have access to welfare, career planning and placement services, and accommodation in well-equipped and functional halls of residence.

* Opportunities will be provided for staff and facilities evaluation exclusively by students.

1.6. Name of Faculty/Programme

College of Engineering/Electrical/Electronics Engineering

1.7. Name and Designation of the Provost

Engr. Prof. Moses Oluwafemi ONIBONOJE *MNSE, MIEEE, R.Eng*.

1.8. COREN Registration Number of the Provost

COREN: R30783

1.8.1. Vision of the College of Engineering Afe Babalola University

To be a world-class College of Engineering that nurtures innovation, fosters technological excellence, and produces ethical, globally competitive engineers dedicated to solving societal challenges and advancing sustainable development.

1.8.2. Mission of the College of Engineering Afe Babalola University

1. *To provide a dynamic and research-driven engineering education that integrates cutting-edge technology, hands-on learning, and industry collaboration.*

2. *To develop highly skilled engineers who uphold integrity, professionalism, and entrepreneurial spirit in addressing real-world challenges.*

3. *To promote interdisciplinary research and innovation that contributes to national and global technological advancements.*

4. *To develop engineers with critical thinking, creativity, and leadership skills for engineering solutions that drive sustainable development.*

5. *To uphold the highest academic and ethical standards in training engineers*

who will lead in various sectors of the economy.

1.9. Programme to be Accredited

Electrical/Electronics Engineering

1.10. Degree to be Awarded and Abbreviation

Bachelor of Engineering (B.Eng.)

1.11. Duration of the Programme

Five (5) years

1.12. Name and Designation of the Head of Programme

Engr. Dr. Adedayo Olukayode OJO *MNSE, R.Eng., MIAENG*

1.13. COREN Registration Number of the Head of Programme

COREN: R33,712

1.14. History of Electrical/Electronics Engineering Degree Programme

Established in September 2010 alongside three other departments within the College of Engineering, the Electrical/Electronics and Computer Engineering department proudly offers robust programs, one of which is B. Eng. degree in Electrical/Electronics Engineering. The pioneering professors of the programme include Professor I. E. OWOLABI and Professor OLOWOLAFE who led the programme to amiable heights, including medals at the Microsoft® ImagineCup™ global competition event held in USA. This five-year program is grounded in a strong foundation of mathematics, basic sciences, general studies, and specialized engineering courses, designed to foster both technical expertise and ethical responsibility. Our commitment to integrity is woven into every facet of our academic culture. We emphasize ethical engineering practices, encouraging students to uphold honesty, accountability, and professionalism in their work. Our seasoned staff members are equipped with diverse specializations, bringing a wealth of expertise to the classroom, thereby guiding students toward academic excellence and principled innovation. The programme prioritizes hands-on training to prepare students for real-world challenges. Our well-equipped laboratories, staffed by skilled technologists, provide practical learning environments where students engage in experiments, projects, and simulations to deepen their technical skills.

1.15. Philosophy of Electrical/Electronic Engineering

The programme strongly believes in the production of graduates who appreciate the value of the dignity of labor and who are problem solvers, self-reliant, employers of labor, highly productive and globally relevant in every sphere of human endeavor.

1.16. Vision and Mission of the Programme

1.16.1. Vision of the Programme

To be a world-class center of excellence in Electrical and Electronic Engineering education, fostering academic innovation, ethical leadership, and holistic development in character, research, and vocational competence.

1.16.2. Mission of the Programme

To produce highly competent Electrical and Electronic Engineers equipped with strong technical expertise, practical skills, and professional ethics. This will be achieved through rigorous academic training, hands-on experience, and the development of innovative solutions to global engineering challenges, empowering graduates to excel in their professional endeavors of applying scientific knowledge and electrical engineering principles for the resolution of technical and social problems.

1.17. Institution Academic Sessions

Approved Academic Calendar for 2024/2025 Academic Session

First Semester, 2024/2025

S/N	Events	Dates
1.	Returning Students Resumption	Monday 23rd September 2024
2.	- College of Engineering	Tuesday 24th September 2024
3.	- College of Law	Tuesday 24th September 2024
4.	- College of MHS	Wednesday 25th September 2024
5.	- College of Sciences/Pharmacy	Thursday 26th September 2024
6.	- College of SMS	Friday 27th September 2024
7.	Lectures Begin (Fresh Students)	Monday 30th September - Friday 18th October 2024
8.	Course Registration Online (Returning Students)	Monday 30th September 2024
9.	**Fresh Students Resumption**	Monday 30th September 2024
10.	- College of Engineering	Monday 30th September 2024
11.	- College of Law	Tuesday 1st October 2024
12.	- College of MHS	Wednesday 2nd October

		2024
13.	- College of Sciences/Pharmacy	Thursday 3rd October 2024
14.	- College of SMS	Friday 4th October 2024
15.	Orientation	Monday 7th - Friday 11th October 2024
16.	Course Registration Online (Fresh Students)	Monday 14th October 2024
17.	Lectures Begin (Fresh Students)	Monday 14th October 2024
18.	Late Registration with Penalty (Returning Students)	Monday 21st - 31st October 2024
19.	Convocation	Monday 21st October 2024
20.	End of Year Break	Friday 20th December 2024 - Saturday 4th January 2025
21.	ABUAD Academic Commencement Anniversary	Saturday 4th January 2025
22.	Lectures Continue	Monday 6th January 2025
23.	Matriculation	Tuesday 21st January 2025
24.	First Semester Examinations	Monday 27th January - Friday 14th February 2025
25.	End of First Semester Break	Saturday 15th February - Monday 17th February 2025
26.	**Marking and Processing of Results**	Monday 17th February - Friday 7th March 2025
27.	- (a) Marking of Scripts	Monday 17th February 2025
28.	- (b) Departmental Examination Board	Wednesday 12th March 2025
29.	- (c) College Examination Board	Friday 14th March 2025
30.	- (d) Provost Board Meeting - CPDD	Wednesday 19th March 2025
31.	- (e) Senate Meeting	Monday 24th March 2025
32.	- (f) Online Results	Monday 24th March 2025

Second Semester, 2024/2025.

S/N	**Events**	**Dates**
1.	Students Resumption	Thursday 6th March 2025
2.	- College of Engineering	Friday 7th March 2025
3.	- College of Law/SMS	Saturday 8th March 2025
4.	- College of MHS	Sunday 9th March 2025
5.	- College of Sciences/Pharmacy	Monday 10th March 2025
6.	Course Registration Online	Monday 10th - 21st March

		2025
7.	Lectures Begin	Monday 24th March - Friday 20th June 2025
8.	Late Registration with Penalty	Monday 24th March - Friday 28th March 2025
9.	Second Semester Exams (Final Year) Commence	Monday 9th June - Friday 13th June 2025
10.	Second Semester Exams all other Students	Monday 16th June - Friday 27th June 2025
11.	Students' Vacation	Saturday 28th June 2025
12.	**Marking & Processing of Results**	Monday 30th June - Friday 11th July 2025
13.	- (a) Marking of Scripts	Monday 14th July 2025
14.	- (b) College Examination Board	Wednesday 16th July 2025
15.	- (c) Provost Board Meeting - CPDD	Friday 18th July 2025
16.	- (d) Senate Meeting	Wednesday 23rd July 2025
17.	- (e) Online Results	Monday 28th July 2025
18.	- (f) End of Session	Tuesday 29th July 2025
19.	**New Session for 2024/2025 Begin**	**September 22nd 2025**

Approved Academic Calendar for 2025/2026 Academic Session

First Semester 2025/2026 Academic Session

S/N	**Activity/Description**	**Date**
1.	Fresh Students Resumption – College of Engineering	Monday 22nd September, 2025
2.	Fresh Students Resumption – College of Law	Tuesday 23rd September, 2025
3.	Fresh Students Resumption – College of MHS	Wednesday 24th September, 2025
4.	Fresh Students Resumption – College of Sciences/Pharmacy	Thursday 25th September, 2025
5.	Fresh Students Resumption – College of SMS	Friday 26th September, 2025
6.	Course Registration Online (Fresh Students)	Monday 6th – Friday 17th

		October, 2025
7.	Lectures Begin (Fresh Students)	Monday 29th September, 2025
8.	Returning Students Resumption – College of Engineering	Monday 29th September, 2025
9.	Returning Students Resumption – College of Law	Tuesday 30th September, 2025
10.	Public Holiday (National Day)	Wednesday 1st October, 2025
11.	Returning Students Resumption – College of MHS	Wednesday 1st October, 2025
12.	Returning Students Resumption – College of Sciences/Pharmacy	Thursday 2nd October, 2025
13.	Returning Students Resumption – College of SMS	Friday 3rd October, 2025
14.	Orientation	Wednesday 8th – Friday 10th October, 2025
15.	Course Registration Online (Returning Students)	Monday 6th – Friday 17th October, 2025
16.	Lectures Begin (Returning Students)	Monday 6th October, 2025
17.	Late Registration with Penalty (Returning Students)	Monday 20th – Friday 31st October, 2025
18.	Convocation	Tuesday 21st October, 2025
19.	End of Year Break	Wednesday 17th December, 2025 – Monday 5th January, 2026
20.	Christmas Break	Thursday 25th December, 2025
21.	ABUAD Academic Commencement Anniversary	Sunday 4th January, 2026
22.	Lectures Continue	Monday 5th January, 2026
23.	Matriculation	Wednesday 21st January, 2026
24.	First Semester Examinations	Monday 26th January – Friday

		13th February, 2026
25.	End of First Semester Break	Saturday 14th – Saturday 21st February, 2026
26.	Marking & Processing of Results – Marking of Scripts	Monday 16th February – Friday 6th March, 2026
27.	Departmental Examination Board	Monday 9th March, 2026
28.	College Examination Board	Thursday 12th March, 2026
29.	Provost Board Meeting – CPDD	Monday 16th March, 2026
30.	Senate Meeting	Wednesday 25th March, 2026
31.	Online Results	Friday 27th March, 2026

Second Semester 2025/2026 Session

1.	Students Resumption	Monday 2nd March, 2026
2.	Eid El-Fitr (Tentative)	Friday 20th – Sunday 21st March, 2026
3.	Course Registration Online	Monday 2nd – Friday 20th March, 2026
4.	Lectures Begin	Monday 2nd March, 2026
5.	Late Registration with Penalty	Monday 23rd March – Friday 3rd April, 2026
6.	Easter Break (Good Friday & Easter Monday)	Friday 3rd – Monday 6th April, 2026
7.	Id-El-Kabir (Tentative)	Wednesday 27th – Thursday 28th May, 2026
8.	Second Semester Exams (Final Year) Commence	Monday 1st – Friday 5th June, 2026
9.	Second Semester Exams (All Other Students)	Monday 8th – Friday 19th June, 2026

10.	Students' Vacation	Saturday 20th June, 2026

Marking and Processing of Results 2025/2026 Session

1.	Marking Scripts	Monday 22nd June – Friday 10th July, 2026
2.	Departmental Examination Board	Monday 13th July, 2026
3.	College Examination Board	Thursday 16th July, 2026
4.	Provost Board Meeting – CPDD	Monday 20th July, 2026
5.	Senate	Wednesday 29th July, 2026
6.	End of Session	Friday 31st July, 2026
7.	New Session (2026/2027) Begins	Monday 21st September, 2026

1.18. Programme History of Accreditation

Table 1.X: Last Accreditation: Areas of concern and recommendation

Year	**Programme Score**	**Areas of Concern**	**Actions taken**
2021	88%	1. Inability of students to attend to laboratory, and workshop practicals cannot be controlled by lecturers.	1. The laboratories and practical sessions have been re-structured such that each laboratory is now coordinated by a lecturer in conjunction with the technologists.
		2. The number of the laboratory technologists and	2. More technologists have since been

		technicians is not adequate.	employed as can be seen in the staff list.

1.19. Self-initiated Improvements of the Programme

The following are some of the measures employed for self-improvement initiative carried out by the programme over the years:

1. Regularly assess the curriculum to ensure it aligns with current industry needs and technological advancements.
2. Actively soliciting feedback from students on the programme, including coursework, teaching methods, and resources.
3. Invite industry professionals to share their knowledge and experiences with students during the career week.
4. Organize seminars on emerging technologies and industry trends to keep faculty and students updated.
5. Encourage collaboration and knowledge sharing among students through group projects and peer-to-peer learning activities.
6. Utilize online platforms for delivering course content, simulations, and self-paced learning opportunities.
7. Introduce students to open-source hardware platforms like Arduino and Raspberry Pi, and software tools for simulation and design.
8. Increase the focus on hands-on projects that require applying theoretical knowledge to solve real-world problems. Encourage collaboration and design thinking.

SECTION TWO:
THE PROGRAMME EDUCATIONAL OBJECTIVES

2.3. The Programme Educational Objectives (PEOs)

The Programme Educational Objectives (PEOs) of the Electrical/Electronics Engineering of Afe Babalola University Ado-Ekiti is a list of concise objectives that drives the activities and direction of the programme. These PEOs are tabulated as follows:

2.4. Programme Educational Objectives

PEO1 *To produce highly skilled and competent Electrical/Electronics engineers that will contribute to academia, industrial and professional Engineering practice.*

PEO2 *To produce proficient, self-reliant, and excellent Electrical/Electronics Engineers equipped with strong technical skills, a solid understanding of fundamental concepts, and the ability to address contemporary challenges in line with modern technological developments.*

PEO3 *To produce graduates who will actively engage in continuous professional development and lifelong learning, pursue advanced studies, and obtain relevant professional certifications, thereby remaining competent and relevant in a rapidly evolving engineering environment.*

PEO4 *To produce graduates who can provide practical and cost-effective solutions to real-world problems by applying their modeling knowledge, innovative thinking, and entrepreneurial skills, as well as their ability to design, develop, and solve complex engineering challenges across various fields of Electrical/Electronics Engineering.*

PEO5 *To produce graduates that will uphold ethical standards, acquire knowledge, demonstrate integrity, leadership and practice Electrical/Electronics engineering responsibly in compliance with national and international professional codes and sustainable development principles.*

2.5. Publication of the Programme Educational Objectives (PEOs)

The Programme Educational Objectives (PEOs) of Electrical and Electronic Engineering at Afe Babalola University are documented in the official programme handbook, on the notice boards, and on the University website www.abuad.edu.ng. They are also communicated on a regular basis during faculty and departmental meetings, student awareness sessions, orientation programmes, placement activities, training exercises, and through selected social media platforms. To further ensure that graduates are well-informed, competent, and equipped with the knowledge, skills, and mindset required in the fast-evolving world of technology, related

documents such as the vision, mission, motto, and Program Outcomes (POs) are shared in the same way. These materials are also made available to all programme stakeholders for reference and engagement.

2.5.1. Consistency of the PEOs with the Vision and Mission of the University

The Programme Educational Objectives (PEOs) are fully aligned with the Vision and Mission of Afe Babalola University. The emphasis on producing academically minded graduates with outstanding results in Electrical and Electronics Engineering directly supports the vision of the University of being a world-class educational centre of excellence in academics. By fostering technical competence, leadership, innovation, and entrepreneurship, the PEOs reflect the mission of the University on producing highly skilled and socially relevant graduates capable of applying scientific knowledge to solve societal problems. The integration of vocational development and entrepreneurship skills corresponds with the University's vision of promoting vocational excellence and its mission to transform students into innovative managers and resourceful technocrats. Furthermore, the focus on effective communication, collaboration, sportsmanship, professionalism, and ethical responsibility mirrors the University's vision of excellence in character and sports. Finally, the PEOs' commitment to lifelong learning, professional development, and contributing to academia and industry reinforces the mission of developing expert thinkers who sustain the growth and advancement of the engineering profession both within and beyond Nigeria.

2.5.2. Consistency of PEOs with University/Programme's Vision and Mission

Table 2.1: Mapping of PEOs to Vision of University/Programme

PEOs	University's Vision		
	Excellence in Academics	**Excellence in Character**	**Excellence in Vocational Development**
PEO1	x		x
PEO2	x	x	x
PEO3	x	x	x
PEO4	x	x	x
PEO5	x		x
PEOs	**Programme's Vision**		
	Excellence in Education and Innovation	**Ethical Leadership**	**Holistic Development in Character, research, and vocational competence**

PEO1	x		x
PEO2	x	x	x
PEO3	x		x
PEO4	x	x	x
PEO5	x	x	

Table 2.2: Mapping of PEOs to Mission of University/Programme

PEOs	**University's Mission**		
	Highly Skilled and Socially Relevant Graduates	**Scientific Knowledge for the Resolution of Problems**	**Expert Thinkers, Innovative Managers and Resourceful Technocrats in all Fields**
PEO1	x	x	
PEO2	x		x
PEO3		x	x
PEO4	x		
PEO5	x		x
	Programme's Mission		
	Competent Engineers with Ethics	**Training and Innovation**	**Applying Knowledge to Problems**
PEO1		x	x
PEO2	x	x	
PEO3			x
PEO4	x		
PEO5	x		

2.5.3. Consistency of PEO with stakeholders of the Programme

The PEOs have been developed in inconsistency with the different stakeholders of the Electrical/Electronics Engineering programme. These include the students, Faculty members and instructors, Alumni, employers, industry partners, parent forum, administrators, community and advisory board:

1. **Students**

   Students are the direct beneficiaries of the programme. They strive to acquire the knowledge, skills, and competences that prepare them for professional careers. Their progress is influenced by how well the Programme Educational Objectives (PEOs) are achieved. Students are expected to understand and comply with academic regulations, instructions from lecturers and support staff, and policies set by the university administration.

2. **Faculty and Instructors**

   Faculty members and instructors play a central role in delivering the curriculum, designing teaching activities, and evaluating learning outcomes. They supervise, mentor, and guide students while aligning teaching strategies with the PEOs. Their responsibilities also include conducting research, supervising student research projects, and ensuring that teaching remains effective. In addition, technologists and instructors facilitate laboratory demonstrations and provide support for students during practical sessions.

3. **Alumni**

   Alumni consist of graduates who have either advanced their studies or entered the workforce. Their professional and academic experiences help to assess the relevance and effectiveness of the PEOs. By sharing feedback on how well the programme prepared them for future challenges, they contribute to continuous improvement.

4. **Employers**

   Employers have a strong interest in the qualities and competencies of graduates, as they are potential employees. Their assessment of graduate performance in the workplace provides valuable insights into how closely the PEOs align with current industry expectations and professional demands.

5. **Accreditation Bodies**

   Accrediting organizations establish the standards and benchmarks for evaluating the quality and effectiveness of academic programmes. They review the programme for compliance with regulatory requirements and the extent to which the PEOs are achieved. In Nigeria, this includes agencies such as the National Universities Commission (NUC) and the Council for the Regulation of Engineering in Nigeria (COREN).

6. **Industry Partners**

   Industry partners collaborate with the university by offering internships, cooperative placements, research opportunities, and other forms of experiential learning. Their knowledge of emerging technologies and workplace requirements influences the continuous adaptation of the PEOs to industry realities.

7. **Parents' Forum**

Through advocacy, constructive feedback, and active involvement, parents play a vital role in the success of the programme. They foster stronger links between families and the university, which positively affects student achievement and wellbeing. Parents also provide financial support for their wards by ensuring that required fees are promptly paid.

8. **Administrators**
   Programme administrators, including the Head of Department (HOD), university management, governing council, and non-academic staff, oversee the overall functioning of the programme. Their duties include resource management, evaluation, and long-term planning. The governing council makes key policy decisions, while management ensures that daily operations are effectively carried out. Administrative staff further support smooth operations, thereby ensuring that the mission and vision of the institution remain aligned with the PEOs.
9. **Community and Society**
   The wider community benefits from the programme through the contributions of graduates who support social development, economic growth, and innovation. Society is therefore an important stakeholder as it gains from the knowledge and services provided by graduates.
10. **Advisory Board**
    Advisory boards are composed of faculty members, alumni, and industry representatives who provide expert guidance to the programme. They share perspectives on labour market needs, industry trends, and best practices, thereby offering strategies for continuous improvement of the programme.

2.6. Description of how the results obtained from the PEO evaluation will be used to improve the effectiveness of the programme.

The results obtained from the evaluation of the Programme Educational Objectives (PEOs) will be carefully analyzed to identify strengths, weaknesses, and opportunities for improvement within the Electrical and Electronics Engineering programme. The feedback from graduates, employers, and other stakeholders will serve as a guide for refining the curriculum to align with technological advancements and industry needs. Emphasis will be placed on incorporating emerging areas such as Artificial Intelligence, Renewable Energy Systems, Smart Grids, and the Internet of Things, while also improving teaching strategies through regular staff training and exposure to modern pedagogical methods.

Furthermore, the evaluation outcomes will be used to strengthen industry collaboration, internship structures, and research focus to ensure graduates are well-prepared for professional practice. Areas where performance gaps are observed, such as soft skills, ethics, and innovation, will be addressed through mentoring, career development, and experiential learning initiatives. The entire process

will form part of a continuous quality improvement framework aimed at enhancing the relevance, effectiveness, and overall excellence of the Electrical and Electronics Engineering programme.

2.7. Types of Evaluation Surveys

The achievement of the Programme Educational Objectives (PEOs) will be evaluated every five years. This long-term review will be supported by annual data collection through surveys and feedback mechanisms involving students, alumni, employers, and other stakeholders. The process will include the following approaches:

- **Graduate Surveys:** Alumni will be engaged through surveys or interviews to share their experiences and provide insights on how well the PEOs have been reflected in their professional development.

- **Employer Feedback:** Employers who have hired graduates from the programme will be consulted to assess the quality of graduate performance and how effectively the PEOs align with workplace demands.

- **Academic Performance Review:** Student achievements will be analyzed using academic records, project outcomes, and research contributions to evaluate their level of mastery in key areas.

- **Programme and Course Assessments:** Both courses and the overall programme will be periodically reviewed to determine their relevance, quality, and alignment with the established educational objectives.

- **Alumni Tracking:** Records of alumni career progress and professional success will be monitored to evaluate the long-term influence of the programme on graduates and their contributions to industry and society.

- **Advisory Board Input:** Advisory boards, together with a follow-up team that includes faculty members, will use available platforms such as social media to provide guidance, recommendations, and independent evaluations.

- **Student Feedback:** Current students will provide feedback through surveys, interviews, interactive sessions, and continuous course evaluations to capture their perspectives on programme delivery and outcomes.

2.8. Frequency of PEO Review

PEOs are to be reviewed regularly every **FIVE** years based on the reports from various surveys carried out.

2.9. Evaluation of Level of Achievement of PEOs

- **Data Collection:** Information relevant to the achievement of PEOs will be gathered from multiple sources, including the surveys and feedback mechanisms outlined in Section 2.6.
- **Assessment of Alignment**: The collected data will be analyzed to determine how well it reflects the intended outcomes of each PEO. Graduate performance will be compared against predefined indicators to establish the extent of achievement.
- **Strengths and Areas for Improvement:** A SWOT (Strengths, Weaknesses, Opportunities, and Threats) analysis will be employed to highlight areas of strong attainment as well as aspects needing attention. This periodic analysis helps identify patterns of success and opportunities for targeted improvement.
- **Continuous Improvement:** Findings from the evaluation will inform curriculum updates, teaching methods, and programme modifications. Stakeholders such as faculty, students, alumni, and employers will be engaged in this process to ensure the programme remains relevant and effective.
- **Documentation and Communication:** The evaluation procedures and outcomes will be formally documented to ensure accountability. Results will be shared with key stakeholders—including faculty, students, accrediting bodies, parent forum, and industry partners—to reinforce the programme's commitment to quality assurance and ongoing improvement.
- **Advisory Board Meetings:** Quarterly meetings with the departmental advisory board, conducted either virtually through platforms such as Zoom or Google Meet, will be held to review PEO attainment levels and provide strategic input.
- **Student Engagement:** Meetings between academic level advisers and the Head of Programme will be encouraged to gather direct feedback from students on the quality and delivery of lectures.

2.10. Measurable Key Performance Indicators (KPIs)

Table 2.3: Key Performance Indicators (KPI) for the PEOs

S/N	Performance Educational Objectives (PEO)	Key Performance Indicators (KPI)
PEO1	***To produce highly skilled and competent Electrical/Electronics engineers that will contribute to academia, industrial and professional Engineering practice.***	1. Percentage of graduates who enroll in postgraduate programs within 3 years of graduation. 2. Number of student-authored publications, conference papers, or patents per academic

		year. 3. Percentage of final-year students undertaking research-oriented projects aligned with emerging technologies.
PEO2	***To produce proficient, self-reliant, and excellent Electrical/Electronics Engineers equipped with strong technical skills, a solid understanding of fundamental concepts, and the ability to address contemporary challenges in line with modern technological developments.***	1. Graduate employment rate in engineering-related fields within 12 months of graduation. 2. Percentage of alumni in leadership or supervisory roles within 5 years of graduation. 3. Number of alumni-owned startups or entrepreneurial ventures established within 5 years.
PEO3	***To produce graduates who will actively engage in continuous professional development and lifelong learning, pursue advanced studies, and obtain relevant professional certifications, thereby remaining competent and relevant in a rapidly evolving engineering environment.***	**1. Percentage of student projects addressing real-world or industry-related problems.** **2. Employer satisfaction ratings on graduates' problem-solving and technical application skills.** **3. Number of industry-linked projects, consultancies, or internships completed annually.**
PEO4	***To produce graduates who can provide practical and cost-effective solutions to real-world problems by applying their modeling knowledge, innovative thinking, and entrepreneurial skills, as well as their ability to design, develop, and solve complex engineering challenges across various fields of Electrical/Electronics Engineering.***	1. Percentage of graduates who participate in professional societies, workshops, or certifications within 3 years of graduation. 2. Employer ratings of graduates' teamwork, communication, and ethical standards. 3. Number of interdisciplinary or collaborative projects completed successfully by students each year.
PEO5	***To produce graduates that will uphold ethical standards, acquire***	1. Percentage of graduates employed in academia, research institutions, or higher

	knowledge, demonstrate integrity, leadership and practice Electrical/Electronics engineering responsibly in compliance with national and international professional codes and sustainable development principles.	education. 2. Number of graduates licensed/registered with COREN or other professional bodies within 5 years. 3. Percentage of graduates contributing to industry through professional engineering roles within 3 years of graduation.

2.11. Overall Survey Rating (OSR)

OSR provides a concise and standardized representation of stakeholders' perceptions and experiences, offering valuable insights for evaluation, improvement, and decision-making purposes.

2.12. Review of the academic curriculum – Continuous Quality Improvement (CQI) Strategies

The CQI strategies to be implemented in relationship with PEOs are as follows.

1. **Regular Assessment and Data Collection**: The programme will put in place a methodical procedure to gather information regarding PEO achievement. This could include the previously mentioned surveys, interviews, employer assessments, comments from alumni, academic performance indicators, and other pertinent information sources. The degree of success attained by each PEO will be determined by analyzing and assessing this data. Based on the evaluation results, trends, strengths, weaknesses, and opportunities for improvement will be closely followed.

2. **Stakeholder Engagement**: The assessment process will engage stakeholders including employers, industrial partners, educators, students, alumni, and community representatives. Get their opinion on the PEOs' applicability and efficacy as well as suggestions for development.

3. **Curriculum Review and Revision**: The assessment findings lead to an inform curriculum review and revision efforts. This will lead to identification of opportunities to enhance the curriculum, instructional methods, learning resources, and assessment

tools to better align with the PEOs and meet the evolving needs of stakeholders.

4. **Faculty/Faculty Development**: Provide professional development opportunities for Faculty/faculty to enhance their teaching effectiveness, incorporate best practices, and align instructional strategies with the PEOs. Encourage collaboration and sharing of expertise among Faculty/faculty members. Organize more conferences, seminars, and workshops. Staff are also encouraged and to attend yearly COREN assembly, Nigeria Society of Engineers yearly workshop both locally and nationally to learn about current progress in Engineering profession.

5. **Student Support and Advising**: Implement support mechanisms to help students achieve the PEOs, such as academic level advising, mentoring programs, career counseling, and opportunities for experiential learning. Foster a supportive and inclusive learning environment that encourages student engagement and success.

6. **Continuous Monitoring through Quality Assurance (QA):** The University has put in place a quality assurance team in collaboration with the programme to establish a feedback loop to monitor the effectiveness of quality improvement efforts and adjust as needed. Regularly review assessment data, solicit feedback from stakeholders, and track progress towards achieving the PEOs. It is a committee committed to fostering good ethics and best practices.

7. **Documentation and Reporting**: The results of quality improvement efforts, including assessment findings, action plans, and outcomes, will be documented and communicated transparently to stakeholders about progress, challenges, and accomplishments related to the PEOs.

8. **Continuous Improvement**: These are done through thorough analysis in regular board meetings. This will foster a culture of continuous improvement within the programme emphasizing the importance of ongoing assessment, reflection, and adaptation, to ensure that quality and relevancy of the programme is not tampered with.

SECTION THREE:

THE PROGRAMME OUTCOMES (POs)

3.1. Programme Outcomes (POs)

The Programme Outcomes (POs) for the Electrical/Electronics Engineering Programme at Afe Babalola University are presented in Table 3.1. They are carefully structured statements that represent the proficiencies we expect of our students by the time they graduate.

Consistency of PO's with COREN Twelve (12) POs

The Programme Outcomes of the Electrical/Electronics Engineering program (twelve in number) is consistent with COREN's Programme Outcomes in the area of engineering knowledge, problem analysis, design/development of solutions, investigation and modern tool usage. Other areas of consistency include the engineer and society, environment and sustainability, ethics, individual and team work, communication, project management and lifelong learning.

Table 3.1: Programme Outcomes

PO1	**Engineering Knowledge**	Apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of developmental and complex engineering problems.
PO2	**Problem Analysis**	Identify, formulate, research literature, and analyze developmental and complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences
PO3	**Design/Development of Solutions**	Proffer solutions for developmental or complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations
PO4	**Investigation**	Conduct investigation into developmental or complex problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions
PO5	**Modern Tool Usage**	Create, select, and apply appropriate techniques, resources and modern engineering and ICT tools, including prediction, modeling, and optimization to developmental and complex engineering activities, with an understanding of the limitations

PO6	**The Engineer and Society:**	Apply reasoning informed by contextual knowledge including Humanities and Social Sciences to assess societal, health, safety, legal and cultural issues, and the consequent responsibilities relevant to professional engineering practice
PO7	**Environment and Sustainability**	Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development
PO8	**Ethics**	Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice, including adherence to the COREN Engineers Code of Conducts
PO9	**Individual and Teamwork**	Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings
PO10	**Communication**	Communicate effectively on developmental or complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions
PO11	**Project Management**	Demonstrate knowledge and understanding of engineering, management and financial principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multi-disciplinary environments
PO12	**Lifelong Learning**	Recognize the need for and have the preparations and ability to engage in independent and lifelong learning in the broadest context of technological and social changes.

3.2. Publication of the Programme Objectives (POs)

The Objectives and Outcomes of the Electrical/Electronics Engineering Programme are published on the university website www.abuad.edu.ng .

Student Handbook: The POs are also published in the handbook which is made available in the university website.

Notice Board: The POs are pasted on the notice board for the students to see often so as to be always aware.

3.3. Relationship of Programme Outcomes to Programme Educational Objectives:

1. The programme outcomes equip students with the basic principles of Electrical and Electronic Engineering, serving as a strong foundation for building a career at the highest level of professional expertise.
2. It enhances the ability of students to apply technical knowledge while fostering analytical thinking, innovative design skills, creativity, adaptability, and leadership capacity.
3. It offers students the chance to expand their knowledge after gaining a general understanding of all aspects of Electrical and Electronic Engineering practice, with opportunities to specialize in fields such as Telecommunication Systems, Electronics, Embedded Systems, Control Systems, Power Systems, Renewable Energy, Power Electronics, Energy Management and Technology, Electrical Machines, and Electromagnetic Fields and Waves.
4. It delivers hands-on training within industries and Electrical and Electronic Engineering establishments, preparing students for effective professional practice. All the twelve programme outcomes are coded **PO1 – PO12** and used to map the programme educational objectives coded **PEO1 – PEO5**. This is presented in Table 3.3. The programme outcomes were developed to link that of the regulatory body for engineering education in Nigeria (COREN) Furthermore, these programme outcomes have been adequately directed to achieve the desired impact of the curriculum in fulfilling the programme educational objectives.

Table 3.3: Link between the Program Outcomes (PO) and the Program Educational Objectives (PEOs)

Programme Educational Outcomes	PEO1	PEO2	PEO3	PEO4	PEO5
Engineering Knowledge	x	x	x	x	x
Problem Analysis	x	x	x		x
Design/Development of Solutions	x	x	x	x	x
Investigation	x		x		
Modern Tool Usage	x	x	x	x	
The Engineer and Society:	x	x		x	x
Environment and Sustainability				x	
Ethics	x		x		x
Individual and Teamwork	x		x	x	
Communication	x			x	
Project Management			x		x
Lifelong Learning		x	x	x	x

3.4. Description of the PO definition or elements/performance indicators

i. Direct approaches for assessing Program Outcomes (POs) focus on clearly evaluating the knowledge and abilities of students in relation to defined performance indicators. These methods provide tangible proof of how well students have achieved the expected learning goals and how prepared they are to meet professional requirements. Such direct assessments include examinations, quizzes, assignments, laboratory work, project reports, performance appraisals, and final year or capstone projects.

ii. The assessment process is designed to mirror as closely as possible the real tasks that students are expected to handle as engineers or professionals, including research projects, laboratory activities, oral presentations, and related academic exercises.

3.5. Description of the processes used to establish and review the POs, and the extent to which the program's various stakeholders are involved in these processes (where applicable). This includes describing the tools used in the processes (survey, meetings, interviews, etc.) and frequency of activities and timelines

The feedback and evaluation tools at the end of a four-year cycle are hereafter described:

i. Employer Feedback Survey: This survey collects insights from employers regarding the performance, competencies, and workforce readiness of graduates. It highlights how effectively the program aligns with industry expectations and professional requirements.

ii. Alumni Feedback Survey: This survey gathers responses from graduates about their learning experiences, career development, and the practical relevance of their academic training. Alumni input helps measure how well the program meets its long-term educational objectives.

iii. Employment Statistics: This record tracks the employment status and career progress of graduates, including job placements, advanced studies, and professional achievements. It provides quantitative evidence of the program's success in supporting graduates' career growth and academic advancement.

iv. Industry Advisory Board: This board, consisting of professionals and industry stakeholders, offers advice, recommendations, and feedback to ensure the program remains current and relevant. Their expertise in industry trends, new technologies, and skill requirements contributes to ongoing curriculum updates and enhancements.

- **Effectiveness Criteria**

  Survey Satisfaction: A minimum of 50 percent of survey responses must score 3 or above on a 5-point scale, showing that participants are satisfied with program outcomes and graduate performance.

  Graduate Success: At least 50 percent of graduates should either be employed or pursuing advanced studies, demonstrating the program's effectiveness in preparing students for professional careers or further education.

3.6. Description of the mapping of courses with POs

The connection between Course Learning Outcomes (CLOs) and Program Outcomes (POs) is structured to effectively support the overall objectives of the program. To ensure adequate coverage of competencies, at least half of the mapped courses must demonstrate attainment of

the related POs. In addition, a minimum of 60 percent of student feedback should score 3 or above on a five-point scale, showing satisfactory achievement of the expected learning outcomes.

This linkage of CLOs to POs is incorporated into the course compact for every course. Through this process, instructors clearly specify how the learning objectives of their courses contribute to the wider goals of the program. Embedding this mapping into course documentation promotes transparency and ensures proper alignment between course content, intended learning outcomes, and program outcomes.

3.7 Application of assessment results to further develop and improve the POs.

The results from assessments are used for the continuous improvement of the following:

- Curriculum
- Lecture delivery methods
- Student participation levels in class

3.8. Description of the materials that demonstrate achievement of the POs

SWEP and SIWES logbooks, laboratory reports, and project reports, along with assessments such as assignments, tests, and examination booklets, serve as tangible evidence demonstrating the fulfilment or achievement of the Program Outcomes (POs).

3.9. Description of the extent to which the programme's various stakeholders are involved in the processes.

The roles of the stakeholder in the program processes include:

i. Alumni and Graduate Feedback: Graduates provide meaningful perspectives that help refine both the curriculum and institutional practices. Their input highlights strengths and areas for improvement, ensuring the program stays relevant and continues to prepare students effectively for professional practice.

ii. Institutional Participation: The institution plays a central role in shaping and updating the curriculum. Adjustments are guided by industry developments, evolving academic standards, and recommendations from key stakeholders, thereby enriching the overall learning experience.

iii. Industry Advisory Boards: These boards ensure that the Electrical/Electronics Engineering curriculum reflects current industry expectations. Their contributions support alignment between academic training and workplace requirements, equipping students with essential knowledge and skills.

iv. Employer Assessments: Feedback from employers offers an external measure of how well academic preparation translates into workplace performance. These evaluations reveal strengths and gaps in graduate competencies, guiding curriculum adjustments that enhance employability.

v. Faculty Development: Continuous professional growth among faculty members strengthens teaching quality and directly benefits students. By advancing their knowledge and skills, faculty ensure that learning outcomes remain consistent with program goals and industry standards.

3.10. Describe how the results obtained from the assessment will be applied to further develop and improve the POs

The results obtained from the assessment of the Programme Outcomes (POs) will be systematically analyzed to determine the extent to which students are achieving the expected knowledge, skills, and attitudes outlined in the Electrical/Electronics Engineering curriculum. Areas of strong performance will be consolidated and used as benchmarks, while identified gaps will guide targeted curriculum revisions, teaching enhancements, and laboratory upgrades. Feedback from students, faculty, alumni, and industry partners will also be integrated to ensure that the POs remain relevant to current engineering practices, emerging technologies, and accreditation standards.

Furthermore, the assessment outcomes will inform continuous improvement strategies such as faculty development programmes, pedagogical innovations, and interdisciplinary project integration to strengthen the linkage between theoretical knowledge and practical applications. The results will also support the realignment of course objectives and assessment methods to better reflect the POs, ensuring that graduates are well-prepared for professional practice, research, and innovation. Overall, the findings will drive a culture of evidence-based improvement, sustaining the quality and global competitiveness of the Electrical and Electronics Engineering programme.

3.11. Description of Continuous Quality Improvement (CQI) strategies to be implemented in relation to POs

Continuous Quality Improvement (CQI) strategies include:

i. External Examiner Evaluation: Outcome-Based Education (OBE) appraisal forms are employed by external examiners to evaluate examination questions and prepare detailed reports. This practice upholds assessment standards and promotes continuous program enhancement.

ii. Lecture Monitoring: Lectures are regularly observed as part of a structured quality improvement strategy. Although specific processes may vary, the monitoring typically involves systematic review of teaching methods to highlight strengths and identify areas needing refinement.

iii. Curriculum Review: The program curriculum is reviewed every five years. Two reviews have been completed to date, and the current third review (CCMAS) allows the department to contribute thirty percent of its content. This cycle ensures that the curriculum stays updated, relevant, and aligned with contemporary academic and industry requirements.

iv. Facilities Upgrades: Continuous investments are made in upgrading infrastructure, including the acquisition of smart boards and advanced laboratory equipment. These improvements create a modern learning environment that enhances both teaching and student learning experiences.

v. Faculty Development: Faculty members engage in ongoing professional growth through workshops, seminars, and training sessions. These activities keep them informed about new teaching methods, innovations in pedagogy, and technological advances, ultimately improving instructional quality and student outcomes.

vi. End of semester surveys: Students are asked to submit a survey and ratings of key indices for the performances of their lecturers.

3.12. Policies, Internal Processes and Practices that are in Place at all Levels Within the Institution Relating to the Accreditation Criteria

The following are some of the policies and processes put in place by the programme to ensure compliance with COREN requirements:

a) **Quality Assurance Committee:** The Electrical/Electronics programme utilizes a dedicated committee composed of professors and senior academic staff to oversee instructional quality. This committee evaluates the performance of lecturers and technologists every semester and recommends improvements to university management.

b) **Exam Vetting Committee:** Standardized assessment is maintained by a vetting committee that reviews all examination questions for 100 to 400-level courses through senior co-examiners. Final approval is conducted by the Head of Department as Chief Examiner.

c) **External Examiner Implementation:** The department appoints an approved External Examiner, currently Prof. Jide J. Popoola, to moderate final-year examination questions, marking guides, and projects. Feedback from moderation reports is analyzed during board meetings to implement remedial actions.

d) **COREN Recommendations:** Following the previous accreditation exercise, the department successfully implemented key recommendations by restructuring laboratory sessions so that each is coordinated by a lecturer alongside technologists and by increasing the number of registered technical staff.

Criterion 1: Programme Educational Objectives (PEOs)

The PEOs (PEO1–PEO5) define the long-term goals for graduates, such as contributing to professional engineering practice, solving contemporary technological challenges, and upholding high ethical standards. These objectives are published in handbooks, on notice boards, and on the

university website. They are reviewed every five years using feedback from alumni, employers, and industry partners.

Criterion 2: Programme Outcomes (POs)

The programme has adopted 12 POs that are fully aligned with the COREN and Washington Accord attributes. These outcomes ensure graduates possess the necessary engineering knowledge, problem-analysis skills, and modern tool usage capabilities required for professional practice.

Criterion 3: Course Learning Outcomes (CLOs)

Every course is defined by specific, measurable CLOs that are mapped to the programme outcomes within individual Course Compacts. This mapping ensures that the mastery of course-specific content directly contributes to the achievement of the broader graduation requirements.

Criterion 4: Curriculum and Learning Process

The curriculum is built on an Outcome-Based Education (OBE) framework and uses a Modularised Course Unit System. It integrates 30% local content specialization through "signature courses" designed to address regional engineering needs. The learning process combines traditional lectures with Problem-Based Learning (PBL) and mandatory practical exposure.

Criterion 5: Students

Admission follows COREN regulation, requiring five O'Level credits (including Mathematics, Physics, and Chemistry) for UTME or relevant diplomas for Direct Entry. Students are guided by Academic Level Advisers and must maintain a minimum of 80% attendance to be eligible for final examinations.

Criterion 6: Continuous Quality Improvement (CQI)

CQI strategies involve systematic curriculum reviews every five years and annual data collection from various stakeholders. The department uses employer satisfaction ratings and graduate performance data to refine laboratory facilities and teaching methods.

Criterion 7: Staffing

The department maintains a strong academic faculty, with 11 out of 15 (73.3%) dedicated staff holding doctoral degrees. Over 90% of the academic staff are registered professional engineers with COREN. The university supports staff development through Ph.D. sponsorship and sponsorship for the annual COREN assembly.

Criterion 8: Physical Facilities and Infrastructures

Learning is supported by nine specialized laboratories, including the Electrical Machine, Power and Energy Lab, and the National Instruments Lab, all equipped with advanced workstations. Facilities include smart boards, high-speed WiFi, and a departmental library with a capacity for 80 students.

Criterion 9: Institutional Linkage and Community Service

Strong industrial linkages are managed through the Industry Advisory Board, which provides feedback on curriculum relevance. Students gain mandatory real-world experience through 24 weeks of SIWES industrial exposure at recognized firms such as the Transmission Company of Nigeria or TransAfam Power.

Criterion 10: Institutional Support and Funding

The university provides consistent financial commitment for faculty salaries, research grants, and regular infrastructure maintenance. Resources are allocated annually for laboratory re-tooling and the acquisition of modern engineering software (e.g., MATLAB, Altium) to maintain accreditation readiness.

3.13. Involvement of industry in discussions and fora, professional practice exposure, and collaborative projects/research for the solutions to engineering problems.

The Electrical/Electronics Engineering programme maintains strong industry ties to ensure that its students are well-prepared for professional practice and equipped with current technological skills.

3.13.1. Involvement in Discussions and Fora

The department actively engages with industry stakeholders to keep the curriculum and teaching methods aligned with modern engineering requirements:

a) Industry Advisory Board: This board, comprised of seasoned professionals and industry experts, provides guidance on labor market needs, emerging technological trends, and curriculum enhancements to ensure graduates are workplace-ready.

b) Career Weeks and Seminars: Industry professionals are invited annually to share real-world knowledge and industrial experiences during Career Week. Additionally, guest seminars and invited lectures from various engineering professional associations are integrated into the Engineer in Society (GET 101) course.

3.13.2. Professional Practice Exposure and Certifications

Exposure to professional practice is facilitated through specialized certifications and extensive industrial placements:

a) **Professional Certifications (New Horizons):** Through a partnership with New Horizons, the department integrates professional IT and engineering certifications directly into the curriculum:

   i) CCNA (ABUAD-EEE-102) at the 100 level.

   ii) Python Programming (ABUAD-EEE-202) at the 200 level.

   iii) Machine Learning (ABUAD-EEE-302) at the 300 level.

   iv) Embedded System/IoT (ABUAD-EEE-502) at the 500 level.

b) **Industrial Placements (United Nigeria Airlines):** Students gain deep industrial exposure through the Students Industrial Work Experience Scheme (SIWES). A notable example of this partnership is United Nigeria Airlines, which has provided industrial training and exposure for as many as 25 students from the programme.

3.13.3. Collaborative Projects, Research, and Awards

Industry collaboration is central to the department's research and student recognition efforts:

a) **Industry-Sponsored Projects:** Final-year students are encouraged to undertake projects that tackle complex engineering challenges sourced directly from industrial practice. This is managed by the departmental Research and Workshop Committee to foster applied research and innovation.

b) **Excellence Awards (Dorman Long Engineering):** To motivate students and promote professional standards, Dorman Long Engineering Nigeria Limited provides several awards for different categories of graduating students, recognizing academic excellence and innovative engineering potential.

CHAPTER FOUR: COURSE LEARNING OUTCOMES (CLOs)

4.1. Course Learning Outcomes

This section contains the Course Learning Outcomes (CLOs) for courses in the programme, and also include the course outline, lecture plans and the mapping of each course the programme outcomes. The CLO information can all be viewed at the university website at www.abuad.edu.ng



AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code:	COE 302	**Course Title:**	Object Oriented Programming
Semester:	First	**Credit Hours**:	**3**
Contact Hours:	Lecture - Two hours lectures per Week	**Location**:	B-22
Lecturer(s):	Engr. Dr. J. F. Orimolade		

Textbooks and other Materials:

i. Cay Horstmann (2013). Java for Everyone, 2nd Edition. John Wiley & Sons, Inc.
ii. Murali Chemuturi (2019). Computer Programming for Beginners, A step-by Step Guide. Taylor & Francis Group, LLC

Course Overview and description

This course covers the basics of programming language such as variables, data types, expressions assignment statements, conditional and iterative control structures with emphases to Java programming language. Other key aspects such as Control structures, arrays, methods; pointers, and memory allocation, functions and parameter passing, are also covered. The course has been tailored towards Object Oriented (OOP) paradigm in Java. Object Oriented Programming concepts such objects, classes and subclasses, Inheritance, encapsulation, data abstraction and Polymorphism are well covered in the course. The course addresses the structure of graphical user interface (GUI), with specific application to java programming language.

b) **Pre-requisites:** Nil

c) **Co-requisite(s):** Nil

d) **Role in Curriculum:** Core

e) **Course Learning Outcome (CLO) and Programme Outcomes (PO)**

Course Learning Outcomes (CLO): upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**POs**	**Assessment Tools**
Understand the fundamental principles of programming language and the general software development techniques	2	1, 2	Assignment, quizzes and Exams
Understand the top down approach to software development techniques used in Java programming language, and analysis of java software programs and modification for new applications	2	1, 2, 4,	Assignment, quizzes and Exams
Design and transform simple real-life problem into software solutions using the general programming language techniques with emphasis on java syntax and methods	4	2, 3, 6	Assignment, quizzes and Exams
Apply Object Oriented programming to analyze, design and implement java-based OOP to develop applications to real life problems.	4,5	3, 8	Assignment, quizzes and Exams
Understand the Graphical User Interface (GUI) development packages such as JavaFx and apply same to programming projects in Java.	1,2	2, ,8	Assignment, quizzes and Exams

f) **Mapping of CLOs to relevant POs**

Course Learning Outcomes	Programme Outcomes											
	1	2	3	4	5	6	7	8	9	10	11	12
Understand the fundamental principles of programming language and the general software development techniques	3		2									
Understand the top down approach to software development techniques used in Java programming language, and analysis of java software programs and modification for new applications	2	3		2								
Design and transform simple real life problem into software solutions using the general programming language techniques with emphasis on java syntax and methods		3	3			2						
Apply Object Oriented programming to analyze, design and implement java-based OOP to develop applications to real life problems.			3					2				
Understand the Graphical User Interface (GUI) development packages such as JavaFx and apply same to programming projects in Java.		3						2				3

Keys: 1 = slightly related, 2 = moderately related, 3 = highly related

g) Expectations of Students

a. Attend 2 hours of lectures per week

b. Turn in assignments considered suitable by the tutors

c. Complete continuous assessment which involves mid-semester test

d. Complete a $2^{1}/_{2}$-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of Power Point slides, Lecture templates-handouts)

i) Course Content or Outline

Topics	Weeks
Introduction to Procedural and OOP in Java: Definition of alphabet, syntax and semantics as used in selected higher-level language. Explanation of the terms; Variables, data types, expressions and assignment statements, Simple I/O, conditional and iterative control structures.	1-3
Multidimensional arrays pointers: Multidimensional arrays; pointers, pointer to pointer, dynamic data structures and memory allocation. Functions and parameter passing, structured decomposition. Development of programmes for solving basic economic, engineering and scientific problems.	4-6
Implementation of object-oriented concepts: (week 7-10) Implementation of object-oriented concepts; encapsulation and information hiding, Separation of behaviour and implementation, classes and subclasses, Inheritance (overriding, dynamic dispatch), Polymorphism (subtype polymorphism vs. inheritance), Class hierarchies. Control structures, Arrays, Methods, Applets.	7-10
Graphics and event handling: (week 11-12) Software design and construction in context of large OOP libraries taught in Java. Review of OOP, the structure of graphical user interface (GUI), OOP libraries, GUI application design and construction, OOP software development strategies, approaches to programming in teams.	11-12
Lecture free week	13
Revision & Examination	14 - 15

j) Method of Grading Continuous Assessment

Continuous Assessment (Assignments): 20%

Continuous Assessment (Test):20%

End of Semester Examination: 60%

k) Tutorials

Tutorials are taken at the end of each module.

l) Term Paper and Assignments

Practical and real life assignments will be given to students on various topics. This will help students understanding of the course contents.

m) Contemporary Issues and Industrial Relevance

The course equips students with the general knowledge of Object Oriented Programming (OOP) with specific emphasis on java programming syntax. The advantages and coding efficiency using OOP as compare with the procedural approach has been exposed to the students. Different real life problems are analyzed and translated into software solutions using OOP in java programming language.



AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code: EEE 221 **Course Title:** Basic Electrical Engineering I

Semester: First **Credit Hours**: 3 hours

Contact Hours: Lecture - Two hours lectures per week

Location: Engineering Auditiorium **Lecturer(s)**: Engr. Prof. M.O Onibonoje, Mr. Victor Akinpelu

Textbooks and other Materials:

- Elements of Electromagnetics by N. O. Sadiku
- Network Theory by C. K. Alexander and M. N. O. Sadiku Jean-Pierre
- Measurement and Instrumentation Principle by Alan S. Morris

a) Course Overview and description

Basic Electrical Engineering I introduces students to the fundamental principles and applications of electrical engineering, serving as a foundation for advanced courses in circuits, electromagnetics, machines, and power systems. The course begins with an exploration of electric and magnetic fields, covering Coulomb's law, Gauss's law, Biot–Savart law, and Ampere's law,

before progressing into the concepts of magnetic circuits, magnetomotive force, permeability, and B–H curves. Students also study lumped circuit elements, resistors, capacitors, and inductors, alongside their voltage–current characteristics and roles in energy storage and dissipation. Core circuit analysis techniques are introduced through network theorems such as Kirchhoff's laws, Thevenin's and Norton's theorems, Superposition, and Delta–Star transformations. The course further develops analytical skills through the study of AC circuits, including single-phase and three-phase systems, impedance, phasors, resonance, power factor, and load balancing under both balanced and unbalanced conditions. Emphasis is placed on applying mathematical tools to real engineering problems, interpreting results, and evaluating circuit performance. By combining theoretical concepts with problem-solving practice, the course equips students with the skills needed to analyze, model, and assess electrical systems while building competencies in engineering reasoning, teamwork, communication, and professional growth as outlined in the programme outcomes.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Core

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcome: Upon completion of the course, the student should be able to:	**Suitable Bloom Taxonomy Relevant to the CLO's**	**POs**	**Assessment Tools**
Evaluate the interactions between electric fields of charge and magnetic fields of current, assessing their implications in basic engineering applications.	1,2,5	1,2,4,9	Assignment, quizzes and Exams
Explain the characteristics of lumped circuit elements (resistance, capacitance, inductance) and their terminal behavior.	2,3,4	1,2	Assignment, quizzes and Exams
Apply network theorems (Kirchhoff's laws, Superposition, Thevenin's, Norton's, and Star/Delta transformations) to solve practical electrical circuit problems.	3,4,6	1,2,3,4	Assignment, quizzes and Exams
Analyze single-phase and three-phase AC circuits, including active loads, apparent power, and both balanced and unbalanced conditions.	1,3,4	1,2,3,4,5,10	Assignment, quizzes and Exams
Apply and interpret features of magnetic circuits, including magnetomotive force, permeability, and B–H curves of materials, in solving practical problems.	2,3,6	1,2,4,5,8,9,10	Assignment, quizzes and Exams

f) Mapping of CLOs to relevant POs

Course Learning Outcomes	Programme Outcomes (POs):											
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
Evaluate the interactions between electric fields of charge and magnetic fields of current, assessing their implications in basic engineering applications.	**3**	**2**		**2**					**2**			
Explain the characteristics of lumped circuit elements (resistance, capacitance, inductance) and their terminal behavior.	**3**	**2**										
Apply network theorems (Kirchhoff's laws, Superposition, Thevenin's, Norton's, and Star/Delta transformations) to solve practical electrical circuit problems.	**3**	**3**	**2**	**2**								
Analyze single-phase and three-phase AC circuits, including active loads, apparent power, and both balanced and unbalanced conditions.	**3**	**3**	**3**	**2**	**2**					**2**		
Apply and interpret features of magnetic circuits, including magnetomotive force, permeability, and B–H curves of materials, in solving practical problems.	**3**	**2**		**1**	**1**			**1**	**1**	**1**		

Keys: 1 = Slightly related, 2 = Moderately related, 3 = Highly related

g) Expectations of Students

a. Attend 3 hours of lectures per week

b. Turn in assignments and projects as considered suitable by the tutors

c. Complete continuous assessment which involves mid-semester test

d. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of PowerPoint slides, Lecture templates-textbooks)

12) **Course Content or Outline**

Week	Topics / Subtopics
1	**Introduction to Electrical Engineering Concepts** • Course overview and expectations • Physical quantities: charge, current, voltage, power • Relationship between electricity and magnetism
2	**Electric Field of Charge (Part 1)** • Coulomb's law • Electric field intensity • Electric potential and potential difference • Work and energy in an electric field
3	**Electric Field of Charge (Part 2)** • Gauss's law (basic introduction) • Capacitance of simple configurations (parallel plate, spherical, cylindrical) • Dielectrics and permittivity
4	**Magnetic Field of Current (Part 1)** • Biot–Savart law • Ampere's circuital law • Magnetic flux and flux density • Magnetic force on current-carrying conductors
5	**Magnetic Field of Current (Part 2)** • Magnetomotive force (MMF) • Magnetic field strength (H)

	• Permeability (μ_0, μr) • B–H curves of materials
6	**Magnetic Circuits** • Reluctance and analogy with Ohm's law • Series and parallel magnetic circuits • Leakage and fringing effects • Practical applications: transformer cores, electrical machines
7	**Lumped Circuit Elements** • Resistance: Ohm's law, temperature dependence, conductance • Capacitance: charging/discharging, energy storage • Inductance: self and mutual inductance, energy in inductors • I–V relationships for R, L, C
8	**Network Theorems (Part 1)** • Kirchhoff's current law (KCL) and voltage law (KVL) • Linearity • Superposition theorem • Source transformation
9	**Network Theorems (Part 2)** • Thevenin's theorem • Norton's theorem • Reciprocity theorem • Star/Delta (Y–Δ) transformations
10	**Introduction to AC Circuits (Single-Phase)** • Sinusoids and phasors • RMS and average values • Impedance and admittance of R, L, C • Power in AC circuits: real, reactive, apparent
11	**Single-Phase AC Circuit Analysis** • Series and parallel RLC circuits • Resonance (series & parallel) • Power factor and correction methods • Practical load analysis

12	**Three-Phase AC Circuits** • Balanced 3-phase systems (Y and Δ connections) • Line and phase relationships • Power measurement in 3-phase systems • Advantages of 3-phase supply
13	**Unbalanced Three-Phase Loads & Course Review** • Analysis of unbalanced star and delta loads • Neutral shift in unbalanced systems • Case studies: power distribution applications • Comprehensive course review and exam preparation
14	**Revision and Lecture Free Week**
15	**Examination**

Method of Grading Continuous Assessment

Continuous Assessment

- Assignment: 10%
- Quizzes/attendance: 10%
- Test: 20%

End of Semester Examination: 60%

k) Tutorials

Tutorial will be offered at the end of each module.

l) Group works and Assignments

The group works and assignments are designed to challenge students to think critically and solve problems in the world of electromagnetism, analyze real-world scenarios and propose solutions using the acquired knowledge of electromagnetic principles. The course also includes assignments designed to solidify your understanding through calculations, problem-solving exercises, and potentially, simulations or experiments involving electromagnetic phenomena.

m) Contemporary Issues and Industrial Relevance

This course is highly relevant to today's rapidly evolving power and electronics industry. A solid grasp of electric and magnetic fields, network theorems, and AC power systems underpins modern technologies such as renewable energy integration, smart grids, and efficient industrial power distribution. The course equips students with analytical and problem-solving skills essential for addressing challenges like energy efficiency, sustainable power generation, and reliable communication systems. Its concepts directly apply in industries ranging from power

engineering and electronics manufacturing to automation, ICT, and emerging renewable technologies.

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AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code: EEE 222 **Course Title:** Basic Electrical Engineering II

Semester: Second **Credit Hours**: 3 hours

Contact Hours: Lecture - Two hours lectures per week

Location: Engineering Auditiorium **Lecturer(s)**: Engr. Prof. M. O. Onibonoje, Mr. Victor Akinpelu

Textbooks and other Materials:

- Electricity and Magnetism by T. Matsushita
- Basic Network Theory by J. Vlach
- Integrated Electronics by J. Millman and C. Holkias

a) Course Overview and description

Basic Electrical Engineering II builds upon the foundational concepts introduced in Basic Electrical Engineering I, extending students' understanding from classical circuit theory into the domain of semiconductor physics and electronic devices. The course begins with an introduction to the atomic structure and the fundamental principles governing semiconductors, including the behavior of charge carriers, effective mass, mobility, conductivity, carrier lifetime, and recombination phenomena. Students are then introduced to transport mechanisms such as drift, diffusion, continuity equations, and the Hall effect, which provide the basis for analyzing semiconductor behavior in practical applications. The course progresses to the study of PN junction materials, diodes, and their practical uses in rectification, filtering, and voltage regulation circuits, with emphasis on Zener diode applications. Building on this, transistor theory is explored in depth, covering bipolar junction transistors (BJT) in common-base, common-emitter, and common-collector configurations, as well as field-effect transistors (FETs). Their static and dynamic characteristics are analyzed, and their role as switching devices and amplifiers

is emphasized. Students will also engage with small-signal amplifier design and AC/DC load line analysis to strengthen both analytical and design skills. By the end of the course, students are expected to not only understand the theoretical principles of semiconductor devices but also apply them in designing, analyzing, and interpreting basic electronic circuits, thereby bridging the gap between circuit theory and modern electronics. This prepares students for advanced courses in electronics and equips them with problem-solving and design competencies relevant to real-world electrical engineering applications.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Core

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcome: Upon completion of the course, the student should be able to:	**Suitable Bloom Taxonomy Relevant to the CLO's**	**POs**	**Assessment Tools**
Explain the atomic structure and basic principles of semiconductors, including intrinsic and extrinsic materials, charge carriers, and energy bands.	1,2	1,2	Assignment, quizzes and Exams
Describe and calculate key semiconductor properties such as effective mass, carrier mobility, conductivity, carrier lifetime, and recombination.	2,3,4	1,2	Assignment, quizzes and Exams
Apply carrier transport equations and analyze semiconductor behavior using continuity equations, drift and diffusion models, and the Hall effect.	3,4,5	1,2	Assignment, quizzes and Exams
Analyze the structure and function of PN junction diodes, and evaluate their applications in rectification, filtering, and regulation circuits.	4,5	1,2,5	Assignment, quizzes and Exams
Explain and compare the operation of bipolar junction transistors (BJT) in CB, CE, and CC configurations, including input/output characteristics.	2,4,5	2,3,6,7	Assignment, quizzes and Exams
Design and interpret transistor-based circuits such as small-signal	3,6	1,2,3,5	Assignment, quizzes and Exams

amplifiers and switching devices using AC/DC load line techniques.			

f) Mapping of CLOs to relevant POs

Course Learning Outcomes	Programme Outcomes (POs):											
	1	2	3	4	5	6	7	8	9	10	11	12
Explain the atomic structure and basic principles of semiconductors, including intrinsic and extrinsic materials, charge carriers, and energy bands.	3	2										
Describe and calculate key semiconductor properties such as effective mass, carrier mobility, conductivity, carrier lifetime, and recombination.	3	3										
Apply carrier transport equations and analyze semiconductor behavior using continuity equations, drift and diffusion models, and the Hall effect.	3	3										
Analyze the structure and function of PN junction diodes, and evaluate their applications in rectification, filtering, and regulation circuits.	3	3			2							
Explain and compare the operation of bipolar junction transistors (BJT) in CB, CE, and CC configurations, including input/output characteristics.		3	2			2	1					
Design and interpret transistor-based circuits such as small-signal amplifiers and switching devices using AC/DC load line techniques.	3	3	3		3							

Keys: 1 = Slightly related, 2 = Moderately related, 3 = Highly related

g) Expectations of Students

a. Attend 3 hours of lectures per week
b. Turn in assignments and projects as considered suitable by the tutors
c. Complete continuous assessment which involves mid-semester test
d. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of PowerPoint slides, Lecture templates-textbooks)

13) **Course Content or Outline**

Weeks	Topics & Subtopics
Week 1	**Introduction – Nature of Atom** • Structure of matter: protons, neutrons, electrons • Energy levels and shells • Valence electrons and bonding in solids
Week 2	**Basic Concepts of Semiconductors (Part 1)** • Intrinsic semiconductors • Charge carriers: electrons and holes • Energy bands: conduction band, valence band, band gap
Week 3	**Basic Concepts of Semiconductors (Part 2)** • Extrinsic semiconductors: n-type and p-type doping • Effective mass of charge carriers • Carrier concentration and conductivity
Week 4	**Carrier Transport Phenomena** • Carrier mobility • Drift current and diffusion current • Continuity equation

Week 5	**Carrier Transport & Effects** • Carrier lifetime and recombination • Flow equations in semiconductors • Hall effect and applications
Week 6	**PN Junctions (Part 1)** • Formation of PN junction • Depletion region and barrier potential • Junction capacitance
Week 7	**PN Junctions (Part 2)** • PN junction under forward bias • PN junction under reverse bias • Breakdown mechanisms
Week 8	**Rectification and Filtering** • Half-wave rectifiers • Full-wave rectifiers (center-tap and bridge) • Capacitor and LC filters
Week 9	**Regulation (Zener Diodes)** • Zener breakdown and avalanche breakdown • Zener diode as voltage regulator • Practical Zener regulator circuits
Week 10	**Bipolar Junction Transistors (BJT) – Part 1** • BJT structure and operation principles • CB, CE, and CC configurations • Input and output characteristics

Week 11	**Bipolar Junction Transistors (BJT) – Part 2** • DC load line analysis • Operating point (Q-point) • BJT biasing methods
Week 12	**Field Effect Transistors (FETs)** • JFET structure and characteristics • MOSFET structure and operation • FET applications
Week 13	**Amplifiers & Switching Devices** • Small signal amplifiers • AC/DC load line analysis • Transistors as switches
14	**Revision and Lecture Free Week**
15	**Examination**

Method of Grading Continuous Assessment

Continuous Assessment

- Assignment: 10%
- Quizzes/attendance: 10%
- Test: 20%

End of Semester Examination: 60%

k) Tutorials

Tutorial will be offered at the end of each module.

l) Group works and Assignments

The group works and assignments are designed to challenge students to think critically and solve problems in the world of electromagnetism, analyze real-world scenarios and propose solutions using the acquired knowledge of electromagnetic principles. The course also includes assignments designed to solidify your understanding through calculations, problem-solving exercises, and potentially, simulations or experiments involving electromagnetic phenomena.

m) Contemporary Issues and Industrial Relevance

Basic Electrical Engineering II is highly relevant in today's technology-driven world, where semiconductors and electronic devices form the backbone of modern industries. From consumer

electronics and renewable energy systems to telecommunications and automation, the principles of semiconductor physics, PN junctions, transistors, and amplifiers are central to innovation. With the rapid growth of microelectronics, VLSI, and nanotechnology, engineers require strong foundational knowledge to address issues of miniaturization, efficiency, and sustainability. This course equips students with the skills to contribute to advancements in electronics, power systems, and emerging fields like IoT and smart devices, ensuring industrial and societal impact.



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AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code:	EEE 325	**Course Title:**	Electrical Machines I
Semester:	First	**Credit Hours:**	3
Contact Hours:	Lecture – Three hours lectures per Week	**Location:**	D3-9
Lecturer(s):	Engr. Dr. O. N. Omogbai		

Textbooks and other Materials:

1. Electrical Machines with MATLAB® 2nd Edition by Turan Gonen.
2. Basics Of Electrical Drives 4th Revised edition by S K Pillai.
3. Electric machinery and transformers by Bhag S. Guru. Huseyin R. Hiziroglu.-3rd. edition
4. Hughes Electrical & Electronic Technology 10th Edition by Edward Hughes, John Hiley, Keith Brown, Ian McKenzie Smith.

a) Course Overview and description

This course covers the fundamentals of electromechanical devices and transformers, including energy conversion principles, transformer performance and types, DC machine operation, excitation, characteristics, and applications. It emphasizes practical aspects such as motor/generator selection, speed control, and braking, preparing students to analyze and apply machines and transformers in real engineering systems.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Core

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcomes: upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**Pos**	**Assessment Tools**
Explain the basic principles of relays, actuators, and electromechanical energy conversion.	1, 2	1, 2, 3	Assignment, quizzes and Exams
Describe the operation, performance parameters, and equivalent circuits of transformers.	1, 2	1, 2, 4, 5	Assignment, quizzes and Exams
Apply concepts of efficiency, regulation, and per-unit values in transformer analysis.	3, 4	2, 3, 10	Assignment, quizzes and Exams
Differentiate between types of transformers, including autotransformers and instrument transformers, and discuss their design elements.	2, 4, 5	2, 3, 11	Assignment, quizzes and Exams
Analyze transformer behavior in polyphase circuits and conditions for parallel operation.	3, 4, 5	2, 3, 6, 10	Assignment, quizzes and Exams
Examine the construction, excitation methods, armature reaction, and performance characteristics of DC generators and motors.	2, 4	1, 2, 5	Assignment, quizzes and Exams
Evaluate and select suitable motors and generators for practical applications, including speed control,	3, 4, 5	2, 3, 4, 10, 11	Assignment, quizzes and Exams

electric braking, cross-field, and commutator machines.			

f) Mapping of CLOs to relevant POs

Course Learning Outcomes	**Programme Outcomes (POs):**											
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
Explain the basic principles of relays, actuators, and electromechanical energy conversion.	**3**	**2**	**1**	**1**	**1**							
Describe the operation, performance parameters, and equivalent circuits of transformers.	**3**	**3**	**2**	**2**	**2**							
Apply concepts of efficiency, regulation, and per-unit values in transformer analysis.	**2**	**3**	**2**	**2**	**2**			**1**		**2**	**1**	
Differentiate between types of transformers, including autotransformers and instrument transformers, and discuss their design elements.	**2**	**3**	**3**	**1**						**1**	**2**	
Analyze transformer behavior in polyphase circuits and conditions for parallel operation.	**2**	**3**	**3**	**1**	**1**	**1**				**2**	**2**	
Examine the construction, excitation methods, armature reaction, and performance characteristics of DC generators and motors.	**3**	**3**	**2**	**1**	**3**		**1**					
Evaluate and select suitable motors and generators for practical applications, including speed control, electric braking, cross-field, and commutator machines.	**2**	**3**	**3**	**2**	**2**	**1**		**1**		**2**	**3**	**1**

Keys: 1 = Slightly related, 2 = Moderately related, 3 = Highly related

g) Expectations of Students

a. Attend 3 hours of lectures per week

b. Turn in assignments and projects as considered suitable by the tutor

c. Complete continuous assessment which involves mid-semester test

d. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of PowerPoint slides, Lecture templates-textbooks)

14) **Course Content or Outline**

Topics	Weeks
Fundamentals of Relays, Actuators & Energy Conversion • Principles of electromagnetic relays and actuators • Electromechanical energy conversion mechanisms	1
Transformer Theory & Modeling • Construction and operation • Performance parameters and equivalent circuit models	2
Transformer Analysis Techniques • Efficiency and voltage regulation • Per-unit system applications	3 – 4
Transformer Types & Design • Autotransformers, instrument transformers • Core design and winding configurations	5 – 6
Polyphase & Parallel Transformer Operations • Behavior in multi-phase systems • Conditions for successful parallel operation	7 – 8
DC Machines: Construction & Performance • Generator and motor components • Excitation methods and armature reaction • Key performance characteristics	9 – 10
Motor & Generator Selection for Applications • Speed control and electric braking • Cross-field and commutator machine applications	11 – 12
Lecture free week	13

Revision & Examination	14 – 15

j) Method of Grading Continuous Assessment

Continuous Assessment

- Assignment: 10%
- Quizzes/attendance: 10%
- Test: 20%

End of Semester Examination: 60%

k) Tutorials

Tutorial will be offered at the end of each module.

l) Group works and Assignments

Coursework requires students to analyze the behavior of transformers and DC machines, differentiating their types and examining their characteristics. Through collaborative and individual assignments, students apply principles of electromechanical energy conversion to evaluate performance and select appropriate machines for practical applications, thereby solidifying their ability to describe and explain fundamental operations.

m) Contemporary Issues and Industrial Relevance

The course equips students to analyze the performance of transformers and DC machines and evaluate their suitability for real-world applications, emphasizing reliable, efficient, and safe operation. Through the application of design and selection principles, students learn to propose robust solutions that meet operational constraints, supporting sustainable engineering practices.

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code: EEE 315 Course Title: Electromagnetic Fields and Waves I

Semester: First **Credit Hours**: 3 hours

Contact Hours: Lecture - Three hours lectures per week

Location: Hall C3-6 **Lecturer(s)**: Engr. Dr. A. O. Ojo

Textbooks and other Materials:

- Owolabi & Ojo (2022), Electromagnetic Fields and Waves – Principles and Applications Engineering Electromagnetics, 1st edition, Matprints Inc.

- Gunther Lehner (2010), Electromagnetic Field Theory for Engineers and Physicists, Springer Heidelberg Dordrecht London New York

a) Course Overview and description

This course covers the fundamental concepts of electromagnetics, emphasizing the mathematical tools and physical principles essential for analyzing and solving problems in electrical engineering. Topics include vector analysis, integral theorems, and the behavior of electric and magnetic fields governed by laws such as Coulomb's Law, Gauss's Law, Faraday's Law, Biot-Savart and Ampere's Law. The course covers electric fields and electric potentials for various charge distributions, including lines, rings, surfaces, volumes, and electric dipoles. A key focus is on Maxwell's Equations and their role in understanding electromagnetic wave propagation in various media, as well as practical applications such as wave behavior, reflection, refraction, and transmission, and modern communication systems and their societal and ethical impacts.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Core

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcomes: upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**POs**	**Assessment Tools**
Understand the fundamental mathematical tools such as vector analysis, calculus, integra theorems and dimensional analysis	2, 4, 5	1	Assignment, quizzes and Exams
Understand the behaviour of electric and potential around different structures; surfaces and volumes.	2, 3	1, 2, 5	Assignment, quizzes and Exams
Analyze the fundamental laws governing electric and magnetic fields.	2, 3, 4	1, 2	Assignment, quizzes and Exams
Apply the concept of Maxwell's Equations and their role in analyzing electromagnetic fields and waves.	2, 3	1, 2, 5	Assignment, quizzes and Exams
Compare the behavior of wave propagation mechanism in different media and their effect.	2, 3, 4	2, 5	Assignment, quizzes and Exams
Justify the concept of electromagnetic application in the modern society.	2, 6	8, 10, 11	Assignment, quizzes and Exams

f) Mapping of CLOs to relevant POs

Course Learning Outcomes	**Programme Outcomes (POs):**											
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
Understand the fundamental mathematical tools such as vector analysis, calculus, integra theorems and dimensional analysis	**3**											
Understand the behaviour of electric and potential around different structures; surfaces and volumes.	**3**	**2**			**2**							
Understand the fundamental laws governing electric and magnetic fields.	**3**	**2**										
Apply the concept of Maxwell's Equations and their role in analyzing electromagnetic fields and waves.	**3**	**3**			**3**							
Compare the behavior of wave propagation mechanism in different media and their effect.		**3**			**3**							
Justify the concept of electromagnetic application in the modern society.								**3**		**3**	**3**	

Keys: 1 = Slightly related, 2 = Moderately related, 3 = Highly related

g) Expectations of Students

a. Attend 3 hours of lectures per week

b. Turn in assignments and projects as considered suitable by the tutors

c. Complete continuous assessment which involves mid-semester test

d. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of PowerPoint slides, Lecture templates-textbooks)

15) **Course Content or Outline**

Topics	**Weeks**
Introduction to Vector Analysis and Applications: • Scalars and Vectors, Vector Operations: Addition, Subtraction, Dot Product, Cross Product, Coordinate Systems and Transformations • Gradient of Scalar Fields, Divergence and Curl of Vector Fields, Applications in Electric Fields • Statement and Proof of Divergence Theorem, Statement and Proof of Stokes' Theorem, Applications in Electromagnetic Fields	1 – 3
Electric Field and Electric Potential in conductors and dielectrics • Behavior Around Point Charges and Continuous Charge Distributions, Electric Potential and Electric Field Relationship.	4 – 5

• Electric Field Inside Conductors and Dielectrics, Boundary Conditions and Discontinuities in Electric Fields	
Fundamental Laws: Coulomb's Law, Gauss's Law, Ampere's Law, Biot-Savart Law and Their Applications • Coulomb's Law and Its Applications, Gauss's Law and Its Applications • Biot-Savart Law, Ampere's Law, Magnetic Potential and Applications	6 – 7
Maxwell's Equations and Time-Varying Fields: Derivation and Physical Interpretation • Gauss's Law for Electricity and Magnetism, Faraday's Law of Induction, Ampère's Law with Maxwell's Correction • Displacement Current and Time-Varying Fields, Derivation and Physical Interpretation of Maxwell's Equations in Time-Varying Fields	8 – 9
Wave Propagation in Free Space, Dielectrics, and Conductors • Electromagnetic Wave Propagation in Free Space, Wave Behavior in Dielectrics and Conductors	10 – 11

• Reflection and Refraction of Electromagnetic Waves, Transmission of Electromagnetic Waves	
Electromagnetic Applications: Modern Communication Systems, Power Systems and Societal Impact • Role of Electromagnetic Waves in Communication Systems, Societal Impact and Ethical Considerations of Electromagnetic Applications	12
Lecture free week	13
Revision & Examination	14 - 15

j) Method of Grading Continuous Assessment

Continuous Assessment

- Assignment: 10%
- Quizzes/attendance: 10%
- Test: 20%

End of Semester Examination: 60%

k) Tutorials

Tutorial will be offered at the end of each module.

l) Group works and Assignments

The group works and assignments are designed to challenge students to think critically and solve problems in the world of electromagnetism, analyze real-world scenarios and propose solutions using the acquired knowledge of electromagnetic principles. The course also includes assignments designed to solidify your understanding through calculations, problem-solving exercises, and potentially, simulations or experiments involving electromagnetic phenomena.

m) Contemporary Issues and Industrial Relevance

This course equips students with the mathematical foundation and electromagnetism principles crucial for understanding and developing advanced technologies like microwave devices and communication systems. By mastering these concepts, the knowledge and tools sought after in the engineering industry is obtained.



AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code: EEE 319 **Course Title:** Measurement and Instrumentation

Semester: First **Credit Hours**: 3 hours

Contact Hours: Lecture - Two hours lectures per week

Location: Engineering Auditorium **Lecturer(s)**: Engr. Prof. M.O Onibonoje, Mr. Victor Akinpelu

Textbooks and other Materials:

- Measurement and Instrumentation Principle by Alan S. Morris
- Electrical Technology by B. L. Theraja and A.K. Theraja
- A Course in Electronic and Electrical Measurement & Instrumentation by Gupta
- Basic Concepts of Medical Instrumentation by Walter H. Olson

a) Course Overview and description

Measurement and Instrumentation introduces students to the fundamental principles, techniques, and tools essential for accurate measurement in electrical and electronic engineering. The course provides a solid grounding in units, symbols, and standards of measurement, followed by a study of the theory of errors, distinguishing between systematic and random errors and methods for minimizing their impact on results. Students will explore a wide range of instruments, beginning with traditional indicating devices such as moving coil, moving iron, dynamometer, and electrostatic instruments, before progressing to advanced measuring systems

for DC and AC quantities, including impedance, power, and energy measurements. The use of instrument potentiometers, oscilloscopes, and cathode-ray tubes (CRT) for measurement applications forms a crucial part of the curriculum, along with the study of pen recorders and modern digital test instruments. In addition, the course introduces students to practical issues such as signal interference, screening, and grounding techniques that ensure accuracy and reliability in measurement systems. Through this comprehensive coverage, students develop the ability to identify and understand key electrical quantities, interpret measurement results, and effectively apply modern test equipment in experimental and industrial contexts. The course outcomes emphasize both conceptual understanding and practical application, aligning with broader programme learning objectives that encourage problem-solving, communication, ethical responsibility, and the integration of technology in engineering practice. Ultimately, EEE 319 equips students with the essential skills to analyze, interpret, and communicate measurement data, preparing them for advanced studies and professional practice in electrical and electronic engineering.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Core

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

f) Mapping of CLOs to relevant POs

Keys: 1 = Slightly related, 2 = Moderately related, 3 = Highly related

Course Learning Outcome: Upon completion of the course, the student should be able to:	**Suitable Bloom Taxonomy Relevant to the CLO's**	**POs**	**Assessment Tools**
Identify and understand the various electrical quantities, units, and symbols	1,2,4	1,2,5,8	Assignment, quizzes and Exams
Understand the theory of errors, the different types of errors and the means of overcoming the errors.	2,3,5	2,5,6,10	Assignment, quizzes and Exams
Understand the A.C and D.C measurements	2,3,6	1,2,3,4,5,6	Assignment, quizzes and Exams
Know the features and calculations involved in using various indicating instruments and oscilloscope	1,3,4	1,2,3,4,5,6	Assignment, quizzes and Exams
Understand the concept of digital test instruments	2,4,6	1,2,3,4,5,7	Assignment, quizzes and Exams

Course Learning Outcomes	**Programme Outcomes (POs):**

	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
Identify and understand the various electrical quantities, units, and symbols	**3**	**2**			**2**			**2**				
Understand the theory of errors, the different types of errors and the means of overcoming the errors.	**2**				**3**	**2**				**2**		
Understand the A.C and D.C measurements	**3**	**3**	**2**	**2**	**3**	**2**						
Know the features and calculations involved in using various indicating instruments and oscilloscope	**3**	**2**	**2**	**2**	**3**	**2**						
Understand the concept of digital test instruments	**2**	**2**	**3**	**3**	**2**		**2**					

g) Expectations of Students

a. Attend 3 hours of lectures per week

b. Turn in assignments and projects as considered suitable by the tutors

c. Complete continuous assessment which involves mid-semester test

d. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of PowerPoint slides, Lecture templates-textbooks)

16) **Course Content or Outline**

Week	Topics & Subtopics
1	**Units and Symbols** • SI units and derived units • Conversion of units • Common electrical symbols and notations
2	**Theory of Errors (Part I)** • Types of errors: systematic vs random • Sources of errors in measurements • Statistical treatment of data
3	**Theory of Errors (Part II)** • Accuracy, precision, and tolerance • Limiting errors and probable errors • Error reduction techniques
4	**Indicating Instruments (Part I)** • Moving coil instruments: construction & operation • Moving iron instruments • Calibration and applications
5	**Indicating Instruments (Part II)** • Dynamo-meter type instruments • Comparison of indicating instruments • Limitations and accuracy considerations
6	**Electrostatic Indicating Instruments** • Principle of operation

	• Applications in voltage measurement • Advantages and disadvantages
7	**DC & AC Measurements (Part I)** • Measurement of resistance, current, and voltage • AC bridges for impedance measurement • Watt-meter principles
8	**DC & AC Measurements (Part II)** • Power factor measurement • Energy meters (DC & AC) • Calibration of meters
9	**Measurement of Power and Energy** • Dynamometer wattmeter • Single-phase and three-phase power measurement • Errors in power measurement
10	**Instrument Potentiometers** • DC potentiometer: construction & applications • AC potentiometer: working principle • Calibration of instruments using potentiometers
11	**Oscilloscope and CRT Applications** • Oscilloscope basic construction & controls • Measurement of voltage, current, frequency, phase • Applications of CRT in measurement
12	**Pen Recorders & Digital Test Instruments**

	• Types of recorders (strip chart, X–Y recorders) • Working principle of pen recorders • Digital voltmeters and multimeters • Frequency counters
13	**Signal Interference, Screening & Grounding Techniques** • Sources of electrical interference • Shielding and grounding practices • Signal conditioning techniques • Course review and revision
14	**REVISION**
15	**EXAMINATION**

Method of Grading Continuous Assessment

Continuous Assessment

- Assignment: 10%
- Quizzes/attendance: 10%
- Test: 20%

End of Semester Examination: 60%

k) Tutorials

Tutorial will be offered at the end of each module.

l) Group works and Assignments

The group works and assignments are designed to challenge students to think critically and solve problems in the world of electromagnetism, analyze real-world scenarios and propose solutions using the acquired knowledge of electromagnetic principles. The course also includes assignments designed to solidify your understanding through calculations, problem-solving exercises, and potentially, simulations or experiments involving electromagnetic phenomena.

m) Contemporary Issues and Industrial Relevance

Measurement and Instrumentation is highly relevant to contemporary engineering practice, as accurate measurement underpins technological innovation, industrial automation, and quality

assurance. In modern industries such as power systems, telecommunications, and electronics manufacturing, reliable instrumentation ensures efficiency, safety, and compliance with international standards. With the growing integration of digital test instruments, smart sensors, and real-time monitoring systems, engineers must adapt to rapidly evolving technologies. This course equips students with the skills to address these industrial needs, fostering competence in error analysis, data interpretation, and advanced measurement tools essential for tackling present and future engineering challenges.



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AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code: EEE 324 **Course Title:** Linear systems

Semester: Second **Credit Hours**: 3 hours

Contact Hours: Lecture - Two hours lectures per week

Location: Hall B2-5 **Lecturer(s)**: Engr. Prof. M. O. Onibonoje, Mr. Victor Akinpelu

Textbooks and other Materials:

- Linear System by H. Bourles
- Control System by Norman Nise
- Feedback and Control Systems by Joseph J. DiStefano, Allen R. Stubberud and Ivan J. Williams

a) Course Overview and description

Linear Systems provides students with a comprehensive foundation in the modeling, analysis, and evaluation of dynamic systems across electrical, mechanical, and thermal domains. The course begins by introducing mathematical models of physical systems and develops the concept of system analogies, enabling learners to translate problems from one physical domain to another. Emphasis is placed on control system classifications and feedback concepts, highlighting their role in shaping system performance and ensuring desired outputs under varying conditions.

Students explore transfer functions, block diagrams, and signal flow graphs, building the skills to represent, simplify, and interconnect systems efficiently. The course further investigates system responses, covering both transient and steady-state behaviors, and equips students with frequency-domain analysis tools such as root-locus, Bode plots, Nyquist diagrams, and polar plots for performance characterization and design insights. Stability analysis forms a crucial component, with techniques such as Routh-Hurwitz, Nyquist, and Bode criteria applied to determine robustness and reliability of systems. Finally, an introduction to analogue computer simulation enhances practical understanding by demonstrating system behavior and control strategies. By the end of the course, students will be able to model real-world systems, analyze responses, assess stability, and apply feedback principles, thereby gaining a strong analytical base for advanced courses in control engineering, signal processing, and system design. The knowledge gained equips learners to tackle contemporary industrial challenges where system modeling and control are vital, from automation and robotics to energy systems and communication networks.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Core

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcome: Upon completion of the course, the student should be able to:	**Suitable Bloom Taxonomy Relevant to the CLO's**	**POs**	**Assessment Tools**
Develop mathematical models of electrical, mechanical, and thermal systems using system analogies and differential equations.	2,3,4	1,2	Assignment, quizzes and Exams
Explain the classifications of control systems, and describe feedback concepts including their impact on system performance.	1,2,5	2	Assignment, quizzes and Exams
Analyze the transient and frequency responses of linear systems using time-domain and frequency-domain methods	3,4,6	1,2,5	Assignment, quizzes and Exams
Assess the stability of control systems using techniques such as Routh-Hurwitz, Bode, and Nyquist criteria	2,5,6	1,2,3	Assignment, quizzes and Exams
Construct and simplify transfer functions, block diagrams, and signal flow graphs to describe and analyze control systems	1,3,4	1,2,3	Assignment, quizzes and Exams

f) Mapping of CLOs to relevant POs

Course Learning Outcomes	Programme Outcomes (POs):											
	1	2	3	4	5	6	7	8	9	10	11	12
Develop mathematical models of electrical, mechanical, and thermal systems using system analogies and differential equations.	3	3										
Explain the classifications of control systems, and describe feedback concepts including their impact on system performance.		3										
Analyze the transient and frequency responses of linear systems using time-domain and frequency-domain methods	3	3			2							
Assess the stability of control systems using techniques such as Routh-Hurwitz, Bode, and Nyquist criteria	3	3	2									
Construct and simplify transfer functions, block diagrams, and signal flow graphs to describe and analyze control systems	3	3	2									

Keys: 1 = Slightly related, 2 = Moderately related, 3 = Highly related

g) Expectations of Students

a. Attend 3 hours of lectures per week

b. Turn in assignments and projects as considered suitable by the tutors

c. Complete continuous assessment which involves mid-semester test

d. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of PowerPoint slides, Lecture templates-textbooks)

17) **Course Content or Outline**

Week	Topics & Subtopics
1	**Mathematical models of physical systems** • Introduction to dynamic systems • Review of differential equations • Modeling principles and assumptions
2	**Mathematical models of physical systems (cont.)** • Mechanical system modeling (translational & rotational) • Electrical system modeling (RLC circuits) • Thermal and fluid systems modeling
3	**Analogous concepts in systems** • Force–voltage and force–current analogies • Translational–rotational analogies • Applications of analogies to mixed systems
4	**Control systems: introduction and classification** • Open-loop vs closed-loop control • Types of control systems (servo, regulatory, adaptive) • Real-world control system examples

5	**Feedback concepts in control systems** • Principles of feedback • Advantages of feedback (accuracy, disturbance rejection) • Effects of feedback on sensitivity and stability
6	**Transfer functions**• Definition and derivation from differential equations • Standard forms of transfer functions • Time constants and system order
7	**Block diagrams and signal flow graphs** • Representation of systems • Rules for block diagram reduction • Mason's Gain Formula and signal flow analysis
8	**Transient response of systems** • Time-domain specifications (rise time, overshoot, settling time) • First- and second-order system responses • Effect of system parameters on transient behavior
9	**Frequency response of systems – Root locus** • Root locus concepts • Rules for root locus construction • System performance from root locus
10	**Frequency response of systems – Bode plots** • Bode magnitude and phase plots • Standard factors and asymptotic approximations • Gain margin and phase margin

11	**Frequency response of systems – Nyquist and Polar plots** • Nyquist stability criterion • Polar plot interpretation • Relationship between Nyquist and Bode analyses
12	**System stability analysis** • Routh-Hurwitz criterion and stability condition • Stability margins using Nyquist and Bode plots• Relative vs absolute stability
13	**Introduction to analogue computer simulation** • Basics of analogue computer • Simulation of differential equations • Case studies in system modeling and control
14	**Revision**
15	**Examination**

Method of Grading Continuous Assessment

Continuous Assessment

- Assignment: 10%
- Quizzes/attendance: 10%
- Test: 20%

End of Semester Examination: 60%

k) Tutorials

Tutorial will be offered at the end of each module.

l) Group works and Assignments

The group works and assignments are designed to challenge students to think critically and solve problems in the world of electromagnetism, analyze real-world scenarios and propose solutions using the acquired knowledge of electromagnetic principles. The course also includes assignments designed to solidify your understanding through calculations, problem-solving exercises, and potentially, simulations or experiments involving electromagnetic phenomena.

m) Contemporary Issues and Industrial Relevance

Linear systems form the backbone of modern engineering applications, from automation and robotics to power and communication networks. With industries increasingly reliant on control, modeling, and stability analysis for reliable and efficient operations, knowledge of system dynamics is crucial. This course equips students with tools to analyze performance, ensure stability, and optimize feedback in real-world systems. Contemporary challenges such as renewable energy integration, smart grids, autonomous vehicles, and advanced manufacturing highlight the industrial relevance of linear systems, making the course essential for addressing emerging technological and sustainability demands.



AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code: EEE 321 Course Title: Electric Circuit Theory I

Semester: First **Credit Hours**: 3 hours

Contact Hours: Lecture - Three hours lectures per week

Location: Hall C3-6 **Lecturer(s)**: Engr. Dr. A. O. Ojo & Engr O. Imahe

Textbooks and other Materials:

- Charles Alexander & Matthew Sadiku (2021), Circuit-Fundamentals of Electric Circuits 7th Ed (1).

a) Course Overview and description

Circuit elements, circuit theorems, applications. Network response to steps, ramp, impulses. Network functions, response to exponential, sinusoidal sources. Laplace transform, polezero analysis, network synthesis, resonance, two-point analysis, ladder network. Star-Delta transformation, T, P Networks.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Core

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcomes: upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**POs**	**Assessment Tools**
Analyze the characteristics and applications of various circuit elements using fundamental circuit theorems	4	1, 2, 4, 12	Assignments, Quizzes, Exams
Evaluate the transient and steady-state network responses to step, ramp, and impulse inputs in electrical circuits	5	1, 2	Assignments, Quizzes, Exams
Assess the behavior of network functions and their responses to exponential and sinusoidal sources in dynamic systems	4	1	Assignments, Quizzes, Exams
Design and synthesize electrical networks using Laplace transform, pole-zero analysis, and resonance techniques, including ladder and two-point configurations	5	1, 2	Assignments, Quizzes, Exams
Apply Star-Delta transformation and T/Pi network configurations to optimize circuit performance and analyze complex network topologies	4	1, 8, 11, 12	Assignments, Quizzes, Exams

f) Mapping of CLOs to relevant POs

Course Learning Outcomes	Programme Outcomes (POs):											
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
Analyze the characteristics and applications of various circuit elements using fundamental circuit theorems	**3**	**3**		**2**								**3**
Evaluate the transient and steady-state network responses to step, ramp, and impulse inputs in electrical circuits	**3**	**3**										
Assess the behavior of network functions and their responses to exponential and sinusoidal sources in dynamic systems	**3**											
Design and synthesize electrical networks using Laplace transform, pole-zero analysis, and resonance techniques, including ladder and two-point configurations	**3**	**3**										
Apply Star-Delta transformation and T/Pi network configurations to optimize circuit performance and analyze complex network topologies	**3**							**2**			**3**	**3**

Keys: 1 = Slightly related, 2 = Moderately related, 3 = Highly related

g) Expectations of Students

a. Attend 3 hours of lectures per week
b. Turn in assignments and projects as considered suitable by the tutors
c. Complete continuous assessment which involves mid-semester test
d. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of PowerPoint slides, Lecture templates-textbooks)

12) **Course Content or Outline**

Topics	Weeks
Fundamentals of Circuit Elements and Theorems • Passive and active circuit elements: resistors, capacitors, inductors, sources • Circuit laws: Ohm's Law, Kirchhoff's Voltage and Current Laws • Thevenin and Norton equivalents, Superposition and Maximum Power Transfer Theorems • Applications of circuit theorems in DC and AC analysis	1 – 3
Network Response to Excitations • Transient and steady-state analysis of circuits • Response to step, ramp, and impulse inputs • Time-domain behavior of RL, RC, and RLC circuits • Initial and final conditions, natural and forced responses	4 – 5
Network Functions and Sinusoidal Analysis • Introduction to network functions and transfer functions • Frequency response and Bode plots • Response to exponential and sinusoidal sources • Phasor analysis and complex impedance	6 – 7
Laplace Transform and Network Synthesis • Laplace transform fundamentals and inverse techniques • Application of Laplace transform in circuit analysis • Pole-zero analysis and system stability • Resonance in electrical networks • Ladder networks and two-point configurations	8 – 11
Advanced Network Configurations and Transformations • Star-Delta transformation and its applications • T and Pi network configurations • Optimization of circuit performance using transformations • Analysis of complex network topologies.	12 – 13
Lecture free week	14
Revision & Examination	15

j) Method of Grading Continuous Assessment

Continuous Assessment

- Assignment: 10%
- Quizzes/attendance: 10%
- Test: 20%

End of Semester Examination: 60%

k) Tutorials

Tutorial will be offered at the end of each module.

l) Group works and Assignments

The group works and assignments in Circuit Theory I are designed to challenge students to think critically and solve problems within the domain of electrical circuits, encouraging analysis of real-world scenarios and the application of circuit principles to propose effective solutions. These tasks also reinforce students' understanding through calculations, analytical exercises, and potentially, simulations or experiments involving modern circuit configurations and network behaviors.

m) Contemporary Issues and Industrial Relevance

Electrical circuit theory faces modern challenges such as miniaturization, energy efficiency, and integration with smart technologies like IoT and intelligent control systems. Engineers must also address cybersecurity in embedded systems and rely on advanced simulation tools to model increasingly complex circuit behaviors. This course is vital across industries including power, telecommunications, consumer electronics, and automotive, where circuit design directly impacts system performance and reliability. From optimizing energy distribution to enabling smart devices and automation, circuit theory remains a cornerstone of innovation and engineering practice.

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AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code:	EEE 514	**Course Title:**	Reliability Engineering
Semester:	Second	**Credit Hours**:	**2**
Contact Hours:	Lecture - Two hours lectures per Week	**Location**:	B2-3
Lecturer(s):	Engr. Dr. O. N. Omogbai		

Textbooks and other Materials:

1. Reliability, Maintainability and Risk Practical methods for engineers. Eighth Edition. Dr David J Smith. Elsevier Ltd
2. Reliability Engineering Theory and Practice. 8th Edition. Springer.

a) Course Overview and description

This course introduces the fundamental principles of reliability, maintainability, and availability (RMA) with a focus on electrical and electronic systems, including power systems. Students will learn elementary reliability theory and its practical applications to the design, testing, and maintenance of electronic components and systems. Topics include, fault types and analysis, design strategies for enhanced reliability, and environmental considerations. The course emphasizes testing methods and real-world applications to prepare students for challenges in engineering design and system maintenance.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Core

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcomes: upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**Pos**	**Assessment Tools**
Explain the basic concepts of reliability, maintainability, availability in the field of Engineering	2	1, 2, 7, 8, 9	Assignment, quizzes and Exams
Assess the probability of system and component failures with elementary reliability theory.	4, 5	1, 2, 5	Assignment, quizzes and Exams
Interpret types of faults/failures for reliability improvement.	4, 5, 6	1, 2, 5, 7, 8, 9, 10	Assignment, quizzes and Exams
Apply reliability engineering to power systems and electronics systems.	3, 4, 5	1, 2, 3, 6, 4, 10	Assignment, quizzes and Exams
Assess the test characteristics of electrical and electronic components for validity.	2, 3, 4, 5	1, 2, 4, 5, 6, 10	Assignment, quizzes and Exams
Design of electrical and electronic systems and components for higher reliability.	4, 5, 6	1, 2, 3, 4, 5, 6, 9, 10, 11, 12	Assignment, quizzes and Exams

f) Mapping of CLOs to relevant POs

Course Learning Outcomes	Programme Outcomes (POs):											
	1	2	3	4	5	6	7	8	9	10	11	12
Explain the basic concepts of reliability, maintainability, availability in the field of Engineering	3	2					1	2	2			
Assess the probability of system and component failures with elementary reliability theory.	3	3			2							
Interpret types of faults/failures for reliability improvement.	2	3			2		1	2	2	1		
Apply reliability engineering to power systems and electronics systems.	3	3	2	2		1				1		
Assess the test characteristics of electrical and electronic components for validity.	3	2		3	3	1				2		
Design of electrical and electronic systems and components for higher reliability.	3	3	3	2	2	2			2	2	3	2

Keys: 1 = Slightly related, 2 = Moderately related, 3 = Highly related

g) Expectations of Students

a. Attend 2 hours of lectures per week

b. Turn in assignments and projects as considered suitable by the tutors

c. Complete continuous assessment which involves mid-semester test

d. Complete a 2-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of PowerPoint slides, Lecture templates-textbooks)

18) **Course Content or Outline**

Topics	Weeks
Introduction to Reliability, maintainability, availability in the field of Engineering. ▪ Why Engineering Products Fail ▪ Quantifying Reliability, maintainability, availability. ▪ Pattern of Failures with Time ▪ Design Factors	1
Reliability Theory ▪ Theory of First Principles ▪ General and Specific Reliability Functions ▪ Reliability of Simple and Complex Systems ▪ Reliability Block Diagram	2
Failure Mechanism and Fault Detection/Analysis ▪ Failure Modes and Effects Analysis (FMEA) ▪ Fault Tree Analysis (FTA)	3 – 4
Failure Rate of Electronic Components ▪ Resistors, Capacitors, Inductors etc. ▪ Introduction to Robot Reliability	5 – 6
Power system reliability ▪ The Renewal Process (Repairable System) Recognizing the Run-Fail-Repair-Run Cycle. ▪ Repairable Components in Series and Parallel ▪ Failure Bunching ▪ Key Indices of Power System Reliability.	7 – 8
Reliability Testing ▪ Reliability Test Categories ▪ Accelerated Tests	9 – 10
Design for Reliability DfR ▪ Demonstration of the DfR Process in Electronic products ▪ Demonstration of the DfR Process in Power Systems.	11 – 12
Lecture free week	13
Revision & Examination	14 – 15

j) Method of Grading Continuous Assessment

Continuous Assessment

- Assignment: 10%
- Quizzes/attendance: 10%
- Test: 20%

End of Semester Examination: 60%

k) Tutorials

Tutorial will be offered at the end of each module.

l) Group works and Assignments

Group work and assignments in this course are structured to deepen students' understanding of reliability engineering concepts through collaborative analysis and hands-on application. Teams evaluate system and component failure probabilities using reliability theory, interpret fault types for improvement strategies, and apply reliability principles to power and electronic systems. Assignments include testing and validating electrical components, and designing systems for enhanced reliability. This approach fosters teamwork, critical thinking, and practical problem-solving aligned with industry standards.

m) Contemporary Issues and Industrial Relevance

This course helps develop in the students, essential skills for designing reliable, maintainable, and available systems; equipping would be engineers to predict failures, analyze faults, and apply reliability principles to power and electronic systems. Valid testing and robust design practices ensure compliance, safety, and performance. Together, the outcomes support sustainable engineering and foster lifelong learning in a rapidly evolving field.

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AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code: EEE 543 **Course Title:** Renewable Energy Systems

Semester:	First	**Credit Hours:**	2
Contact Hours:	Lecture - Two hours lectures per Week	**Location:**	B2-3
Lecturer(s):	Engr. Dr. O. N. Omogbai		

Textbooks and other Materials:

1. Renewable Energy: Power for a Sustainable Future" by Godfrey Boyle (Oxford University Press)
2. Renewable and Efficient Electric Power Systems" by Gilbert M. Masters (Wiley)
3. Introduction to Renewable Energy" by Vaughn Nelson (CRC Press)

a) Course Overview and description

This course provides an overview of the modern energy landscape, covering conventional and non-conventional energy sources, energy efficiency, energy security, and distributed generation. It explores solar energy systems—both thermal and photovoltaic—focusing on PV characteristics, equivalent circuits, efficiency, sizing, economics, and peak power operation. Wind energy is addressed through wind power relations, turbine types and ratings, generator selection, variable speed operation, and control strategies. The course also examines grid integration of renewable systems, including stand-alone, hybrid, and micro-grid configurations, with emphasis on economics, interface requirements, stable operation, and impacts on power quality such as harmonics, transients, flicker, and reactive power.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Elective

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcomes: upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**Pos**	**Assessment Tools**
Classify energy sources and evaluate conventional vs. nonconventional energy resources.	4, 5	1, 2, 8, 10, 11	Assignment, quizzes and Exams
Analyze energy efficiency, energy security, and the role of distributed generation in modern power systems.	4	1, 2, 3, 10, 11	Assignment, quizzes and Exams
Explain the principles, components, and performance metrics of solar thermal and photovoltaic (PV) systems.	2	1, 3, 4, 7	Assignment, quizzes and Exams
Perform design calculations related to PV system sizing, efficiency, economics, and peak power tracking.	3, 6	1, 2, 3, 4, 5, 10, 11	Assignment, quizzes and Exams
Assess wind energy systems, including turbine types, power extraction, generator selection, and control strategies.	5	1, 2, 3, 6	Assignment, quizzes and Exams
Evaluate grid integration of renewable systems, including micro-grids, hybrid configurations, and interface requirements.	5	2, 3, 4, 6, 11, 12	Assignment, quizzes and Exams
Analyze the impact of renewable energy systems on power quality, including harmonics, voltage stability, and reactive power support.	4	1, 2, 3, 5, 9, 10, 11	Assignment, quizzes and Exams

f) Mapping of CLOs to relevant POs

Course Learning Outcomes	Programme Outcomes (POs):											
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
Classify energy sources and evaluate conventional vs. nonconventional energy resources.	**3**	**2**						**2**		**2**	**1**	
Analyze energy efficiency, energy security, and the role of distributed generation in modern power systems.	**3**	**3**	**2**		**2**					**3**	**2**	
Explain the principles, components, and performance metrics of solar thermal and photovoltaic (PV) systems.	**3**		**2**	**2**			**2**					
Perform design calculations related to PV system sizing, efficiency, economics, and peak power tracking.	**3**	**3**	**2**	**2**	**2**					**1**	**1**	
Assess wind energy systems, including turbine types, power extraction, generator selection, and control strategies.	**3**	**3**	**3**			**2**						
Evaluate grid integration of renewable systems, including micro-grids, hybrid configurations, and interface requirements.		**3**	**3**	**3**		**3**				**2**	**1**	**2**
Analyze the impact of renewable energy systems on power quality, including harmonics, voltage stability, and reactive power support.	**3**	**3**	**2**		**2**					**3**	**2**	

Keys: 1 = Slightly related, 2 = Moderately related, 3 = Highly related

g) Expectations of Students

a. Attend 2 hours of lectures per week

b. Turn in assignments and projects as considered suitable by the tutor

c. Complete continuous assessment which involves mid-semester test

d. Complete a 2-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of PowerPoint slides, Lecture templates-textbooks)

19) **Course Content or Outline**

Topics	Weeks
Energy Sources and Classification • Classification of energy sources: Renewable vs. Non-renewable • Conventional energy resources: Coal, oil, natural gas, nuclear • Nonconventional energy resources: Solar, wind, biomass, geothermal, tidal • Comparative evaluation: Availability, sustainability, environmental impact	1
Energy Efficiency and Security • Energy efficiency metrics and improvement strategies • Energy auditing and demand-side management • Energy security: Definitions, indicators, geopolitical factors • Role of distributed generation in enhancing reliability and resilience	2
Solar Energy Systems • Solar radiation fundamentals and site assessment • Solar thermal systems: Principles, collectors, storage, applications • Photovoltaic (PV) systems: Components, configurations, technologies • Performance metrics: Efficiency, fill factor, IV curves, degradation	3 – 4
PV System Design and Analysis • PV system sizing: Load estimation, panel selection, inverter matching • Efficiency calculations and loss analysis • Economic evaluation: Payback period, LCOE, incentives • Maximum Power Point Tracking (MPPT) techniques	5 – 6
Wind Energy Systems • Wind resource assessment and site selection • Turbine types: Horizontal vs. vertical axis • Power extraction: Betz limit, Cp curves • Generator selection: Synchronous, asynchronous, DFIG	7 – 8

• Control strategies: Pitch, yaw, torque control	
Grid Integration of Renewables • Grid-connected vs. off-grid systems • Micro-grids: Architecture, control, islanding • Hybrid systems: PV-wind-diesel, storage integration • Interface requirements: Inverters, protection, synchronization	9 – 10
Power Quality in Renewable Systems • Power quality parameters: Harmonics, flicker, voltage stability • Reactive power support and compensation • Impact of intermittent generation on grid stability • Standards and mitigation techniques	11 – 12
Lecture free week	13
Revision & Examination	14 – 15

j) Method of Grading Continuous Assessment

Continuous Assessment

- Assignment: 10%
- Quizzes/attendance: 10%
- Test: 20%

End of Semester Examination: 60%

k) Tutorials

Tutorial will be offered at the end of each module.

l) Group works and Assignments

Coursework engages students in group projects and individual assignments to classify energy sources, evaluate conventional and renewable technologies, and analyze energy efficiency, security, and distributed generation. Through collaborative tasks, they assess system performance, grid integration, and power quality impacts using practical and analytical approaches.

m) Contemporary Issues and Industrial Relevance

The course enables students to classify energy sources, analyze renewable technologies

and power systems, and apply design principles to evaluate performance, grid integration, and power quality—supporting efficient, secure, and sustainable energy solutions.

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AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code:	EEE 558	**Course Title:**	Energy Conversion and Storage
Semester:	Second	**Credit Hours**:	**2**
Contact Hours:	Lecture - Two hours lectures per Week	**Location**:	B2-3
Lecturer(s):	Engr. Dr. O. N. Omogbai		

Textbooks and other Materials:

1. Robert A. Huggins, Energy Storage: Fundamentals, Materials and Applications. Second Edition.
2. Renewable Energy: Power for a Sustainable Future" (5th Edition) by Godfrey Boyle, Bob Everett, and Janet Ramage (Oxford University Press).
3. Energy Conversion Systems" (2nd Edition). by S.S. Murthy (Universities Press)

a) Course Overview and description

This course provides a comprehensive introduction to electromechanical energy conversion and various sources of motive power. It covers battery technologies, testing methods, fault diagnosis, repair techniques, and the effects of environmental conditions on battery performance and lifespan. Students will also explore small-scale power systems and the concept of distributed generation, with a focus on practical applications and sustainability in modern energy systems.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Core

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcomes: upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**Pos**	**Assessment Tools**
Assess the various sources of energy, towards identifying sustainable solutions.	5	1, 3, 10, 11	Assignment, quizzes and Exams
Explain the energy transformations in Electromechanical energy conversion	2	1, 2, 3, 7	Assignment, quizzes and Exams
Solve energy inefficiency issues by Waste heat recovery.	3	1, 2, 3, 10	Assignment, quizzes and Exams
Describe Energy Storage options to identify sustainable solutions.	2, 5	1, 3, 4, 7, 11	Assignment, quizzes and Exams
Explain Electrical Energy Conversion and Storage with the aid of primary and secondary cells	2	1, 2, 3, 7	Assignment, quizzes and Exams
Diagnose battery fault and assess battery life via testing.	4, 5	2, 4, 5, 6	Assignment, quizzes and Exams
Explain the concept of distributed generation and propose a conceptual design for a small-scale distributed generation system.	2, 6	1, 2, 3, 6, 7, 10, 12	Assignment, quizzes and Exams

f) Mapping of CLOs to relevant POs

Course Learning Outcomes	Programme Outcomes (POs):											
	1	2	3	4	5	6	7	8	9	10	11	12
Assess the various sources of energy, towards identifying sustainable solutions.	2		2							3	3	
Explain the energy transformations in Electromechanical energy conversion	3	3	2				1					
Solve energy inefficiency issues by Waste heat recovery.	2	3	3							3		
Describe Energy Storage options to identify sustainable solutions.	2		2	2			1				2	
Explain Electrical Energy Conversion and Storage with the aid of primary and secondary cells	3	2	2				1					
Diagnose battery fault and assess battery life via testing.		3		2	3	2						
Explain the concept of distributed generation and propose a conceptual design for a small-scale distributed generation system.	2	2	3			1	1			3		1

Keys: 1 = Slightly related, 2 = Moderately related, 3 = Highly related

g) Expectations of Students

a. Attend 2 hours of lectures per week

b. Turn in assignments and projects as considered suitable by the tutors

c. Complete continuous assessment which involves mid-semester test

d. Complete a 2-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of PowerPoint slides, Lecture templates-textbooks)

20) **Course Content or Outline**

Topics	Weeks
Energy and Energy Types ▪ Non-Renewable Energy Sources ▪ Renewable Energy Resources ▪ Electric Energy ▪ Sources of Electrical Energy	1
Electromechanical Energy Conversion ▪ Devices For Electromechanical Energy Conversion ▪ Energy Balance ▪ Simple Translatory Electromechanical system	2
Waste heat ▪ Waste heat recovery ▪ Waste Heat to Power Technologies ▪ Challenges Associated with Waste Heat to Power Technologies	3 – 4
Energy Storage Systems ▪ Services offered by Energy Storage Systems ▪ Electrical Energy Storage Technologies	5 – 6
Cells And Batteries ▪ Characteristics of Batteries ▪ Smart Battery System ▪ Batteries for Electric and Hybrid Electric Vehicles ▪ Electric-Utility and Other Applications of Batteries	7 – 8
Battery Testing ▪ Battery Test Approaches ▪ Battery Maintenance ▪ Faults in Battery Energy Storage Systems	9 – 10
Distributed Generation (DG) ▪ Merits of DG Integration into the Grid ▪ DG and Energy Efficiency ▪ Role of Energy Storage in DG ▪ DG Challenges	11 – 12
Lecture free week	13
Revision & Examination	14 – 15

j) Method of Grading Continuous Assessment

Continuous Assessment

- Assignment: 10%
- Quizzes/attendance: 10%
- Test: 20%

End of Semester Examination: 60%

k) Tutorials

Tutorial will be offered at the end of each module.

l) Group works and Assignments

The group works and assignments in this course are central to achieving the course learning outcome: assessing energy sources for sustainable solutions. Through collaborative projects, students learn to diagnose real-world energy inefficiency issues, conceptualize distributed generation systems, and evaluate energy storage options. The individual assignments help solidify understanding through calculations in electromechanical conversion, analysis of battery life data, and the principles of electrical energy storage, building a comprehensive skill set for sustainable energy problem-solving.

m) Contemporary Issues and Industrial Relevance

This course equips students to address critical contemporary issues in the energy sector by providing a deep understanding of electromechanical systems and sustainable solutions. The curriculum is designed around industry-relevant challenges, focusing on diagnosing energy inefficiencies, designing distributed generation systems for grid resilience, and evaluating advanced storage technologies. Students will learn to propose viable solutions for industrial decarbonization, waste heat recovery, and the integration of renewables, preparing them to contribute directly to the global energy transition..

Course Code: EEE 566 **Course Title:** Non-Linear systems

Semester: Second **Credit Hours**: 3 hours

Contact Hours: Lecture - Two hours lectures per week

Location: Hall B2-5 **Lecturer(s)**: Engr. Prof. M.O Onibonoje, Mr. Victor Akinpelu

Textbooks and other Materials:

- Modern control theories: Non linear, optimal and Adaptive systems, Csaki.F (1972)
- Applied Non linear control, Slotine, J.E (1991)
- Control Theory: Multivariable and Non linear Methods, Khali, H
- Non-linear Control systems, Marcel Dekker Inc, Vukic, Z, Kulija (2003)

a) Course Overview and description

Non Linear Systems provides an advanced study of the principles, analysis, and applications of nonlinear control systems, which are fundamental to modern engineering practice. Unlike linear systems, nonlinear systems exhibit complex behaviors such as limit cycles, bifurcations, and instability, making their analysis critical in real-world scenarios where ideal assumptions of linearity no longer apply. The course introduces students to nonlinear differential equations, common sources of nonlinearities in engineering systems, and approximation techniques including linearization, piecewise linear methods, and describing functions for analyzing system stability and predicting limit cycles. Students will gain practical skills in evaluating system dynamics using phase-plane methods and Lyapunov's stability theory, which are powerful tools for assessing nonlinear responses and ensuring robust control. In addition, the course bridges theory and practice by covering sampled-data systems with z-transform analysis and by incorporating programmable logic controllers (PLCs) for process automation applications, thus equipping learners with both analytical and industrially relevant competencies. Emphasis is placed on developing the ability to model, analyze, and design nonlinear control strategies for complex systems encountered in robotics, power systems, aerospace, and industrial automation. By the end of the course, students will be able to formulate nonlinear system models, apply appropriate approximation and stability techniques, and implement control logic for practical applications. Overall, this course builds a solid foundation in nonlinear control, preparing students for advanced research, professional practice, and innovation in electrical, electronic, and control engineering fields.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Core

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcome: Upon completion of the course, the student should be able to:	**Suitable Bloom Taxonomy Relevant to the CLO's**	**POs**	**Assessment Tools**
Describe the fundamental characteristics of nonlinear systems and common sources of nonlinearities	1,2,4	1,2,8	Assignment, quizzes and Exams
Formulate and analyze nonlinear differential equations relevant to control systems	3,4,5	1,2	Assignment, quizzes and Exams
Apply linearization and approximation methods to simplify and analyze nonlinear systems near equilibrium points	2,3,6	1,2	Assignment, quizzes and Exams
Evaluate the behavior of nonlinear control systems through phase-plane analysis, describing function methods, or simulations	3,4,5	1,2,3,4	Assignment, quizzes and Exams
Develop and test control logic for simple industrial process automation using PLCs	2,3,6	3,5,6,11,12	Assignment, quizzes and Exams

Course Learning Outcomes	Programme Outcomes (POs):											
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
Describe the fundamental characteristics of nonlinear systems and common sources of nonlinearities	**3**	**2**						**2**				
Formulate and analyze nonlinear differential equations relevant to control systems	**3**	**3**										
Apply linearization and approximation methods to simplify and analyze nonlinear systems near equilibrium points	**3**	**3**										
Evaluate the behavior of nonlinear control systems through phase-plane analysis, describing function methods, or simulations	**3**	**3**	**3**	**3**								
Develop and test control logic for simple industrial process automation using PLCs			**3**		**3**	**3**					**1**	**2**

f) Mapping of CLOs to relevant POs

Keys: 1 = Slightly related, 2 = Moderately related, 3 = Highly related

g) Expectations of Students

a. Attend 3 hours of lectures per week

b. Turn in assignments and projects as considered suitable by the tutors

c. Complete continuous assessment which involves mid-semester test

d. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of PowerPoint slides, Lecture templates-textbooks)

21) **Course Content or Outline**

Weeks	Topics & Subtopics
1	**Introduction to Nonlinear Systems** • Definition and importance of nonlinear systems in control engineering • Differences between linear and nonlinear systems • Examples of nonlinearities in physical and engineering systems
2	**Nonlinear Differential Equations** • Formulation of nonlinear differential equations • Solution methods (analytical vs numerical) • Stability considerations in nonlinear models
3	**Characteristics of Nonlinear Systems** • Common nonlinear behaviors (saturation, dead-zone, hysteresis) • Nonlinear system responses (bifurcation, chaos, limit cycles) • Real-world examples in power, robotics, and communication systems
4	**Linearization Approximations**

	• Concept of equilibrium points • Taylor series expansion and Jacobian method • Applications of linearization in simplifying system analysis
5	**Piecewise Linear Approximation** • Principles of piecewise linearization • Graphical interpretation of nonlinearities • Examples in relay and saturation systems
6	**Describing Function Analysis (Part 1)** • Concept of the describing function • Derivation for common nonlinearities (relay, saturation, dead-zone) • Limit cycle prediction basics
7	**Describing Function Analysis (Part 2)** • Dual input describing function • Application to system stability • Practical examples and case studies
8	**Phase-Plane Method (Part 1)** • Introduction to phase-plane analysis • Construction of phase trajectories • Isoclines and singular points
9	**Phase-Plane Method (Part 2)** • Classification of singular points • Limit cycles in phase-plane • Transient analysis using phase-plane techniques

10	**Lyapunov's Indirect Method** • Introduction to Lyapunov stability theory • Direct vs indirect method • Applications to nonlinear systems
11	**Lyapunov Stability Applications** • Stability analysis using Lyapunov functions • Constructing Lyapunov functions • Engineering examples in electrical and mechanical systems
12	**Sampled Data Systems (Part 1)** • Introduction to sampled data and discrete-time systems • Z-transform basics • Pulse transfer functions
13	**Sampled Data Systems (Part 2)** • Stability analysis in the z-plane • Digital control of nonlinear systems • Industrial applications and case studies
14	**Revision**
15	**Examination**

Method of Grading Continuous Assessment

Continuous Assessment

- Assignment: 10%
- Quizzes/attendance: 10%
- Test: 20%

End of Semester Examination: 60%

k) Tutorials

Tutorial will be offered at the end of each module.

l) Group works and Assignments

The group works and assignments are designed to challenge students to think critically and solve problems in the world of electromagnetism, analyze real-world scenarios and propose solutions using the acquired knowledge of electromagnetic principles. The course also includes assignments designed to solidify your understanding through calculations, problem-solving exercises, and potentially, simulations or experiments involving electromagnetic phenomena.

m) Contemporary Issues and Industrial Relevance

Nonlinear systems play a critical role in addressing modern engineering challenges where real-world processes rarely behave linearly. From renewable energy integration and smart grids to robotics, aerospace, and autonomous vehicles, nonlinear dynamics govern system behavior and stability. Contemporary issues such as sustainability, efficiency, and safety demand advanced nonlinear modeling, analysis, and control strategies. Industrially, the course equips students with tools like phase-plane methods, Lyapunov stability, and PLC-based automation, directly relevant to manufacturing, power systems, and process industries. Mastery of nonlinear system analysis prepares engineers to innovate and solve complex, evolving problems in modern technology-driven industries.



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AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code:	EEE 431	**Course Title:**	Electric Power Principles
Semester:	First	**Credit Hours**:	**3**
Contact Hours:	Lecture - Three hours lectures per Week	**Location**:	B2-3
Lecturer(s):	Engr. Prof. O. J. Dada/Dr. Omogbai		

Textbooks and other Materials:

1. Modern Power System Analysis –D. P. Kothari & I. J. Nagarath
2. Electrical Power Systems –C. L. Wadhwa
3. A Course in Power Systems –J. B. Gupta.

a) Course Overview and description

This course provides a solid foundation in power system engineering, covering the structure and operation of electrical networks. Topics include power stations and their roles, economic operation, and power factor correction. Students will study polyphase theory, voltage profiles in D.C. and A.C. distribution, and transmission line characteristics such as inductance, capacitance, skin effect, corona discharge, and bundled conductors. The course also examines transmission materials and introduces load forecasting methods for effective planning and reliability.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Core

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcomes: upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**Pos**	**Assessment Tools**
Represent the structure of power systems and explain different types of power station	1, 2	1, 2, 8, 10	Assignment, quizzes and Exams
Assess the economic operation of power systems under various operating conditions	5	1, 2, 10, 11	Assignment, quizzes and Exams
Analyze power factor correction	3, 4	1, 2, 3	Assignment, quizzes and Exams
Explain the polyphase theory and evaluate the voltage profile in D.C. and A.C. distribution circuits	1, 2, 4, 5	1, 2, 3, 4, 5	Assignment, quizzes and Exams
Describe different types of material for overhead transmission lines and solve inductance and capacitance of transmission lines	1, 3	1, 2, 3, 5, 10	Assignment, quizzes and Exams
Explain the concept of skin effect, corona effect, bundled conductors	2	1, 2, 3, 4, 8, 10	Assignment, quizzes and Exams
Describe different types of load forecasting methods	1, 2	1, 2, 4, 6, 8, 11	Assignment, quizzes and Exams

f) Mapping of CLOs to relevant POs

Course Learning Outcomes	Programme Outcomes (POs):											
	1	2	3	4	5	6	7	8	9	10	11	12
Represent the structure of power systems and explain different types of power station	3	2						2		2		
Assess the economic operation of power systems under various operating conditions	3	3								3	2	
Analyze power factor correction	3	3	3									
Explain the polyphase theory and evaluate the voltage profile in D.C. and A.C. distribution circuits	3	3	2		3					10		
Describe different types of material for overhead transmission lines and solve inductance and capacitance of transmission lines	3	3	2		3					2		
Explain the concept of skin effect, corona effect, bundled conductors	3	2	3	2				8		2		
Describe different types of load forecasting methods	3	3		2		1		2				

Keys: 1 = Slightly related, 2 = Moderately related, 3 = Highly related

g) Expectations of Students

a. Attend 3 hours of lectures per week

b. Turn in assignments and projects as considered suitable by the tutors

c. Complete continuous assessment which involves mid-semester test

d. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of PowerPoint slides, Lecture templates-textbooks)

22) **Course Content or Outline**

Topics	Weeks
Introduction to Power Systems • Structure and components of power systems • Generation, transmission, and distribution • Types of power stations	1
Economic Operation of Power Systems • Load curves • Economic dispatch	2
Power Factor Correction • Definition and significance of power factor • Causes of low power factor • Methods of correction: • Economic benefits	3 – 4
Polyphase Systems and Distribution Circuits • Polyphase theory and balanced/unbalanced systems • Voltage profile analysis in D.C./A.C. distribution circuits • Voltage drop and regulation	5 – 6
Transmission Line Materials and Parameters • Conductors used in overhead lines: • Mechanical and electrical properties • Calculation of Inductance/Capacitance of transmission line	7 – 8
Advanced Transmission Concepts • Skin effect and Corona discharge • Bundled conductors: advantages and applications	9 – 10
Load Forecasting Techniques • Importance of load forecasting	11 - 13

• Short-term, medium-term, and long-term forecasting • Methods	
Lecture free week	13
Revision & Examination	14 – 15

j) Method of Grading Continuous Assessment

Continuous Assessment

- Assignment: 10%
- Quizzes/attendance: 10%
- Test: 20%

End of Semester Examination: 60%

k) Tutorials

Tutorial will be offered at the end of each module.

l) Group works and Assignments

The group assignments in this course are designed to translate power system theory into practical competency. Through problem-based learning (PBL), students will analyze real-world power system scenarios—such as economic dispatch, voltage profile management, and transmission line design—and propose viable engineering solutions. These activities are structured to solidify understanding of core principles, including polyphase systems, power factor correction, and electromagnetic phenomena like skin and corona effects, thereby achieving the course's learning outcomes.

m) Contemporary Issues and Industrial Relevance

This course equips students with the essential skills to address modern challenges in the evolving electrical power industry. It focuses on the analysis of power systems with a touch on economic operation. Key topics include transmission line analytic calculations, load forecasting in power systems, and industrial power factor correction for cost and carbon reduction. Ultimately, the course prepares students to contribute to developing reliable, economical, and sustainable power systems for the future.

AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code: EEE 471 **Course Title:** Digital Systems and Electronics

Semester: First **Credit Hours**: 3 hours

Contact Hours: Lecture - Two hours lectures per week

Location: Hall C-36 **Lecturer(s)**: Engr. Prof. M. O. Onibonoje, Mr. Victor Akinpelu

Textbooks and other Materials:

- Electronic Instrumentation, second edition – H. S. Kalsi, Tata McGraw Hill, 2004.
- Modern Electronic Instrumentation and Measurement Techniques –A.D. Helfrick and W.D. Cooper, PHI, 5th Edition, 2002.
- Basic Sensors and Principles -Robert A. Peura and John G. Webster
- Digital Systems (Principles and Applications)-Ronald J. Tocci, Neal S. Widmer & Gregory L. Moss)

a) Course Overview and description

Digital Systems and Electronics provides students with a comprehensive understanding of the principles and applications of modern instrumentation, sensors, and electronic systems used in measurement and industrial automation. The course introduces the fundamental concepts of instrumentation science, focusing on the role of sensors and transducers in capturing various physical parameters such as temperature, pressure, force, light, and flow. Students learn signal processing and interfacing techniques to improve data accuracy, reduce noise, and ensure effective communication between hardware and software components. Emphasis is placed on digital logic, microcontroller programming, and system integration, enabling learners to design and implement data acquisition and control systems. The course also explores noise reduction strategies, logic gate operations, and the application of expert system instrumentation, preparing students for real-world challenges in electronics and digital measurement. By analyzing industrial case studies, learners gain practical insights into how instrumentation supports efficiency, safety, and innovation in diverse engineering fields such as manufacturing, healthcare, and automation. With a balance of theoretical principles and hands-on application, EEE 471 equips students with the analytical, technical, and problem-solving skills necessary to design, evaluate, and deploy effective digital instrumentation solutions. Ultimately, the course not only builds proficiency in modern electronic systems but also fosters adaptability to evolving

technologies, bridging the gap between academic knowledge and contemporary industrial practices.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Core

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcome: Upon completion of the course, the student should be able to:	**Suitable Bloom Taxonomy Relevant to the CLO's**	**POs**	**Assessment Tools**
Explain the basic principles of instrumentation science, including the role of sensors and transducers in measuring physical quantities.	1,2,4	1,2,3	Assignment, quizzes and Exams
Demonstrate the ability to select and utilize appropriate sensors and transducers for various measurement applications, understanding their operational principles and characteristics.	3,4,5	2,4,5,6	Assignment, quizzes and Exams
Apply the signal processing techniques to filter noise to enhance signal quality in measurement systems and ensure accurate data interpretation	2,3,6	1,3,5	Assignment, quizzes and Exams
Demonstrate proficiency in programming microcontrollers for data acquisition and control applications and integrate hardware with software solutions.	3,5,6	2,3,4,5	Assignment, quizzes and Exams
Analyse industrial applications of instrumentation, identify challenges and propose effective solutions using learned concepts.	4,5,6	2,5,10,12	Assignment, quizzes and Exams

f) Mapping of CLOs to relevant POs

Course Learning Outcomes	Programme Outcomes (POs):											
	1	2	3	4	5	6	7	8	9	10	11	12
Explain the basic principles of instrumentation science, including the role of sensors and transducers in measuring physical quantities.	3	2	2									
Demonstrate the ability to select and utilize appropriate sensors and transducers for various measurement applications, understanding their operational principles and characteristics.		3		2	2	1						
Apply the signal processing techniques to filter noise to enhance signal quality in measurement systems and ensure accurate data interpretation	2		3		3							
Demonstrate proficiency in programming microcontrollers for data acquisition and control applications and integrate hardware with software solutions.		2	3	3	2							
Analyse industrial applications of instrumentation, identify challenges and propose effective solutions using learned concepts.		2			2					2		2

Keys: 1 = Slightly related, 2 = Moderately related, 3 = Highly related

g) Expectations of Students

a. Attend 3 hours of lectures per week

b. Turn in assignments and projects as considered suitable by the tutors

c. Complete continuous assessment which involves mid-semester test

d. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of PowerPoint slides, Lecture templates-textbooks)

23) **Course Content or Outline**

Week	Topics & Subtopics
1	**Basic Principles of Instrumentation Science** • Definition and scope of instrumentation • Measurement standards and units • Characteristics of instruments (accuracy, precision, sensitivity, resolution) • Classification of instruments (analog vs. digital)
2	**Sensors and Transducers – Introduction** • Principles of sensing and transduction • Types of transducers (active vs. passive) • Static and dynamic characteristics of transducers • Selection criteria for sensors
3	**Temperature Measurement Sensors** • Thermocouples and thermistors • Resistance Temperature Detectors (RTDs)

	• Infrared sensors • Applications in industry
4	**Pressure and Force Measurement Sensors** • Strain gauges and piezoelectric sensors • Bourdon tubes, diaphragms, and bellows • Load cells for force measurement • Calibration and errors
5	**Light Intensity and Flow Measurement Sensors** • Photodiodes, phototransistors, LDRs • Optical encoders • Flow sensors (electromagnetic, ultrasonic, turbine) • Industrial uses
6	**Signal Processing and Interfacing Techniques – I** • Introduction to signal conditioning • Amplifiers and filters • Analog-to-Digital (ADC) and Digital-to-Analog Conversion (DAC)
7	**Signal Processing and Interfacing Techniques – II** • Interfacing sensors to microcontrollers • Isolation and protection techniques • Data acquisition systems • Real-world interfacing examples
8	**Noise and Signal Interference – I** • Sources of noise in electronic systems

	• Electromagnetic interference (EMI) and crosstalk • Shielding and grounding techniques
9	**Noise and Signal Interference – II / Logic Gates and Operations** • Noise filtering methods (active and passive filters, digital filtering) • Logic gates (AND, OR, NOT, NAND, NOR, XOR, XNOR) • Boolean algebra and truth tables
10	**Introduction to Microcontrollers – I** • Architecture of microcontrollers • Programming basics (C/Assembly for microcontrollers) • GPIO operations and peripheral interfacing
11	**Introduction to Microcontrollers – II** • Data acquisition with microcontrollers • Control applications (PWM, timers, interrupts) • Case study: simple measurement/control project
12	**Introduction to Expert System Instrumentation** • Concept of expert systems in instrumentation • Artificial intelligence in measurement and automation • Decision-making and rule-based systems • Examples in smart sensors and smart industry
13	**Industrial Applications of Instrumentation** • Instrumentation in manufacturing • Healthcare and biomedical instrumentation • Process automation

	• Challenges and emerging trends (IoT, Industry 4.0, smart factories)
14	**REVISION**
15	**EXAMINATION**

Method of Grading Continuous Assessment

Continuous Assessment

- Assignment: 10%
- Quizzes/attendance: 10%
- Test: 20%

End of Semester Examination: 60%

k) Tutorials

Tutorial will be offered at the end of each module.

l) Group works and Assignments

The group works and assignments are designed to challenge students to think critically and solve problems in the world of electromagnetism, analyze real-world scenarios and propose solutions using the acquired knowledge of electromagnetic principles. The course also includes assignments designed to solidify your understanding through calculations, problem-solving exercises, and potentially, simulations or experiments involving electromagnetic phenomena.

m) Contemporary Issues and Industrial Relevance

Digital Systems and Electronics is highly relevant in addressing contemporary challenges driven by rapid technological advancements in automation, smart manufacturing, and data-driven industries. With the growing demand for precision, efficiency, and reliability, modern systems increasingly rely on advanced sensors, microcontrollers, and signal processing for accurate measurement and control. The course equips students to tackle industrial issues such as noise interference, system integration, and intelligent instrumentation, which are vital in sectors like healthcare, energy, and manufacturing. Its focus on digital and expert systems also aligns with Industry 4.0, IoT applications, and the push toward sustainable and innovative industrial solutions.

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AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code: EEE 502 **Course Title:** Electrical Design and Services

Semester: Second **Credit Hours**: 3 hours

Contact Hours: Lecture - Two hours lectures per week

Location: Hall B2-3 **Lecturer(s)**: Engr. Prof. M.O Onibonoje, Mr. Victor Akinpelu

Textbooks and other Materials:

- A textbook of Electrical Technology Volume I-III by B.L and A.K Theraja (2005)
-Handbook of Electrical Design Details Neil Sclater and John E.Traister (2003)
-Design of Electrical Services for Buildings Fourth Edition Barrie Rigby (2005)

a) Course Overview and description

This advanced-level course provides a comprehensive exploration of the principles and practices involved in the design and implementation of electrical services in residential, commercial, and industrial buildings. The course emphasizes both theoretical foundations and practical design considerations essential for professional electrical engineers. Key topics include the design and installation of lighting and power systems, analysis of energy supply and distribution networks, and the selection of appropriate cables, wiring methods, and accessories to ensure efficient and safe operation. Students will examine outdoor low voltage lines and cables, with a focus on technical and environmental challenges in overhead and underground installations. The course also covers protection systems and their characteristics, equipping students with the skills to design protective schemes that comply with safety standards and operational reliability. Emphasis is placed on proper earthing methods and the testing of electrical installations to ensure compliance with national and international codes. An in-depth study of illumination principles complements the course, enabling students to design effective lighting systems based on factors such as lux levels, uniformity, energy efficiency, and user comfort. By the end of the course, students will be capable of producing detailed electrical service designs that are compliant, efficient, and tailored to specific building requirements, preparing them for roles in consultancy, construction, or further specialization in electrical systems engineering.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Core

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcome: Upon completion of the course, the student should be able to:	**Suitable Bloom Taxonomy Relevant to the CLO's**	**POs**	**Assessment Tools**
Design and specify lighting and power distribution systems for various building types, considering energy efficiency and compliance with relevant standards.	5,6	2,3,10	Assignment, quizzes and Exams
Analyze and plan energy supply and distribution schemes, including transformer sizing, load estimation, and distribution panel arrangements.	4,5,6	2,6,10	Assignment, quizzes and Exams
Compute the details of appropriate cables, wiring systems, and accessories based on load characteristics, environmental conditions, and installation requirements for building	3,5	1,2,9,10	Assignment, quizzes and Exams
Demonstrate protection principles to electrical installations, including circuit protection, fault detection, and coordination of protective devices.	2,3,5,6	8,9,11	Assignment, quizzes and Exams
Implement lighting designs according to given specifications to ensure the safety, reliability, and compliance of electrical installations, aesthetics, and practicality	5,6	2,3,4,10	Assignment, quizzes and Exams

f) Mapping of CLOs to relevant POs

Course Learning Outcomes	Programme Outcomes (POs):											
	1	2	3	4	5	6	7	8	9	10	11	12
Design and specify power and lighting distribution systems for various building types, considering energy efficiency and compliance with relevant standards		3	3							3		
Analyze and plan energy supply and distribution schemes, including transformer sizing, load estimation, and distribution panel arrangements.		3				3				3		
Compute the details of appropriate cables, wiring systems, and accessories based on load characteristics, environmental conditions, and installation requirements for building	2	3							3	3		
Demonstrate protection principles to electrical installations, including circuit protection, fault detection, and coordination of protective devices.								3	3		3	
Implement lighting designs according to given specifications to ensure the safety, reliability, and compliance of electrical installations, aesthetics, and practicality		3	3	3						3		

Keys: 1 = Slightly related, 2 = Moderately related, 3 = Highly related

g) Expectations of Students

a. Attend 3 hours of lectures per week

b. Turn in assignments and projects as considered suitable by the tutors

c. Complete continuous assessment which involves mid-semester test

d. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of PowerPoint slides, Lecture templates-textbooks)

24) **Course Content or Outline**

Topics	Weeks
Lighting and Power Installation: • Considerations necessary for Electrical installation • Electrical project documents and industry standards • Stages involved in power and lighting installation. • Accessories	1 – 3
Energy Supply and Distribution: • Electrical Power Distribution systems • Energy Vending and contracts • Implication of power factor on single phase and three phase systems	4 – 5
Choice of Cable, Wiring systems and Accessories: • Requirements for Building Wiring • Electrical cable types, color codes and cable rating • Standard wiring templates • Building wiring systems and their required components	6 – 7
Low Voltage Line and Cables (Load Calculation): • Introduction to concept of simultaneity • Using simultaneity concepts in load calculations in both lighting and power switchgears.	8 – 9
Protection and Characteristics of low voltage equipment: • Wiring Protection systems • Types of Faults common to building electrical systems and protection methods to mitigate them	10 – 11
Illumination: • Considerations in lighting design and basic parameters used to describe lighting. • Luminaires and protection classes of lighting equipment • Lighting design calculations and modern lighting design methods	12
Lecture free week	13
Revision & Examination	14 - 15

Method of Grading Continuous Assessment

Continuous Assessment

- Assignment: 10%
- Quizzes/attendance: 10%
- Test: 20%

End of Semester Examination: 60%

k) Tutorials

Tutorial will be offered at the end of each module.

l) Group works and Assignments

The group works and assignments are designed to challenge students to think critically and solve problems in the world of electromagnetism, analyze real-world scenarios and propose solutions using the acquired knowledge of electromagnetic principles. The course also includes assignments designed to solidify your understanding through calculations, problem-solving exercises, and potentially, simulations or experiments involving electromagnetic phenomena.

m) Contemporary Issues and Industrial Relevance

The course in electrical services design is highly relevant today as it equips professionals to create efficient, safe, and innovative electrical systems that meet modern demands in both residential and industrial settings. The course also emphasizes future-proofing electrical infrastructure to accommodate flexible, interconnected environments such as modern workplaces and smart buildings, enhancing sustainability, reliability, and user experience. In industry, it supports the design of robust power distribution, automation, and backup systems critical for operational efficiency and scalability, aligning with trends in energy efficiency, digitalization, and predictive maintenance.

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AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code: EEE 506 **Course Title:** Digital Systems Design

			with VHDL
Semester:	Second	**Credit Hours**:	3
Contact Hours:	Lecture - Two hours lectures per Week	**Location**:	B0-4
Lecturer(s):	Engr. Dr. O. T. Ibitoye and Engr. O. Imahe		

Textbooks and other Materials:

i. Introduction to VLSI design flow. By Sneh Saurabh. Cambridge, 2023.
ii. Free Range VHDL. B. Mealy and F. Tappero, 2023.

a) Course Overview and description

This course covers the fundamental concepts of Finite State Machine: definition, mealy and Moore models, state diagram, state table, transition table. Sequential circuits design using flip-flops, asynchronous, and synchronous circuit design. Algorithm State Machine. Design examples and exercises. Structured Design: Design constructs, Design Levels, Geometry-based interchange formats, Computer aided electronic system design tools, Schematic circuit capture, Hardware description languages, Design process (simulation, synthesis), Structural design decomposition. Introduction to VHDL: VHDL language abstractions, Design hierarchies, VHDL component, Lexical description, VHDL source file, Data types, Data objects, Language statements, Concurrent VHDL, Sequential VHDL, Advanced features of VHDL (library, package and subprograms). Structural level modelling, Register-Transfer level modelling, FSM with datapath level modelling, Algorithmic level modelling. Introduction of ASIC, Types of ASIC, ASIC design process, Standard cell ASIC synthesis, FPGA Design Paradigm, FPGA synthesis, FPGA/CPLD Architectures. VHDL Design: Top-down design flow, Verification, simulation alternatives, simulation speed, Formal verification, Recommendations for verification, Writing RTL VHDL code for synthesis, top-down design with FPGA. VHDL synthesis, optimization and mapping, constraints, technology library, delay calculation, synthesis tool, synthesis directives. Computer-aided design of logic circuits.

b) **Pre-requisites:** Nil

c) **Co-requisite(s):** Nil

d) **Role in Curriculum:** Compulsory

e) **Course Learning Outcome (CLO) and Programme Outcomes (PO)**

Course Learning Outcomes: upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**POs**	**Assessment Tools**
Develop combinational and sequential circuits with VHDL; utilizing various data types and objects, language statements, and libraries and packages.	4/5	1, 2, 11	Assignment, quizzes and Exams
Demonstrate an understanding of asynchronous and synchronous design with VHDL.	3	2, 3, 6, 10	Assignment, quizzes and Exams
Design finite state machines, and differentiate between Mealy and Moore finite state machines. Demonstrate understanding of structured design architecture to build larger digital systems/modules.	4/5	1, 4, 6	Assignment, quizzes and Exams
Demonstrate an understanding of various VHDL models, their key components and use-cases.	3	1, 3, 4	Assignment, quizzes and Exams
Propose Application Specific Integrated Circuits (ASIC) and Field-Programmable Gate Arrays (FPGA). Design ASIC and FPGA using various programming tools.	4/5	5, 9, 12	Assignment, quizzes and Exams
Evaluate VHDL design, synthesis and optimization using various simulation and verification tools.	5	5, 8	Assignment, quizzes and Exams

f) Mapping of CLOs to relevant POs

Course Learning Outcomes	Programme Outcomes											
	1	2	3	4	5	6	7	8	9	10	11	12
Develop combinational and sequential circuits with VHDL; utilizing various data types and objects, language statements, and libraries and packages.	2	2									2	
Demonstrate an understanding of asynchronous and synchronous design with VHDL.		2	2			2				2		
Design finite state machines, and differentiate between Mealy and Moore finite state machines. Demonstrate understanding of structured design architecture to build larger digital systems/modules.	2			2		2						
Demonstrate an understanding of various VHDL models, their key components and use-cases.	3		2	2								
Propose Application Specific Integrated Circuits (ASIC) and					2				2			2

Field-Programmable Gate Arrays (FPGA). Design ASIC and FPGA using various programming tools.												
Evaluate VHDL design, synthesis and optimization using various simulation and verification tools.					**2**			**2**				

Keys: 1 = slightly related, 2 = moderately related, 3 = highly related

g) Expectations of Students

a. Attend 2 hours of lectures per week

b. Turn in assignments considered suitable by the tutors

c. Complete continuous assessment which involves mid-semester test

d. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of Power Point slides, Lecture templates-handouts)

i) Course Content or Outline

Topics	Weeks
General digital system concept	1
Introduction to VHDL	2
VHDL language abstractions, Design hierarchies, VHDL component, Lexical description, VHDL source file, Data types, Data objects, Language statements, Concurrent VHDL, Sequential VHDL, Advanced features of VHDL (library, package and subprograms).	3 - 6
Structural level modeling, Register-Transfer level modeling, FSM with data path level modeling, and Algorithmic level modeling	7 - 8
Introduction of ASIC, Types of ASIC, ASIC design process, Standard cell ASIC synthesis, FPGA Design Paradigm, FPGA synthesis, FPGA/CPLD Architectures.	9-10
VHDL Design: Top-down design flow, Verification, simulation alternatives, simulation speed, Formal verification, Recommendations for verification, Writing RTL VHDL code for synthesis, top-down design with FPGA. VHDL synthesis, optimization and mapping, constraints, technology library, delay calculation, synthesis tool, synthesis directives.	11-12
Lecture free week	13
Revision & Examination	14 - 15

j) Method of Grading Continuous Assessment

Continuous Assessment (Assignments): 20%

Continuous Assessment (Test):20%

End of Semester Examination: 60%

k) Tutorials

Tutorials are taken at the end of each module.

l) Term Paper and Assignments

A term paper focusing on the analysis of techniques and tools for the development of ASIC and FPGA.

m) Contemporary Issues and Industrial Relevance

The course equips students with the knowledge to develop and deploy a medium scale ASIC.



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AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code:	EEE 511	**Course Title:**	Digital Signal Processing
Semester:	First	**Credit Hours**:	**3**
Contact Hours:	Lecture - Two hours lectures per Week	**Location**:	B2-3
Lecturer(s):	Engr. Dr. A. O. Salau		

Textbooks and other Materials:

1. S. Esakkirajan , T. Veerakumar, Badri N Subudhi, Digital Signal Processing, Springer Singapore, XVIII, 523p, ISBN: 978-981-99-6751-3. DOI: 10.1007/978-981-99-6752-0
2. Maurice Bellanger; Benjamin A. Engel, Digital Signal Processing: Theory and Practice, Published by IEEE, 370p.

a) Course Overview and description

This course discusses the advantages of digital over analogue signal processing, highlights the problems of digitization, and presents an overview of the applications of digital signal processing (DSP), basic elements of DSP system. In addition, the course covers the fundamental concepts of digital processing of analogue signals: Sampling of analogue signals, sampling theorem, aliasing, quantization, noise, and coding, types and selection of ADC/DAC, Sigma-delta ADC. Furthermore, analytical tools such as z-transform, properties, transfer function, inverse z-transform, z-plane poles and zeros, analysis of linear time-invariant in z-domain, system stability are studied. The concept of discrete Fourier analysis and its various categories are studied. Lastly, the digital filter definition

and types are presented.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Compulsory

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcomes: upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**POs**	**Assessment Tools**
Understand the evolution from analogue to digital systems and the benefits of digital over analogue signal processing.	2	1, 2	Assignment, quizzes and Exams
Understand the use of mathematical analytical tools such as z-transform, properties, transfer function, inverse z-transform, z-plane poles and zeros, analysis of linear time-invariant in z-domain for system stability.	2, 3	1, 2, 3, 6	Assignment, quizzes and Exams
Analyze the Fourier analysis, discrete Fourier analysis and their various categories.	4	2,3,4,5	Assignment, quizzes and Exams
Understand the concept of digital filters, types, and application.	2, 3	1, 6	Assignment, quizzes and Exams
Practicalise the application of DSP in audio and video systems.	3, 5	1, 2, 3, 5	Assignment, quizzes and Exams

f) Mapping of CLOs to relevant POs

Course Learning Outcomes	Programme Outcomes											
	1	2	3	4	5	6	7	8	9	10	11	12
Understand the evolution from analogue to digital systems and the benefits of digital over analogue signal processing.	3	2										
Understand the use of mathematical analytical tools such as z-transform, properties, transfer function, inverse z-transform, z-plane poles and zeros, analysis of linear time-invariant in z-domain for system stability.	3	3	2			3						
Analyze the Fourier analysis, discrete Fourier analysis and their various categories.		3	3	2	2							
Understand the concept of digital filters, types, and application.	3					3						
Practicalise the application of DSP in audio and video systems.	3	2	3		3							

Keys: 1 = slightly related, 2 = moderately related, 3 = highly related

g) Expectations of Students

a. Attend 2 hours of lectures per week

b. Turn in assignments considered suitable by the tutors

c. Complete continuous assessment which involves mid-semester test

d. Complete a $2^1/_2$-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of Power Point slides, Lecture templates-handouts)

i) Course Content or Outline

Topics	Weeks
• Signal representation in the time domain. • Basic signal processing concepts.	1-2
Fourier Transform: • Discrete Fourier Transform	3 - 5
• Sampling theorem	6 - 7
Linear time invariant systems: • Continuous-Time systems • Discrete-Time systems.	8 - 9
Discrete convolution: • Properties of associativity, commutativity, and distributivity. • Concept of circular convolution	10
• Z-Transform	11-12
Lecture free week	13
Revision & Examination	14 - 15

j) Method of Grading Continuous Assessment

Continuous Assessment (Assignments): 20%

Continuous Assessment (Test): 20%

End of Semester Examination: 60%

k) Tutorials

Tutorials are taken at the end of each module.

l) Term Paper and Assignments

A term paper focus on comparative analysis of digital and analog signal processing in modern applications. This will focus on the advantages, limitations, and relevance in areas like communications, imaging, and healthcare.

m) Contemporary Issues and Industrial Relevance

The course equips students with the knowledge to address critical challenges digital filter design, conversion of analog signals to digital. Contemporary issues in Digital Signal Processing (DSP) are shaped by the rapid growth of data-driven technologies, the need for real-time processing, and the demand for energy-efficient systems in modern devices. With the proliferation of IoT, 5G/6G

networks, and multimedia applications, DSP must address challenges such as quantization noise, aliasing, and digitization errors while maintaining high accuracy and speed. Emerging trends also highlight the integration of DSP with artificial intelligence and machine learning for advanced pattern recognition and adaptive filtering, while cybersecurity and ethical concerns around signal manipulation, surveillance, and privacy further emphasize the need for robust and secure DSP techniques. Additionally, the evolution of specialized hardware platforms such as GPUs, FPGAs, and dedicated DSP processors underscores the importance of balancing computational efficiency with flexibility in implementation. The industrial relevance of DSP spans across telecommunications, consumer electronics, healthcare, automotive systems, defense, and entertainment. In telecommunications, DSP enables reliable data transmission through modulation, compression, and error correction, while in healthcare it underpins diagnostic imaging, biomedical signal analysis, and hearing aids. Modern automobiles and autonomous systems rely on DSP for radar, LiDAR, and real-time sensor fusion, whereas defense applications use it for secure communication, radar, and sonar systems. Consumer technologies such as smartphones, digital cameras, and streaming platforms also depend heavily on DSP for noise reduction, image enhancement, and data compression. By providing the analytical and practical tools to process, analyze, and interpret signals effectively, DSP remains central to advancing innovations in multiple industries and addressing the technological needs of a data-driven world.



AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code:	EEE 554	**Course Title:**	Power Systems Communication and Control
Semester:	Second	**Credit Hours**:	**3**
Contact Hours:	Lecture - Three hours lectures per Week	**Location**:	B2-3
Lecturer(s):	Engr. Prof. O. J. Dada/Dr. Omogbai		

Textbooks and other Materials:

1. D. P. Kothari & I. J. Nagarath (2009), Modern Power System Analysis, 3rd Edition, Tata McGraw Hill Education Private Ltd.
2. C. L. Wadhwa (2009), Electrical Power Systems, 4th Edition, New Age International Ltd.
3. J. B. Gupta, (2021), A Course in Power Systems, 11th Edition, S k Kataria & Sons
4. Journal and Conference Papers.

a) Course Overview and description

This course provides a comprehensive exploration of electric power systems, focusing on the theoretical and practical aspects of transmission line performance through Kirchhoff's laws and Ohm's law. Students will analyze power generation, voltage control mechanisms, and the phenomena of voltage collapse within large-scale grids. The curriculum delves into the formulation of mathematical models for power system stability and the interpretation of system responses under various disturbances. It also covers the principles and justification for using power line carrier systems for high-frequency communication. Finally, the course critically assesses the impact of deregulation, smart grid, and micro-grid technologies on the Nigerian power sector, emphasizing the integration of distributed generation for enhanced reliability and efficiency.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Elective

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcomes: upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**Pos**	**Assessment Tools**
Evaluate the performance of transmission lines using Kirchhoff's voltage and current, and Ohm's law, etc.	5	1, 2	Assignment, quizzes and Exams
Analyse the power generation and voltage control, and voltage collapses of electricity power grid.	4	1, 2, 5	Assignment, quizzes and Exams
Formulate the mathematical equations governing power system stability studies.	6	1, 2	Assignment, quizzes and Exams
Interpret the stability of power systems under different disturbances	2, 4	2, 5	Assignment, quizzes and Exams
Understand and justify the use of power line carrier systems for high-frequency communication in the power industry	2, 5	1, 4, 6	Assignment, quizzes and Exams
Assess the application of deregulation and smart/micro-grid technologies to the Nigerian power grid and the integration of distributed generation	5	3, 10, 11	Assignment, quizzes and Exams

f) Mapping of CLOs to relevant POs

Course Learning Outcomes	Programme Outcomes (POs):											
	1	2	3	4	5	6	7	8	9	10	11	12
Evaluate the performance of transmission lines using Kirchhoff's voltage and current, and Ohm's law, etc.	3	3										
Analyse the power generation and voltage control, and voltage collapses of electricity power grid.	3	3			2							
Formulate the mathematical equations governing power system stability studies.	3	3										
Interpret the stability of power systems under different disturbances		3			2	1	1	1			1	1
Understand and justify the use of power line carrier systems for high-frequency communication in the power industry	2			3		2						
Assess the application of deregulation and smart/micro-grid technologies to the Nigerian power grid and the integration of distributed generation			3							3	3	

Keys: 1 = Slightly related, 2 = Moderately related, 3 = Highly related

g) Expectations of Students

a. Attend 3 hours of lectures per week

b. Turn in assignments and projects as considered suitable by the tutors

c. Complete continuous assessment which involves mid-semester test

d. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of PowerPoint slides, Lecture templates-textbooks)

25) **Course Content or Outline**

Topics	**Weeks**
Transmission line theory ▪ Short line. ▪ Medium line, ▪ Long line, ▪ Two Port Network	1 - 4
Power systems studies ▪ Power generation & voltage control ▪ Voltage collapse, contingency planning ▪ Automatic voltage regulators & regulating transformers ▪ Power system stability studies	5 – 8
Power line system communication ▪ Data transmission and power line carrier	9
Grid and Distributed Generations ▪ Electricity deregulation and market operations ▪ Integration of distributed generations into the grid ▪ Smart Grids	10 – 12
Lecture free week	13
Revision & Examination	14 – 15

j) Method of Grading Continuous Assessment

Continuous Assessment

- Assignment: 10%
- Quizzes/attendance: 10%
- Test: 20%

End of Semester Examination: 60%

k) Tutorials

Tutorial will be offered at the end of each module.

l) Group works and Assignments

Group work and assignments in this course are designed to reinforce the core learning outcomes by encouraging collaborative analysis and practical application. Students work in teams to evaluate transmission line performance using Kirchhoff's laws and Ohm's law, simulate power system stability under disturbances, and model voltage control scenarios. Assignments include case studies on voltage collapse, mathematical formulation of stability equations, and the use of power line carrier systems for communication. Groups also assess the impact of deregulation and smart grid technologies on Nigeria's power sector, proposing integration strategies for distributed generation. This approach fosters critical thinking, teamwork, and real-world problem-solving.

m) Contemporary Issues and Industrial Relevance

Contemporary issues and industrial relevance are integrated into the course through targeted group projects and assignments that reflect real-world challenges in the power sector. Students critically examine voltage collapse incidents, power system disturbances, and the role of high-frequency communication technologies in modern grid operations. Assignments also explore the impact of deregulation, smart grid innovations, and distributed generation on the Nigerian electricity market, fostering awareness of evolving industry practices. By engaging with current trends and technologies, students develop practical insights and problem-solving skills aligned with the demands of today's power industry.

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AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code:	EEE 553	**Course Title:**	Power System Engineering
Semester:	First	**Credit Hours**:	**3**
Contact Hours:	Lecture - Three hours lectures per Week	**Location**:	B2-3
Lecturer(s):	Engr. Prof. O. J. Dada/Dr. Omogbai		

Textbooks and other Materials:

1. D. P. Kothari & I. J. Nagarath, Modern Power System Analysis
2. J. B. Gupta, A Course in Power Systems.
3. Electrical Power Systems - C. L. Wadhwa

a) Course Overview and description

This course provides a comprehensive foundation in power system analysis, equipping students to model, analyze, and solve key operational challenges. Students will learn to represent complex power networks using one-line diagrams and per-unit systems, leading to the application of load flow methods like Gauss-Seidel and Newton-Raphson to determine steady-state voltages and power flows. The curriculum extends to both symmetrical and unsymmetrical fault analysis, enabling the calculation of fault currents, the design of mitigation strategies, and the modeling of sequence networks. Finally, the course covers the fundamental principles of power system protection, including fault clearing sequences and discrimination methods, ensuring a holistic understanding of system reliability and security.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Core

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcomes: upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**Pos**	**Assessment Tools**
Explain and represent power systems using one-line diagrams, per-unit analysis, and equivalent impedance models.	1, 2	1, 2, 4	Assignment, quizzes and Exams
Apply load flow methods to determine bus voltages, power injections, and system operating conditions.	3	1, 2, 4, 5	Assignment, quizzes and Exams
Construct and analyze bus admittance and impedance matrices for power system studies.	4	1, 2, 4	Assignment, quizzes and Exams
Evaluate short-circuit conditions by calculating fault currents, fault levels, and the impact of protective devices.	3, 4	1, 2, 5	Assignment, quizzes and Exams
Compute fault levels for symmetrical faults and design mitigation strategies, such as current limiting reactors, to protect power system equipment.	5, 6	1, 2, 3, 10	Assignment, quizzes and Exams
Model sequence networks for unsymmetrical fault analysis and calculate fault currents in different parts of the system.	4, 5	1, 2, 4, 5	Assignment, quizzes and Exams
Explain the principles of power system protection, including the operational	1, 2	1, 2, 7, 9, 10	Assignment, quizzes and Exams

sequence of fault clearing and the methods used for fault discrimination.			

f) Mapping of CLOs to relevant POs

Course Learning Outcomes	Programme Outcomes (POs):											
	1	2	3	4	5	6	7	8	9	10	11	12
Explain and represent power systems using one-line diagrams, per-unit analysis, and equivalent impedance models.	3	2		2								
Apply load flow methods to determine bus voltages, power injections, and system operating conditions.	3	3		2	2							
Construct and analyze bus admittance and impedance matrices for power system studies.	3	3		2								
Evaluate short-circuit conditions by calculating fault currents, fault levels, and the impact of protective devices.	2	3	1							2		
Compute fault levels for symmetrical faults and design mitigation strategies, such as current limiting reactors, to protect power system equipment.	2	3	3							3	1	2
Model sequence networks for unsymmetrical fault analysis and calculate fault currents in different parts of the system.	3	3		2	2							
Explain the principles of power system protection, including the operational sequence of fault clearing and the methods used for fault discrimination.	3	2					2		3	3		

Keys: 1 = Slightly related, 2 = Moderately related, 3 = Highly related

g) Expectations of Students

a. Attend 3 hours of lectures per week

b. Turn in assignments and projects as considered suitable by the tutors

c. Complete continuous assessment which involves mid-semester test

d. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of PowerPoint slides, Lecture templates-textbooks)

26) **Course Content or Outline**

Topics	Weeks
Power System Representation One-Line Diagrams Per-Unit System Equivalent Impedance Models	1
Load Flow Analysis Fundamentals of Load Flow Load Flow Methods System Operating Conditions	2
Network Matrices Bus Admittance Matrix (Ybus) Bus Impedance Matrix (Zbus)	3 – 4
Short-Circuit Analysis Symmetrical Faults Protective Device Impact	5 – 6
Fault Mitigation Strategies Current Limiting Reactors System Hardening Techniques	7 – 8
Unsymmetrical Fault Analysis Sequence Networks Fault Current Calculations	9 – 10
Power System Protection Principles Fault Clearing Sequence Fault Discrimination Methods Protection Schemes	11-12
Lecture free week	13
Revision & Examination	14 – 15

j) Method of Grading Continuous Assessment

Continuous Assessment

- Assignment: 10%
- Quizzes/attendance: 10%
- Test: 20%

End of Semester Examination: 60%

k) Tutorials

Tutorial will be offered at the end of each module.

l) Group works and Assignments

Group assignments in this course is emphasized through problem-based learning (PBL) to apply power system analysis and protection concepts. Students will use one-line diagrams, per-unit analysis, load flow methods, and fault studies to evaluate system conditions, design mitigation strategies, and analyze protection schemes, thereby reinforcing the course learning outcomes.

m) Contemporary Issues and Industrial Relevance

This course develops practical skills in modern power system analysis and protection, addressing industrial needs for reliability, security, and efficiency. Students learn to apply load flow methods, fault analysis, and protection principles to real-world challenges such as grid stability, equipment protection, and sustainable system operation. By integrating theoretical tools with industry-relevant practices, the course prepares graduates to tackle contemporary issues in power system planning and operation.

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AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code: EEE 561 **Course Title:** Control Engineering

Semester: First **Credit Hours**: 3 hours

Contact Hours: Lecture - Two hours lectures per week

Location: Hall B-33 **Lecturer(s)**: Engr. Prof. M.O Onibonoje, Mr. Victor Akinpelu

Textbooks and other Materials:

· Linear State-Space Control Systems – Robert L. Williams

· Modern Control Engineering – Katsuhiko Ogata

a) Course Overview and description

Control Engineering provides students with an advanced understanding of modern control theory through the state-space approach, equipping them with both theoretical foundations and practical skills for the analysis and design of dynamic systems. The course begins with the formulation of state-space models and their relationship to transfer functions, enabling learners to bridge classical and modern control representations. Students will explore the role of eigenvalues and eigenvectors in determining system response and stability, while also mastering critical concepts such as controllability and observability that form the backbone of system design. Building on this foundation, the course emphasizes the design and implementation of state feedback controllers, observers, and compensators, with a strong focus on optimal control through the Linear Quadratic Regulator (LQR) framework. Real-world applications, including robotics, circuit synthesis, and signal processing, are integrated to illustrate the industrial relevance of control strategies. Additionally, students will gain hands-on experience with computer-aided tools for simulation, modeling, and optimization, preparing them for contemporary engineering challenges. By the end of the course, learners are expected to demonstrate the ability to analyze, design, and optimize control systems that meet performance criteria under practical constraints, while also appreciating the interdisciplinary applications of control engineering across electrical, mechanical, and electronic domains. This course thus fosters problem-solving, innovation, and lifelong learning, while strengthening students' readiness to contribute effectively in research, industry, and advanced studies.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Core

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcome: Upon completion of the course, the student should be able to:	**Suitable Bloom Taxonomy Relevant to the CLO's**	**POs**	**Assessment Tools**
Analyse state-space representations for linear systems and understand their significance in control theory.	2,4,6	1	Assignment, quizzes and Exams
Demonstrate proficiency in converting between transfer functions and state-space representations and recognize implications for system behavior.	2,3,4	1,2	Assignment, quizzes and Exams
Analyse eigenvalues and eigenvectors to assess system stability and response, applying concepts to real-world scenarios.	1,4,6	2,3	Assignment, quizzes and Exams
Design and implement state feedback control systems and observers, including compensators and Linear Quadratic Regulators	3,5,6	2,3,5	Assignment, quizzes and Exams
Optimize control strategies in practical applications such as robotics and signal processing using computer-aided tools.	2,3,6	3,4,6	Assignment, quizzes and Exams

f) Mapping of CLOs to relevant POs

Course Learning Outcomes	Programme Outcomes (POs):											
	1	2	3	4	5	6	7	8	9	10	11	12
Analyse state-space representations for linear systems and understand their significance in control theory.	3											
Demonstrate proficiency in converting between transfer functions and state-space representations and recognize implications for system behavior.	3	3										
Analyse eigenvalues and eigenvectors to assess system stability and response, applying concepts to real-world scenarios.		3	2									
Design and implement state feedback control systems and observers, including compensators and Linear Quadratic Regulators		3	3		2							
Optimize control strategies in practical applications such as robotics and signal processing using computer-aided tools.			3	3		2						

Keys: 1 = Slightly related, 2 = Moderately related, 3 = Highly related

g) Expectations of Students

a. Attend 3 hours of lectures per week
b. Turn in assignments and projects as considered suitable by the tutors
c. Complete continuous assessment which involves mid-semester test
d. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of PowerPoint slides, Lecture templates-textbooks)

12) **Course Content or Outline**

Week	Topics & Subtopics
1	**State-space description of linear systems** • Introduction to state variables and vectors • Derivation of state equations from physical systems • State-space representation of electrical and mechanical systems
2	**State-space description of linear systems (cont'd)** • Matrix representation of dynamic systems • Homogeneous vs non-homogeneous equations • State transition matrix and solution of state equations
3	**Transfer functions and state-space realizations** • Transfer function derivation from state equations • State-space realization of transfer functions • Canonical forms (controllable, observable, diagonal)
4	**Transfer functions and realizations (cont'd)** • Minimal realizations

	• Equivalence of transfer function and state-space models • Examples in electrical and mechanical systems
5	**Eigenvalues, eigenvectors, and system response** • Eigenvalue/eigenvector computation • Relation to modes of system response • Natural response and stability interpretation
6	**System stability analysis** • Stability criteria (internal, BIBO) • Role of eigenvalues in stability • Examples of stable vs unstable systems
7	**Controllability and observability** • Definitions and concepts • Kalman controllability and observability tests • Practical implications in system design
8	**State feedback control** • Pole placement method • Design of controllers using state feedback • Limitations of feedback control
9	**Observers and modal control** • Full-order and reduced-order observers • Modal control design • Observer-based system design
10	**Combined control–observer compensator**

	• Concept and structure • Separation principle • Practical applications and examples
11	**Linear Quadratic Regulator (LQR)** • Optimal control formulation • Quadratic cost function • Solution of the Riccati equation • LQR design examples
12	**Applications of control engineering** • Robotics systems control • Circuit synthesis and design • Signal processing applications
13	**Computer-aided analysis and design** • MATLAB/Simulink for control design • Simulation and performance evaluation • Case studies and project presentations

Method of Grading Continuous Assessment

Continuous Assessment

- Assignment: 10%
- Quizzes/attendance: 10%
- Test: 20%

End of Semester Examination: 60%

k) Tutorials

Tutorial will be offered at the end of each module.

l) Group works and Assignments

The group works and assignments are designed to challenge students to think critically and solve problems in the world of electromagnetism, analyze real-world scenarios and propose solutions using the acquired knowledge of electromagnetic principles. The course also includes assignments designed to solidify your understanding through calculations, problem-solving exercises, and potentially, simulations or experiments involving electromagnetic phenomena.

m) Contemporary Issues and Industrial Relevance

Control Engineering remains at the forefront of contemporary technological advancement, addressing critical issues such as automation, system optimization, and energy efficiency in modern industries. With the increasing reliance on robotics, autonomous systems, and intelligent manufacturing, mastery of state-space methods and optimal control strategies equips engineers to design reliable and adaptable solutions. In fields like aerospace, automotive, power systems, and signal processing, control techniques ensure safety, stability, and performance under complex constraints. The course's emphasis on computer-aided design tools further aligns with industry demands for digitalization, smart technologies, and innovative problem-solving in rapidly evolving engineering environments..

Course Code: EEE 513 Course Title: Communication Systems Engineering

Semester: First **Credit Hours**: 3 hours

Contact Hours: Lecture - Three hours lectures per week

Location: Hall B2-5 **Lecturer(s)**: Engr. Dr. A. O. Ojo & Engr. A. Akinola

Textbooks and other Materials:

- Owolabi & Ojo (2022), Electrical Communication Principles, Systems and Applications, 1st Edition, Matprints Inc.

- Marcelo & Valdemar (2005), Communication Systems, 3rd Edition, Springer.

a) Course Overview and description

This course provides a comprehensive introduction to the principles and applications of modern communication systems, focusing on both foundational and advanced topics. It covers signal representation, Fourier analysis, and the distinction between analog and digital signals, laying the groundwork for understanding modulation techniques like AM, FM, and digital schemes such as BASK and M-ary modulation. The course explores wireless communication, including cellular systems, smart antennas, and the internet's role in global connectivity. The concluding part of this course covers topics that include electromagnetic wave propagation, radar systems, and satellite communication, with emphasis on real-world challenges like atmospheric effects. Emerging technologies such as software-defined radio (SDR) and cognitive radio (CR) are highlighted, focusing on their roles in dynamic spectrum management and efficient spectrum utilization.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Core

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcomes: upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**POs**	**Assessment Tools**
Develop a comprehensive understanding of signal representation, including analog and	2, 3, 4	1, 2, 5	Assignment, quizzes and Exams

digital signals, Fourier series, Fourier transforms, and their applications in analyzing communication signals.			
Explore the principles and applications of analog modulation techniques like AM and FM, and digital modulation schemes such as BASK, BFSK, and M-ary modulation, including their roles in communication systems.	2, 3	1, 2	Assignment, quizzes and Exams
Investigate line and channel coding techniques for error prevention and correction in communication systems to ensure reliable signal transmission and reception.	2, 3, 4	2, 4, 5	Assignment, quizzes and Exams
Analyze the principles of electromagnetic wave propagation, the impact of atmospheric conditions, satellite communication, radar systems, and the role of antennas, including smart antennas, in modern communication.	2, 4, 5	1, 5	Assignment, quizzes and Exams
Understand the functionalities of software-defined radio (SDR) and cognitive radio (CR) for dynamic spectrum management, emphasizing their significance in addressing spectrum scarcity and improving wireless communication efficiency.	4, 5, 6	2, 4, 10	Assignment, quizzes and Exams
Study the architecture and technologies underlying the internet and cellular communication systems, including spectrum regulation, GSM generations, and the role of mobile	2, 3, 4	1, 4	Assignment, quizzes and Exams

switching centers in ensuring effective connectivity.			

f) Mapping of CLOs to relevant POs

Course Learning Outcomes	Programme Outcomes (POs):											
	1	2	3	4	5	6	7	8	9	10	11	12
Develop a comprehensive understanding of signal representation, including analog and digital signals, Fourier series, Fourier transforms, and their applications in analyzing communication signals.	3	2			3							
Explore the principles and applications of analog modulation techniques like AM and FM, and digital modulation schemes such as BASK, BFSK, and M-ary modulation, including their roles in communication systems.	3	2										
Investigate line and channel coding techniques for error prevention and correction in communication systems to ensure reliable signal transmission and reception.		3		2	3							
Analyze the principles of electromagnetic wave propagation, the impact of atmospheric conditions, satellite communication, radar systems, and the role of antennas, including smart antennas, in modern communication.	3				3							
Understand the functionalities of software-defined radio (SDR) and cognitive radio (CR) for dynamic spectrum management, emphasizing their significance in addressing spectrum scarcity and improving wireless communication efficiency.		3		3						3		

Study the architecture and technologies underlying the internet and cellular communication systems, including spectrum regulation, GSM generations, and the role of mobile switching centers in ensuring effective connectivity.	**3**			**3**								

Keys: 1 = Slightly related, 2 = Moderately related, 3 = Highly related

g) Expectations of Students

a. Attend 3 hours of lectures per week

b. Turn in assignments and projects as considered suitable by the tutors

c. Complete continuous assessment which involves mid-semester test

d. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of PowerPoint slides, Lecture templates-textbooks)

13) **Course Content or Outline**

Topics	Weeks
Analyze Signals and Fourier Transformations: • Analog and Digital Signals: Characteristics, types, and differences, Time Domain and Frequency Domain Representation. • Fourier Series and its application to periodic signals, Fourier Transform and its role in signal analysis, Signal Processing using Fourier Transforms. • Signal Spectra and Bandwidth, Sampling Theorem and its importance in signal conversion, Sinusoidal Signals and their Phasor Representation.	1 – 3
Understand Analog and Digital Modulation Techniques • Analog Modulation Techniques: Amplitude Modulation (AM), Frequency Modulation (FM), Principles and Applications of AM and FM, Digital Modulation: Binary Amplitude Shift Keying (BASK), Binary Frequency Shift Keying (BFSK). • M-ary Modulation Schemes: M-ary Amplitude Shift Keying (M-ASK), M-ary Frequency Shift Keying (M-FSK), Phase Modulation and Phase Shift Keying (PSK), Modulation Demodulation Processes, Bandwidth and Power Requirements in Modulated Signals, Applications of Modulation in Communication Systems.	4 – 5
Examine Error Coding and Reduction Techniques • Representation of digital data as electrical signals, ensuring synchronization and reducing error. Non-Return to Zero (NRZ), Manchester encoding, and Bipolar AMI. Channel Coding Techniques, Redundant bits for error detection and correction. Block codes like Hamming code, Reed-Solomon, and convolutional codes.	6 – 7

• Error Detection and Correction: Detection and correction of errors in transmitted data, parity check, checksums, and Cyclic Redundancy Check (CRC), Data integrity. Hamming codes and Reed-Solomon codes, Reliability in communication systems.	
Explore Electromagnetic Wave Propagation and Antenna Systems • Electromagnetic Wave Propagation: Basic Principles and Free-Space Propagation, Impact of Atmospheric Conditions on Wave Propagation: Troposphere, Ionosphere, Propagation in Different Media: Line of Sight, Ground Wave, Sky Wave • Satellite Communication: Propagation, Link Budget, Satellite Orbits, Radar Systems: Working Principles, Radar Range Equation, Applications, Role of Antennas: Types of Antennas, Antenna Arrays, Smart Antennas and Adaptive Beamforming. Propagation Effects on Communication Systems	8 – 9
Evaluate Emerging Communication Technologies • Introduction to Software-Defined Radio (SDR), Key Components of SDR: Transceivers, Filters, and Modulators, Cognitive Radio (CR) and its Role in Spectrum Management. • Spectrum Sensing and Spectrum Sharing Techniques in CR, SDR in Wireless Communication: Flexibility and Efficiency, Cognitive Radio Applications: Dynamic Spectrum Access and Management, CR and its Role in Addressing Spectrum Scarcity, Challenges in Implementing CR and SDR	10 – 11
Understand Internet and Cellular Communication Systems • Internet Architecture: Protocols, IP Addressing, Routing, and TCP/IP, Cellular Communication Systems: Architecture, Frequency Reuse, GSM	12

Architecture and Technology: Base Stations, MSC, and HLREvolution of Mobile Networks: From 2G to 5G, Mobile Switching Centers (MSC) and their Role in Call Management, Spectrum Regulation and Frequency Allocation, Cellular Network Topologies and Design, Internet of Things (IoT) and its Integration with Cellular Networks	
Lecture free week	13
Revision & Examination	14 - 15

j) Method of Grading Continuous Assessment

Continuous Assessment

- Assignment: 10%
- Quizzes/attendance: 10%
- Test: 20%

End of Semester Examination: 60%

k) Tutorials

Tutorial will be offered at the end of each module.

l) Group works and Assignments

The group works and assignments are designed to challenge students to think critically and solve problems in the world of communication, analyze real-world scenarios and propose solutions using the acquired knowledge of communication principles and system. The course also includes assignments designed to solidify your understanding through calculations, problem-solving exercises, and potentially, simulations or experiments involving modern communication systems.

m) Contemporary Issues and Industrial Relevance

The communication course addresses issues like spectrum scarcity, cybersecurity, 5G rollout, energy efficiency, and IoT integration in smart cities. It is relevant to industries such as telecommunications, satellite communication, radar systems, and consumer electronics, focusing on technologies like SDR, CR, signal modulation, channel coding, and electromagnetic wave propagation. The course also covers spectrum regulation, mobile switching centers, and cellular systems, all critical for advancing communication networks and addressing global challenges.

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AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code: EEE 322 Course Title: **Electric Circuit Theory II**

Semester: Second **Credit Hours**: 3 hour

Contact Hours: Lecture - Three hours lecture per week

Location: C3-6 **Lecturer(s)**: Engr. Dr. A. O. Ojo & Engr. O. Imahe

Textbooks and other Materials:

Charles Alexander and Matthew Sadiku (2021), Circuit-Fundamentals of Electric Circuits 7th Ed, McGraw-Hill

a) Course Overview and description

Networks; Node, loop analysis, Realization of transfer functions, Filters: design, operation, low, high, Band pass. Butterworth, Chebychev, filters, Non-linear circuit analysis Locus diagrams. Network functions. Active network synthesis and analysis: Foster and Cauers.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Core

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcomes: upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**POs**	**Assessment Tools**
Apply node and loop analysis techniques to solve complex electrical networks and validate circuit behavior under various conditions.	3, 4	1, 2, 4, 12	Assignment, quizzes and Exams

Design and realize transfer functions for electrical systems, including the synthesis and implementation of active networks using Foster and Cauer forms.	6	1, 2, 3, 11, 12	Assignment, quizzes and Exams
Analyze and interpret the characteristics and performance of analog filters such as low-pass, high-pass, band-pass, Butterworth, and Chebyshev filters.	4, 5	1, 2, 3, 11, 12	Assignment, quizzes and Exams
Evaluate non-linear circuit responses using locus diagrams and assess the impact of non-linearity on system performance.	5	1, 2	Assignment, quizzes and Exams
Assess and synthesize network functions for both passive and active systems, demonstrating proficiency in filter design and network optimization.	5, 6	1, 2, 3, 9, 12	Assignment, quizzes and Exams

f) Mapping of CLOs to relevant POs

Course Learning Outcomes	Programme Outcomes (POs):											
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
Apply node and loop analysis techniques to solve complex electrical networks and validate circuit behavior under various conditions.	**3**	**3**		**2**								**2**
Design and realize transfer functions for electrical systems, including the synthesis and implementation of active networks using Foster and Cauer forms.	**3**	**3**	**2**								**2**	**2**
Analyze and interpret the characteristics and performance of analog filters such as low-pass, high-pass, band-pass, Butterworth, and Chebyshev filters.	3	**3**	**2**								**2**	**2**
Evaluate non-linear circuit responses using locus diagrams and assess the impact of non-linearity on system performance.	**3**	**2**										
Assess and synthesize network functions for both passive and active systems, demonstrating proficiency in filter design and network optimization.	**3**	**3**	**2**						**2**			**2**

Keys: 1 = Slightly related, 2 = Moderately related, 3 = Highly related

g) Expectations of Students

a. Attend 3 hours of lectures per week

b. Turn in assignments and projects as considered suitable by the tutors

c. Complete continuous assessment which involves mid-semester test

d. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of PowerPoint slides, Lecture templates-textbooks)

14) **Course Content or Outline**

Topics	**Weeks**
Advanced Network Analysis Techniques • Node and loop analysis methods for complex circuits • Matrix formulation and solution of network equations • Application of network theorems in multi-loop systems	1-2
Network Functions and Transfer Function Realization • Definition and properties of network functions • Realization of transfer functions using passive and active components • Foster and Cauer forms of network synthesis	3-4
Filter Design and Analysis • Classification and characteristics of filters: low-pass, high-pass, band-pass • Butterworth and Chebyshev filter design principles • Frequency response and bandwidth considerations	5-6
Non-Linear Circuit Analysis and Locus Diagrams • Behavior of non-linear components in circuits • Graphical analysis using locus diagrams • Stability and response of non-linear networks	7-8
Active Network Synthesis and Applications • Design of active networks using operational amplifiers • Realization of complex transfer functions • Practical considerations in active filter implementation	9-10
Advanced Filter Structures and Optimization • M-derived and constant-k filters • Filter cascading and impedance matching • Design for minimal distortion and optimal power handling	11-12
Lecture free week	13
Revision & Examination	14 - 15

j) Method of Grading Continuous Assessment

Continuous Assessment

- Assignment: 10%
- Quizzes/attendance: 10%
- Test: 20%

End of Semester Examination: 60%

k) Tutorials

Tutorial will be offered at the end of each module.

l) Group works and Assignments

The group works and assignments in Circuit Theory II are designed to challenge students to think critically and solve complex problems in electrical network analysis and synthesis. Students will engage with real-world circuit scenarios, applying advanced concepts such as transfer function realization, filter design, and non-linear analysis to propose effective engineering solutions. The course also includes assignments aimed at reinforcing understanding through rigorous calculations, analytical exercises, and potentially, simulations or experiments involving modern circuit configurations and active network synthesis.

m) Contemporary Issues and Industrial Relevance

Circuit Theory II addresses modern challenges such as the need for high-performance filter design, efficient network synthesis, and the analysis of non-linear systems in increasingly complex electrical environments. It also engages with emerging demands in signal integrity, active circuit optimization, and the integration of analog-digital hybrid systems in smart technologies. This course is highly relevant to industries such as power electronics, embedded systems, telecommunications hardware, and control engineering, where precise circuit behavior and network design are critical. It supports technologies like active filter implementation, transfer function realization, and operational amplifier-based systems, all essential for developing reliable, scalable, and efficient electronic solutions in modern industrial applications.

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AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code: EEE 316 Course Title: **Electromagnetic Field and Waves II**

Semester: Second **Credit Hours**: 3 hour

Contact Hours: Lecture - Three hours lecture per week

Location: Hall B2-5 **Lecturer(s)**: Engr. Dr. A. O. Ojo & Engr. A. Akinola

Textbooks and other Materials:

- Owolabi I. E. and Ojo A. O., Electromagnetic Field and Waves, Principles and Applications, Matprints Nigeria.
- Kraus, T. D., Electromagnetics, 3rd edition, McGraw Hill, New York.

a) Course Overview and description

This course provides an in-depth exploration of electromagnetic theory, focusing on Maxwell's equations, their derivations, and applications in wave propagation, transmission lines, waveguides, and antennae. Students will learn to analyze electromagnetic phenomena using vector calculus, study wave behavior in various media (conductors, ionized media, and across boundaries), and apply wave theory to communication systems. Practical exercises and simulations will reinforce theoretical concepts, preparing students for advanced topics in electrical and communication engineering.

b) Pre-requisites: EEE321

c) Co-requisite(s): Nil

d) Role in Curriculum: Core

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcomes: upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**POs**	**Assessment Tools**

Derive and interpret Maxwell's equations in both differential and integral forms using vector calculus across various coordinate systems.	2, 4	1, 2, 4	Assignment, quizzes and Exams
Apply scalar and vector potential formulations to solve electromagnetic field problems and calculate associated potentials.	3, 4	1, 2, 5	Assignment, quizzes and Exams
Analyze electromagnetic wave properties such as phase velocity, polarization, energy flow, and wave behavior at media interfaces using boundary conditions and the Poynting vector	4, 5	1, 2, 5, 11	Assignment, quizzes and Exams
Evaluate wave propagation in conductors and ionized media, including phenomena like skin effect and plasma frequency, and assess their impact on signal transmission.	5	1, 2, 6	Assignment, quizzes and Exams
Design and analyze transmission lines, waveguides, and antennas, calculating key parameters and applying electromagnetic theory to practical communication systems.	5, 6	1, 2, 3, 5, 12	Assignment, quizzes and Exams

Course Learning Outcomes	Programme Outcomes (POs):											
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
Derive and interpret Maxwell's equations in both differential and integral forms using vector calculus across various coordinate systems.	**3**	**3**		**2**								
Apply scalar and vector potential formulations to solve electromagnetic field problems and calculate associated potentials.	**3**	**3**			**2**							**2**
Analyze electromagnetic wave properties such as phase velocity, polarization, energy flow, and wave behavior at media interfaces using boundary conditions and the Poynting vector	3	**3**			**2**							
Evaluate wave propagation in conductors and ionized media, including phenomena like skin effect and plasma frequency, and assess their impact on signal transmission.	**3**	**2**				**2**						
Design and analyze transmission lines, waveguides, and antennas, calculating key parameters and applying electromagnetic theory to practical communication systems.	**3**	**3**	**2**		**3**	**2**		**2**			**2**	**3**

Keys: 1 = Slightly related, 2 = Moderately related, 3 = Highly related

g) Expectations of Students

a. Attend 3 hours of lectures per week

b. Turn in assignments and projects as considered suitable by the tutors

c. Complete continuous assessment which involves mid-semester test

d. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of PowerPoint slides, Lecture templates-textbooks)

Course Content or Outline

Topics	Weeks
Review of Vector Calculus and Coordinate Systems • Gradient, divergence, curl • Cartesian, cylindrical, and spherical coordinates	1
Introduction to Maxwell's Equations • Physical meaning of each equation • Historical context and derivation overview	2
Maxwell's Equations in Differential Form • Derivation using vector calculus • Application in static and dynamic fields **Maxwell's Equations in Integral Form** • Gauss's law, Faraday's law, Ampère's law • Interpretation across coordinate systems	3 - 4
Scalar and Vector Potentials • Definitions and physical meaning • Lorenz and Coulomb gauges	5
Solving Field Problems with Potentials • Poisson's and Laplace's equations • Boundary conditions and uniqueness theorem	6
Electromagnetic Wave Properties • Phase velocity and group velocity • Polarization types • Energy flow	7
Wave Behavior at Media Interfaces • Reflection, refraction, transmission • Boundary conditions • Snell's law	8
Poynting Vector and Energy Conservation • Derivation and interpretation • Applications in power flow analysis	9
Wave Propagation in Conductors and Ionized Media • Skin effect and attenuation • Plasma frequency • Dispersion	10

Transmission Lines and Waveguides • Characteristic impedance and VSWR • TE, TM, TEM modes	11
Antennas and Communication Systems • Radiation patterns • Gain and directivity • Practical design considerations	12
Lecture free week	13
Revision & Examination	14 - 15

j) Method of Grading Continuous Assessment

Continuous Assessment

- Assignment: 10%
- Quizzes/attendance: 10%
- Test: 20%

End of Semester Examination: 60%

k) Tutorials

Tutorial will be offered at the end of each module.

l) Group works and Assignments

The group works and assignments in Electromagnetic Field and Waves II are crafted to deepen students' analytical skills and foster mastery in solving complex electromagnetic field problems. Learners will engage with real-world scenarios involving wave propagation, boundary interactions, and electromagnetic system design. Through rigorous tasks, students will derive and interpret Maxwell's equations across coordinate systems, apply scalar and vector potentials to field modeling, and analyze wave behavior using boundary conditions and the Poynting vector.

Assignments will challenge students to evaluate phenomena such as skin effect and plasma frequency in conductors and ionized media, assessing their impact on signal transmission. Practical components may include simulations or experiments involving transmission lines, waveguides, and antenna systems, reinforcing theoretical understanding through hands-on exploration. These activities are designed to cultivate critical thinking, precision in mathematical modeling, and the ability to translate electromagnetic theory into effective engineering solutions.

m) Contemporary Issues and Industrial Relevance

Electromagnetic Field and Waves II addresses contemporary challenges in the modeling, analysis, and application of electromagnetic fields across complex environments. The course equips students with the ability to derive and interpret Maxwell's equations in multiple coordinate systems, apply scalar and vector potentials, and analyze wave behavior at media interfaces using boundary conditions and energy flow principles.

It responds to emerging demands in high-frequency communication, electromagnetic compatibility, and the design of efficient transmission systems. Students engage with advanced topics such as wave propagation in conductors and ionized media, antenna theory, and the optimization of waveguides and transmission lines for modern communication networks. This course is highly relevant to industries including wireless communications, radar systems, satellite technology, and RF engineering, where precise electromagnetic modeling and system design are essential.

Technologies supported include antenna array design, impedance matching, energy-efficient waveguide systems, and electromagnetic simulation tools—critical for developing scalable, high-performance solutions in today's interconnected and signal-sensitive world.



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AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code: EEE 564

Course Title: Seminars on the Use of Software for Electrical Electronic Engineering

Semester: Second **Credit Hours**: 1 hour

Contact Hours: Lecture - One hour lecture per week

Location: Hall B2-5 **Lecturer(s)**: Engr. Dr. A. O. Ojo

Textbooks and other Materials:

- Rashid, M. H. (2017), SPICE for Circuits and Electronics Using PSpice, 4th Edition, CRC Press.
- Moore, H. (2018), MATLAB for Engineers, 5th Edition, Pearson.

a) Course Overview and description

This seminar course introduces students to the application of software tools in solving real-world problems in electrical and electronics engineering. It covers software for circuit design, simulation, analysis, control systems, and project management. Through lectures, hands-on sessions, case studies, and student-led seminars, participants will explore industry-standard tools, their applications, and emerging trends. Students will develop skills in selecting and using appropriate software for engineering tasks and present a seminar on a chosen software tool.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Core

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcomes: upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**POs**	**Assessment Tools**
Identify key software tools used in electrical and electronics engineering for circuit design, simulation, analysis, and control systems.	1, 2	1, 4, 10	Assignment, quizzes and Exams
Explain the applications and limitations of specific software tools in solving real-world electrical and electronics engineering problems.	2, 3	1, 2, 10	Assignment, quizzes and Exams
Utilize selected software tools to design and simulate a basic electrical or electronic system, such as a circuit or control model.	3, 4	1, 3, 5	Assignment, quizzes and Exams
Compare the features and performance of at least two software tools used for a specific electrical engineering task (e.g., circuit simulation or power system analysis).	4, 5	1, 5, 11	Assignment, quizzes and Exams

Develop and present a comprehensive seminar on current trends in electrical electronic engineering and the application of a chosen software tool, including a practical demonstration or case study.	5, 6	7, 8, 12	

Course Learning Outcomes	**Programme Outcomes (POs):**											
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
Identify key software tools used in electrical and electronics engineering for circuit design, simulation, analysis, and control systems.	**3**			**3**						**2**		
Explain the applications and limitations of specific software tools in solving real-world electrical and electronics engineering problems.	**3**	**2**								**2**		
Utilize selected software tools to design and simulate a basic electrical or electronic system, such as a circuit or control model.	3		**2**		**2**	**2**						
Compare the features and performance of at least two software tools used for a specific electrical engineering task (e.g., circuit simulation or power system analysis).	**3**				**2**						**2**	
Develop and present a comprehensive seminar on current trends in electrical electronic engineering and the application of a chosen software tool, including a practical demonstration or case study.							**3**	**2**				**3**

j) Method of Grading Continuous Assessment

Continuous Assessment

- Assignment: 10%

- Quizzes/attendance: 10%

- Test: 20%

End of Semester Examination: 60%

k) Tutorials

Tutorial will be offered at the end of each module.

l) Group Works and Assignments

The group works and assignments in *Seminars on the Use of Software for Electrical Electronic Engineering* are designed to cultivate students' proficiency in applying digital tools to solve practical engineering problems. Learners will engage with real-world scenarios involving circuit design, control systems, and signal analysis, using industry-standard software platforms to simulate, optimize, and evaluate system performance.

Assignments will challenge students to identify appropriate software tools for specific engineering tasks, compare their features and limitations, and implement basic electrical or electronic systems through simulation. Group projects may include collaborative design exercises, comparative tool evaluations, and seminar presentations that integrate technical analysis with practical demonstrations. These activities are structured to enhance critical thinking, digital fluency, and the ability to translate software capabilities into effective engineering solutions.

m) Contemporary Issues and Industrial Relevance

Seminars on the Use of Software for Electrical Electronic Engineering addresses the growing importance of software-driven design and analysis in modern electrical engineering practice. The course responds to contemporary challenges such as the integration of analog-digital hybrid systems, signal integrity in high-speed circuits, and the need for efficient simulation in complex embedded environments.

Students explore current trends in software applications for power systems, control engineering, and telecommunications hardware, gaining insight into how digital tools support innovation in smart technologies and automation. The course is highly relevant to industries such as embedded

systems, power electronics, industrial automation, and electronic product development, where software proficiency is essential for reliability, scalability, and performance.

Technologies supported include circuit simulation platforms, control system modeling tools, and power system analysis software which are critical for developing intelligent, adaptive, and efficient engineering solutions in today's digitally transformed landscape.



AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code:	EEE 405	**Course Title:**	Numerical Computations
Semester:	First	**Credit Hours**:	3
Contact Hours:	Lecture - Two hours lectures per Week	**Location**:	C3-6
Lecturer(s):	Engr. Dr. O. T. Ibitoye		

Textbooks and other Materials:

i. Introduction to Numerical Methods and MATLAP Programming for Engineers. Todd Young and Martin J. Mohlenkamp. Department of Mathematics Ohio University Athens, OH 45701 youngt@ohio.ed, December 13, 2023.

ii. An Introduction to Numerical Methods and Analysis. James F. Epperson. John Wiley & Sons, 10 Aug 2021.

a) Course Overview and description

This course covers the fundamental concepts of numerical analysis with specific emphasis on the various numerical methods and tools essential for analyzing and solving engineering problems with minimal errors. Topics include errors in numerical analysis, convergence theory of computational solution to problem: numerical algorithm formulation and design, numerical software systems. Iterative solutions of non-linear systems: (Newton's Method). Numerical solutions of linear systems. Numerical computation of Eigen values and Eigen vectors. Curve fitting; function approximation.

Numerical differentiation and integration (Simpson's Rule etc.). Explicit and implicit methods. Differential equation (Euler's method etc.). Numerical integration and differentiation. Linear algebra. Finite differences. Introduction to the use of Octave, MATLAB or LabVIEW in numerical computation and engineering applications.

b) **Pre-requisites:** Nil

c) **Co-requisite(s):** Nil

d) **Role in Curriculum:** Compulsory

e) **Course Learning Outcome (CLO) and Programme Outcomes (PO)**

Course Learning Outcomes: upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**POs**	**Assessment Tools**
Outline the fundamental methods of analyzing errors in numerical systems	4	1, 2, 6	Assignment, quizzes and Exams
Propose the convergence theory of computational solution to engineering problems	4, 5	1, 2, 3	Assignment, quizzes and Exams
Design numerical algorithms using numerical software systems	4	1, 3, 10, 12	Assignment, quizzes and Exams
Compare various numerical methods such as Newton's method, numerical computation of Eigen values and Eigen vectors, Simpson's rules and Euler's method	4	1, 3, 4, 5	Assignment, quizzes and Exams
Analyze the concept of numerical integration and differentiation	4	1, 7, 8, 12	Assignment, quizzes and Exams
Develop various models using MATLAB for numerical computation and engineering applications	4, 5	1, 9, 11, 12	Assignment, quizzes and Exams

f) Mapping of CLOs to relevant POs

Course Learning Outcomes	Programme Outcomes											
	1	2	3	4	5	6	7	8	9	10	11	12
Outline the fundamental methods of analyzing errors in numerical systems	3	3				2						
Propose the convergence theory of computational solution to engineering problems	2	3	3									
Design numerical algorithms using numerical software systems	3		3							2		2
Compare various numerical methods such as Newton's method, numerical computation of Eigen values and Eigen vectors, Simpson's rules and Euler's method	3		2	3	3							
Analyze the concept of numerical integration and differentiation	3						2	2				3
Develop various models using MATLAB for numerical computation and engineering applications	3								2		2	3

Keys: 1 = slightly related, 2 = moderately related, 3 = highly related

g) Expectations of Students

a. Attend 2 hours of lectures per week

b. Turn in assignments considered suitable by the tutors

c. Complete continuous assessment which involves mid-semester test

d. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of Power Point slides, Lecture templates-handouts)

i) Course Content or Outline

Topics	Weeks
Analysis of errors in numerical systems	1-2
Convergence theory of computational solution to engineering problems	3 - 4
Numerical algorithms	5 - 6
Numerical methods: Newton's method, numerical computation of Eigen values and Eigen vectors, Simpson's rules and Euler's method	7 - 9
Numerical integration and differentiation	10
MATLAB models for numerical computation and engineering applications	11-12
Lecture free week	13
Revision & Examination	14 - 15

j) Method of Grading Continuous Assessment

Continuous Assessment (Assignments): 20%

Continuous Assessment (Test):20%

End of Semester Examination: 60%

k) Tutorials

Tutorials are taken at the end of each module.

l) Term Paper and Assignments

A term paper focusing on the general applications of numerical computation in various engineering fields. The assignment involves highlighting the importance of different models and algorithms in numerical systems.

m) Contemporary Issues and Industrial Relevance

The course equips students with the knowledge to develop various models using MATLAB for numerical computation in industrial applications.



AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code: EEE 427 Course Title: Data Communication and Networks

Semester: First **Credit Hours**: 3 hours

Contact Hours: Lecture - Three hours lectures per week

Location: Hall B2-5 **Lecturer(s)**: Engr. Dr. A. O. Ojo & Engr. Olawale

Textbooks and other Materials:

- Owolabi & Ojo (2022), Electrical Communication Principles, Systems and Applications, 1st Edition, Matprints Inc.

- Prakash C. Gupta, (2014), Data Communication and Computer Networks, 2nd Edition, PHI Learning Private Limited.

a) Course Overview and description

Introduction to Data communications: the Development of Data Communications; types and sources of data, simple communications network, transmission definitions, one way transmission, half duplex transmission, transmission codes, transmission modes, parallel transmission, serial transmission, bit synchronization, character synchronization, character

synchronization, synchronous transmission, asynchronous transmission, efficiency of transmission, error detection methods and data compression. Protocols: Introduction to network protocol. Seven Layer ISO-OSI standard protocols and network architecture. Transport protocols, session services protocols, and other protocols. Institute of Electrical and Electronics Engineering 802 standards. Error control and Data Compression: Forward Error Control; error detection methods; parity checking; linear block codes, cyclic redundancy checking; feedback error control, data compression, Huffman coding and dynamic Huffman coding. Local Area Networks: medium access control techniques – Ethernet, token bus and token ring; LAN standards; fibre distributed data interface, metropolitan area network. Peer-to-peer, Client Server. Client-Server Requirements: GUI design standards, interface independence, platform independence, transaction processing, and connectivity, reliability, backup and recovery mechanisms. Information Network Software; Features and benefits of major recovery mechanisms. Information Network Software: features and benefits of major Network Operating Systems. Network OS: (e.g. Novell NetWare, UNIX/LINUX, OS/2 & Windows NT). TCP/IP and Network OS. INTERNET: Definition, architecture, services, Internet addressing. Internet protocol, IPv4, IPv6. Internet programming, Intranet. System administration, and security issues.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Core

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcomes: upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**POs**	**Assessment Tools**
Explain fundamental concepts of data transmission and synchronization techniques in communication systems.	1, 2	1, 2, 4	Assignment, quizzes and Exams
Analyze error detection and correction methods, including block codes, CRC, and data compression algorithms.	2, 3	1, 2, 5	Assignment, quizzes and Exams

Evaluate the structure and function of network protocols, including the OSI model and IEEE 802 standards.	3, 4	1, 2, 3, 4, 11	Assignment, quizzes and Exams
Design and simulate basic LAN architectures using medium access control techniques and client-server models.	4, 5	1, 2, 3, 5, 6, 11, 12	Assignment, quizzes and Exams
Assess the role of network operating systems and Internet protocols in modern communication systems, including IPv4/IPv6 and system security.	5, 6	1, 2, 7, 8, 11, 12	Assignment, quizzes and Exams

f) Mapping of CLOs to relevant POs

Course Learning Outcomes	Programme Outcomes (POs):											
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
Explain fundamental concepts of data transmission and synchronization techniques in communication systems.	**3**	**3**		**2**								
Analyze error detection and correction methods, including block codes, CRC, and data compression algorithms.	**3**	**3**			**2**							
Evaluate the structure and function of network protocols, including the OSI model and IEEE 802 standards.	3	**3**	**2**	**2**							**2**	
Design and simulate basic LAN architectures using medium access control techniques and client-server models.	**3**	**2**	**3**		**3**	**2**					**2**	**2**
Assess the role of network operating systems and Internet protocols in modern communication systems, including IPv4/IPv6 and system security.	**2**	**2**					**3**	**3**			**2**	**3**

Keys: 1 = Slightly related, 2 = Moderately related, 3 = Highly related

g) Expectations of Students

a. Attend 3 hours of lectures per week

b. Turn in assignments and projects as considered suitable by the tutors

c. Complete continuous assessment which involves mid-semester test

d. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of PowerPoint slides, Lecture templates-textbooks)

15) **Course Content or Outline**

Topics	Weeks
Introduction to Data Communications • Development of Data Communications • Types and sources of data • Simple communication networks • Transmission definitions and modes	1 – 2
Transmission Techniques and Synchronization • One-way, half-duplex, full-duplex • Parallel vs. serial transmission • Bit and character synchronization • Synchronous vs. asynchronous transmission • Efficiency of transmission	3 – 4
Error Control and Data Compression • Forward error control • Parity checking, block codes, CRC • Feedback error control • Huffman and dynamic Huffman coding	5 – 6
Network Protocols and Architecture • Introduction to protocols • OSI model and ISO standards • Transport and session protocols • IEEE 802 standards	7 – 8
Local Area Networks and Client-Server Models • MAC techniques: Ethernet, token bus/ring • LAN standards and FDDI	

• Peer-to-peer vs. client-server • GUI design, platform independence • Reliability and recovery mechanisms	9 – 10
Network Operating Systems and Internet Protocols • Features of major network OS: Novell, UNIX, Windows NT • TCP/IP stack • IPv4 vs. IPv6 • Internet architecture and addressing • Intranet, system administration, security	11
Seminar Presentations • Student-led seminars on selected topics • Case studies and practical demonstrations	12
Lecture free week	13
Revision & Examination	14 - 15

j) Method of Grading Continuous Assessment

Continuous Assessment

- Assignment: 10%
- Quizzes/attendance: 10%
- Test: 20%

End of Semester Examination: 60%

k) Tutorials

Tutorial will be offered at the end of each module.

l) Group works and Assignments

The group works and assignments in *EEE 427: Data Communication and Networks* are designed to build students' technical fluency and collaborative problem-solving skills in digital

communication systems. Learners will work on real-world scenarios such as designing LAN topologies, configuring network protocols, and simulating data transmission models.

Assignments will include protocol analysis, error control coding exercises, and comparative evaluations of network operating systems. Group projects may involve building client-server models, simulating packet transmission, and presenting case studies on Internet architecture and security. These activities foster analytical thinking, teamwork, and the ability to apply theoretical knowledge to practical networking challenges.

m) Contemporary Issues and Industrial Relevance

Data Communication and Networks addresses the evolving landscape of digital communication, where speed, reliability, and security are paramount. The course responds to contemporary demands in areas such as high-speed data transmission, secure Internet protocols, and scalable network architectures.

Students explore current trends in IPv6 deployment, cloud-based networking, and cybersecurity frameworks. The course is highly relevant to industries including telecommunications, enterprise IT, embedded systems, and Internet infrastructure—where robust data communication is foundational.

Technologies supported include TCP/IP stack implementation, LAN simulation tools, network OS platforms (e.g., Linux, Windows Server), and Internet protocol analysis—critical for developing resilient, efficient, and secure communication systems in today's connected world.

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AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code:	EEE 562	**Course Title:**	Power Electronics
Semester:	Second	**Credit Hours**:	**2**
Contact Hours:	Lecture – Three hours lectures per Week	**Location**:	B2-5
Lecturer(s):	Engr. A. Akinola		

Textbooks and other Materials:

1. M. H. Rashid, (2011), Power electronics handbook: devices, circuits, and applications handbook, 3rd Edition, Butterworth-Heinemann.
2. P. T. Krein, (1998) Elements of Power Electronics. New York: Oxford University Press,
3. P. Horowitz and W. Hill, (2015) The art of electronics. Cambridge University Press, 3rd Edition.

a) Course Overview and description

This course explores the history and technical aspects of mobile communication systems, from early cellular concepts to modern digital standards like GSM. It covers challenges like signal interference and fading, while also covering network capacity, traffic management, and call processing protocols.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Compulsory

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcomes: upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**POs**	**Assessment Tools**
Understand the working principles of power semiconductor devices	2	1, 2	Assignment, quizzes and Exams
Apply the knowledge of the working principles of power semiconductor to analyze their switching characteristics.	3, 4	1, 2, 3, 5	Assignment, quizzes and Exams
Analyze switched circuits with different loads (RLC, RC, RL etc.).	4	1, 2, 3	Assignment, quizzes and Exams
Analyze the design of power converters (AC-DC, DC-DC).	4	1, 2, 3, 4	Assignment, quizzes and Exams
Develop protection circuits for power semiconductor switching devices.	6	1, 2, 3, 4, 5, 6	Assignment, quizzes and Exams

f) Mapping of CLOs to relevant POs

Course Learning Outcomes	Programme Outcomes											
	1	2	3	4	5	6	7	8	9	10	11	12
Understand the working principles of power semiconductor devices	3	2										
Apply the knowledge of the working principles of power semiconductor to analyze their switching characteristics.	3	3	2		3							
Analyze switched circuits with different loads (RLC, RC, RL etc.).	3	3	2									
Analyze the design of power converters (AC-DC, DC-DC).	3	3	3	3								
Develop protection circuits for power semiconductor switching devices.	3	3	3	3	3	2						

Keys: 1 = slightly related, 2 = moderately related, 3 = highly related

g) Expectations of Students

a. Attend 3 hours of lectures per week

b. Turn in assignments considered suitable by the tutors

c. Complete continuous assessment which involves mid-semester test

d. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of Power Point slides, Lecture templates-handouts)

i) Course Content or Outline

Topics	Weeks
Introduction to Power Electronics	1
Power Semiconductor Devices: • Diodes • MOSFET • IGBT • Thyristor	2 - 4
Analysis of Switched Circuits	5 - 7
AC-DC Converters	8
DC-DC Converters	9 -10
Inverters (DC-AC Conversion)	11
Industrial Applications (EVs, Renewable Energy, Power Supplies)	12
Lecture free week	13
Revision & Examination	14 - 15

j) Method of Grading Continuous Assessment

Continuous Assessment (Assignments): 15%

Continuous Assessment (Test):20%

Attendance (5%)

End of Semester Examination: 60%

k) Tutorials

Tutorials are taken at the end of each module.

l) Term Paper and Assignments

A term paper focus on understanding a specific aspect of power semiconductor, such as analyzing commutation methods and strategies in Thyristors for specific applications.

m) Contemporary Issues and Industrial Relevance

The course equips students with the knowledge of power electronics for energy conversion. The challenges of meeting global energy challenges cannot be addressed without appropriate, reliable, and efficient power supply. Therefore, the teaching is focused on equipping the students, with knowledge that will enable them use power semiconductor technology to improve power supplies, industrial automation, energy storage, industrial drives, and electric power transmission and distribution.

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AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code:	EEE 523	**Course Title:**	Industrial Electronics Design
Semester:	First	**Credit Hours**:	3
Contact Hours:	Lecture – Three hours lectures per Week	**Location**:	B2-3
Lecturer(s):	Engr. A. Akinola		

Textbooks and other Materials:

3. A. K. Gupta, S. K. Arora and J. R. Westcott. (2017), Industrial Automation and Robotics. Publisher: David Pallai Mercury Learning and Information
4. P Malindi (2021) Advanced Practical Electronics – Circuits & Systems. MLD Technologies, Gonubie, East London. ISBN: 978-0-620-91554-0
5. S. Manesis and G. Nikolakopoulos. (2019), Introduction to Industrial Automation. CRC Press Taylor & Francis Group

a) Course Overview and description

This course explores the Characteristics and industrial applications of thyristors and other SCR devices. Transducers and their applications in sensing light, voltage pressure, motion, current temperature, etc. Mechanical relays, solid state relays and stepping motors. Real time control and remote-control concepts in instrumentation. Micro-processor and micro-computer-based systems and general home and industrial instrumentation.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Elective

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcomes: upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**POs**	**Assessment Tools**
Explain the operation and characteristics of thyristors and SCR devices, and evaluate their roles in industrial power control and switching applications.	2,5	1, 2, 3, 5	Assignment, quizzes and Exams
Classify various types of transducers and describe their principles of operation for sensing physical quantities.	2	1, 2	Assignment, quizzes and Exams
Select appropriate transducers for specific measurement and control applications in industrial and domestic environments.	3	1, 2, 3, 4	Assignment, quizzes and Exams
Compare mechanical relays, solid-state relays, and stepping motors in terms of operation.	4	1, 2, 3	Assignment, quizzes and Exams
Design basic control systems using PLC, microprocessor and microcomputer-based systems for real-time and remote-control techniques in instrumentation systems.	6	1, 2, 3, 4	Assignment, quizzes and Exams

f) Mapping of CLOs to relevant POs

Course Learning Outcomes	Programme Outcomes											
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
Explain the operation and characteristics of thyristors and SCR devices, and evaluate their roles in industrial power control and switching applications.	**3**	**3**	**2**		**2**							
Classify various types of transducers and describe their principles of operation for sensing physical quantities.	**3**	**2**										
Select appropriate transducers for specific measurement and control applications in industrial and domestic environments.	**3**	**2**	**3**	**2**								
Compare mechanical relays, solid-state relays, and stepping motors in terms of operation.	**2**	**3**	**2**									
Design basic control systems using PLC, microprocessor and microcomputer-based systems for real-time and remote-control techniques in instrumentation systems.	**3**	**3**	**3**	**3**								

Keys: 1 = slightly related, 2 = moderately related, 3 = highly related

g) Expectations of Students

a. Attend 3 hours of lectures per week

b. Turn in assignments considered suitable by the tutors

c. Complete continuous assessment which involves quiz and mid-semester test

d. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of Power Point slides, Lecture templates-handouts)

i) Course Content or Outline

Topics	Weeks
• Introduction to industrial electronics and solid-state devices	1
• Thyristors and SCR Devices	2- 3
Introduction Transducers: • Sensors and Actuators • Transducers for Electrical Quantities • Transducers for Physical Quantities • Light and Optical Sensors	4 - 6
Mechanical and Solid-State Relays:	7
Stepping Motors and Control Mechanisms	8
Real-Time Control Concepts and Remote Control in Instrumentation	9 -10
PLC and Microprocessor/Microcomputer-Based Systems	11 - 13
Revision & Examination	14 - 15

j) Method of Grading Continuous Assessment

Continuous Assessment (Assignments): 10%

Continuous Assessment (Test):20%

Continuous Assessment (Quiz): 5%

Attendance (5%)

End of Semester Examination: 60%

k) Tutorials

Tutorials are taken at the end of each module.

l. Design Assignments report

An assignment focused on automating a production process. Involving programming PLCs, and selecting appropriate sensor for specific applications. Student would submit a report of the design.

m) Contemporary Issues and Industrial Relevance

The course equips students with the knowledge to understand the methods of signal acquisition and conversions for device applications in control systems. By understanding transducers, sensor technology, actuator and PLCs, students can contribute to the development and optimization of industrial control systems.

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AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code:	COE 427	**Course Title:**	Data Communication and Network
Semester:	First	**Credit Hours**:	**3**
Contact Hours:	Lecture - Three hours lectures per week	**Location**:	LAB. B0-4
Lecturer(s):	Engr. A. Akinola		

Textbooks and other Materials:

I. Israel E. Owolabi and Adedayo O. Ojo (2023), Electrical Communication Principles: System and Applications. 1st edition, Matprint, Ibadan.
II. Prakash C. Gupta (2014), Data Communications and Computer Networks, 2nd edition, PHI Learning Private Limited, Delhi.
III. William Stallings (2014), Data and Computer Communications, 10th edition (International Edition) Pearson Education Limited, Edinburgh Gate, Harlow, Essex.

a) Course Overview and description

This course covers the fundamentals of computer network data communication concepts, data and signals, data transmission and network types. Topics include; Introduction to data communications, Network protocols and architecture, Error control and Data Compression, Local Area Networks and the Internet:

b) **Pre-requisites:** Nil

c) **Co-requisite(s):** Nil

d) **Role in Curriculum:** Compulsory

e) **Course Learning Outcome (CLO) and Programme Outcomes (PO)**

Course Learning Outcomes: upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**POs**	**Assessment Tools**
Explain the fundamental concepts of data communications	2	1, 2	Assignment, quizzes and Exams
Describe and analyze network protocols and architecture.	4	1, 2, 3	Assignment, quizzes and Exams
Apply error detection, error control, and data compression techniques to digital communication systems.	3	2, 3, 4	Assignment, quizzes and Exams
Compare and evaluate local area network (LAN) technologies and architectures in terms of access methods, performance, and standards.	5	2,3,10	Assignment, quizzes and Exams
Demonstrate and assess Internet technologies and services, and basic security mechanisms.	3, 5	3, 4,5,7, 8	Assignment, quizzes and Exams

f) **Mapping of CLOs to relevant POs**

Course Learning Outcomes	Programme Outcomes											
	1	2	3	4	5	6	7	8	9	10	11	12
Explain the fundamental concepts of data communications	3	2										
Describe and analyze network protocols and architecture.	3	3	2									
Apply error detection, error control, and data compression techniques to digital communication systems.		3	3	3								
Compare and evaluate local area network (LAN) technologies and architectures in terms of access methods, performance, and standards.	3			2								
Demonstrate and assess Internet technologies and services, and basic security mechanisms.		3	3	2			2	3				

Keys: 1 = slightly related, 2 = moderately related, 3 = highly related

g) Expectations of Students

a. Attend 3 hours of lectures per week

b. Turn in assignments considered suitable by the tutors

c. Complete continuous assessment which involves mid-semester test

d. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of Power Point slides, Lecture templates-handouts and Physical network equipment and tools)

i) Course Content or Outline

Topics	Weeks
Introduction to Data communications • the Development of Data Communications; • types and sources of data, • simple communications network, • transmission definitions	1-2
Network Protocols: • Introduction to network protocol. • Seven Layer ISO-OSI standard protocols and network architecture. • TCP/IP, Transport protocols, session services protocols. • IEEE 802 standards.	3 - 5
Error control and Data Compression • Error Detection Methods; Parity Checking; Linear Block Codes, Cyclic Redundancy Checking • Feedback Error Control. • Data Compression, Huffman Coding and Dynamic Huffman Coding.	6 - 8
Local Area Networks: • medium access control techniques. • LAN standards; fibre distributed data interface, metropolitan area network. • Peer-to-peer, Client-Server Requirements.	8 - 11
INTERNET: • Definition, architecture, services, • Internet addressing. • Internet protocol, IPv4, IPv6. • System administration, and Network Security issues.	11 - 12
Revision & Examination	14 - 15

j) Method of Grading Continuous Assessment

Continuous Assessment (Assignments): 10%

Continuous Assessment (Test):20%

Continuous Assessment (Quiz): 5%

Attendance (5%)

End of Semester Examination: 60%

k) Tutorials

Tutorials are taken at the end of each module.

l) Contemporary Issues and Industrial Relevance

The course exposes students to the principles underlying the exchange of data between two or more systems that are directly or indirectly connected over short or long distances. It equips students with the knowledge necessary for efficient and effective management of computer communication network.



AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code:	EEE 326	**Course Title:**	Electrical Machines II
Semester:	Second	**Credit Hours**:	**3**
Contact Hours:	Lecture - Three hours lectures per Week	**Location**:	B-04
Lecturer(s):	Engr. A. Akinola		

Textbooks and other Materials:

1. S. J. Chapman, (1999), Electric machinery fundamentals. New York: McGraw.
2. S. K. Sahdev, (2018) Electrical Machines, Cambridge University Press
3. T. Wildi, (2014) Electrical Machines, Drives, and Power Systems Pearson Education Limited Edinburgh Gate. Sixth Edition

a) Course Overview and description

This course explores the production of rotating magnetic fields in AC machines. Simple theory of three-phase induction motors; torque speed characteristics, Single-phase induction motors-applications. Selection of motors, for practical applications. Synchronous machines.

b) Pre-requisites: EEE 325: Electrical Machines I

c) Co-requisite(s): Nil

d) Role in Curriculum: Compulsory

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcomes: upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**POs**	**Assessment Tools**
Explain the principle of operation of A.C. machines, including the production of rotating magnetic fields in three-phase systems.	2	1, 2	Assignment, quizzes and Exams
Analyze the construction and working principles of three-phase induction motors	4	1, 2, 3	Assignment, quizzes and Exams
Compare and contrast the performance and applications of single-phase and three-phase motors.	5	2, 3, 10	Assignment, quizzes and Exams
Describe the basic operating principles and applications of synchronous machines.	2	1, 2	Assignment, quizzes and Exams
Apply knowledge of A.C. motor characteristics to select appropriate motors for practical applications.	3	2, 3, 10, 11	Assignment, quizzes and Exams

f) Mapping of CLOs to relevant POs

Course Learning Outcomes	Programme Outcomes											
	1	2	3	4	5	6	7	8	9	10	11	12
Explain the principle of operation of A.C. machines, including the production of rotating magnetic fields in three-phase systems.	3	2										
Analyze the construction and working principles of three-phase induction motors	3	3	2									
Compare and contrast the performance and applications of single-phase and three-phase motors.		3	2							2		
Describe the basic operating principles and applications of synchronous machines.	3	2										
Apply knowledge of A.C. motor characteristics to select appropriate motors for practical applications.		3	3					2		3	2	

Keys: 1 = slightly related, 2 = moderately related, 3 = highly related

g) Expectations of Students

a. Attend 3 hours of lectures per week

b. Turn in assignments considered suitable by the tutors

c. Complete continuous assessment which involves mid-semester test

d. Complete a 3 -hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of Power Point slides, Lecture templates-handouts)

i) Course Content or Outline

Topics	Weeks
Introduction to AC Machines • Overview of electric machines (DC vs AC). • Role of AC machines in power systems and industry. • Basic review of magnetic circuits and electromagnetic induction. • Principle of rotating magnetic field (RMF). • Mathematical representation of RMF in three-phase systems. • Phasor diagrams and RMF speed relation	1 -2
Introduction to Three-Phase Induction Motors: • Constructional features (stator, rotor: squirrel cage & wound type). • Principle of operation: induction principle. • Concept of slip and rotor frequency. • Equivalent Circuit of Induction Motors. • Power flow diagram of induction motors. • Torque-Speed Characteristics of Induction Motors • Speed control: stator voltage control, rotor resistance control, pole changing, frequency control. • Three-phase induction motor tests • Industrial and commercial applications. • Selection criteria: torque, speed, power rating.	3 - 6
Single-Phase Induction Motors – Principles • Need for single-phase motors. • Double revolving field theory. • Types: split-phase, capacitor-start, capacitor-run, shaded pole motors. • Applications of Single-Phase Induction Motors • Performance comparison with 3-phase induction motors.	7 - 8
Synchronous Machines – (Alternators and Motors) • Construction of synchronous machines. • Principle of operation: synchronous speed, excitation. • Types: cylindrical rotor, salient pole. • Equivalent circuit of synchronous machines (Alternators and Motors) • Applications of Synchronous Machines • Comparing AC machine types (induction vs synchronous vs single-phase).	10 - 12
Revision & Examination	13 - 15

j) Method of Grading Continuous Assessment

Continuous Assessment (Assignments): 10%

Continuous Assessment (Test):25%

Attendance (5%)

End of Semester Examination: 60%

k) Tutorials

Tutorials are taken at the end of each module.

l) Assignments

An assignment focused on production of rotating magnetic field in three phase machine analyzing slip, speed and torque production.

m) Contemporary Issues and Industrial Relevance

The course equips students with the knowledge to address critical challenges in AC machine selection for application. By understanding the construction, working principle and operation of these machines, students can contribute to the selection and installation for practical industry application.

AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code:	EEE 535	**Course Title:**	High Frequency and Microwave Electronics
Semester:	First	**Credit Hours**:	3
Contact Hours:	Lecture - Three hours lectures per Week	**Location**:	B2-1
Lecturer(s):	Engr. T. E. Fabunmi		

Textbooks and other Materials

Pozar, D. M. (2012). Microwave Engineering. Wiley, 4th Edition.
Collin, R. E. (2001). Foundations for Microwave Engineering. Wiley-IEEE Press.
Gupta, K. C., & Verma, R. (2014). Microwave Engineering. S. Chand Publications.
Liao, S. Y. (2010). Microwave Devices and Circuits. Pearson, 3rd Edition.
Lecture notes, PowerPoint slides

a) Course Overview and Description

This course surveys high-frequency and microwave electronics. It introduces wave propagation in waveguides and resonators, passive and active components, and modern semiconductor microwave devices. Students will learn about directional couplers, isolators, circulators, klystrons, magnetrons, traveling-wave tubes, parametric amplifiers, and solid-state microwave sources such as Gunn diodes, PIN diodes, and varactors. The course also explores optoelectronic devices such as photodiodes and phototransistors and their applications in communication and radar systems.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Elective

e) Course Learning Outcomes (CLOs) and Programme Outcomes (POs)

Course Learning Outcomes: upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**POs**	**Assessment Tools**
Explain the principles of wave propagation in waveguides, resonators, and passive components.	2	1, 8	Assignment, Quizzes, Exams
Analyze the characteristics of microwave passive devices such as couplers, isolators, and circulators.	4	1, 2, 3	Assignment, Quizzes, Exams
Evaluate the operation of active microwave devices such as klystrons, magnetrons, and traveling-wave tubes.	5	1, 2, 3, 4	Assignment, Quizzes, Exams
Apply knowledge of semiconductor microwave devices (varactor, PIN, Gunn diodes, photodiodes, phototransistors) to practical circuit design.	3	2, 3, 6	Assignment, Quizzes, Exams
Demonstrate the role of microwave electronics in modern communication and radar applications.	3	6, 10, 11, 12	Assignment, Quizzes, Exams

f) Mapping of CLOs to Programme Outcomes (POs)

Course Learning Outcomes	**Programme Outcomes**											
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**

Explain wave propagation and passive components	3							2				
Analyze microwave passive devices	3	2	2									
Evaluate active microwave devices	3	2	2	2								
Apply semiconductor microwave devices		2	2			2						
Demonstrate microwave electronics in communication/radar						2				2	2	2

Keys: 1 = Slightly related, 2 = Moderately related, 3 = Highly related

g) Expectations of Students

a. Attend lectures regularly.
b. Complete all assignments and quizzes.
c. Participate actively in tutorials and group work.
d. Complete continuous assessment which involves mid-semester test
e. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

a. PowerPoint-based lectures and whiteboard explanations.
b. Group discussions and interactive problem-solving sessions.

i) Course Content or Outline

Topics	**Weeks**
Introduction to microwave engineering and electromagnetic spectrum	1
Wave propagation in waveguides and resonators	2
Passive components: reactive and resistive elements	3
Directional couplers and tees	4
Ferrite devices: isolators and circulators	5
Active microwave devices: klystrons, magnetrons	6

Traveling-wave tubes and parametric amplifiers	7
Solid-state microwave sources and amplifiers	8
Varactor and PIN diodes	9
Gunn-effect diodes and their applications	10
Photodiodes and phototransistors in microwave systems	11
Applications in communication, radar, and satellite systems	12
Lecture free week	13
Revision & Examination	14–15

j) Method of Grading Continuous Assessment

Continuous Assessment (Assignments): 20%
Continuous Assessment (Test/Quizzes): 20%
End of Semester Examination: 60%

k) Tutorials

Tutorials are taken at the end of each module.

l) Term Paper and Assignments

Each student will prepare a term paper on an advanced microwave device (e.g., Gunn diode oscillator, magnetron in radar, or photonic microwave systems). Assignments will involve simulation and design tasks on passive and active components.

m) Contemporary Issues and Industrial Relevance

Microwave electronics is vital for modern telecommunications, radar, satellite communication, and defense systems. This course equips students with knowledge relevant to 5G/6G wireless technologies, IoT connectivity, and aerospace applications. Graduates with this training are highly valued in industries developing sustainable and high-speed communication systems.



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AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code:	EEE 441	**Course Title:**	Control Theory/Systems and Servomechanism
Semester:	First	**Credit Hours**:	3
Contact Hours:	Lecture - Three hours lectures per Week	**Location**:	B2-1
Lecturer(s):	Engr. T. E. Fabunmi		

Textbooks and other Materials:

Ogata, K. (2010). Modern Control Engineering. Prentice Hall, 5th Edition.
Dorf, R. C., & Bishop, R. H. (2022). Modern Control Systems. Pearson, 14th Edition.
Nise, N. S. (2020). Control Systems Engineering. Wiley, 8th Edition.
Bolton, W. (2015). Mechatronics: Electronic Control Systems in Mechanical and Electrical Engineering. Pearson.
Lecture notes, PowerPoint slides

a) Course Overview and Description

This course provides a foundation in feedback control theory and servomechanisms. Students will study system modeling (electrical, mechanical, hydraulic, pneumatic, and thermal), frequency response techniques, stability analysis, compensation methods, and controller design. The course integrates practical servomechanism applications with theoretical analysis to prepare students for modern control engineering challenges in industry.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Compulsory

e) Course Learning Outcomes (CLOs) and Programme Outcomes (POs)

Course Learning Outcomes: upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**POs**	**Assessment Tools**
Explain the fundamentals of feedback control systems and their components.	2	1, 8	Assignment, Quizzes, Exams
Model dynamic systems (electrical, hydraulic, pneumatic, thermal) using transfer functions.	3	1, 2, 3	Assignment, Quizzes, Exams
Analyze system response and stability using time-domain and frequency-domain methods (Bode, Nyquist, Root Locus).	4	1, 2, 3, 4	Assignment, Quizzes, Exams
Design compensators and controllers (PID, Lag, Lead, Lead-Lag) for stability, tracking, and regulation objectives.	5	2, 3, 4, 10, 12	Assignment, Quizzes, Exams
Demonstrate practical application of servomechanisms (motors, tachogenerators, actuators, error detectors).	3	6, 11, 12	Assignment, Quizzes, Exams

f) Mapping of CLOs to Programme Outcomes (POs)

Course Learning Outcomes	**Programme Outcomes**											
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**

Explain fundamentals of feedback control systems	3							2				
Model dynamic systems with transfer functions	3	2	2									
Analyze system response and stability	3	2	2	2								
Design compensators and controllers		2	3	2						2		2
Demonstrate servomechanism applications						2					2	2

Keys: 1 = Slightly related, 2 = Moderately related, 3 = Highly related

g) Expectations of Students

f. Attend lectures regularly.
g. Complete all assignments and quizzes.
h. Participate actively in tutorials and group work.
i. Complete continuous assessment which involves mid-semester test
j. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

c. PowerPoint-based lectures and whiteboard explanations.
d. Group discussions and interactive problem-solving sessions.

i) Course Content or Outline

Topics	Weeks

Introduction to feedback control systems and servomechanisms	1
Transfer functions and system modeling: electrical, mechanical, hydraulic, pneumatic, thermal	2
Time response analysis and system classification	3
Stability analysis: Routh-Hurwitz criterion Frequency response: Polar, Bode, Nyquist, Nichols	4
Nyquist stability criterion, M- and N-circles, inverse Nyquist plots	5
Relative stability, root locus method	6
Bode diagrams and determination of transfer function from plots	7
Feedback compensation: objectives of tracking and regulation	8
Controller design: PID, Lead, Lag, Lead-Lag compensation	9
Servomechanical systems: motors, tachogenerators, error detectors, actuators	10
Applications of servomechanisms: position & velocity control systems, voltage regulators	11
Introduction to nonlinear systems and case studies	12
Lecture free week	13
Revision & Examination	14–15

j) Method of Grading Continuous Assessment

Continuous Assessment (Assignments): 20%
Continuous Assessment (Test/Quizzes): 20%
End of Semester Examination: 60%

k) Tutorials

Tutorials are scheduled at the end of each module.

l) Term Paper and Assignments

Students will submit a term paper analyzing a real-world servomechanism (e.g., robotics, aerospace control, automotive control). Assignments will include transfer function modeling, stability analysis, and compensator design problems.

m) Contemporary Issues and Industrial Relevance

This course prepares students for careers in automation, robotics, aerospace, and power systems. Servomechanism applications are key in robotics, renewable energy, automotive systems (ABS, cruise control), and process industries. By learning modern control methods, students gain skills directly relevant to emerging technologies and sustainable automation.

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AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code:	EEE 556	**Course Title:**	Electric Drives and Applications
Semester:	First	**Credit Hours**:	3
Contact Hours:	Lecture - Three hours lectures per Week	**Location**:	B2-1
Lecturer(s):	Engr. T. E. Fabunmi		

Textbooks and other Materials

Gopal K. Dubey, Fundamentals of Electric Drives, Narosa Publishing, Latest Edition.
R. Krishnan, Electric Motor Drives: Modeling, Analysis and Control, Prentice Hall.
Vedam Subrahmanyam, Electric Drives: Concepts and Applications, Tata McGraw-Hill.

Class notes, PowerPoint slides, and online resources provided by the lecturer.

a) Course Overview and Description

This course introduces students to the fundamental principles and applications of electric drives. It covers various motor types, speed–torque characteristics, speed control techniques, braking, motor rating and selection, protection, and control. Emphasis is placed on the industrial and commercial applications of electric drives, their role in automation, and their relevance in modern engineering practice.

b) Pre-requisites: EEE 326

c) Co-requisite(s): Nil

d) Role in Curriculum: Elective

e) Course Learning Outcomes (CLOs) and Programme Outcomes (POs)

Course Learning Outcomes: upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**POs**	**Assessment Tools**
Explain the historical development and classifications of electric motors including DC, AC, synchronous, and special motors.	2	1, 8	Assignment, Quizzes, Exams
Analyze speed–torque characteristics and methods of speed control in electric drives.	4	1, 2, 3	Assignment, Quizzes, Exams
Apply braking techniques, power rating selection, and protection methods in electric drive systems.	3	2, 3, 10	Assignment, Quizzes, Exams

Design and evaluate control schemes for electric drive applications under realistic constraints.	5	3, 4, 10, 12	Assignment, Quizzes, Exams
Demonstrate the application of electric drives in industrial and commercial systems.	3	6, 11, 12	Assignment, Quizzes, Exams

f) Mapping of CLOs to Programme Outcomes (POs)

Course Learning Outcomes	**Programme Outcomes**											
	1	2	3	4	5	6	7	8	9	10	11	12
Explain motor types and development	3							2				
Analyze speed–torque and control methods	3	2	2									
Apply braking, rating and protection methods		2	2							2		
Design and evaluate control schemes			3	2						2		2
Demonstrate industrial applications						2					2	2

Keys: 1 = Slightly related, 2 = Moderately related, 3 = Highly related

g) Expectations of Students

k. Attend lectures regularly.
l. Complete all assignments and quizzes.
m. Participate actively in tutorials and group work.
n. Complete continuous assessment which involves mid-semester test
o. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

e. PowerPoint-based lectures and whiteboard explanations.
f. Group discussions and interactive problem-solving sessions.

i) Course Content or Outline (Topics vs Weeks)

Topics	Weeks
Historical development of electric drives; Overview of motor types	1
DC Motors Shunt, Series, and Compound Wound: principles and applications	2
AC Induction Motors construction, speed torque characteristics	3
Special Motors Repulsion, Reluctance, Hysteresis, Synchronous motors	4
Speed torque characteristics of various motors	5
Speed control methods of electric drives	6
Braking methods: regenerative, dynamic, plugging	7
Motor power rating selection for industrial applications	8
Motor protection methods (thermal, overload, fault)	9
Control of drive motors conventional and modern methods	10
Applications of electric drives in industry (case studies)	11
Applications of electric drives in commercial systems (case studies)	12
Lecture free week	13
Revision & Examination	14–15

j) Method of Grading Continuous Assessment

Continuous Assessment (Assignments): 20%
Continuous Assessment (Test/Quizzes): 20%
End of Semester Examination: 60%

k) Tutorials

Tutorials are taken at the end of each module.

l) Term Paper and Assignments

Each student will complete a term paper on an application of electric drives (e.g., automation, renewable energy integration, electric vehicles). Assignments will include problem sets on speed–torque analysis, braking, and motor control strategies.

m) Contemporary Issues and Industrial Relevance

This course equips students with the ability to address industrial challenges in automation, renewable energy, and electric vehicles. By learning drive applications, students can design energy-efficient, safe, and sustainable motor control systems. The knowledge is directly relevant to industries such as manufacturing, oil & gas, power systems, and transportation.



AFE BABALOLA UNIVERSITY
ADO-EKITI (ABUAD)
KM. 8.5, AFE BABALOLA WAY,
ADO-EKITI, EKITI STATE, NIGERIA.
P.M.B 5454 ADO-EKITI.

AFE BABALOLA UNIVERSITY ADO-EKITI (ABUAD)

College of Engineering

Department of Electrical/Electronics and Computer Engineering

Course Code:	EEE 509	**Course Title:**	Advanced Electronics
Semester:	First	**Credit Hours**:	**3**
Contact Hours:	Lecture - Three hours lectures per Week	**Location**:	B2-5

Lecturer(s): Engr. A. Akinola

Textbooks and other Materials:

IV. O. N. Pandey (2022), Electronics Engineering. 2nd edition, Springer, Switzerland

V. Daniel A. Steck, Analog and Digital Electronics, available online at http://steck.us/teaching (revision 1.0.6, 22 June 2021).

VI. Kevin Karplus (2020), Applied Analog Electronics: a first course in electronics, 2nd edition, Leanpub, California, USA
VII. Anil K. Maini (2007), Digital Electronics: Principles, Devices and Applications 1st edition, John Wiley & Sons Ltd, The Atrium, Southern Gate, Chichester, West Sussex, England

a) Course Overview and description

This course covers the operations and applications operational amplifiers (Op-amp) for linear and non-linear operations, dynamic logic systems and digital circuits in instrumentation. Topics include; Linear I.C. op-amp, linear and non-linear operations, logarithmic amplifiers, Analog-to-Digital (ADC) and Digital-to- Analog (DAC) converters, gyrators and negative impedance converters, the 555-timer structure and applications; four-quadrant multipliers, dynamic logic systems and RAM memory circuits.

b) Pre-requisites: Nil

c) Co-requisite(s): Nil

d) Role in Curriculum: Compulsory

e) Course Learning Outcome (CLO) and Programme Outcomes (PO)

Course Learning Outcomes: upon successful completion of the course, students will be able to:	**Suitable Bloom Taxonomy Relevant to the CLOs**	**POs**	**Assessment Tools**
Analyze and apply operational amplifiers in linear and non-linear circuit design.	3, 4	1, 2, 3	Assignment, quizzes and Exams
Design and evaluate advanced analog circuits including, logarithmic amplifiers, multipliers, gyrators and negative impedance converters.	5, 6	1, 2, 3, 4	Assignment, quizzes and Exams
Compare and implement ADC/DAC systems in instrumentation	3, 4	1, 2, 3, 4, 5	Assignment, quizzes and Exams
Apply 555 timer circuits in various timing applications	4	1, 2, 3	Assignment, quizzes and Exams
Demonstrate digital circuit applications in instrumentation	4	1, 2, 3, 4, 5 ,6, 7	Assignment, quizzes and Exams

f) Mapping of CLOs to relevant POs

Course Learning Outcomes	Programme Outcomes											
	1	2	3	4	5	6	7	8	9	10	11	12
Analyze and apply operational amplifiers in linear and non-linear circuit design.	3	3	2									
Design and evaluate advanced analog circuits (log amplifiers, multipliers, gyrators, NICs).	3	3	3	2								
Compare and implement ADC/DAC systems in instrumentation.	3	3	3	3	2							
Apply 555 timer circuits in various timing applications.	2	2	2									
Demonstrate digital circuit applications in instrumentation	3	3	3	2	2	2	2					

Keys: 1 = slightly related, 2 = moderately related, 3 = highly related

g) Expectations of Students

a. Attend 3 hours of lectures per week

b. Turn in assignments considered suitable by the tutors

c. Complete continuous assessment which involves mid-semester test

d. Complete a 3-hour comprehensive final examination at the end of the semester.

h) Methods of Lecture Delivery/Teaching Aids

Lecture Delivery Methods

Educator-Student Interaction (Discussion method and analogies)

Teaching Aids

Visual Aids (Use of Power Point slides, Lecture templates-handouts)

i) Course Content or Outline

Topics	Weeks
• Introduction to Op-Amp IC	1
• Linear Op-Amp and Non-linear Op-Amp Circuits	2-3
• Logarithmic Amplifiers	4
• 555 Timer Structure/ Applications	5 - 6
• ADC /DAC Types	7 - 8
• Gyrators and NICs	9
• Four-Quadrant Multipliers	10
• Dynamic Logic Systems • RAM Memory Circuits • Application digital Circuits in Instrumentation	11-12
• **Revision & Examination**	13 - 15

j) Method of Grading Continuous Assessment

Continuous Assessment (Assignments): 10%

Continuous Assessment (Test):20%

Continuous Assessment (Quiz): 5%

Attendance (5%)

End of Semester Examination: 60%

k) Tutorials

Tutorials are taken at the end of each module.

l) Term Paper and Assignments

A term paper focused on specific aspect of a Digital instrumentation, such as analyzing digital circuit for specific application in digital instrumentation or a report of circuit detailing design of specific application using 555 timer.

m) Contemporary Issues and Industrial Relevance

The course equips students with the knowledge of design of electronic systems. By understanding linear and nonlinear operations of operational amplifies, students would be able to design circuit for specific applications as in analog to digital converters, digital to analog converters.

SECTION FIVE: CURRICULUM AND LEARNING PROCESS

The Curriculum

The Curriculum

Discuss the program structure and course contents to show how they are appropriate to, consistent with, and support the development of the range of intellectual and practical skills and attainment or achievement of the PLOs:

The program structure and course contents are designed to be appropriate, consistent, and supportive of developing a range of intellectual and practical skills, ensuring the attainment of the Programme Learning Outcomes (PLOs). The curriculum is carefully crafted to provide a comprehensive education in Electrical/Electronic Engineering, integrating both theoretical knowledge and hands-on experience. Each course is aligned with the PLOs, contributing to the overall educational objectives of the programme. Core courses cover fundamental concepts, advanced theories, and practical applications, ensuring that students gain a deep understanding of the field. Elective courses allow students to specialize in areas of interest, further enhancing their expertise and skills. Laboratory work, projects, and internships are integral components of the programme, providing students with opportunities to apply their knowledge in real-world scenarios. These practical experiences are essential for developing problem-solving abilities, technical proficiency, and professional competencies. The curriculum includes courses on ethics, communication, and management to ensure that graduates are well-rounded professionals capable of addressing complex challenges in their careers. The programme structure and course contents are meticulously designed to support the holistic development of students, equipping them with the necessary skills and knowledge to achieve the PLOs and excel in the field of Electrical/Electronic Engineering.

Distribution of the Engineering Courses

Visit departmental handbook for details on the course structure and course contents at departmental website

COURSE CURRICULUM

Categorization of Courses

Courses are categorized as follows:

<table>
<tr><th rowspan="2">Domain</th><th rowspan="2">Knowledge Area</th><th colspan="2">COREN/NUC Recommended</th><th colspan="2">Institute's Program Breakup</th></tr>
<tr><th></th><th>Overall</th><th>Total</th><th>Overall</th></tr>
<tr><th colspan="2"></th><th>Total Credit</th><th>%</th><th>Credits</th><th>%</th></tr>
<tr><td rowspan="3">Non-Engineering</td><td>Humanity</td><td rowspan="3">As per discipline specific COREN BMAS guidelines</td><td rowspan="3">25% - 40%</td><td>21</td><td rowspan="3">29%</td></tr>
<tr><td>Management Sciences</td><td>8</td></tr>
<tr><td>Natural Sciences</td><td>28</td></tr>
<tr><td rowspan="7">Engineering</td><td>Computing</td><td rowspan="5">As per discipline specific COREN BMAS guidelines</td><td rowspan="7">60% - 75%</td><td>13</td><td rowspan="7">71%</td></tr>
<tr><td>Engineering Foundation</td><td>36</td></tr>
<tr><td>Major Based Core (Breadth)</td><td>20</td></tr>
<tr><td>Major Based Core (Depth)</td><td>40</td></tr>
<tr><td>Inter-Disciplinary Engineering Breadth (Electives)</td><td>12</td></tr>
<tr><td>Final Year Design Project</td><td>6</td><td>6</td></tr>
<tr><td>Industrial Training SIWES</td><td>13</td><td>12</td></tr>
<tr><td colspan="2">Total</td><td>130-138</td><td></td><td>197</td><td></td></tr>
</table>

i. Compulsory Courses: Students must register for and pass before graduation.

ii. Elective Courses: Courses students may offer and pass to obtain the minimum units required for graduation. Such courses are normally selected from within or outside the programme with the approval of the department.

iii. Required Course: A course specified by a programme which a student shall take and pass.

iv. Pre-requisite Courses: Courses students must take and pass before taking another specified course.

v. Institutional Courses: Courses whose knowledge is essential for the preparation of Afe Babalola University for future employment after graduation.

vi. Carry-over Courses: On the account of any failure by a student to satisfy a pass requirement in any course, such student will be required to repeat only the course/courses the following academic session or re-write the prescribed examination in such course at the end of the academic session.

Students carrying over courses shall not be allowed to register for more than the approved maximum number of credits for the semester.

B.ENG. ELECTRICAL AND ELECTRONICS (5-YEAR PROGRAMME)

100 LEVEL FIRST SEMESTER

Course Code	Course Title	Units	LH	PH	Status
GST 111	Communication in English	2	15	45	Compulsory
GET 101	Engineer in Society	1	15	-	Compulsory
MTH 101	Elementary Mathematics I: Algebra and Trigonometry	2	30	-	Compulsory
PHY 101	General Physics I: Mechanics	2	30	-	Compulsory
PHY 107	General Practical Physics I	1	-	45	Compulsory
EEE 102	Introduction to Electrical and Electronics Engineering	2	30	-	Compulsory
ABUAD-CHM 101	General Chemistry I	2	45	-	Compulsory
ABUAD-CHM 107	General Practical Chemistry I	1	-	45	Compulsory
AFE 101	Health Awareness Education	1	15	-	Compulsory
AFE 103	Leadership	1	15	-	Compulsory
TOTAL		**15**			

100 LEVEL SECOND SEMESTER

Course Code	Course Title	Units		LH	PH	

GST 112	Nigerian Peoples and Culture	2		30	-	Compulsory
GET 102	Engineering Graphics and Solid Modelling I	2		15	45	Compulsory
MTH 102	Elementary Mathematics II: Calculus	2		30	-	Compulsory
PHY 102	General Physics II: Behaviour of Matter	2		30	-	Compulsory
PHY 108	General Practical Physics II	1		-	45	Compulsory
ABUAD-CHM 102	General Chemistry II	2		45	-	Compulsory
ABUAD-CHM 108	General Practical Chemistry II	1		-	45	Compulsory
FOREIGN LANGUAGE ELECTIVE (SELECTION OF ONE)						
AFE 124	French Language I	1		15	-	Elective
AFE 126	Chinese Language	1		15	-	Elective
NIGERIAN LANGUAGE ELECTIVE (SELECTION OF ONE)						
AFE 128	Igbo Language	1		15	-	Elective
AFE 130	Yoruba Language	1		15	-	Elective
AFE 132	Hausa Language	1		15	-	Elective
TOTAL		**14**				

SECOND YEAR (200 LEVEL): FIRST SEMESTER COURSE STRUCTURE

Course Code	**Course Title**	**Units**	**Status**	**LH**	**PH**
ENT 211	Entrepreneurship and Innovation	2	C	30	-
GET 201	Applied Electricity I	3	C	45	-
GET 203	Engineering Graphics and Solid Modeling II	2	E	15	45
GET 205	Fundamentals of Fluid Mechanics	3	C	45	-
GET 207	Applied Mechanics	3	C	45	-
GET 209	Engineering Mathematics I	3	C	45	-
GET 211	Computing and Software Engineering	3	C	30	45
One foreign language is required					
AFE 203*	French Language Acquisition II	1	E		
AFE 205*	Chinese Language II	1	E		
	One Nigerian Language is required				

AFE 207**	Igbo Language II	1	E		
AFE 209**	Yoruba Language II	1	E		
AFE 211**	Hausa Language II	1	E		
Total		**23**			

SECOND YEAR (200 LEVEL): SECOND SEMESTER COURSE STRUCTURE

Course Code	**Course Title**	**Units**	**Status**	**LH**	**PH**
GST 212	Philosophy, Logic and Human Existence	2	C	30	-
GET 204	Students Workshop Practice	2	C	15	45
GET 206	Fundamentals of Thermodynamics	2	C	45	-
GET 210	Engineering Mathematics II	3	C	45	-
EEE 202	Applied Electricity II	3	C	30	45
EEE 204	Electrical Engineering Materials	3	E	45	-
ABUAD-GET 208	Strength of Materials	3	C	45	-
AFE 202	Food Production and Health Awareness	1			
*GET 299	SIWES I	3*	C	9 Weeks	
Total		**20**			

300 LEVEL FIRST SEMESTER

Course Code	**Course Title**	**Units**	**Duration (Hours)**			**Pre-Req.**	**Status**
			L	**T**	**P**		
EEE 313	Electronics I (Physical)	3	2	1	-	-	Compulsory
EEE 315	Electromagnetic Field and Wave Theory I	3	2	1	-	-	Compulsory
EEE 317	Electrical/Electronic Laboratory I	1	-	-	3	-	Compulsory
EEE 319	Measurements and Instrumentation	3	2	1	-	-	Compulsory
EEE 321	Electric Circuit Theory I	3	2	1	-	-	Compulsory
EEE 325	Electrical Machines I	3	2	1	-	-	Compulsory

EEE 341	Software Development Techniques	3	2	-	3	-	Compulsory
ENG 381	Engineering Mathematics III	3	2	1	-	-	Compulsory
AFE 301	Practical Entrepreneurship	1	1	-	3	-	Compulsory
		23					

300 LEVEL SECOND SEMESTER

Course Code	**Course Title**	**Units**	**Duration (Hours)**			**Pre-Req.**	**Status**
			L	**T**	**P**		
EEE 314	Electronics II (Analogue)	3	2	1	-	EEE 313	Compulsory
EEE 316	Electromagnetic Field and Wave Theory II	3	2	1	-	EEE 315	Compulsory
EEE 318	Electrical/Electronic Laboratory II	1	-	-	3	-	Compulsory
EEE 322	Electric Circuit Theory II	3	2	1	-	EEE 321	Compulsory
EEE 324	Linear Systems	3	2	1	-	-	Compulsory
EEE 326	Electrical Machines II	3	2	1	-	EEE 325	Compulsory
EEE 328	Communication Principles	3	2	1	-		Compulsory
ENG 382	Engineering Mathematics IV	3	2	1	-		Compulsory
ENG 384	Engineering Economics	2	2	-	-		Compulsory
		24					

400 LEVEL FIRST SEMESTER

Course Code	**Course Title**	**Units**	**Duration (Hours)**			**Pre-Req**	**Status**
			L	**T**	**P**		
EEE 405	Numerical Computations	3	2	1	-	-	Compulsory
EEE 415	Assembly Language Programming	2	2	-	-	-	Compulsory
EEE 423	Electrical/Electronic Laboratory III	1	-	-	3	EEE 318	Compulsory
EEE 427	Data Communication and Networks	3	2	1	-	-	Compulsory
EEE 431	Electric Power Principles	3	2	1	-	-	Compulsory

EEE 441	Control Theory and Servomechanism	3	2	1	-	-	Compulsory
EEE 471	Digital Electronics and Systems	3	2	1	-	EEE 314	Compulsory
ENG 485	Technical Communications (Report Writing)	2	2	-	-	--	Compulsory
		20					

400 LEVEL SECOND SEMESTER

Course Code	**Course Title**	**Units**	**Duration (Hours)**			**Pre-Req**	**Status**
			L	**T**	**P**		
ENG 290	Student Work Experience Programme I (SWEP I)	3				-	Compulsory
ENG 390	Student Work Experience Programme II (SWEP II)	3				ENG 290	Compulsory
ENG 490	Student Industrial Work Experience Scheme (SIWES)	6				ENG 390	Compulsory
		12					

500 LEVEL FIRST SEMESTER

COURSE CODE	**COURSE TITLE**	**L**	**T**	**P**	**U**	**PRE-REQ.**	**STATUS**
EEE 507	Modelling and Simulation of Electrical and Electronics Systems	1	-	3	2	-	C
EEE 509	Advanced Electronics	2	-	3	3	-	C
EEE 511	Digital Signal Processing	2	1	-	3	-	C
EEE 561	Control Engineering	2	1	-	3	EEE 441	C
ENG 583	Engineering Management and Law	2	-	-	2	-	C
Electives - Minimum of two courses to be selected							
ELECTRONICS/ COMMUNICATION							
EEE 513	Communication Systems Engineering	2	1	-	3		E
EEE 523	Industrial Electronics Designs	2	1	-	3	-	E
EEE 535	High Frequency & Microwave Electronics	2	1	-	3	-	E
POWER AND MACHINES							

EEE 553	Power System Engineering	2	1	-	3	-	E
EEE 555	High Voltage Engineering	2	-	-	2	-	E
EEE 543	Renewable Energy System	2	-	-	2	-	E
CONTROL AND COMPUTER							
EEE 565	Modern Control Engineering (Modern)	2	1	-	3	-	E
EEE 523	Industrial Electronics Designs	2	1	-	3	-	E
EEE 517	Analogue and Digital Computers	2	-	-	2		E

SECOND SEMESTER 500 LEVEL COURSES

COURSE CODE	COURSE TITLE	L	T	P	U	PRE-REQ.	STATUS
EEE 502	Electrical Services Design	2	-	-	2	-	C
EEE 504	Embedded System	2	1	-	3		C
EEE 514	Reliability Engineering	1	1	-	2	-	C
EEE 562	Power Electronics	2	1	-	3	EEE 326	C
EEE 564	Seminars on Use of Engineering Software Packages in Electrical/Electronics Field	-	-	3	1	-	C
EEE 599	Student Project	-	-	18	6	-	C
Electives - Minimum of two courses to be selected							
ELECTRONICS/ COMMUNICATION							
EEE 512	Digital Communication	2	-	-	2	EEE 328	E
EEE 506	Digital System Design with VHDL Programming	2	1	-	3	-	E
EEE 578	Broadcasting and Internet Technology	1	-	3	2	-	E
EEE 576	Microcomputer Technology	3	-	-	3	-	E
POWER AND MACHINES							
EEE 554	Power Systems Communication and Control	2	1	-	3	EEE 553	E
EEE 556	Electric Drives and Applications	2	-	3	3	EEE 326	E
EEE 558	Electrical Energy Conversion and Storage	2	-	-	2	-	E
CONTROL AND COMPUTER							
EEE 566	Control Engineering III (Non-Linear & Process Automation)	2	1	-	3	EEE 324	E
EEE 506	Digital System Design with VHDL Programming	2	1	-	3	-	E
EEE 576	Microcomputer Technology	3	-	-	3	-	E

EEE 512	Digital Communication	2	-	-	2	EEE 413	E

5.1. Course Description

100 LEVEL

GST 111: Communication in English (2 Units C: LH=15; PH 45)

Learning Outcomes

At the end of this course, students should be able to:

1. identify possible sound patterns in English Language;
2. list notable language skills;
3. classify word formation processes;
4. construct simple and fairly complex sentences in English;
5. apply logical and critical reasoning skills for meaningful presentations;
6. demonstrate an appreciable level of the art of public speaking and listening; and
7. write simple and technical reports.

Course Contents

Sounds and sound patterns in English Language (vowels and consonants, phonetics and phonology). English word classes (lexical and grammatical words, definitions, forms, functions, usages, collocations). major word formation processes; the sentence in English (types: structural and functional). grammar and usage (tense, concord and modality). Reading and types of reading, comprehension skills, 3RsQ. Logical and critical thinking; reasoning methods (logic and syllogism, inductive and deductive argument, analogy, generalisation and explanations). Ethical considerations, copyright rules and infringements. Writing activities: pre-writing (brainstorming and outlining). writing (paragraphing, punctuation and expression). post- writing (editing and proofreading). Types of writing (summary, essays, letter, curriculum vitae, report writing, note-making) etc. Mechanics of writing. Information and Communication Technology in modern language learning. Language skills for effective communication. The art of public speaking.

GET 101: Engineer in Society (1 Unit C: LH=15)

Learning Outcomes

At the end of this course, the students should be able to:

1. differentiate between science, engineering and technology, and relate them to innovation;
2. distinguish between the different cadres of engineering – engineers, technologists, technicians and craftsmen and their respective roles and competencies;
3. identify and distinguish between the relevant professional bodies in engineering;
4. categorise the goals of global development or sustainable development goals (SDGs); and

5. identify and evaluate safety and risk in engineering practice.

Course Contents

History, evolution and philosophy of science. engineering and technology. The engineering profession – engineering family (engineers, technologists, technicians and craftsmen), professional bodies and societies. Engineers' code of conduct and ethics, and engineering literacy. Sustainable development goals (SDGs), innovation, infrastructures and nation building - economy, politics, business. Safety and risk analysis in engineering practice. Engineering competency skills – curriculum overview, technical, soft and digital skills. Guest seminars and invited lectures from different engineering professional associations.

MTH 101: Elementary Mathematics I (Algebra and Trigonometry)
(2 Units C: LH=30)

Learning Outcomes

At the end of the course students should be able to:
1. define and explain set, subset, union, intersection, complements, and demonstrate the use of Venn diagrams;
2. solve quadratic equations;
3. solve trigonometric functions;
4. identify various types of numbers; and
5. solve some problems using binomial theorem.

Course Contents

Elementary set theory, subsets, union, intersection, complements, Venn diagrams. Real numbers, integers, rational and irrational numbers. Mathematical induction, real sequences and series, theory of quadratic equations, binomial theorem, complex numbers, algebra of complex numbers, the argand diagram. De-Moiré's theorem, nth roots of unity. Circular measure, trigonometric functions of angles of any magnitude, addition and factor formulae.

PHY 101: General Physics I (Mechanics) (2 Units C: LH=30)
Learning Outcomes

On completion, the students should be able to:
1. identify and deduce the physical quantities and their units;
2. differentiate between vectors and scalars;
3. describe and evaluate motion of systems on the basis of the fundamental laws of mechanics;
4. apply Newton's laws to describe and solve simple problems of motion;

5. evaluate work, energy, velocity, momentum, acceleration, and torque of moving or rotating objects;
6. explain and apply the principles of conservation of energy, linear and angular momentum;
7. describe the laws governing motion under gravity; and
8. explain motion under gravity and quantitatively determine behaviour of objects moving under gravity.

Course Contents

Space and time; units and dimension, vectors and scalars, differentiation of vectors: displacement, velocity and acceleration; kinematics; Newton's laws of motion (inertial frames, impulse, force and action at a distance, momentum conservation); relative motion; application of Newtonian mechanics; equations of motion; conservation principles in physics, conservative forces, conservation of linear momentum, kinetic energy and work, potential energy, system of particles, centre of mass; rotational motion; torque, vector product, moment, rotation of coordinate axes and angular momentum. Polar coordinates; conservation of angular momentum; circular motion; moments of inertia, gyroscopes and precession; gravitation: Newton's law of gravitation, Kepler's laws of planetary motion, gravitational potential energy, escape velocity, satellites motion and orbits.

PHY 107: General Practical Physics I (1 Unit C: PH 45)

Learning Outcomes

On completion, the student should be able to:
1. conduct measurements of some physical quantities;
2. make observations of events, collect and tabulate data;
3. identify and evaluate some common experimental errors;
4. plot and analyse graphs; and
5. draw conclusions from numerical and graphical analysis of data.

Course Contents

This introductory course emphasizes quantitative measurements. Experimental techniques. The treatment of measurement errors. Graphical analysis. The experiments include studies of meters, the oscilloscope, mechanical systems, electrical and mechanical resonant systems, light, heat, viscosity, etc. (covered in PHY 101, 102, 103 and PHY 104). However, emphasis should be placed on the basic physical techniques for observation, measurements, data collection, analysis, and deduction.

EEE 102: Introduction to Electrical and Electronics Engineering (2 units C: LH=15)

Learning Outcomes

Students will be able to:

1. comprehend the duties and functions of an Electrical and Electronics Engineer (EEE);
2. state the requirements for the profession and career opportunities;
3. state the careers related to EEE; and
4. explain the future of EEE.

Course Contents

History of Electrical Engineering. Evolution of EEE. Duties of EE Engineers. Areas of specialisation and work environment. Skill requirements (soft and hard). Qualities for EE Engineers. Careers related to EEE. Typical course modules. Job outlook/opportunities for EE Engineers. Future of EEE. Professional registration (NSE, COREN, IEEE, IET, etc.). Passive components (R, L, C, transformers): descriptive features, including values and colour codes, uses in electrical circuits.DC and AC signal parameters

ABUAD-CHM 101: General Chemistry I (2 Units C: LH=30)

Senate-approved relevance

This course is essential for students at our university as it builds a strong foundation in chemistry, aligning with Sustainable Development Goals (SDGs) 4 (Quality Education) and 9 (Industry, Innovation, and Infrastructure). A deep understanding of chemical principles is crucial for advancing in scientific research, environmental sustainability, and technological innovation. By mastering the content, students will be equipped to contribute to sustainable industrial practices, environmental conservation, and scientific advancements, directly impacting global challenges. This course fosters critical thinking and problem-solving skills, preparing students to be leaders in science, technology, and sustainability.

Overview

This course introduces the fundamental concepts of chemistry, including atoms, molecules, elements, and compounds. It explores modern electronic theory, electronic configuration, and periodicity, and examines molecular structures, chemical bonding, and stoichiometry. Students will also study the kinetic theory of matter, thermochemistry, reaction rates, equilibrium, and thermodynamics. Additionally, the course covers the properties of gases, acids, bases, redox

reactions, electrochemistry, and radioactivity, providing a comprehensive understanding of essential chemical principles.

Objectives

1. understand and describe the basic structure of atoms, molecules, elements, and compounds.
2. apply electronic configuration and periodicity to explain chemical properties.
3. analyze and predict the shapes and structures of molecules and solids.
4. solve chemical equations and apply principles of stoichiometry, thermodynamics, and kinetics.
5. understand and apply concepts of redox reactions, electrochemistry, and radioactivity.

Learning Outcomes

At the end of this course, the students should be able to:

1. define atom, molecules and chemical reactions;
2. discuss the modern electronic theory of atoms;
3. write electronic configurations of elements on the periodic table;
4. rationalise the trends of atomic radii, ionisation energies, electronegativity of the elements, based on their position in the periodic table;
5. identify and balance oxidation–reduction equation and solve redox titration problems;
6. draw shapes of simple molecules and hybridised orbitals;
7. identify the characteristics of acids, bases and salts, and solve problems based on their quantitative relationship;
8. apply the principles of equilibrium to aqueous systems using LeChatelier's principle to predict the effect of concentration, pressure and temperature changes on equilibrium mixtures;
9. analyse and perform calculations with the thermodynamic functions, enthalpy, entropy and free energy; and
10. determine rates of reactions and its dependence on concentration, time and temperature.

Course Contents

Atoms, molecules, elements and compounds, and chemical reactions. Modern electronic theory of atoms. Electronic configuration, periodicity and building up of the periodic table. Hybridisation and shapes of simple molecules. Valence forces; Structure of solids. Chemical equations and stoichiometry; chemical bonding and intermolecular forces, kinetic theory of matter. Elementary thermochemistry; rates of reaction, equilibrium and thermodynamics. Acids, bases and salts. Properties of gases. Redox reactions and introduction to electrochemistry. Radioactivity.

ABUAD-CHM 107: General Practical Chemistry I (1 Unit C: PH 45)

Senate-approved relevance

This laboratory course is integral to our university's commitment to Quality Education (SDG 4) and Responsible Consumption and Production (SDG 12). It provides hands-on experience in fundamental chemical techniques, reinforcing theoretical knowledge from CHM 101. By mastering these practical skills, students develop the ability to conduct accurate experiments, analyze data, and present findings effectively—essential competencies for careers in science, industry, and research. The course also promotes safe and sustainable laboratory practices, preparing students to contribute to sustainable industrial processes and environmental stewardship.

Overview

This laboratory course offers practical experience in key chemical techniques, complementing the theoretical content of CHM 101. Students will engage in acid-base titrations, qualitative analysis, redox reactions, and gravimetric analysis. Emphasis is placed on accurate data collection, analysis, and presentation, fostering essential laboratory skills. The course prepares students to apply these techniques in scientific research and industry, ensuring a comprehensive understanding of experimental chemistry.

Objectives

The objectives of the course are to:

1. perform acid-base titrations with precision and accuracy.
2. conduct qualitative and quantitative analysis of chemical substances.
3. understand and execute redox reactions and gravimetric analysis.
4. analyze and interpret experimental data effectively.
5. present laboratory findings clearly and professionally.

Learning Outcomes

At the end of this course, the students should be able to:

1. state the general laboratory rules and safety procedures;
2. collect scientific data and correct carry out chemical experiments;
3. identify the basic glassware and equipment in the laboratory;
4. state the differences between primary and secondary standards;
5. perform redox titration;
6. record observations and measurements in the laboratory notebooks; and
7. analyse the data to arrive at scientific conclusions.

Course Contents

Laboratory experiments designed to reflect topics presented in courses CHM 101 and CHM 102. These include acid-base titrations, qualitative analysis, redox reactions, gravimetric analysis, data analysis and presentation.

GST 112: Nigerian Peoples and Cultures (2 Units C: LH=30)

Learning Outcomes

At the completion of the course, the students are expected to be able:

1. analyse the historical foundation of Nigerian cultures and arts in pre-colonial times;
2. identify and list the major linguistic groups in Nigeria;
3. explain the gradual evolution of Nigeria as a political entity;
4. analyse the concepts of trade and economic self-reliance of Nigerian peoples in relation to national development;
5. enumerate the challenges of the Nigerian state regarding nation building;
6. analyse the role of the Judiciary in upholding fundamental human rights
7. identify the acceptable norms and values of the major ethnic groups in Nigeria; and
8. list possible solutions to identifiable Nigerian environmental, moral and value problems.

Course Contents

Nigerian history, culture and art up to 1800 (Yoruba, Hausa and Igbo peoples and cultures; peoples and cultures of the minority ethnic groups). Nigeria under colonial rule (advent of colonial rule in Nigeria; colonial administration of Nigeria). Evolution of Nigeria as a political unit (amalgamation of Nigeria in 1914; formation of political parties in Nigeria; nationalist movement and struggle for independence). Nigeria and challenges of nation building (military intervention in Nigerian politics; Nigerian Civil War). Concepts of trade and economics of self-reliance (indigenous trade and market system; indigenous apprenticeship system among Nigerian peoples; trade, skill acquisition and self-reliance). Social justice and national development (definition and classification of law); Judiciary and fundamental rights. Individuals, norms and values (basic Nigerian norms and values, patterns of citizenship acquisition; citizenship and civic responsibilities; indigenous languages, usage and development; negative attitudes and conducts [Cultism, kidnapping and other related social vices]). Re-orientation, moral and national values (The 3Rs – Reconstruction, Rehabilitation and Re-orientation; re-orientation strategies: Operation Feed the Nation (OFN), Green Revolution, Austerity Measures, War Against Indiscipline and Corruption (WAIC), Mass Mobilization for Self-Reliance, Social Justice and Economic Recovery (MAMSER), National Orientation Agency (NOA). Current socio-political and cultural developments in Nigeria.

GET 102: Engineering Graphics and Solid Modelling I (2 Units C: LH=15; PH 45)

Learning Outcomes

At the end of this course, the students should be able to:

1. have a good grasp of design thinking and be obsessed with the determination to apply such to solving simple every day and also complex problems;
2. recognize the fundamental concepts of engineering drawing and graphics;
3. show skills to represent the world of engineering objects in actionable solid models, and put such models in a form where they can be inputs for simulation and analyses;
4. analyse such models for strength and cost.
5. prepare the objects for modern production and manufacturing techniques of additive and subtractive manufacturing;
6. recognise that engineering is multidisciplinary in the sense that mechanical, electrical and other parts of physical structures are modelled in context as opposed to the analytical nature of the courses they take;and
7. analyse and master the basics of mechanical and thermal loads in engineering systems.

Course Contents

Introduction to design thinking and engineering graphics. First and third angle orthogonal projections. Isometric projections; sectioning, conventional practices, conic sections and development. Freehand and guided sketching – pictorial and orthographic. Visualisation and solid modelling in design, prototyping and product-making. User interfaces in concrete terms. Design, drawing, animation, rendering and simulation work spaces. Sketching of 3D objects. Viewports and sectioning to shop drawings in orthographic projections and perspectives. Automated viewports. Sheet metal and surface modelling. Material selection and rendering. This course will use latest professional design tools such as fusion 360, solid works, solid edge or equivalent.

MTH 102: Elementary Mathematics II (Calculus) (2 Units C: LH=30)

Learning Outcomes

At the end of the course, students should be able to:

1. identify the types of rules in differentiation and integration;
2. recognise and understand the meaning of function of a real variable, graphs, limits and continuity;
3. solve some applications of definite integrals in areas and volumes;
4. solve function of a real variable, plot relevant graphs, identify limits and idea of continuity;
5. identify the derivative as limit of rate of change;
6. identify techniques of differentiation and perform extreme curve sketching;

7. identify integration as an inverse of differentiation;
8. identify methods of integration and definite integrals; and
9. perform integration application to areas, volumes.

Course Contents

Functions of a real variable, graphs, limits and idea of continuity. The derivative, as limit of rate of change. Techniques of differentiation, maxima and minima. Extreme curve sketching, integration, definite integrals, reduction formulae, application to areas, volumes (including approximate integration: Trapezium and Simpson's rule).

PHY 102: General Physics II (Behaviour of Matter) (2 Units C: LH=30)

Learning Outcomes

On completion, the students should be able to:

1. explain the concepts of heat and temperature and relate the temperature scales;
2. define, derive and apply the fundamental thermodynamic relations to thermal systems;
3. describe and explain the first and second laws of thermodynamics, and the concept of entropy;
4. state the assumptions of the kinetic theory and apply techniques of describing macroscopic behaviour;
5. deduce the formalism of thermodynamics and apply it to simple systems in thermal equilibrium; and
6. describe and determine the effect of forces and deformation of materials and surfaces.

Course Contents

Heat and temperature, temperature scales; gas laws; general gas equation; thermal conductivity; first Law of thermodynamics; heat, work and internal energy, reversibility; thermodynamic processes; adiabatic, isothermal, isobaric; second law of thermodynamics; heat engines and entropy, Zero's law of thermodynamics; kinetic theory of gases; molecular collisions and mean free path; elasticity; Hooke's law, Young's shear and bulk moduli; hydrostatics; pressure, buoyancy, Archimedes' principles; Bernoullis equation and incompressible fluid flow; surface tension; adhesion, cohesion, viscosity, capillarity, drops and bubbles.

PHY 108: General Practical Physics II (1 Unit C: PH 45)

Learning Outcomes

On completion, the student should be able to:

1. conduct measurements of some physical quantities;
2. make observations of events, collect and tabulate data;
3. identify and evaluate some common experimental errors;

4. plot and analyse graphs;
5. draw conclusions from numerical and graphical analysis of data; and
6. prepare and present practical reports.

Course Contents

This practical course is a continuation of PHY 107 and is intended to be taught during the second semester of the 100 level to cover the practical aspect of the theoretical courses that have been covered with emphasis on quantitative measurements, the treatment of measurement errors, and graphical analysis. However, emphasis should be placed on the basic physical techniques for observation, measurements, data collection, analysis and deduction.

ABUAD-CHM 102: General Chemistry II (2 Units C: LH=30)

Senate-approved relevance

This course is crucial to our university's mission of advancing Quality Education (SDG 4) and fostering Innovation (SDG 9). It provides a comprehensive understanding of organic chemistry, which is fundamental to multiple scientific disciplines, including pharmaceuticals, materials science, and environmental science. The course highlights the role of organic chemistry in developing new materials like fullerenes, which are essential for nanotechnology. By mastering the concepts and techniques in organic chemistry, students are prepared to contribute to innovative solutions in health, industry, and sustainability, addressing global challenges through scientific advancement.

Overview

This course offers an in-depth study of organic chemistry, beginning with its historical development and significance. It covers the electronic theory, structure determination, nomenclature, and functional groups of organic compounds. Students will explore reaction mechanisms, kinetics, stereochemistry, and the chemistry of hydrocarbons, alcohols, ethers, amines, and other key organic compounds. Additionally, the course introduces the chemistry of selected metals and non-metals, including comparative chemistry of group IA, IIA, and IVA elements, and transition metal chemistry, with a focus on their practical applications in various industries.

Objectives

1. understand the historical development and significance of organic chemistry.
2. describe the structure, nomenclature, and functional groups of organic compounds.

3. analyze reaction mechanisms, kinetics, and stereochemistry in organic reactions.
4. explore the chemistry of hydrocarbons, alcohols, ethers, amines, and other key organic compounds.
5. examine the chemistry of selected metals, non-metals, and transition metals, including their industrial applications.

Learning Outcomes

At the end of this course, the students should be able to:

1. state the importance and development of organic chemistry;
2. define fullerenes and its applications;
3. discuss electronic theory;
4. determine the qualitative and quantitative of structures in organic chemistry;
5. state rules guiding nomenclature and functional group classes of organic chemistry;
6. determine the rate of reaction to predict mechanisms of reaction;
7. identify classes of organic functional group with brief description of their chemistry;
8. discuss comparative chemistry of group 1A, IIA and IVA elements; and
9. describe basic properties of transition metals.

Course Contents

Historical survey of the development and importance of organic chemistry; fullerenes as fourth allotrope of carbon, uses as nanotubules, nanostructures, nanochemistry. Electronic theory in organic chemistry. Isolation and purification of organic compounds; determination of structures of organic compounds including qualitative and quantitative analysis in organic chemistry; nomenclature and functional group classes of organic compounds. Introductory reaction mechanism and kinetics. Stereochemistry. The chemistry of alkanes, alkenes, alkynes, alcohols, ethers, amines, alkyl halides, nitriles, aldehydes, ketones, carboxylic acids and derivatives. The chemistry of selected metals and non-metals. Comparative chemistry of group IA, IIA and IVA elements. Introduction to transition metal chemistry.

ABUAD- CHM 108: General Practical Chemistry II (1 Unit C: PH 45)

Senate-approved relevance

This practical course is integral to our university's focus on Quality Education (SDG 4) and Responsible Consumption and Production (SDG 12). It builds on the foundation established in CHM 107, enhancing students' proficiency in functional group analysis and quantitative analysis using volumetric methods. These skills are essential for accurate chemical analysis in research and industry, promoting sustainable practices in chemical production and environmental monitoring. By mastering these techniques, students are better prepared to contribute to scientific innovation and responsible resource management, aligning with global sustainability goals.

Overview

This course continues from CHM 107, offering advanced laboratory experiments in functional group analysis and quantitative analysis through volumetric methods. Students will refine their skills in identifying functional groups and conducting precise quantitative measurements. The course emphasizes accurate data collection, analysis, and interpretation, which are critical for success in research and industrial applications. By the end of the course, students will have a deeper understanding of practical chemical analysis and its importance in various scientific fields.

Objectives

The objectives of the course are to:

1. perform advanced functional group analysis accurately.
2. conduct quantitative analysis using volumetric methods with precision.
3. analyze and interpret experimental data effectively.
4. develop proficiency in laboratory techniques for chemical analysis.
5. apply practical skills to real-world chemical problems and research.

Learning Outcomes

At the end of this course, the students should be able to:

1. state the general laboratory rules and safety procedures;
2. collect scientific data and correctly carry out chemical experiments;
3. identify the basic glassware and equipment in the laboratory;
4. identify and carry out preliminary tests which include ignition, boiling point, melting point, test on known and unknown organic compounds;
5. carry out solubility tests on known and unknown organic compounds;
6. carry out elemental tests on known and unknown compounds; and
7. carry out functional group/confirmatory test on known and unknown compounds which could be acidic/basic/ neutral organic compounds.

Course Contents

Continuation of CHM 107. Additional laboratory experiments to include functional group analysis, quantitative analysis using volumetric methods.

200 LEVEL FIRST SEMESTER

ENT 211: Entrepreneurship and Innovation (2 Units C: LH=30)

Learning Outcomes

At the end of this course, students should be able to:

1. explain the concepts and theories of entrepreneurship, intrapreneurship, opportunity seeking, new value creation and risk-taking;
2. state the characteristics of an entrepreneur;
3. analyse the importance of micro and small businesses in wealth creation, employment generation and financial independence;
4. engage in entrepreneurial thinking;
5. identify key elements in innovation;
6. describe the stages in enterprise formation, partnership and networking, including business planning;
7. describe contemporary entrepreneurial issues in Nigeria, Africa and the rest of the world; and
8. state the basic principles of e-commerce.

The concept of entrepreneurship (entrepreneurship, intrapreneurship/corporate entrepreneurship); theories, rationale and relevance of entrepreneurship (Schumpeterian and other perspectives, risk-taking, necessity and opportunity-based entrepreneurship, and creative destruction); characteristics of entrepreneurs (opportunity seeker, risk-taker, natural and nurtured, problem solver and change agent, innovator and creative thinker); entrepreneurial thinking (critical thinking, reflective thinking and creative thinking). Innovation (The concept of innovation, dimensions of innovation, change and innovation, knowledge and innovation). Enterprise formation, partnership and networking (basics of business plan, forms of business ownership, business registration and alliance formation, and joint ventures). Contemporary entrepreneurship issues (knowledge, skills and technology, intellectual property, virtual office and networking). Entrepreneurship in Nigeria (biography of inspirational entrepreneurs, youth and women entrepreneurship, entrepreneurship support institutions, youth enterprise networks and environmental and cultural barriers to entrepreneurship). Basic principles of e-commerce.

GET 201: Applied Electricity I (3 Units C: LH=45)

Learning Outcomes

Students will be able to:

1. discuss the fundamental concepts of electricity and electrical d.c. circuits;
2. state, explain and apply the basic d.c. circuit theorems;
3. explain the basic a.c. circuit theory and
4. apply to solution of simple circuits.

Course contents

Fundamental concepts: Electric fields, charges, magnetic fields. current, B-H curves Kirchhoff's laws, superposition. Thevenin Norton theorems, Reciprocity, RL, RC, RLC circuits. DC, AC bridges, Resistance, Capacitance, Inductance measurement, Transducers, Single phase circuits, Complex j - notation, AC circuits, impedance, admittance, and susceptance.

GET 203: Engineering Graphics and Solid Modeling II (3 Units E: LH=30; PH=45)

Learning Outcomes

Students should be able to:

1. apply mastery of the use of projections to prepare detailed working drawing of objects and designs;
2. develop skills in parametric design to aid their ability to see design in the optimal specification of materials and systems to meet needs;
3. be able to analyze and optimize designs on the basis of strength and material minimization;
4. get their appetites wet in seeing the need for the theoretical perspectives that create the basis for the analysis that are possible in design and optimization, and recognize/understand the practical link to excite their creativity and ability to innovate; and
5. be able to translate their thoughts and excitements to produce shop drawings for multi physical, multidisciplinary design.

Course Contents

Projection of lines, auxiliary views and mixed projection. Preparation of detailed working production drawing; semi-detailed drawings, conventional presentation methods. Solid, surface and shell modeling. Faces, bodies and surface intersections. Component-based design. Component assembly and motion constraints. Constrained motions and animation. Introduction to electronics modeling. Electronics board layout preparation, Component libraries and Schematic design. Parametric modeling and adaptive design. Simulation for material optimization. Designing for manufacturing. Additive and subtractive manufacturing. Production for 3-D printing, Laser cutting and CNC machinery. Arrangement of engineering components to form a working plant (Assembly Drawing of a Plant).

GET 205: Fundamentals of Fluid Mechanics (3 Units C: LH=45)

Learning Outcomes

At the end of this course, the students should be able to:

1. explain the properties of fluids;
2. determine forces in static fluids and fluids in motion;
3. determine whether a floating body will be stable;
4. determine the effect of various instruments, (valves, orifices, bends and elbows) on fluid f low in pipes;
5. measure flow parameters with venturi meters, orifice meters, weirs and others;
6. perform calculations based on principles of mass, momentum and energy conservation;
7. perform dimensional analysis and simple fluid modelling problems; and
8. specify the type and capacity of pumps and turbines for engineering applications.

Course Contents

Fluid properties, hydrostatics, fluid dynamics using principles of mass, momentum and energy conservation from a control volume approach. Flow measurements in pipes, dimensional analysis, and similitude, 2-dimensional flows. Hydropower systems.

GET 207: Applied Mechanics (3 Units C: LH=45)

Learning Outcomes

Students will acquire the ability to:

1. explain the fundamental principles of applied mechanics, particularly equilibrium analysis, friction, kinematics and momentum;
2. identify, formulate, and solve complex engineering problems by applying principles of engineering, science, mathematics and applied mechanics;
3. synthesize Newtonian Physics with static analysis to determine the complete load impact (net forces, shears, torques, and bending moments) on all components (members and joints) of a given structure with a load; and
4. apply engineering design principles to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

Course Contents

Forces, moments, couples. Equilibrium of simple structures and machine parts. Friction. First and second moments of area; centroids. Kinematics of particles and rigid bodies in plane motion. Newton's laws of motion. Kinetic energy and momentum analyse

GET 209: Engineering Mathematics I (3 Units C: LH=45)

Learning Outcomes

At the end of the course, the students should be able to:

1. solve qualitative problems based on vector and matrix analyses such as linear independence and dependence of vectors, rank etc;
2. describe the concepts of limit theory and nth order differential equations and their applications to physical phenomena;
3. solve the problems of differentiation of functions of two variables and know about the maximization and minimization of functions of several variables;
4. describe the applications of double and triple integration in finding the area and volume of engineering solids, and explain the qualitative applications of Gauss, Stoke's and Green's theorem;
5. explain ordinary differential equations and applications, and develop a mathematical model of linear differential equations, as well as appreciate the necessary and sufficient conditions for total differential equations; and
6. analyse basic engineering models through partial differential equations such as wave equation, heat conduction equation, etc., as well as fourier series, initial conditions and its applications to different engineering processes.

Course Contents

Limits, continuity, differentiation, introduction to linear first order differential equations, partial and total derivatives, composite functions, matrices and determinants, vector algebra, vector calculus, directional derivatives.

GET 211: Computing and Software Engineering (3 Units C: LH=30; PH=45)

Learning Outcomes

At the end of the course, the students should be able to:

1. describe and apply computing, software engineering knowledge, best practices, and standards appropriate for complex engineering software systems;
2. develop competence in designing, evaluating, and adapting software processes and software development tools to meet the needs of an advanced development project through practical object-oriented programming exposure taught in concrete terms with a specific modern language – preferable selected from Python, Java or C++;
3. use widely available libraries to prepare them for machine learning, graphics and design simulations;
4. develop skills in eliciting user needs and designing an effective software solution;

5. recognise human, security, social, and entrepreneurial issues and responsibilities relevant to engineering software and the digitalisation of services; and
6. acquire capabilities that can further be developed to make them productively employable by means of short Internet courses in specific areas.

Course Contents
Introduction to computers and computing; computer organisation – data processing, memory, registers and addressing schemes; Boolean algebra; floating-point arithmetic; representation of non-numeric information; problem-solving and algorithm development; coding (solution design using flowcharts and pseudo codes). Data models and data structures; computer software and operating system; computer operators and operators' precedence; components of computer programs; introduction to object oriented, structured and visual programming; use of MATLAB in engineering applications. ICT fundamentals, Internet of Things (IoT). Elements of software engineering.

AFE 201: Fitness for Life (1, 0, 0) 1 Unit
AFE 203: Language Acquisition II (1, 0, 0) 1 Unit
AFE 205: Basic Skill Acquisition (1, 0, 0) 1 Unit

200 LEVEL SECOND SEMESTER
GST 212: Philosophy, Logic and Human Existence (2 Units C: LH=30)
Learning Outcomes
At the end of the course, students should be able to:
1. know the basic features of philosophy as an academic discipline;
2. identify the main branches of philosophy & the centrality of logic in philosophical discourse;
3. know the elementary rules of reasoning;
4. distinguish between valid and invalid arguments;
5. think critically and assess arguments in texts, conversations and day-to-day discussions; 6. critically asses the rationality or otherwise of human conduct under different existential conditions;
7. develop the capacity to extrapolate and deploy expertise in logic to other areas of knowledge, and 8. guide his or her actions, using the knowledge and expertise acquired in philosophy and logic.

Course Content
Scope of philosophy; notions, meanings, branches and problems of philosophy. Logic as an indispensable tool of philosophy. Elements of syllogism, symbolic logic— the first nine rules of

inference. Informal fallacies, laws of thought, nature of arguments. Valid and invalid arguments, logic of form and logic of content — deduction, induction and inferences. Creative and critical thinking. Impact of philosophy on human existence. Philosophy and politics, philosophy and human conduct, philosophy and religion, philosophy and human values, philosophy and character molding, etc.

GET 204: Students Workshop Practice (2 Units C: LH=15; PH=45)

Learning Outcomes

At the end of this course, the students should be able to:

1. identify various basic hands and machine tools, analogue and digital measurement devices and instruments, and acquire skills in their effective use and maintenance;
2. practically apply basic engineering technologies, including metrology, casting, metal forming and joining, materials removal, machine tooling (classification, cutting tool action, cutting forces, non-cutting production) and CNC machining technology;
3. master workshop and industrial safety practices, accident prevention and ergonomics;
4. physically recognize different electrical & electronic components like resistances, inductances, capacitances, diodes, transistors and their ratings;
5. connect electric circuits, understand different wiring schemes, and check ratings of common household electrical appliances and their basic maintenance; and
6. determine household and industrial energy consumption, and understand practical energy conservation measures.

Course Contents

The course comprises general, mechanical and electrical components: supervised hands-on experience in safe usage of tools and machines for selected tasks; Use of measuring instruments (calipers, micrometers, gauges, sine bar, wood planners, saws, sanders, and pattern making). Machine shop: lathe work shaping, milling, grinding, reaming, metal spinning. Hand tools, gas and arc welding, cutting, brazing and soldering. Foundry practice. Industrial safety and accident prevention, ergonomics, metrology. Casting processes. Metal forming processes: hot-working and cold-working processes (forging, press-tool work, spinning, etc.). Metal joining processes (welding, brazing and soldering). Heat treatment. Material removal processes. machine tools and classification. Simple theory of metal cutting. Tool action and cutting forces. Introduction to CNC machines. Supervised identification, use and care of various electrical and electronic components such as resistors, inductors, capacitors, diodes and transistors. Exposure to different electric circuits, wiring schemes, analogue and digital electrical and electronic measurements. Household and industrial energy consumption measurements. Practical energy conservation principles.

GET 206: Fundamentals of Engineering Thermodynamics (3 Units C: LH=45)

Learning Outcomes

At the end of this course, the students should be able to:

1. describe basic concepts of thermodynamics, quantitative relations of Zeroth, first, second and third laws;
2. define and explain system (surrounding, closed and open system), control volume and control mass, extensive and intensive properties;
3. calculate absolute and gage pressure, and absolute temperature, calculate changes in kinetic, potential, enthalpy and internal energy;
4. evaluate the properties of pure substances i.e. evaluate the state of the pure substances such as compressed liquid, saturated liquid-vapour mixture and superheated vapour using property diagrams and tables; arrange the ideal and real gas equations of state,
5. formulate the first law of thermodynamics for a closed system i.e. organize the change in energy in the closed systems via heat and work transfer;
6. distinguish heat transfer by conduction, convection and radiation, and calculate the amount of heat energy transferred;
7. calculate the changes in moving boundary work, spring work, electrical work and shaft work in closed systems;
8. apply the first law of thermodynamics for closed systems and construct conservation of mass and energy equations;
9. formulate the first law of thermodynamics to the open systems i.e. describe steady-flow open system, apply the first law of thermodynamics to the nozzles, diffusers, turbines, compressors, throttling valves, mixing chambers, heat exchangers, pipe and duct flow;
10. construct energy and mass balance for unsteady-flow processes;
11. evaluate thermodynamic applications using second law of thermodynamics;
12. calculate thermal efficiency and coefficient of performance for heat engine, refrigerators and heat pumps; and
13. restate perpetual-motion machines, reversible and irreversible processes.

Course Contents

Basic concepts, definitions and laws (quantitative relations of Zeroth, first, second and third laws of thermodynamics). Properties of pure substances: the two-property rule (P-V-T behaviour of pure substances and perfect gases); state diagrams. The principle of corresponding state; compressibility relations; reduced pressure; reduced volume; temperature; pseudo-critical constants. The ideal gas: specific heat, polytropic processes. Ideal gas cycles; Carnot; thermodynamic cycles, turbines, steam and gas, refrigeration. The first law of thermodynamics –

heat and work, applications to open and closed systems. The steady f low energy equation (Bernoulli's equation) and application. Second law of thermodynamics, heat cycles and efficiencies.

GET 210: Engineering Mathematics II (3 Units C: LH=45)

Learning Outcomes

At the end of the course, the students should be able to:

1. describe physical systems using ordinary differential equations (ODEs);
2. explain the practical importance of solving ODEs, solution methods, and analytically solve a wide range of ODEs, including linear constant coefficient types;
3. numerically solve differential equations using MATLAB and other emerging applications; 4. perform calculus operations on vector-valued functions, including derivatives, integrals, curvature, displacement, velocity, acceleration, and torsion, as well as on functions of several variables, including directional derivatives and multiple integrals;
5. solve problems using the fundamental theorem of line integrals, Green's theorem, the divergence theorem, and Stokes' theorem, and perform operations with complex numbers; 6. apply the concept and consequences of analyticity and the Cauchy-Riemann equations and of results on harmonic and entire functions of complex variables, as well as the theory of conformal mapping to solve problems from various fields of engineering; and
7. evaluate complex contour integrals directly and by the fundamental theorem, apply the Cauchy integral theorem in its various versions, and the Cauchy integral formula.

Course Contents

Introduction to ordinary differential equations (ODEs); theory, applications, methods of solution; second order differential equations. Advanced topics in calculus (vectors and vector valued function, line integral, multiple integral and their applications). Elementary complex analysis including functions of complex variables, limits and continuity. Derivatives, differentiation rules and differentiation of integrals. Cauchy-Riemann equation, harmonic functions, basic theory of conformal mapping, transformation and mapping and its applications to engineering problems. Special functions.

EEE 202: Applied Electricity II (3 Units C: LH=45)

Learning outcomes

Students will be able to:

1. differentiate between various d.c. and a.c. machines;
2. explain the principles of operation of machines;
3. explain the operation of basic semiconductor devices and their basic applications; and

4. explain the principle of operation of communication systems with examples.

Course contents

Basic machines – DC, synchronous alternators, transformers, equivalent circuits. Three- phase balanced circuits, PN junction diode, BJTs, FETs, thyristors, communications fundamentals, introduction of TV, Radio, Telephone systems.

EEE 204 Electrical Engineering Materials (3 Units E: LH=45)

Learning Outcomes

Students will be able to:

1. discuss electron conduction mechanisms in semiconductors;
2. explain transport phenomena in semiconductors; and
3. describe semiconductors device fabrication techniques.

Course contents

Free electron motion in static electric and magnetic fields, electronic structure of matter, conductivity in crystalline solids. Theory of energy bands in conductors, insulators and semiconductors: electrons in metals and electron emissions; carriers and transport phenomena in semiconductors, characteristics of some electron and resistors, diodes, transistors, photo cell and light emitting diode. Elementary discrete devices fabrication techniques and IC technology.

ABUAD-GET 208: Strength of Materials (3 Units C: LH=45)

Senate-approved relevance

This course is integral to our university's commitment to Quality Education (SDG 4) and Industry, Innovation, and Infrastructure (SDG 9). It provides a deep understanding of the mechanical behavior of materials under various loading conditions, which is essential for designing safe and efficient structures. Mastery of these concepts is critical for students pursuing careers in civil, mechanical, and aerospace engineering. The course equips students with the skills needed to analyze and design structural components, contributing to innovative and sustainable infrastructure development, essential for addressing global engineering challenges.

Overview

This course explores the principles of equilibrium, stress-strain relationships, and the application of Hooke's law to various materials. It covers the effects of loading and temperature changes on stresses and strains, torsion in circular members, and shear forces in beams. The course also examines bending moments, bending stresses in beams with symmetrical and combined

loadings, and stress analysis. Students will gain the knowledge necessary to analyze and design structural components under different conditions.

Objectives

The objectives of the course are to:

1. understand the principles of equilibrium in structural systems.
2. analyze stress-strain relationships in materials using generalized Hooke's law.
3. evaluate stresses and strains due to different loading conditions and temperature changes.
4. understand and analyze torsion in circular members.
5. calculate shear forces and bending moments in beams under various loadings.
6. analyze bending stresses in beams with symmetrical and combined loadings.
7. apply stress analysis techniques to structural components.
8. design safe and efficient structural components considering various mechanical factors.

Learning Outcomes

At the end of this course, the students should be able to:

1. recognize a structural system that is stable and in equilibrium;
2. determine the stress-strain relation for single and composite members based on Hooke's law;
3. estimate the stresses and strains in single and composite members due to temperature changes;
4. evaluate the distribution of shear forces and bending moments in beams with distributed and concentrated loads;
5. determine bending stresses and their use in identifying slopes and deflections in beams;
6. use Mohr's circle to evaluate the normal and shear stresses in a multi-dimensional stress system and transformation of these stresses into strains;
7. evaluate the stresses and strains due to torsion on circular members; and
8. determine the buckling loads of columns under various fixity conditions at the ends.

Course Contents

Consideration of equilibrium; composite members, stress-strain relation. Generalized Hooke's law. Stresses and strains due to loading and temperature changes. Torsion of circular members. Shear force, bending moments and bending stresses in beams with symmetrical and combined loadings. Stress

Minimum Academic Standards

As stated in the CCMAS.

GET 299: Students Industrial Work Experience I (SIWES I) (3 Units C: 9 weeks)

Learning Outcomes

SIWES I should provide opportunity for the students to:
1. acquire industrial workplace perceptions, ethics, health and safety consciousness, inter personal skills and technical capabilities needed to give them a sound engineering foundation;
2. learn and practise basic engineering techniques and processes applicable to their specialisations; 3. build machines, devices, structures or facilities relevant to their specific engineering programmes and applications; and
4. acquire competence in technical documentation (log-book) and presentation (report) of their practical experiences.

Course Contents
Practical experience in a workshop or industrial production facility, construction site or special centres in the university environment, considered suitable for relevant practical/industrial working experience but not necessarily limited to the student's major. The students are exposed to hands-on activities on workshop safety and ethics, maintenance of tools, equipment and machines, welding, fabrication and foundry equipment, production of simple devices; electrical circuits, wiring and installation, etc. (8-10 weeks during the long vacation following 200 level).

300 LEVEL FIRST SEMESTER

EEE 313: Electronics I (Physical) (2, 1, 0) 3 UNITS
Free electron motion in static electric and magnetic fields, electronic structure of matter, conductivity in crystalline solids. Theory of energy hands in conductors, insulators and semi-conductors: electrons in metals and electron emissions; carriers and transport phenomena in semi-conductors, characteristics of some electron and resistors, diodes, transistors, photo cell and light emitting diode. Elementary discrete devices fabrication techniques and IC technology.

EEE 315: Electromagnetic Field and Wave Theory I (2, 1, 0) 3 UNITS
Review of electromagnetic laws, Gauss' law, Ampere's and Faraday's laws, Electric potential, Electrostatic fields due to distribution of charges, Energy stored in Electric field, magnetic fields in and around current carrying conductors of different shapes, magnetic vector potential. Time varying magnetic and electric fields, conduction and displacement current in magnetic vector potential.
EEE 317: Electrical/Electronics Laboratory 1 (0, 0, 3) 1 UNIT
Laboratory work involves soldering techniques, measuring instrument, identification of semiconductors, application of circuit and transformer testing. A Laboratory work designed to illustrate topics covered in measurement and instrumentation, electric circuits theory transformers and DC machines.

EEE 319: Measurements and Instrumentation (2, 1, 0) 3 UNITS

Units. Theory of Errors. Direct and Indirect Measurements. General Instrumentation, Moving Coil Instruments (PMMC). Moving Iron Instruments. Basic Meter in DC measurement. Basic meter in AC measurements; rectifier voltmeter, electro-dynamometer and Wattmeter, instrument transformers; DC and AC bridges and their applications; general form of AC bridge universal impedance bridge; Electronic instruments for the measurement of voltage, current resistance and other circuit parameter, electronic voltmeters, AC voltmeters using rectifiers, electronic multimeter, digital volumeters; oscilloscope: vertical deflection system, horizontal deflection system, probes, sampling CRO, Instruments for generating and analyzing waveforms; square-wave and pulse generator, signal generators, function generators, wave analysers, Electronic counters and their applications: time base circuitry, universal counter measurement modes; Analog and digital data acquisition systems: tape recorders, D/A and A/D conversions, sample and hold circuits.

EEE 321: Electric Circuit Theory I (2, 1, 0) 3 UNITS

Circuit elements, circuit theorems, applications. Network response to steps, ramp, impulses. Network functions, response to exponential, sinusoidal sources. Laplace transform, polezero analysis, network synthesis, resonance, two-point analysis, ladder network. Star-Delta transformation, T, P Networks.

EEE 325: Electrical Machines I (2, 1, 0) 3 UNITS

Basic principles of relays and actuators. Transformation of Electric energy. Transformer performance; equivalent circuits, efficiency, regulation, per unit values. Types of Transformers: Auto transformer, Instrument Transformer, Elements of Transformer design. Transformer in Polyphase circuits. Parallel operation of transformers. Basic principles of electromechanical energy conversion. Direct current machines: armature windings, internal torque, and methods of excitation. Armature reaction. Characteristics of D.C. generators and Motors. Basic principles of selection of motors and generators for practical application. Speed control and electric braking. Cross-field machines. Commutator machines.

EEE 341: Software Development Techniques (2, 1, 0) 3 UNITS

Software development life cycle. Top-Down design. Program, design using pseudo-code, flowchart. Flowchart ANSI symbols and usage. Extensive examples, and exercises using pseudo-code/flowchart to solve practical problems in engineering. Debugging and documentation techniques. Programming using a structural language such as C: Symbols, keywords, identifiers, data types, operators, various statements, operator precedence, type conversion, conditional and

control structures, function, recursive functions. Arrays: 1-D, and multi-dimensional arrays, passing elements or whole array to a function. Simple sorting and searching on arrays, pointers, strings, dynamic memory allocation. Structures and Unions: Structure declaration and definition, accessing structures, array of structures, pointers and structures, union declaration, enumerated variables. File Handling: Concept of a file, files and streams, standard file handling functions, binary files, random access files. Advanced Topics: Command line parameters, pointers to functions, creation of header files, stacks, linked lists, bitwise manipulation. Software development in C in MS Windows, UNIX/LINUX environments, header file, preprocessor directives, make, makefile. Static and dynamic linking libraries. Extensive examples, and exercises programming in C to solve practical problems in engineering. Exercises are to be done in the Computer Laboratory.

ENG 381: Engineering Mathematics III (2, 1, 0) 3 UNITS

Linear algebra. Elements of matrices, determinants, inverses of matrices, theory of linear equations, eigen values and eigen vectors. Analytical geometry, coordinate transformation, solid geometry, polar, cylindrical and spherical coordinates. Elements of functions of several variables, surface variables. Ordinary integrals, evaluation of double integrals, triple integrals, line integrals and surface integrals. Derivation and integrals of vectors, the gradient of scalar quantities. Flux of vectors, the curl of a vector field, gauss, greens and stoke's theorems and applications. Singular valued functions. Multivalued functions, analytical functions, cauchy riemann's equations. Singularities and zeroes, contour integration including the use of cauchy's integral theorems, bilinear transformation. Applications of microsoft excel, matlab and/or mathcad.

AFE 301 Practical Entrepreneurship Skills (1, 0, 3) 1 Unit

Profiles of business ventures in the various business sectors such as:
Soap/detergent, tooth brush and tooth paste making; photography; brick making; rope making; brewing; glassware production/ ceramic production, paper production; water treatment/conditioning/packaging; food processing/preservation/packaging; metal fabrication; tanning industry; vegetable oil extraction; farming; fisheries/aquaculture; plastic making; refrigeration/air-conditioning; carving, weaving; bakery; tailoring; printing; carpentry; interior decoration; animal husbandry etc. Case study methodology applied to the development and administration of cases that bring out key issues of business environment, start-up, pains and gains of growth of businesses, etc. With particular reference to nigerian businesses. Experience sharing by business actors in the economy with students during case presentations.

300 LEVEL SECOND SEMESTER COURSES

EEE 314: Electronics II (Analogue) (2, 1, 0) 3 UNITS

Broadband and narrow band amplifiers, power amplifiers, voltage and current stabilizing circuits. Sinusoidal RC and RLC and crystal oscillators. Switching properties of electronic devices. Switching and wave-shaping circuits. Generation of non-sinusoidal waveforms, astable, monostable and bistable multi-vibrators, comparator, Schmitt-trigger and time base generators using discrete transistor, operational amplifier fundamentals, analyses and applications, other analog integrated circuits-Time chips and their applications. Analysis and design of logic gates of various families. Diode logic, RTL, TTL, RCL, MOS and MOS of digital integrated circuits. Interfacing between various logic families. Concepts of small, medium, large and very large integration and their consequences, some digital building blocks, flip-flops, counters, register and decoders. Introduction to D/A and A/D conversion principles.

EEE 316: Electromagnetic Field and Wave Theory II (2, 1, 0) 3 UNITS

Maxwell's equations (in regular coordinates and vector-calculus notation); Derivation of Maxwell's equation, electromagnetic potential and waves, pointing vector, Boundary conditions, wave propagation in good conductors, skin effect, plane waves across boundaries of different media, propagation of electromagnetic waves in ionized media, fundamentals of transmission lines, matching, voltage reflection coefficient standing wave ratio waveguides and antennae. Applications of wave theory in communication.

EEE 318: Electrical/Electronic Laboratory II (0, 0, 3) 1 UNIT

Laboratory work involves winding of Transformer and Protective devices. Confirmation of semiconductor-Op Amp, Timing circuit. A Laboratory work designed to illustrate topics covered in electric circuits theory electronic engineering and AC machines.

EEE 322: Electric Circuit Theory II (2, 1, 0) 3 UNITS

Networks; Node, loop analysis, Realization of transfer functions, Filters: design, operation, low, high, Band pass. Butterworth, Chebychev, filters, Non-linear circuit analysis Locus diagrams. Network functions. Active network synthesis and analysis: Foster and Cauers

EEE 324: Linear Systems (2, 1, 0) 3 UNITS

Mathematical models of physical system. Analogues concepts in electrical mechanical and thermal systems. Transfer functions. Block diagrams and signal flow graphs. Feedback control system; advantages, transient and steady-state response of systems. Pole stability. The root-locus methods. Frequency response of systems; Bode and polar plots system stability. Routh and Nyquist criteria. Introduction to analogue computer simulation.

EEE 326: Electrical Machines II (2, 1, 0) 3 UNITS

Magnetic circuits. Basic principles of relays and activators; Ideal transformer. Equivalent circuits and basic analysis of practical transformers. D.C. machine contraction, characteristics of D.C. generators. Excitation of D.C. machines. Torque-speed characteristics of D.C. motors. A.C. machines; production of rotating magnetic fields. Simple theory of three-phase induction motors; torque speed characteristics, three-phase induction motors Single-phase motors-applications. Selection of motors, for practical applications. Synchronous machines.

EEE 328: Communication Principles (2, 1, 0) 3 UNITS

Introduction to communication systems; General features of point-to-point communication systems over the entire frequency, characteristics of transmitters, receivers and antennae. Radio and TV broadcasting. Telephony. Facsimile. Radar Telemetry. Analogue modulation systems. Amplitude modulation and demodulation methods, DSB, DSBSC, SSB, and VSB. Comparison of AM systems. Angle modulation and demodulation. Frequency and Phase modulation. Wide band and narrow band and FM.

Study of communication systems design and analysis. Topics include signals and spectra, error-control coding, synchronization, multiple-access techniques, spread spectrum signaling, fading channels baseband signaling and detection in noise, digital and analog modulation and demodulation techniques, and communications link budget analysis.

Transmission media; attenuation in open space, air, cable and fibre channels; construction of cables and fibres, sampling theorem, pulse amplitude modulation, pulse width modulation, multiplexing, quantization systems and pulse code modulation, delta modulation, courses and correction of errors in PCM and DM.

ENG 382 Engineering Mathematics IV (2, 1, 0) 3 UNITS

Series solution of second order linear differential equations with variable coefficients. Bessel and legendre equations. Equations with variable coefficients. Sturn-louville boundary value problems. Solutions of equations in two and three dimensions by separation of variables. Eigen value problems. Use of operations in the solution of partial differential equations and linear integral equations. Integral transforms and their inverse including fourier, laplace, mellin and handel transforms. Convolution integrals and hilbert transforms. Calculus of finite differences. Interpolation formulae. Finite difference equations. Runge-kutta and other methods in the solutions of ode and pdes. Numerical integration and differentiation. Applications of microsoft excel, matlab and/or mathcad.

ENG 384 Engineering Economics (2, 0, 0) 2 UNITS

Economic analysis of engineering projects; value systems economic decisions on capital investments and choice of engineering alternatives; new projects, replacement and abandonment policies, risky decisions; corporate financial practices.
Objectives of valuation work/valuer's primary duty and responsibility. Valuer's obligation to his or her client, to other valuers, and to the society. Valuation methods and practices. Valuation reports. Expert witnessing. Ethics in valuation. Valuation i appraisal standards. Price, cost and value. Depreciation and obsolescence. Valuation terminology. Appraisal reporting and review. Real property valuation. Personal property valuation. Machinery and equipment valuation. Oil and gas valuation. Mines and quarries valuation. Cost and schedule management- an engineering function. Supporting skills and knowledge. Role of cost engineer during evaluation phase. Role of cost engineer during the basic design phase. Role of cost engineer in contractor selection. Role of cost engineer during detailed engineering design phase. Role of cost engineer during construction. Cost engineering function as distinct from design engineering function. Canon of ethics for cost engineers. Basic capital cost estimating. Basic operating cost estimating. Basic project planning and scheduling. Cost engineering terminology. Cost engineering standards.

400 LEVEL FIRST SEMESTER COURSES

EEE 405: Numerical Computations (2, 1, 0) 3 Units

Computer as a number crunching tool. Floating-point number representation and arithmetic error, stability, convergence theory of computational solution to problem: numerical algorithm formulation and design, numerical software systems. Iterative solutions of non-linear systems: (Newton's Method). Numerical solutions of linear systems. Numerical computation of Eigen values and Eigen vectors. Curve fitting; function approximation. Numerical differentiation and integration (Simpson's Rule etc). Explicit and implicit methods. Differential equation (Euler's method etc). The Eigen value problem and solution of ordinary differential equations-methods (ODE) of Taylor, Euler predictor-corrector and Runge-kuta. Numerical integration and differentiation. Linear algebra. Finite differences. Introduction to the use of Octave, Matlab or LabView in numerical computation and engineering applications. Emphasis is on the use of software to solve real problems.

EEE 415: Assembly Language Programming (2, 0, 0) 2 Units

Introduction: Language level of abstraction and effect on machine, characteristics of machine code, advantages, justifications of machine code programming, instruction set and dependency on underlying processor. Intel 8086 microprocessor assembly language programming: Programming model as resources available to programmer, addressing modes, instruction

format, instruction set- arithmetic, logical, string, branching, program control, machine control, input/output , etc; assembler directives, hand-assembling, additional 80x86/Pentium instructions. Modular programming. Interrupt and service routine. Interfacing of assembly language to C . Intel 80x87 floating point programming. Introduction to MMX and SSE programming. Motorola 680x0 assembly language programming. Extensive practical engineering problems solving in assembly language using MASM for Intel, and cross-assembler for Motorola.

EEE 423: Electrical/Electronic Laboratory III (0, 0, 3) 1 Unit

Laboratory work involves winding techniques. Computer hardware and software. Maintenance and repairs, communication. A laboratory work set up to illustrate topics covered in control theory, communication principles, electronic engineering, computer techniques and electric power principles.

EEE 427: Data Communication and Network (2, 1, 0) 3 Units

Introduction to Data communications: the Development of Data Communications; types and sources of data, simple communications network, transmission definitions, one way transmission, half duplex transmission, transmission codes, transmission modes, parallel transmission, serial transmission, bit synchronization, character synchronization, character synchronization, synchronous transmission, asynchronous transmission, efficiency of transmission, error detection methods and data compression. Protocols: Introduction to network protocol. Seven Layer ISO-OSI standard protocols and network architecture. Transport protocols, session services protocols, and other protocols. Institute of Electrical and Electronics Engineering 802 standards. Error control and Data Compression: Forward Error Control; error detection methods; parity checking; linear block codes, cyclic redundancy checking; feedback error control, data compression, Huffman coding and dynamic Huffman coding. Local Area Networks: medium access control techniques – Ethernet, token bus and token ring; LAN standards; fibre distributed data interface, metropolitan area network. Peer-to-peer, Client Server. Client-Server Requirements: GUI design standards, interface independence, platform independence, transaction processing, and connectivity, reliability, backup and recovery mechanisms. Information Network Software; Features and benefits of major recovery mechanisms. Information Network Software: features and benefits of major Network Operating Systems. Network OS: (e.g. Novell NetWare, UNIX/LINUX, OS/2 & Windows NT). TCP/IP and Network OS. INTERNET: Definition, architecture, services, Internet addressing. Internet protocol, IPv4, IPv6. Internet programming, Intranet. System administration, and security issues.

EEE 431: Electric Power Principles (2, 1, 0) 3 Units

Types of power station, operation, auxilliaries, economics of operation - stations, substations power supply economics, tariffs. Power factor correction. Polyphase theory. DC, AC power distribution, network calculations, Overhead line conductors. Corona effect, voltage control, circuit breakers, load forecast, siting of generating plants.

EEE 441: Control Theory and Systems (2, 1, 0) 3 Units

Review of feedback control systems. Transfer functions and System modelling (electrical, hydraulic, pneumatic and thermal systems). Review of frequency-response analysis (Polar, Bode, Nyquist, Nichols). Review of pole stability and Routh-Hurwitz's stability criterion. System classification and time response analysis. Nyquist stability criterion, M- and N- circles, inverse Nyquist Plots. Bode diagrams and determination of transfer function from asymptotic plots. Relative stability. Introduction to Feedback compensation. The tracking and regulation objectives. Compensation techniques and controller design: Lag, Lead and Lead-Lag compensation; Three-term (PID) controllers (P, PD, PI and PID); Series and parallel compensation design; controller design using Bode, Nichols and Root locus methods. Servomechanical systems. Servo-motors, tachogenerators, error detectors amplifiers, actuators, valves, etc. and their transfer functions. Servomechanism position and velocity control systems. Voltage regulators. Introduction to nonlinear systems.

EEE 471: Digital Electronics and Systems (2, 1, 0) 3 Units

Electronic instruments for the measurement of voltage, current resistance and other circuit parameter, electronic voltmeters, AC voltmeters using rectifiers, electronic multimeter, digital volumeters; oscilloscope: vertical deflection system, horizontal deflection system, probes, sampling CRO, Instruments for generating and analyzing waveforms; square-wave and pulse generator, signal generators, function generators, wave analysers, Electronic counters and their applications: time base circuitry, universal counter measurement modes; Analog and digital data acquisition systems: tape recorders, D/A and A/D conversions, sample and hold circuits. Digital Test Instruments.

Number Systems and Codes. Logic Gate Simplification of Logic expressions using Boolean Algebra. Simplification of Logic expressions using Karnaugh Method. Design combinational circuit. Flip-Flops. Application of Flip-Flops in the design of counters, registers and timers. Switching and Waves shipping circuit. Generation of non-sinusoidal signal (multi vibrators). Introduction to ADC and DAC. Design of Logic Gates (Diode, DTL, TTL, ECL, CMOS families etc). Memory devices. Latches. Decoders and encoders. Multiplexer circuits. Computer-aided design of Modular Systems. Synchronous and asynchronous operations. Analysis and synthesis of sequential switching circuits, Registers.

Design of combinatorial circuits with emphasis on application. Design of sequential circuits, synchronous and asynchronous. Design of simple CPU, peripheral devices (I/O), memory/storage devices. Stored-program concept.
Introduction to programming of microcontrollers

400 LEVEL SECOND SEMESTER

ENG 490: Students Industrial Work Experience Scheme (6 Units)

A comprehensive internship programme in which students spend a full semester in approved electrical engineering establishments (private and public) and industries. The exposure also provide opportunity for students to sharpen their technical writing skill through field reports, keeping log-books and prepared technical documents under close supervision of competent lecturers. This will be registered alongside ENG 290 (SWEP I) and ENG 390 (SWEP II).

ENG 290: Students Work Experience Programme (SWEP) I (3 UNITS)

Practices and skills in general engineering through instruction in operation of hand and powered tools for wood and metal cutting and fabrication. Supervised hand on experience in safe usage of tools and machine for selected tasks. General practices on automobile repairs, survey, civil, electrical and other related engineering practice. It is regarded as on the job experience in industry chosen for practical working experience but not necessarily limited to the student's major (12 weeks during the long vacation following 200 level).

ENG 390: Students Work Experience Programme (SWEP) II (3 UNITS)

On the job experience in industry chosen for practical working experience but not necessarily limited to the student's major (12 weeks during the long vacation following 300 level).

Eng 490 Students Industrial Work Experience Scheme (Siwes) (6 Units)

On the job experience in industry chosen for practical working experience but not necessarily limited to the student's major (24 weeks from the end of the first semester at 400-level to the beginning of the first semester of the following session. Thus, the second semester at 400-level is spent in industry.)

500 LEVEL FIRST SEMESTER COURSES

EEE 503: Modelling and Simulation of Electrical and Electronics Systems (1, 1, 0) 2 Units

Basic introduction to modelling and simulation. Introduction to MATLAB and AUTOCAD, and their engineering applications. Simulation of modelled systems using MATLAB and others. Application programs in computer aided design (CAD) of Electrical systems. CAD/CAM systems: Introduction: Basic concepts and terminology. Introduction to computer graphic and data base design. Software systems: Interactive languages and data base design for CAD/CAM systems. Description of existing CAD/CAM systems. Application to mechanical, electronic design etc. CAD simulation tools.

Analysis and design of integrated operational amplifiers and advanced circuits such as wideband amplifiers, instrumentation amplifiers, multiplier circuits, voltage controlled oscillators, and phase locked loops, Design techniques for advanced analogue circuits containing transistors and operational amplifiers. Simulation of circuit using appropriate packages e.g. PSPICE, Electronic workbench, Visio technical etc. should be encouraged.

EEE 509: Advanced Electronics (2, 1, 0) 3 Units

Linear I.C. op-amp, linear and non-linear operations, logarithmic amplifiers, A/D and D/A converters, gyrators and negative impedance converters, the 555 timer structure and applications; four-quadrant multipliers, dynamic logic systems and RAM memory circuits, application of digital circuits to instrumentation.

EEE 511: Digital Signal processing (2, 1, 0) 3 Units

Discrete signals and Z-transform, digital Fourier Transform, Fast Fourier Transform. The approximation problem in network theory. Synthesis of low-pass filters. Spectral transforms and their application in synthesis of high-pass and band-pass filters. Digital filtering, digital transfer function aliasing, one-dimensional recursive and non-recursive filters; Computer techniques in filter synthesis, Realisation of filters in hardware and software. Basic image processing concepts.

EEE 561: Control Engineering (2, 1, 0) 3 Units

State space description of linear systems. Transfer functions and their state space realisations. Eigenvalues, eigenvectors, system response and stability. Concepts of controllability and observability. State feedback control, modal control, and observers. Combined control-observer compensator. The Linear Quadratic Regulator (LQR). Applications to include control of robotic systems, circuit synthesis, and signal processing. Computer-aided analysis and design of control systems.

ENG 583 Engineering Management and Law (2, 0, 0) 2 Units

Principles of organization; elements of organization; management by objectives. Financial management, accounting methods, financial statements, cost planning and control, budget and

budgetary control. Depreciation accounting and valuation of assets. Personnel management, selection, recruitment and training, job evaluation and merit rating. Industrial psychology. Resource management; contracts, interest formulae, rate of return. Methods of economic evaluation. Planning decision making; forecasting, scheduling. Production control. Gantt Chart, CPM and PERT. Optimization, linear programming as an aid to decision making, transport and materials handling. Raw materials and equipment. Facility layout and location. Basic principles of work study. Principles of motion economy. Ergonomics in the design of equipment and process.

Common Law: Its history, definition, nature and division. Legislation codification interpretation. Equity: Definition and its main spheres. Law of contracts for Engineers: offer, acceptance, communication termination. General principles of criminal law. Law of torts: definition, classification and liabilities. Patents: requirements, application, and infringement. Registered designs: application, requirements, types and infringement. Company law. Labour law and Industrial Law.

500L FIRST SEMESTER (ELECTIVE COURSES)

EEE 513: Communication Systems Engineering (2, 1, 0) 3 Units

Microwave frequencies and uses; microwave transmission in transmission lines and wave guides, microwave circuits; impedance transformation and matching, microwave circuits; passive microwave devices, resonant and filter circuits, active microwave devices; Klystron and magnetron tubes and semiconductor devices for microwave generation. Antennae: definitions of elementary parameters related to radiation patterns; dipole and operture antennae and the related design parameters; introduction to antennae arrays. Radiowave propagation: propagation in the ionosphere, troposphere and in stratified media; principles of scatter propagation; applications in general broadcast, television and satellite communication systems. Radar systems nature of radar and radar equations; composition of a radar system; application of different types of radars.

EEE 517: Analogue and Digital Computers (2, 1, 0) 3 Units

Analogue computation, electrical analogue of mechanical, electromechanical systems and servomechanisms. Analogue computer elements, potentiometers, operational amplifiers, multipliers, function generators, simulation of systems transfer functions. Digital computer CPU, peripherals, storage. Arithmetic processes. Hybrid computer systems, parallel, vector, RISC's workstations.

EEE 523: Industrial Electronics Design (2, 1, 0) 3 Units

Characteristics and industrial applications of thyristors and other SCR devices. Transducers and their applications in sensing light, voltage pressure, motion, current temperature, etc. Mechanical relays, solid state relays and stepping motors. Real time control and remote control concepts in instrumentation. Micro-processor and micro-computer based systems. Fire alarms, burglar alarms and general home and industrial instrumentation. Fundamentals of Solid State Electronics. Properties of semiconductors and their devices – physical and electrical properties such as band structures; Fermi functions; doping and conductivity. Charge transport properties including drift and diffusion. Presentation of the law of mass action and applications. Junction devices including p-n junction and metal semi-conduction junctions, their I-V characteristics and applications. Other major devices of interest including the bipolar junction transistors (BJTs), metal-oxide field effect transistors (MOSFETs), and C-MOS. Optoelectronic and photonic devices including LED, Lasers,

EEE 535: High Frequency & Microwave Electronics (2, 1, 0) 3 Units

A survey of microwave engineering, models in waveguides and resonators, passive components, reactive and resistive elements, directional couples and teess ferrite isolators and circulators active components. Klystrons, magnetrons, travelling wave tubes, parametric amplifiers and solid state sources, introduction to varactor. PIN, Gunn-effect diode photodiode, phototransistor.

EEE 543: Renewable Energy Systems (2, 1, 0) 3 Units

Energy Scenario: Classification of Energy Sources, Energy resources (Conventional and nonconventional). Energy efficiency and energy security. Distributed generation.

Solar Energy: Solar thermal Systems and Photo voltaic (PV) technology: characteristics of PV, equivalent circuit, efficiency, calculations, applications. sizing and economics. Peak power operation. Wind Energy: Wind speed and power relation, power extracted from wind. Wind power systems: system components, Types of Turbine, Turbine rating. Choice of generators, turbine rating, Variable speed operation, maximum power operation, control systems, system design features. Grid Integration: Stand-alone systems, Concept of Micro-Grid and its components, Hybrid systems. Hybrid system economics, Interface requirements, Stable operation. Effect on power quality, harmonic distortion, voltage transients and sags, voltage flickers, dynamic reactive power support.

EEE 553: Power System Engineering (2, 1, 0) 3 Units

Representation of power systems, one-line diagrams, per-unit quantities, power system equation and analysis, use of network analyser, load flow studies, load forecasting, state flow equations, economic operation of power systems, symmetrical three phase faults on synchronous machines. Bus Impedance matrix in fault calculations, symmetrical components, symmetrical and

unsymmetrical faults, digital calculation of fault currents using the Bus impedance matrix, various types of relays used in power systems, protection systems of power transmission lines, switchgear, circuit breaker, principles of fault detection, discrimination and clearance.

EEE 555: High Voltage Engineering (2, 1, 0) 3 Units

Importance of High voltage generation and transmission. Characteristics and details of high voltage equipment with emphasis on line structure and hardware. Generation of high AC/DC and impulse voltages. Requirement of testing of internal and external insulation system. Propagation of surges in high voltage transmission lines. Protection of transmission lines and substation from direct lighting strokes. Preventive testing of insulation process in multilayer conductors and insulation coordination. Line and substation insulation. Over-head line, bus-bars, isolators and circuit breakers. Corona/Radio interferences and minimization of their effects on the lines. Brief discussion about discharge in gases, breakdown voltages in gases. Electric field calculation for different gases, breakdown voltages in gases. Electric field calculation for different electrode configuration with respect to distance, temperature etc.

EEE 565: Control Engineering (Modern) (2, 1, 0) 3 Units

Digital control: concept of sampling and quantization, z- transforms, inverse z-transforms, zero-order-hold, discrete-time transfer functions and stability analysis; discrete-time state variables of dynamic system, formulation of state vector differential equation, solution of the state equation, the state transition matrix, eigenvalues and eigenvectors; system response and stability. Digital three-term PID design. State feedback control design in discrete time. Digital control implementation and finite word length effect. Introduction to neural network and fuzzy-logic based control systems. Applications to include control of electrical machines, mechatronic and robotic systems. Computer-aided analysis and design of control systems

500 LEVEL SECOND SEMESTER COURSES

EEE 502: Electrical Services Design (2, 0, 0) 2 Units

Lighting installation, power installation, energy supply and distribution, choice of cables and conductors, wiring systems and accessories, outdoor low voltage lines and cables, protection of low voltage installation, and characteristics of low voltage equipment, earthing and testing of electrical installation, illumination.

EEE 504: Embedded Systems Design (2, 1, 0) 3 Units

Introduction to embedded system, components, characteristics, applications. Intel 8051/8031 Micro-controller: Features of the 8051/8031 family, block diagram and definitions of the pin of the

8051, I/O port structure, memory organisation: general purpose RAM, bit addressable RAM, register bank, special function registers, external memory, memory space mapping and decoding, bus control signals timing, a typical 8051 micro-controller based system. Instruction Set and Assembly Language Programming: Addressing modes, the 8051 instruction set and typical examples, assembler operation, assembly language format, assembler directives, operation of assemblers and linkers, programming examples. On-chip Peripheral Devices: I/O ports, operations and uses of port 0, port 1, port 2, port 3, timers: their operations, programming, and applications, serial port: operations and programming, typical applications, serial port interrupt. Interfacing to external memory, keypad, seven-segment LED display, ADC and DAC chips, and input / output port expansion, description and uses of hardware development tools. MOTOROLA M6811 Micro-controller: Features of the M6811 family, block diagram and definitions of the pin of the M6811, I/O port structure, memory organisation: general purpose RAM, bit addressable RAM, register bank, special function registers, external memory, memory space mapping and decoding, bus control signals timing. Instruction Set and Assembly Language Programming. On-chip peripheral devices and I/O interfacing. Introduction to PIC microcontroller: general architecture, applications and selection of microcontroller, advantages, low-end, and high performance PIC. Specific PIC microcontrollers: Features, architecture, block diagram, pin configuration, on-chip memory, and peripheral. Instruction set and Assembly language programming. Serial I/O interfacing: I2C, and SPI interfacing and programming. Memory interfacing: external memory interfacing, EEPROM and Flash memory interfacing. Design exercises using development system.

EEE 514: Reliability Engineering (2, 0, 0) 2 Units

Introduction to reliability, maintainability, availability. Elementary reliability theory. Application to power systems and electronics systems. Test characteristics of electrical and electronic components. Types of faults. Design for higher reliability. Packaging, mounting, ventilation. Protection from humidity, dust.

EEE 562: Power Electronics (2, 1, 0) 3 Units

Switching characteristics of diodes, transistors, thyristors etc. analysis of diode circuit with reactive loads, analysis of circuits using transistors as switches, power control circuits, ACDC converters, characteristics of switching transformers, power semi-conductor device protection, examples of power electronic circuits, solar devices.

EEE 564: Seminars on Use of Engineering Software Packages in Electrical/Electronics Field (1, 0, 0) 1 Unit

This course is to address and discuss current development in the field of electrical & electronics engineering.
Students are expected to present a term paper on the use of Engineering Software packages in implementation and optimization of Electrical/Electronics system within a current trend of interest.

EEE 599: Student Project (0, 0, 18) 6 Units
This course lasts for one academic session. Each student must undertake a project under the supervision of a lecturer, submit a comprehensive project report and present a seminar at the end of the year. A project status report is to be presented at the end of the first semester. Each student must attend Engineering Seminars.

500 LEVEL SECOND SEMESTER (ELECTIVE COURSES)
EEE 506: Digital System Design with VHDL (2, 1, 0) 3 Units
Finite State Machine: definition, mealy and Moore models, state diagram, state table, transition table. Sequential circuits design using flip-flops, asynchronous, and synchronous circuit design. Algorithm State Machine. Design examples and exercises. Structured Design: Design constructs, Design Levels, Geometry-based interchange formats, Computer aided electronic system design tools, Schematic circuit capture, Hardware description languages, Design process (simulation, synthesis), Structural design decomposition. Introduction to VHDL: VHDL language abstractions, Design hierarchies, VHDL component, Lexical description, VHDL source file, Data types, Data objects, Language statements, Concurrent VHDL, Sequential VHDL, Advanced features of VHDL (library, package and subprograms). Structural level modelling, Register-Transfer level modelling, FSM with datapath level modelling, Algorithmic level modelling. Introduction of ASIC, Types of ASIC, ASIC design process, Standard cell ASIC synthesis, FPGA Design Paradigm, FPGA synthesis, FPGA/CPLD Architectures. VHDL Design: Top-down design flow, Verification, simulation alternatives, simulation speed, Formal verification, Recommendations for verification, Writing RTL VHDL code for synthesis, top-down design with FPGA. VHDL synthesis, optimization and mapping, constraints, technology library, delay calculation, synthesis tool, synthesis directives. Computer-aided design of logic circuits.

EEE 512: Digital Communication (2, 1, 0) 3 Units
Block Diagram of digital communication system sampling theorem, Shannm theorem and applications in digital communication system. Advantages of digital signals. Noise in digital system. Filtering and equalisation. Digital modulation techniques: FSK, ASK, QPSK, M-PSK, QAM, etc. Error detection and correction techniques. Encoders/Decoders. Applications of digital

communication system: Satellite communication, telephoning microwave, wireless communication, optical communication, Broadband. Communication. Internet Technology.

EEE 554: Power Systems Communication and Control (2, 1, 0) 3 Units

Review of transmission line theory. High frequency communication on power lines carrier systems and power line carrier applications. Multiplexing, Telemetering, Signal processing and data transmission. Control strategy, control of power generation, voltage control, element of power system stability, automatic voltage regulators, regulating transformers. Grid integration and control of renewable energy sources, smart grid control and monitoring

EEE 556: Electric Drives and Applications (2, 1, 0) 3 Units

Historical development, types of motors shunt, series, compound wound DC, AC induction motors, repulsion, reluctance, hysteresis and synchronous motors. Speed- torque characteristics, speed-control of electric drives. Braking. Motor power rating selection. Protection, control of drive motors. Application of electric drives

EEE 558: Electrical Energy Conversion and Storage (2, 1, 0) 3 Units

Electromechanical energy conversion, sources of motive power. Waste heat recovery. Solar energy nuclear power. Other sources of energy. Wind, geothermal, primary and secondary cells, cars and heavy vehicle batteries, testing, fault diagnosis, repairs. Effect of environmental factors on battery life, small-scale power resources. Concept of distributed generation.

EEE 566: Control Engineering III (Process Automation & Non-Linear) (2, 1, 0) 3 Units

Industrial sensors and actuators. P&ID (Piping and instrumentation) diagrams. Process modelling and identification. PID control and various tuning rules. PLCs, ladder logic and sequential flow charts. Control using PLCs. Industrial control networks. SCADA and DCS systems. The model predictive control concept.

Nonlinear control, common types and effects of nonlinearities, phase-plane and describing function analysis, closed loop response and stability. Linearisation and the concept of gain-scheduling control.

EEE 576: Microcomputer Technology (2, 1, 0) 3 Units

Elements of digital computer design; control unit, micro-programming, bus organization and addressing schemes. Micro-processors, system architecture, bus control, instruction execution and addressing modes. Machine codes, assembly language and high-level language programming, Micro-processors as state machines. Microprocessor interfacing: Input/output. Technique, interrupt systems and direct memory access; interfacing to analogue systems and

applications to D/A and A/D converters. System development tools: simulators, EPROM programming, assemblers and loaders, overview of an available microprocessor application.

EEE 578: Broadcasting and Internet Technology (2, 1, 0) 3 Units

Elements of broadcasting system, studio equipment, microphones, sound, disc, magnetic tape recording, and radio transmitters power amplifiers, FM systems, radiating antenna, television camera, synchronization circuits, video amplifiers. Broadcasting regulations. Frequency spectrum: allocation, assignment and licensing. Regulatory bodies. Design, configuration, and services of CATV, MATV, MMDS systems. Multipath problems. Polarization, field strength, and footprint. Transmitter power rating, beam-width, interference and minimum separation. Digital Television standards: MPEG, DVB, channel coding techniques. HDTV. Digital television/monitor set: LCD, and Plasma technology. Internet Technology: The Internet, definition and services. Internet architecture, OSI layers, TCP/IP, Internet addressing, IPv4, IPv6. Internet broadcasting: principles, components, standards, and applications.

5.2. How the requirements of Complex Problem Solving (CPS) and Complex Engineering Activities (CEA) have been addressed for Electrical Engineering programme

The Electrical Engineering programme addresses the requirements for Complex Problem Solving (CPS) and Complex Engineering Activities (CEA) through an Outcome-Based Education (OBE) framework that integrates these challenges across the curriculum, industrial training, hands-on training and final year projects.

4.2.1. Integration into Programme Outcomes (POs)

The programme's 12 core outcomes are directly aligned with COREN attributes, explicitly defining CPS and CEA as graduation requirements:

For CPS (PO1–PO4): Students must apply fundamental principles to identify, formulate, and analyze complex engineering problems to reach substantiated conclusions and design systems that meet specified needs.

For CEA (PO5, PO10): Students are required to select and apply modern engineering and IT tools for complex activities and communicate effectively about these activities to both technical and general audiences.

2. Coursework and Laboratory Scaffolding

The curriculum progressively increases problem complexity from 200-level to 500-level:

a) Foundational Level: Courses like Applied Mechanics (GET 207) and Basic Electrical Engineering I (EEE 221) introduce students to identifying and solving engineering problems using mathematical modeling and first principles.

b) Advanced Level: 500-level courses such as Power System Engineering (EEE 553) and Control Engineering (EEE 561) require students to solve "complex operational challenges" and analyze dynamic system stability under practical constraints.

c) Modern Tool Usage: The Seminars on Engineering Software (EEE 564) specifically trains students in using industry-standard tools (e.g., MATLAB, Altium, PSCAD) to simulate and optimize complex electronic and power systems, satisfying the "Modern Tool Usage" requirement for CEA.

4.2.2. Industrial Exposure (SIWES)

The Students Industrial Work Experience Scheme (SIWES III) is a critical vehicle for addressing CPS and CEA:

a) Students spend 24 weeks in industry where they are motivated to identify industrial practice engineering challenges and creatively devise impactful solutions.

b) The Engineering Project I (GET 402) course, taken during 400-level, specifically requires students to complete the design phase of a complex engineering problem sourced directly from their industry or community placement.

4.2.3. Final Year Project (EEE 599)

The Final Year Project is the primary assessment point for both CPS and CEA. Students must:

a) Develop a technology or system to solve a known and significant engineering problem.

b) Build or produce hardware/software solutions using the comprehensive skills acquired throughout the five-year programme.

c) Defend their work before a panel that includes an external assessor, demonstrating their ability to communicate complex engineering activities (CEA) effectively

SECTION SIX: STUDENTS

6.1. Admission Requirement

(a) O' Level Subjects

Five O' Level credits to include Mathematics, English Language, Physics, Chemistry and any other subject from Biology, Agriculture, Technical Drawing, Geography.

(b) UTME Subjects

English Language, Mathematics, Physics and Chemistry. Expected duration for UTME candidates shall be 5 years.

(c) Direct Entry

(i) At least two GCE Advanced Level passes which must include Economics, and any other subject from Government, Geography, History, Sociology, Psychology, Mathematics, and Further Mathematics.

(ii) IJMB/JUPEB/NCE/ND or its approved equivalent at an acceptable grade level.

(iii) In addition, candidates must satisfy the O' Level requirements as stated in (a) above.

Table 6.1: Admission requirement

Admission requirements		**Remarks**
SSCE	**Direct Entry**	
Five credit passes in not more than two sittings in English language, Mathematics, Physics, Chemistry, and any other Science subjects in WASSCE, GCE or NECO It is also desirable for candidates to have further Mathematics and Technical Drawing at credit levels. Such candidates shall have added advantage.	Advanced level passes in Mathematics, Chemistry and Physics in addition to UTME requirement. Candidates with National Diploma at Upper credit Level may be considered.	**UTME subjects** The joint Matriculation Examination must be in use of English language, Mathematics, Physics and Chemistry.

6.2. Compliance with Admission Requirements

6.2.1 Inter-University Transfer:

a) Regulations:

Candidates from other accredited institutions can enroll in the Programme. The university rule stipulates that every candidate must complete a minimum of two academic terms at the institution to receive the university degree. Those possessing a Higher National Diploma in a related area may qualify for entry into the 300 level. Each candidate is assessed according to their own qualifications and achievements.

6.3. Academic Regulations

The student handbook, published by the University and the Programme, is distributed to all students. The University handbook includes entry requirements, rules and regulations that govern examination conduct, the grading system, penalties for examination malpractice, probation, expulsion, withdrawals, graduation, and other essential information students need to understand regarding their academic program pursuit. The Programme handbook outlines Programme Educational Objectives, Programme Outcomes, the Curriculum with its respective Course Learning Outcomes and course contents for courses across all levels.

6.4. Standard of Tests and Examinations

The programme adheres to the minimum standards established by the NUC when designing tests and examinations. Throughout the semester, assignments and assessments are conducted to cover the course materials as much as possible. At the conclusion of each semester, individual instructors or a team of lecturers responsible for a course develop examination questions for that subject, integrating Bloom's Taxonomy into the questions. These questions must thoroughly address the course content. Sample examination questions along with marking schemes for all levels of the Programme have been provided. Examination questions from 100 level to 400 level are jointly reviewed by selected co-examiners, consisting of senior academic staff within the Programme. The final review is conducted by the Head of Programme, who is the Chief Examiner. For the 500 level, examination questions undergo external moderation by Senate-approved External Examiner(s). At the end of the session, the External Examiner visits the Programme to moderate the assessments of graduating students, including their final year projects. Examinations take place at the end of each semester.

The final grade for each student consists of the examination score with continuous assessment marks. The continuous assessment includes tasks such as assignments, term papers, mini projects, and laboratory works. These assessments contribute 40% to the total grade, while the

examination score accounts for 60%. The duration of each course examination varies based on the credit value assigned to the course. For a 3-credit course, seven questions are provided, and students must answer five within a three-hour period. Similarly, for a 2-credit course, students are required to answer four out of six questions in two hours. In their final year, students undertake projects worth six units. This project extends throughout the entire session as a continuous effort, with final project defense occurring at the end of the second semester. During this time, students defend of their project through an oral examination. In the final semester, presenting seminars with PowerPoint and demonstrating their projects are mandatory requirements. The seminar plays a key role in enhancing the intellectual growth and confidence of the students.

(a) Admission to the Examination

Every student who is duly registered for courses in a given semester is qualified to take the examinations in those courses, except in the following cases (subject to Senate approval based on College Board recommendation):

(i) a student who stays away from the University for more than three weeks without official approval.

(ii) a student whose attendance at lectures or practical sessions in any course falls below 80 percent.

Before being allowed into the examination hall, each student must:

(i) present a valid University Identity Card along with an Examination Card.

(ii) ensure that no unauthorized or incriminating item is in their possession or among the materials they legitimately brought into the hall, such as rulers, mathematical tables, and similar items.

(iii) maintain orderly conduct and strictly follow all instructions given by the Invigilator or Examiner.

(iv) refrain from any form of malpractice or attempt to engage in examination misconduct.

(b) Examination Marks

The marks awarded in each examination are determined by the internal examiners (Lecturers) and subsequently endorsed by the examination moderators and the Chief Examiner (the HOD). For final year students, the involvement of external examiners is also required during the moderation process. The final stage of moderation takes place at the meeting of the Programme Board of Examiners. Where there is a disagreement, the recommendation of the external

examiners may only be set aside if overruled by a two-thirds majority vote of the Examiners Board.

(c) Board of Examiners

The Programme Board of Examiners is made up of all academic staff within the Programme, with the Head of Programme serving as the Chief Examiner. The College Board of Examiners, on the other hand, comprises all members of the College Board of Studies together with any other individuals assigned the responsibility of examining courses offered within the College.

(d) Semester Load

Each student is required to register in line with the Programme guidelines, carrying not less than 15 credit units and not more than 24 credit units per semester. However, students who have are in special cases may, with the approval of the Vice Chancellor, be allowed to register for more than the maximum credit requirement.

(e) Pass Mark

The pass mark in any course of is 45%.

(f) Grading System

The performance of a student in each course shall be presented in both numerical scores and corresponding letter grades. The overall academic standing of a student for a semester is expressed as the Grade Point Average (*GPA*). The *GPA* is calculated by multiplying the Grade Point (*GP*) earned in each course by the respective course credit unit (*C*) to obtain the Weighted Grade Point (*WGP*). The total WGP for all courses taken in the semester is then divided by the total number of registered credit units to determine the semester *GPA*.

$$GPA = \frac{\sum_i^n (WGP)_i}{\sum_i^n C_i}$$

Where n is the number of courses registered for

Table 6.2: The marks in figures and the corresponding letter grades and Grade Point (GP):

% Score	Letter Grade	Grade Point
70 to 100%	A	5
60 to 69%	B	4
50 to 59%	C	3
45 to 49%	D	2
< 45%	F	0

The Cumulative Grade Point Average (*CGPA*) of a student is the overall average of the Weighted Grade Points (*WGP*) accumulated across all semesters.

The procedure for computing the *CGPA* is as follows:

(a) Multiply the Grade Point (*GP*) obtained in each course by the corresponding course credit unit *C* to determine the Weighted Grade Point (*WGP*).

(b) Add together all the *WGP* values for every course taken throughout the period of study.
(c) Divide the total *WGP* by the total credit units of all registered courses, whether passed or failed, across all semesters.

$$CGPA = \frac{\sum_i^N (WGP)_i}{\sum_i^N C_i}$$

where *N* is the total number of courses registered for all semesters

6.5. Laboratories

The Programme places strong emphasis on laboratory work, as it provides students with the platform to apply the concepts taught in lectures. Practical sessions give students the opportunity to demonstrate experimental skills, confirm theoretical principles and laws of Engineering, and develop hands-on competence. The Programme has the following laboratories:

S/N	Laboratory Name	Associated Courses	Level and Programmes
1.	Electrical Power and Machine/Energy Laboratory	EEE 318	300L Electrical Electronics Engineering
2.	Applied Electricity/Electronics Laboratory	EEE 221 EEE 222 ENG 261 ENG 262	200L Electrical/Electronics Engineering
3.	Microprocessor and Digital System Laboratory	EEE 318	300L Electrical Electronics Engineering
4.	Communication and Network Laboratory	EEE 221 EEE 222 EEE 318 EEE 423	200L Electrical Electronics Engineering 200L Electrical Electronics Engineering 300L Electrical Electronics Engineering 400L Electrical Electronics Engineering
5.	Measurement/ Instrumentation and Process Control Laboratory	EEE 221 EEE 222	200L Electrical Electronics Engineering 200L Electrical Electronics Engineering
6.	Computer Software Engineering Laboratory	COE 307 COE 308	300L Computer Engineering

7.	Prototyping Workshop/Demonstration Laboratory	EEE 221 EEE 222 COE 409	200L Mechatronics Engineering 400L Computer Engineeering
8.	Computer Hardware Laboratory	COE 307 COE 308 COE 411	300L Computer Engineering 400L Computer Engineering 500L Computer Engineering 200L Computer Science
9.	National Instrument Academy and Data Acquisition Laboratory	EEE 221 EEE 222 EEE 318	200L Electrical Electronics Engineering 200L Electrical Electronics Engineering 300L Electrical Electronics Engineering

6.6. Summary of the graduation requirements for the program

This section contains a summary of the process for ensuring and documenting that each graduate completes all graduation requirements for the program
The curriculum for these degree programs is aligned with the approved COREN Benchmark Minimum Academic Standard (BMAS). Comprehensive details are available in the program handbook. To earn a Bachelor of Engineering (B.Eng.) degree in Electrical/Electronic Engineering at Afe Babalola University, students must perform satisfactorily in Core/Compulsory courses, Elective/Option courses, University requirements, Industrial Training, and adherence to National Universities Commission (NUC) and COREN standards.

6.7. External Examination System

The Programme appoints approved External Examiners to moderate final year examination question papers and marking guides. The University regards the external examination system as an essential mechanism for maintaining quality assurance and promoting the highest standards of professionalism. The present External Examiners are:

1. Prof. Jide J. POPOOLA,
Department of Electrical & Electronic Engineering,
Federal University of Technology,
Akure, Ondo State.

Files containing the comments of the external examiner for the examination moderation is provided and available for inspection.

6.8. Describe CQI strategies to be implemented in relation to students

Continuous Quality Improvement (CQI) strategies are designed to enhance the educational experience and outcomes for students by continually assessing and refining various aspects of

the educational process. Several key areas are targeted when implementing CQI strategies related to students:

Student Feedback: We actively gather feedback through surveys, focus groups, and individual interviews to ensure our educational programs continuously evolve to meet student needs. This invaluable input helps us identify areas for improvement, celebrate program strengths, and create the best possible learning experience.

Learning Assessment: By employing a variety of assessments—including formative and summative tasks, standardized tests, project-based work, and performance evaluations—we gain a comprehensive understanding of student learning. Analyzing these results enables us to identify areas needing improvement and design targeted interventions to ensure all students succeed.

Curriculum Review: Our curriculum undergoes regular reviews to ensure alignment with educational goals, industry demands, and evolving trends. This process includes evaluating the curriculum's relevance, coherence, and effectiveness. We involve students by gathering their feedback and incorporating their suggestions for improvement, ensuring the program remains current and provides the best possible learning experience.

Student Engagement: The university fosters a vibrant learning environment by offering a mix of experiential learning opportunities, collaborative projects, and extracurricular activities. We actively track student participation to assess the impact of these initiatives on student satisfaction, motivation, and academic success.

Through CQI strategies such as student feedback mechanisms, diverse assessments, and curriculum reviews, educational institutions can cultivate a learning environment that prioritizes student success and engagement. This ongoing process, fueled by student input and data analysis, ensures programs remain relevant, effective, and ultimately drives student satisfaction.

6.9. Engineering Drawing Practice

Engineering drawing is one of the important courses taken by students and is a centrally administered course by the College of Engineering. The college have drawing studios for the drawing course in Engineering taken by all 200 level students.

6.10. Adequacy of Lectures

There is a provisional lecture timetable and venues for all courses, which every lecturer follows for their lectures. The university have a quality assurance unit that monitors the adequacy of lectures.

6.11. Record of Final Year Students Projects for Previous Sessions

The records of the final year student projects for the programme are available for perusal.

6.12. Mechanism for providing guidance to students of Electrical Engineering on academic, career and aspects pertaining to wellness.

The Electrical/Electronics Engineering programme utilizes a multifaceted approach to student guidance, ensuring that academic progress, professional growth, and personal well-being are supported through structured institutional mechanisms, which include:

6.12.1 Academic Guidance and Mentorship

Academic support is primarily facilitated through the Academic Level Advisers system, where specific faculty members are assigned to each cohort to provide personalized guidance on course registration, academic performance, and adherence to university regulations. For the current session, the advisers for the Electrical programme include Mr. V. Adebajo-Akinpelu (100L), Engr. Dr. N. O. Omogbai (200L), Engr. T. E. Fabunmi (300L), Engr. Dr. T. O. Ejidokun (400L), and Dr. M. A. Ogunlade (500L). Beyond administrative assistance, these advisers and the broader faculty act as mentors, helping students align their learning trajectories with the Programme Educational Objectives (PEOs) and providing instructional support to ensure the mastery of course learning outcomes. Regular departmental and board meetings serve as platforms where student progress is reviewed, and interventions are designed to address academic challenges democratically.

1. **Career Development and Professional Placement**

Career guidance is integrated into both the formal curriculum and the university's administrative services, offering students a roadmap for transitioning into the engineering industry. All students have access to centralized career planning and placement services, which are supplemented by departmental initiatives like "Career Week" and seminars featuring industry professionals. Foundational courses such as Engineer in Society (GET 101) and Introduction to Electrical and Electronics Engineering (EEE 102) introduce students to the diverse specializations and job profiles within the field, ranging from Power and Manufacturing to Research and Software Engineering. Furthermore, the Students Industrial Work Experience Scheme (SIWES) provides critical hands-on exposure across three different levels (SIWES I, II, and III), allowing students to identify industrial challenges and build professional networks before graduation. Professional identity is further fostered through the IEEE and NSE staff advisories, currently coordinated by Engr. Dr. T. O. Ejidokun, which encourage students to engage with global engineering communities.

2. **Student Wellness and Counselling Services**

The department recognizes that holistic student success depends on mental and emotional stability, which is addressed through both departmental and college-level units. A dedicated **Student Counselling Unit** at the College of Engineering, headed by **Mr. Olapade**, serves as the specialized facility for students requiring psychological or emotional support. Within the department, student well-being is monitored by the Welfare Officer, Engr. A. Akinola, who collaborates with level advisers to identify and support students facing personal, health, or social difficulties. These support mechanisms are designed to foster an inclusive and supportive environment that encourages student engagement and success while ensuring that wellness issues do not become barriers to academic achievement. Ethical conduct and personal integrity are also religiously practiced and taught as part of the university's guiding principles, further contributing to the character development and overall wellness of the student body

6.12.2. Students' Workload, Class Sizes for Theory, Laboratory Sessions and Completion of Courses

For the Electrical Engineering programme, the management of student workload, class sizes, and course completion follows the modularised Course Unit System and university regulations as detailed hereafter.

1. **Students' Workload**

The academic workload is determined by credit units assigned to each course within a 15-week semester.

a) Credit Unit Definition: One credit unit is equivalent to one hour of lecture/tutorial per week or three hours of laboratory, workshop, or studio work per week.

b) Semester Registration Limits: Students are required to register for a minimum of 15 credit units and a maximum of 24 credit units per semester.

c) Total Graduation Requirement: To be eligible for the award of the B.Eng. degree, UTME students must complete between 130–138 core units and a total of approximately 197 units over the five-year programme.

2. **Class Sizes and Instructional Ratios**

Class sizes are monitored to ensure effective teaching and access to laboratory facilities. The following table summarizes the enrollment and instructional capacity for the Electrical programme:

Category	Parameter / Detail
Average Theory Class Size	Typically ranges from 29 to 62 students depending on the academic level and session.

Laboratory Capacity	Designed to accommodate students in groups for hands-on activities, such as winding techniques or circuit testing.
Lab Grouping	Students generally work in small groups of 4–5 per station to ensure competence in using equipment like oscilloscopes and power supplies.
Staff-Student Ratio	1:13.2

3. Completion of Courses

Course completion is strictly governed by performance, attendance, and residency regulations.

a) Evaluation Breakdown: Completion of a course is based on a 40% Continuous Assessment (quizzes, assignments, and mid-semester tests) and a 60% Final Examination.

b) Eligibility for Exams: Students must attain a minimum of 75% attendance to be eligible to sit for the final examination in any course.

c) Pass Criteria: The minimum pass mark is an aggregate of 45% in each course. To graduate, students must pass all compulsory and departmental courses with a Cumulative Grade Point Average (CGPA) of not less than 1.00.

d) Residency Limits: The maximum time allowed for completion of the degree is 15 academic semesters for UTME entrants and 13 semesters for Direct Entry students

6.13. Students' activities and involvement in student organizations that provide experience in management and governance, representation in education and related matters and social activities.

The programme provides several structured avenues for students to develop leadership, management, and governance skills while engaging in a vibrant social and technical community.

6.13.1. Leadership and Management Experience

a) IEEE Student Body: Students actively lead the departmental branch of the Institute of Electrical and Electronics Engineers (IEEE), with professional mentorship provided by the IEEE Staff Adviser, Engr. Dr. T. O. Ejidokun. This involvement offers hands-on experience in managing professional engineering activities and networking with global experts.

b) Team Leadership in Projects: The curriculum emphasizes group assignments and capstone design projects, requiring students to rotate through leadership roles to manage project lifecycles, meet deadlines, and coordinate multi-disciplinary teams.

c) Engineering Management Training: Compulsory courses like Engineering Management and Law (ENG 583) and Engineer in Society (GET 101) equip students with formal principles of policy formulation, organizational structure, and financial management.

6.13.2. Educational Representation and Governance

a) Quality Improvement Participation: Students play a direct role in institutional governance by participating in Continuous Quality Improvement (CQI) processes. They provide critical feedback through end-of-semester surveys, focus groups, and interviews, which are used to evaluate lecturer performance and refine the curriculum.

b) Stakeholder Engagement: Students are recognized as key stakeholders and are engaged through interactive sessions and placement activities to ensure the programme meets their evolving professional needs.

6.13.3. Technical Activities and Competitions

a) Innovation and Prototyping: Through the Salt Valley Tech Hub (referenced as the Innovation and Enterprise Centre) and the Engineering Innovation and Fabrication Hub, students engage in hackathons and real-world problem-solving.

b) Robotics: Students participate in advanced robotics activities, utilizing the Prototyping Laboratory to design and implement automated systems, building on knowledge from courses like Robotics and Automation (COE 519).

c) Global Recognition: The programme's focus on competitive technical excellence has led students to win medals at prestigious international events such as the Microsoft ImagineCup.

6.13.4. Social and Recreational Involvement

a) Campus Life and Wellness: The university fosters a holistic environment by integrating social activities like movie nights at the planetarium and recreational events into the student experience.

b) Support Facilities: Modern support facilities, including sports and recreational centers, are provided to facilitate character building and enhance the overall quality of campus life.

SECTION SEVEN:

CONTINUOUS QUALITY IMPROVEMENT

7.1. The CQI Strategies

The process for planning and improving our Electrical/Electronic Engineering programme includes how the curriculum was developed, review of its content, and response to feedback from employers, industry experts, students, and alumni. We also track how individual courses contribute to learning objectives and use assessments to measure programme outcomes. We regularly review these objectives and continuously strive to improve the program's quality by:

- **Quality Teaching:** The programme has a Quality Assurance (QA) committee dedicated to ensuring all lecturers and technologists deliver high-quality instruction. They evaluate lecturers and technologists' performance each semester and recommend improvements for the next semester.
- **Research and Learning:** Another committee encourages both faculty and students to actively participate in research and share their findings. This committee helps find research partners and supports publishing results in respected conferences, journals, and seminars. Students and faculty also participate in seminars, conferences, workshops, and industry visits to stay current and learn new things.

7.2. Employer's Rating of Graduates of the Programme/Department

Our programme has put in place the following to ensure our graduate has a good employers' rating after graduation. Our department has made several recent improvements:

- **Research Focus:** We launched a Research/Seminar/Conference and Workshop Committee to boost collaboration and research output among staff and students. The mandate of this committee includes equipping our final year students with requisite research enhancement tools.
- **Quality Assurance:** We established an Internal Quality Assurance Committee to ensure yearly improvements in faculty development, teaching quality, and student learning.
- **Outcome-Based Education:** We formed an OBE Implementation Committee in 2023 to ensure we meet Outcome-Based Education requirements. The committee lead includes the Head of Programme and a departmental OBE desk officer. Members of the Quality Assurance Committee automatically join the OBE committee. This committee also ensured that EEE 564, a course dedicated to the use of modern software for solving electrical electronic engineering problems, is delivered to satisfaction.
- **Project Vetting:** At the beginning of every academic session, Final Year Project Committee reviewsthe students' research interests and matches their proposed final-year project titles

to staff's expertise to enhance the quality of project and reports. This initiative stemmed from the OBE requirements.

Table 7.1: Employer's Rating of Graduates of Electrical/Electronics Engineering

Student Employement Ratings

S/N	Name of Students	Programme	Name of Employers/ Address	Year of Report	Summary of Employer's Remark
1.	Ezeobidi Clementina	Electrical/Electronic Engineering	Intavalto Nig Ltd, King Meadow Campus, Nottingham	2024	She is a reliable team player who always maintain a positive attitude, even under pressure
2.	Rashida Olomowewe	Electrical/Electronic Engineering	Danelco Limited	2024	Her contributions have been invaluable to our team, and her proactive approach to problem-solving has significantly enhanced our operational efficiency
3.	Bassey Nta	Electrical/Electronic Engineering	Health4Everyone, Abuja, Kukuwa street.	2024	He is doing an amazing job in his position with us, and has been an asset to our organization so far
4.	Tochukwu Diasy	Electrical/Electronic Engineering	Transcorp Hilton, Abuja, Nigeria	2024	She was been recognized this year award in recognition of her professionalism in carrying out assignments and going extra mile to

					accomplish them.
5.	Adebajo Akinpelu	Electrical/Electronic Engineering	Afe Babalola University, Ado Ekiti, Nigeria	2024	He is an excellent staff who has shown admirable level of competence
6.	Oluwadara Toluwani Ojo	Electrical/Electronic Engineering	FCMB PENSIONS LIMITED	2024	Her dedication, strong work ethic and ever-ready to learn attitude has made her an integral part of our team

Rating of Student Industrial Work Experience Scheme

Apart from the employment ratings from the employers of our graduates who are already making waves in the industry, the rating of our some of our students in their SIWES programmes is shown in Table 6.3

Table 7.3: Students rating by Industrial Supervisors for SIWES programme for some students in Electrical and Electronics Engineering for 2024

No.	Name of Students	Programme	Name of Employers /Address	Year of SIWES	Summary of Employers' Remark
1	*Otechekpen Aisosa Emmanuel*	*Electrical/Electronic Engineering (400L)*	*PPC System Engineering Limited, Victorial Island, Lagos.*	*2024*	*Fully participated in the training.*
2	*Agunbiade Oluwadamilare Samuel*	*Electrical/Electronic Engineering (400L)*	*Multidigital Nigeria Limited, Portharcourt, River State.*	*2024*	*Satisfactory*
3	*Tyowua Fater Fadey*	*Electrical/Electronic Engineering (400L)*	*HOTSPOT NETWORK Limited, Abuja.*	*2024*	*Fater has demonstrated exceptional dedication to*

No.	Name of Students	Programme	Name of Employers /Address	Year of SIWES	Summary of Employers' Remark
					work. He shows great initiative and work well independently and collaboratively with the team.
4	*Mokolo Oluchi*	*Electrical/Electronic Engineering (400L)*	*TransAfam Power Limited, River State.*	*2024*	*An excellent student with a zeal for learning and the capacity to grasp new concepts quickly.*
5	*Chinda Denis*	*Electrical/Electronic Engineering (400L)*	*First Power independent Limited, Trans Amadi, Portharcourt.*	*2024*	*Well behaved.*
6	*Sharjeel Ofuaz*	*Electrical/Electronic Engineering (400L)*	*OLAM, Interchange, Shagamu.*	*2024*	*Oliver is very good and understanding boy. He shows interest in every work.*
7	*Abugu Ugonna*	*Electrical/Electronic Engineering (400L)*	*Psriwinkle Residences Limited, Lekki, Lagos.*	*2024*	*Satisfactory performance.*
8	*Mamai Chioma Arid*	*Electrical/Electronic Engineering (400L)*	*National Centre for artificial Intelligence and Robotics, Abuja.*	*2024*	*Very dedicated and hardworking student..*
9	*Wodi Joseph*	*Electrical/Electronic Engineering (400L)*	*Transmission Company of Nigeria, Abuja.*	*2024*	*Joseph is highly intelligent, committed to work and always ready to work.*

Table 7.4: Students rating by Industrial Supervisors for SIWES programme for some students in Electrical/Electronics Engineering for 2025.

No.	Name of Students	Programme	Name of Employers /Address	Year of SIWES	Summary of Employers' Remark
1	*Aki Mary Azu*	*Electrical/Electronic Engineering (300L)*	*Electricity Distribution PLC, Jos.*	*2025*	*An outstanding performance with interest in learning, a team player and hardworking.*
2	*Anjorin Oluwademilade*	*Electrical/Electronic Engineering (300L)*	*Engineering Orritech Nigeria Limited, Lekki Lagos.*	*2025*	*He was committed to duty and always ready to learn.*
4	*Awa Nissan Tojah*	*Electrical/Electronic Engineering (300L)*	*Wimrock Limited Company.*	*2025*	*He is a very hardworking boy, ready to learn and makes himself always available.*
5	*Ogunleye Seth*	*Electrical/Electronic Engineering (300L)*	*RAINOIL LTD, Lekki Lagos.*	*2025*	*Seth has demonstrated good level of attitude to learn . He has shown team work which was very commendable foe an intern.*
6	*Ogor Clinton*	*Electrical/Electronic Engineering (300L)*	*Work and Service, University of Portharcourt, River State.*	*2025*	*He is industrious, ready to take new task, has good initiative, relates cordially with members of the team. Also, asking questions to more facts.*
7	*Aneke Ijeoma Victoria.*	*Electrical/Electronic Engineering (300L)*	*NIIT, Ikeja Lagos.*	*2025*	*She is very eager to learn and her ability to adapt and deliver the given task.*
8	*Kienabere Emmanuel*	*Electrical/Electronic Engineering (300L)*	*Wilamzo Resources Limited, River State.*	*2025*	*He is eager to learn and can see him becoming a good Engineer in the future.*

No.	Name of Students	Programme	Name of Employers /Address	Year of SIWES	Summary of Employers' Remark
9	*Upkai Samuel*	*Electrical/Electronic Engineering (300L)*	*Alpha Oilwell Technologies, River State.*	*2025*	*Samuel did what was required of him during his short stay at the company. He showed good understanding of the concepts shown.*
10	*Ogunrinde Ayomidotunj*	*Electrical/Electronic Engineering (300L)*	*ANOH GAS Processing Plant. Assa North, Imo State.*	*2025*	*Excellent student.*

No.	Name of Students	Programme	Name of Employers /Address	Year of Graduation	Summary of Employers' Remark
1	*Nwogu Johnblasse*	*Electrical/Electronic Engineering (200L)*	*Yugee Technical Solution, Oweri, Imo State.*	*2025*	*He is a good student, always ready to learn without any argument.*
2	*ATIEAN Utsedritselaju*	*Electrical/Electronic Engineering (200L)*	*Total Energies, Delta State.*	*2025*	*He is an interesting personality to work with.*
3	*Ejiofor Emmanuel*	*Electrical/Electronic Engineering (200L)*	*Radigok Nigeria Limite,Agege, Lagos.*	*2025*	*Emmanuel's performance during this period was quite exquisite.*
4	*Dominic Kaosischukwu*	*Electrical/Electronic Engineering (200L)*	*Gaby Sumoh Nigeria Limited, Oweri, Imo State.*	*2025*	*Satisfactory*
5	*Ojo Humble*	*Electrical/Electronic Engineering (200L)*	*Hybitel Computer and Telecommunication, Mushin Lagos.*	*2025*	*Humble is punctual, willing to learn and dedicated to his work.*
6	*Kikiri Godwin*	*Electrical/Electronic Engineering (200L)*	*Honik Engineering Limited, Off*	*2025*	*He pays more attention to details and very*

No.	Name of Students	Programme	Name of Employers /Address	Year of Graduation	Summary of Employers' Remark
			Cheveron Drive, Lekki, Lagos.		*hardworking. I am pleased to have him as a student under me.*
7	*Salami Uassan*	*Electrical/Electronic Engineering (200L)*	*Power Holding Company, 10, Kudirat street, Oke Afa, Lagos.*	*2025*	*Satisfactory.*
8	*Augustine Emmanuel*	*Electrical/Electronic Engineering (200L)*	*Portharcourt Electricity Distribution Company*	*2025*	*He is a brilliant student and his technical knowledge and problem-solving skills are impressive.*
9	*Agbaje Muslim Taiwo*	*Electrical/Electronic Engineering (200L)*	*Anoh Gas Processing Company, Ikoyi Lagos.*	*2025*	*Taiwo is a good listener and always open to new ideas and perspective.*
10	*Anyanwu Ihunanyachi*	*Electrical/Electronic Engineering (200L)*	*Heritage Energy Operational Service Limited, Warri , Delta State.*	*2025*	*He is a very dedicated and hardworking. She is always available to learn new things at any point in time.*

Some letters attesting to the outstanding performance of some of our students during the SIWES programme is presented in **Annex P**

7.3. Response to External Examiner's Comments

Feedback from external experts is taken seriously by the university management. The programme leadership carefully reviews comments from external examiners and discusses the issues raised. We then decide on ways to address any weaknesses identified and ensure the programme's strengths remain strong. Additionally, staff suggestions for improvement are passed on to university management for further action.

COURSE CURRICULUM

Categorization of Courses

Courses are categorized as follows:

i. Compulsory Courses: Students must register for and pass before graduation.

ii. Elective Courses: Courses students may offer and pass to obtain the minimum units required for graduation. Such courses are normally selected from within or outside the programme with the approval of the department.

iii. Required Course: A course specified by a programme which a student shall take and pass.

iv. Pre-requisite Courses: Courses students must take and pass before taking another specified course.

v. Institutional Courses: Courses whose knowledge is essential for the preparation of Afe Babalola University for future employment after graduation.

vi. Carry-over Courses: On the account of any failure by a student to satisfy a pass requirement in any course, such student will be required to repeat only the course/courses the following academic session or re-write the prescribed examination in such course at the end of the academic session.
Students carrying over courses shall not be allowed to register for more than the approved maximum number of credits for the semester.

B.ENG. ELECTRICAL AND ELECTRONICS (5-YEAR PROGRAMME)
100 LEVEL FIRST SEMESTER

Course Code	Course Title	Units	LH	PH	Status
GST 111	Communication in English	2	15	45	Compulsory
GET 101	Engineer in Society	1	15	-	Compulsory
MTH 101	Elementary Mathematics I: Algebra and Trigonometry	2	30	-	Compulsory
PHY 101	General Physics I: Mechanics	2	30	-	Compulsory
PHY 107	General Practical Physics I	1	-	45	Compulsory
EEE 102	Introduction to Electrical and Electronics Engineering	2	30	-	Compulsory
ABUAD-CHM 101	General Chemistry I	2	45	-	Compulsory
ABUAD-CHM 107	General Practical Chemistry I	1	-	45	Compulsory
AFE 101	Health Awareness Education	1	15	-	Compulsory
AFE 103	Leadership	1	15	-	Compulsory
TOTAL		**15**			

100 LEVEL SECOND SEMESTER

Course Code	Course Title	Units		LH	PH	
GST 112	Nigerian Peoples and Culture	2		30	-	Compulsory
GET 102	Engineering Graphics and Solid Modelling I	2		15	45	Compulsory
MTH 102	Elementary Mathematics II: Calculus	2		30	-	Compulsory
PHY 102	General Physics II: Behaviour of Matter	2		30	-	Compulsory
PHY 108	General Practical Physics II	1		-	45	Compulsory
ABUAD-CHM 102	General Chemistry II	2		45	-	Compulsory
ABUAD-CHM 108	General Practical Chemistry II	1		-	45	Compulsory
FOREIGN LANGUAGE ELECTIVE (SELECTION OF ONE)						
AFE 124	French Language I	1		15	-	Elective
AFE 126	Chinese Language	1		15	-	Elective
NIGERIAN LANGUAGE ELECTIVE (SELECTION OF ONE)						
AFE 128	Igbo Language	1		15	-	Elective
AFE 130	Yoruba Language	1		15	-	Elective
AFE 132	Hausa Language	1		15	-	Elective
TOTAL		**14**				

SECOND YEAR (200 LEVEL): FIRST SEMESTER COURSE STRUCTURE

Course Code	Course Title	Units	Status	LH	PH
ENT 211	Entrepreneurship and Innovation	2	C	30	-
GET 201	Applied Electricity I	3	C	45	-
GET 203	Engineering Graphics and Solid Modeling II	2	E	15	45
GET 205	Fundamentals of Fluid Mechanics	3	C	45	-
GET 207	Applied Mechanics	3	C	45	-
GET 209	Engineering Mathematics I	3	C	45	-
GET 211	Computing and Software Engineering	3	C	30	45
	One foreign language is required				
AFE 203*	French Language Acquisition II	1	E		
AFE 205*	Chinese Language II	1	E		
	One Nigerian Language is required				
AFE 207**	Igbo Language II	1	E		
AFE 209**	Yoruba Language II	1	E		

AFE 211**	Hausa Language II	1	E		
Total		**23**			

SECOND YEAR (200 LEVEL): SECOND SEMESTER COURSE STRUCTURE

Course Code	Course Title	Units	Status	LH	PH
GST 212	Philosophy, Logic and Human Existence	2	C	30	-
GET 204	Students Workshop Practice	2	C	15	45
GET 206	Fundamentals of Thermodynamics	2	C	45	-
GET 210	Engineering Mathematics II	3	C	45	-
EEE 202	Applied Electricity II	3	C	30	45
EEE 204	Electrical Engineering Materials	3	E	45	-
ABUAD-GET 208	Strength of Materials	3	C	45	-
AFE 202	Food Production and Health Awareness	1			
*GET 299	SIWES I	3*	C	9 Weeks	
Total		**20**			

300 LEVEL FIRST SEMESTER

Course Code	Course Title	Units	Duration (Hours)			Pre-Req.	Status
			L	T	P		
EEE 313	Electronics I (Physical)	3	2	1	-	-	Compulsory
EEE 315	Electromagnetic Field and Wave Theory I	3	2	1	-	-	Compulsory
EEE 317	Electrical/Electronic Laboratory I	1	-	-	3	-	Compulsory
EEE 319	Measurements and Instrumentation	3	2	1	-	-	Compulsory
EEE 321	Electric Circuit Theory I	3	2	1	-	-	Compulsory
EEE 325	Electrical Machines I	3	2	1	-	-	Compulsory
EEE 341	Software Development Techniques	3	2	-	3	-	Compulsory
ENG 381	Engineering Mathematics III	3	2	1	-	-	Compulsory

AFE 301	Practical Entrepreneurship	1	1	-	3	-	Compulsory
		23					

300 LEVEL SECOND SEMESTER

Course Code	Course Title	Units	Duration (Hours)			Pre-Req.	Status
			L	T	P		
EEE 314	Electronics II (Analogue)	3	2	1	-	EEE 313	Compulsory
EEE 316	Electromagnetic Field and Wave Theory II	3	2	1	-	EEE 315	Compulsory
EEE 318	Electrical/Electronic Laboratory II	1	-	-	3	-	Compulsory
EEE 322	Electric Circuit Theory II	3	2	1	-	EEE 321	Compulsory
EEE 324	Linear Systems	3	2	1	-	-	Compulsory
EEE 326	Electrical Machines II	3	2	1	-	EEE 325	Compulsory
EEE 328	Communication Principles	3	2	1	-		Compulsory
ENG 382	Engineering Mathematics IV	3	2	1	-		Compulsory
ENG 384	Engineering Economics	2	2	-	-		Compulsory
		24					

400 LEVEL FIRST SEMESTER

Course Code	Course Title	Units	Duration (Hours)			Pre-Req	Status
			L	T	P		
EEE 405	Numerical Computations	3	2	1	-	-	Compulsory
EEE 415	Assembly Language Programming	2	2	-	-	-	Compulsory
EEE 423	Electrical/Electronic Laboratory III	1	-	-	3	EEE 318	Compulsory
EEE 427	Data Communication and Networks	3	2	1	-	-	Compulsory
EEE 431	Electric Power Principles	3	2	1	-	-	Compulsory
EEE 441	Control Theory and Servomechanism	3	2	1	-	-	Compulsory
EEE 471	Digital Electronics and Systems	3	2	1	-	EEE 314	Compulsory

ENG 485	Technical Communications (Report Writing)	2	2	-	-	--	Compulsory
		20					

400 LEVEL SECOND SEMESTER

Course Code	Course Title	Units	Duration (Hours)			Pre-Req	Status
			L	T	P		
ENG 290	Student Work Experience Programme I (SWEP I)	3				-	Compulsory
ENG 390	Student Work Experience Programme II (SWEP II)	3				ENG 290	Compulsory
ENG 490	Student Industrial Work Experience Scheme (SIWES)	6				ENG 390	Compulsory
		12					

500 LEVEL FIRST SEMESTER

COURSE CODE	COURSE TITLE	L	T	P	U	PRE-REQ.	STATUS
EEE 507	Modelling and Simulation of Electrical and Electronics Systems	1	-	3	2	-	C
EEE 509	Advanced Electronics	2	-	3	3	-	C
EEE 511	Digital Signal Processing	2	1	-	3	-	C
EEE 561	Control Engineering	2	1	-	3	EEE 441	C
ENG 583	Engineering Management and Law	2	-	-	2	-	C
Electives - Minimum of two courses to be selected							
ELECTRONICS/ COMMUNICATION							
EEE 513	Communication Systems Engineering	2	1	-	3		E
EEE 523	Industrial Electronics Designs	2	1	-	3	-	E
EEE 535	High Frequency & Microwave Electronics	2	1	-	3	-	E
POWER AND MACHINES							
EEE 553	Power System Engineering	2	1	-	3	-	E
EEE 555	High Voltage Engineering	2	-	-	2	-	E
EEE 543	Renewable Energy System	2	-	-	2	-	E
CONTROL AND COMPUTER							
EEE 565	Modern Control Engineering (Modern)	2	1	-	3	-	E
EEE 523	Industrial Electronics Designs	2	1	-	3	-	E
EEE 517	Analogue and Digital Computers	2	-	-	2		E

SECOND SEMESTER 500 LEVEL COURSES

COURSE CODE	COURSE TITLE	L	T	P	U	PRE-REQ.	STATUS
EEE 502	Electrical Services Design	2	-	-	2	-	C
EEE 504	Embedded System	2	1	-	3		C
EEE 514	Reliability Engineering	1	1	-	2	-	C
EEE 562	Power Electronics	2	1	-	3	EEE 326	C
EEE 564	Seminars on Use of Engineering Software Packages in Electrical/Electronics Field	-	-	3	1	-	C
EEE 599	Student Project	-	-	18	6	-	C
Electives - Minimum of two courses to be selected							
ELECTRONICS/ COMMUNICATION							
EEE 512	Digital Communication	2	-	-	2	EEE 328	E
EEE 506	Digital System Design with VHDL Programming	2	1	-	3	-	E
EEE 578	Broadcasting and Internet Technology	1	-	3	2	-	E
EEE 576	Microcomputer Technology	3	-	-	3	-	E
POWER AND MACHINES							
EEE 554	Power Systems Communication and Control	2	1	-	3	EEE 553	E
EEE 556	Electric Drives and Applications	2	-	3	3	EEE 326	E
EEE 558	Electrical Energy Conversion and Storage	2	-	-	2	-	E
CONTROL AND COMPUTER							
EEE 566	Control Engineering III (Non-Linear & Process Automation)	2	1	-	3	EEE 324	E
EEE 506	Digital System Design with VHDL Programming	2	1	-	3	-	E
EEE 576	Microcomputer Technology	3	-	-	3	-	E
EEE 512	Digital Communication	2	-	-	2	EEE 413	E

COURSE DESCRIPTION

100 LEVEL

GST 111: Communication in English (2 Units C: LH=15; PH 45)
Learning Outcomes
At the end of this course, students should be able to:
1. identify possible sound patterns in English Language;
2. list notable language skills;
3. classify word formation processes;

4. construct simple and fairly complex sentences in English;
5. apply logical and critical reasoning skills for meaningful presentations;
6. demonstrate an appreciable level of the art of public speaking and listening; and
7. write simple and technical reports.

Course Contents

Sounds and sound patterns in English Language (vowels and consonants, phonetics and phonology). English word classes (lexical and grammatical words, definitions, forms, functions, usages, collocations). major word formation processes; the sentence in English (types: structural and functional). grammar and usage (tense, concord and modality). Reading and types of reading, comprehension skills, 3RsQ. Logical and critical thinking; reasoning methods (logic and syllogism, inductive and deductive argument, analogy, generalisation and explanations). Ethical considerations, copyright rules and infringements. Writing activities: pre-writing (brainstorming and outlining). writing (paragraphing, punctuation and expression). post- writing (editing and proofreading). Types of writing (summary, essays, letter, curriculum vitae, report writing, note-making) etc. Mechanics of writing. Information and Communication Technology in modern language learning. Language skills for effective communication. The art of public speaking.

GET 101: Engineer in Society (1 Unit C: LH=15)

Learning Outcomes

At the end of this course, the students should be able to:

1. differentiate between science, engineering and technology, and relate them to innovation;
2. distinguish between the different cadres of engineering – engineers, technologists, technicians and craftsmen and their respective roles and competencies;
3. identify and distinguish between the relevant professional bodies in engineering;
4. categorise the goals of global development or sustainable development goals (SDGs); and
5. identify and evaluate safety and risk in engineering practice.

Course Contents

History, evolution and philosophy of science. engineering and technology. The engineering profession – engineering family (engineers, technologists, technicians and craftsmen), professional bodies and societies. Engineers' code of conduct and ethics, and engineering literacy. Sustainable development goals (SDGs), innovation, infrastructures and nation building - economy, politics, business. Safety and risk analysis in engineering practice. Engineering competency skills – curriculum overview, technical, soft and digital skills. Guest seminars and invited lectures from different engineering professional associations.

MTH 101: Elementary Mathematics I (Algebra and Trigonometry)
(2 Units C: LH=30)

Learning Outcomes

At the end of the course students should be able to:

1. define and explain set, subset, union, intersection, complements, and demonstrate the use of Venn diagrams;

2. solve quadratic equations;
3. solve trigonometric functions;
4. identify various types of numbers; and
5. solve some problems using binomial theorem.

Course Contents

Elementary set theory, subsets, union, intersection, complements, Venn diagrams. Real numbers, integers, rational and irrational numbers. Mathematical induction, real sequences and series, theory of quadratic equations, binomial theorem, complex numbers, algebra of complex numbers, the argand diagram. De-Moiré's theorem, nth roots of unity. Circular measure, trigonometric functions of angles of any magnitude, addition and factor formulae.

PHY 101: General Physics I (Mechanics) (2 Units C: LH=30)

Learning Outcomes

On completion, the students should be able to:

1. identify and deduce the physical quantities and their units;
2. differentiate between vectors and scalars;
3. describe and evaluate motion of systems on the basis of the fundamental laws of mechanics;
4. apply Newton's laws to describe and solve simple problems of motion;
5. evaluate work, energy, velocity, momentum, acceleration, and torque of moving or rotating objects;
6. explain and apply the principles of conservation of energy, linear and angular momentum;
7. describe the laws governing motion under gravity; and
8. explain motion under gravity and quantitatively determine behaviour of objects moving under gravity.

Course Contents

Space and time; units and dimension, vectors and scalars, differentiation of vectors: displacement, velocity and acceleration; kinematics; Newton's laws of motion (inertial frames, impulse, force and action at a distance, momentum conservation); relative motion; application of Newtonian mechanics; equations of motion; conservation principles in physics, conservative forces, conservation of linear momentum, kinetic energy and work, potential energy, system of particles, centre of mass; rotational motion; torque, vector product, moment, rotation of coordinate axes and angular momentum. Polar coordinates; conservation of angular momentum; circular motion; moments of inertia, gyroscopes and precession; gravitation: Newton's law of gravitation, Kepler's laws of planetary motion, gravitational potential energy, escape velocity, satellites motion and orbits.

PHY 107: General Practical Physics I (1 Unit C: PH 45)

Learning Outcomes

On completion, the student should be able to:

1. conduct measurements of some physical quantities;
2. make observations of events, collect and tabulate data;

3. identify and evaluate some common experimental errors;
4. plot and analyse graphs; and
5. draw conclusions from numerical and graphical analysis of data.

Course Contents
This introductory course emphasizes quantitative measurements. Experimental techniques. The treatment of measurement errors. Graphical analysis. The experiments include studies of meters, the oscilloscope, mechanical systems, electrical and mechanical resonant systems, light, heat, viscosity, etc. (covered in PHY 101, 102, 103 and PHY 104). However, emphasis should be placed on the basic physical techniques for observation, measurements, data collection, analysis, and deduction.

EEE 102: Introduction to Electrical and Electronics Engineering (2 units C: LH=15)

Learning Outcomes
Students will be able to:
1. comprehend the duties and functions of an Electrical and Electronics Engineer (EEE);
2. state the requirements for the profession and career opportunities;
3. state the careers related to EEE; and
4. explain the future of EEE.

Course Contents
History of Electrical Engineering. Evolution of EEE. Duties of EE Engineers. Areas of specialisation and work environment. Skill requirements (soft and hard). Qualities for EE Engineers. Careers related to EEE. Typical course modules. Job outlook/opportunities for EE Engineers. Future of EEE. Professional registration (NSE, COREN, IEEE, IET, etc.). Passive components (R, L, C, transformers): descriptive features, including values and colour codes, uses in electrical circuits.DC and AC signal parameters

ABUAD-CHM 101: General Chemistry I (2 Units C: LH=30)

Senate-approved relevance
This course is essential for students at our university as it builds a strong foundation in chemistry, aligning with Sustainable Development Goals (SDGs) 4 (Quality Education) and 9 (Industry, Innovation, and Infrastructure). A deep understanding of chemical principles is crucial for advancing in scientific research, environmental sustainability, and technological innovation. By mastering the content, students will be equipped to contribute to sustainable industrial practices, environmental conservation, and scientific advancements, directly impacting global challenges. This course fosters critical thinking and problem-solving skills, preparing students to be leaders in science, technology, and sustainability.

Overview

This course introduces the fundamental concepts of chemistry, including atoms, molecules, elements, and compounds. It explores modern electronic theory, electronic configuration, and periodicity, and examines molecular structures, chemical bonding, and stoichiometry. Students will also study the kinetic theory of matter, thermochemistry, reaction rates, equilibrium, and thermodynamics. Additionally, the course covers the properties of gases, acids, bases, redox reactions, electrochemistry, and radioactivity, providing a comprehensive understanding of essential chemical principles.

Objectives

6. understand and describe the basic structure of atoms, molecules, elements, and compounds.
7. apply electronic configuration and periodicity to explain chemical properties.
8. analyze and predict the shapes and structures of molecules and solids.
9. solve chemical equations and apply principles of stoichiometry, thermodynamics, and kinetics.
10. understand and apply concepts of redox reactions, electrochemistry, and radioactivity.

Learning Outcomes

At the end of this course, the students should be able to:

1. define atom, molecules and chemical reactions;
2. discuss the modern electronic theory of atoms;
3. write electronic configurations of elements on the periodic table;
4. rationalise the trends of atomic radii, ionisation energies, electronegativity of the elements, based on their position in the periodic table;
5. identify and balance oxidation–reduction equation and solve redox titration problems;
6. draw shapes of simple molecules and hybridised orbitals;
7. identify the characteristics of acids, bases and salts, and solve problems based on their quantitative relationship;
8. apply the principles of equilibrium to aqueous systems using LeChatelier's principle to predict the effect of concentration, pressure and temperature changes on equilibrium mixtures;
9. analyse and perform calculations with the thermodynamic functions, enthalpy, entropy and free energy; and
10. determine rates of reactions and its dependence on concentration, time and temperature.

Course Contents

Atoms, molecules, elements and compounds, and chemical reactions. Modern electronic theory of atoms. Electronic configuration, periodicity and building up of the periodic table. Hybridisation and shapes of simple molecules. Valence forces; Structure of solids. Chemical equations and stoichiometry; chemical bonding and intermolecular forces, kinetic theory of matter. Elementary thermochemistry; rates of reaction, equilibrium and thermodynamics. Acids, bases and salts. Properties of gases. Redox reactions and introduction to electrochemistry. Radioactivity.

ABUAD-CHM 107: General Practical Chemistry I (1 Unit C: PH 45)

Senate-approved relevance

This laboratory course is integral to our university's commitment to Quality Education (SDG 4) and Responsible Consumption and Production (SDG 12). It provides hands-on experience in fundamental chemical techniques, reinforcing theoretical knowledge from CHM 101. By mastering these practical skills, students develop the ability to conduct accurate experiments, analyze data, and present findings effectively—essential competencies for careers in science, industry, and research. The course also promotes safe and sustainable laboratory practices, preparing students to contribute to sustainable industrial processes and environmental stewardship.

Overview

This laboratory course offers practical experience in key chemical techniques, complementing the theoretical content of CHM 101. Students will engage in acid-base titrations, qualitative analysis, redox reactions, and gravimetric analysis. Emphasis is placed on accurate data collection, analysis, and presentation, fostering essential laboratory skills. The course prepares students to apply these techniques in scientific research and industry, ensuring a comprehensive understanding of experimental chemistry.

Objectives

The objectives of the course are to:

6. perform acid-base titrations with precision and accuracy.
7. conduct qualitative and quantitative analysis of chemical substances.
8. understand and execute redox reactions and gravimetric analysis.
9. analyze and interpret experimental data effectively.
10. present laboratory findings clearly and professionally.

Learning Outcomes

At the end of this course, the students should be able to:

1. state the general laboratory rules and safety procedures;
2. collect scientific data and correct carry out chemical experiments;
3. identify the basic glassware and equipment in the laboratory;
4. state the differences between primary and secondary standards;
5. perform redox titration;
6. record observations and measurements in the laboratory notebooks; and
7. analyse the data to arrive at scientific conclusions.

Course Contents

Laboratory experiments designed to reflect topics presented in courses CHM 101 and CHM 102. These include acid-base titrations, qualitative analysis, redox reactions, gravimetric analysis, data analysis and presentation.

GST 112: Nigerian Peoples and Cultures (2 Units C: LH=30)

Learning Outcomes

At the completion of the course, the students are expected to be able:

1. analyse the historical foundation of Nigerian cultures and arts in pre-colonial times;
2. identify and list the major linguistic groups in Nigeria;
3. explain the gradual evolution of Nigeria as a political entity;
4. analyse the concepts of trade and economic self-reliance of Nigerian peoples in relation to national development;
5. enumerate the challenges of the Nigerian state regarding nation building;
6. analyse the role of the Judiciary in upholding fundamental human rights
7. identify the acceptable norms and values of the major ethnic groups in Nigeria; and
8. list possible solutions to identifiable Nigerian environmental, moral and value problems.

Course Contents

Nigerian history, culture and art up to 1800 (Yoruba, Hausa and Igbo peoples and cultures; peoples and cultures of the minority ethnic groups). Nigeria under colonial rule (advent of colonial rule in Nigeria; colonial administration of Nigeria). Evolution of Nigeria as a political unit (amalgamation of Nigeria in 1914; formation of political parties in Nigeria; nationalist movement and struggle for independence). Nigeria and challenges of nation building (military intervention in Nigerian politics; Nigerian Civil War). Concepts of trade and economics of self-reliance (indigenous trade and market system; indigenous apprenticeship system among Nigerian peoples; trade, skill acquisition and self-reliance). Social justice and national development (definition and classification of law); Judiciary and fundamental rights. Individuals, norms and values (basic Nigerian norms and values, patterns of citizenship acquisition; citizenship and civic responsibilities; indigenous languages, usage and development; negative attitudes and conducts [Cultism, kidnapping and other related social vices]). Re-orientation, moral and national values (The 3Rs – Reconstruction, Rehabilitation and Re-orientation; re-orientation strategies: Operation Feed the Nation (OFN), Green Revolution, Austerity Measures, War Against Indiscipline and Corruption (WAIC), Mass Mobilization for Self-Reliance, Social Justice and Economic Recovery (MAMSER), National Orientation Agency (NOA). Current socio-political and cultural developments in Nigeria.

GET 102: Engineering Graphics and Solid Modelling I (2 Units C: LH=15; PH 45)

Learning Outcomes

At the end of this course, the students should be able to:

1. have a good grasp of design thinking and be obsessed with the determination to apply such to solving simple every day and also complex problems;
2. recognize the fundamental concepts of engineering drawing and graphics;
3. show skills to represent the world of engineering objects in actionable solid models, and put such models in a form where they can be inputs for simulation and analyses;
4. analyse such models for strength and cost.
5. prepare the objects for modern production and manufacturing techniques of additive and subtractive manufacturing;
6. recognise that engineering is multidisciplinary in the sense that mechanical, electrical and other parts of physical structures are modelled in context as opposed to the analytical nature of the courses they take;and

7. analyse and master the basics of mechanical and thermal loads in engineering systems.

Course Contents

Introduction to design thinking and engineering graphics. First and third angle orthogonal projections. Isometric projections; sectioning, conventional practices, conic sections and development. Freehand and guided sketching – pictorial and orthographic. Visualisation and solid modelling in design, prototyping and product-making. User interfaces in concrete terms. Design, drawing, animation, rendering and simulation work spaces. Sketching of 3D objects. Viewports and sectioning to shop drawings in orthographic projections and perspectives. Automated viewports. Sheet metal and surface modelling. Material selection and rendering. This course will use latest professional design tools such as fusion 360, solid works, solid edge or equivalent.

MTH 102: Elementary Mathematics II (Calculus) (2 Units C: LH=30)

Learning Outcomes

At the end of the course, students should be able to:

1. identify the types of rules in differentiation and integration;
2. recognise and understand the meaning of function of a real variable, graphs, limits and continuity;
3. solve some applications of definite integrals in areas and volumes;
4. solve function of a real variable, plot relevant graphs, identify limits and idea of continuity;
5. identify the derivative as limit of rate of change;
6. identify techniques of differentiation and perform extreme curve sketching;
7. identify integration as an inverse of differentiation;
8. identify methods of integration and definite integrals; and
9. perform integration application to areas, volumes.

Course Contents

Functions of a real variable, graphs, limits and idea of continuity. The derivative, as limit of rate of change. Techniques of differentiation, maxima and minima. Extreme curve sketching, integration, definite integrals, reduction formulae, application to areas, volumes (including approximate integration: Trapezium and Simpson's rule).

PHY 102: General Physics II (Behaviour of Matter) (2 Units C: LH=30)

Learning Outcomes

On completion, the students should be able to:

1. explain the concepts of heat and temperature and relate the temperature scales;
2. define, derive and apply the fundamental thermodynamic relations to thermal systems;
3. describe and explain the first and second laws of thermodynamics, and the concept of entropy;
4. state the assumptions of the kinetic theory and apply techniques of describing macroscopic behaviour;
5. deduce the formalism of thermodynamics and apply it to simple systems in thermal equilibrium; and

6. describe and determine the effect of forces and deformation of materials and surfaces.

Course Contents
Heat and temperature, temperature scales; gas laws; general gas equation; thermal conductivity; first Law of thermodynamics; heat, work and internal energy, reversibility; thermodynamic processes; adiabatic, isothermal, isobaric; second law of thermodynamics; heat engines and entropy, Zero's law of thermodynamics; kinetic theory of gases; molecular collisions and mean free path; elasticity; Hooke's law, Young's shear and bulk moduli; hydrostatics; pressure, buoyancy, Archimedes' principles; Bernoullis equation and incompressible fluid flow; surface tension; adhesion, cohesion, viscosity, capillarity, drops and bubbles.

PHY 108: General Practical Physics II (1 Unit C: PH 45)
Learning Outcomes
On completion, the student should be able to:
1. conduct measurements of some physical quantities;
2. make observations of events, collect and tabulate data;
3. identify and evaluate some common experimental errors;
4. plot and analyse graphs;
5. draw conclusions from numerical and graphical analysis of data; and
6. prepare and present practical reports.

Course Contents
This practical course is a continuation of PHY 107 and is intended to be taught during the second semester of the 100 level to cover the practical aspect of the theoretical courses that have been covered with emphasis on quantitative measurements, the treatment of measurement errors, and graphical analysis. However, emphasis should be placed on the basic physical techniques for observation, measurements, data collection, analysis and deduction.

ABUAD-CHM 102: General Chemistry II (2 Units C: LH=30)

Senate-approved relevance
This course is crucial to our university's mission of advancing Quality Education (SDG 4) and fostering Innovation (SDG 9). It provides a comprehensive understanding of organic chemistry, which is fundamental to multiple scientific disciplines, including pharmaceuticals, materials science, and environmental science. The course highlights the role of organic chemistry in developing new materials like fullerenes, which are essential for nanotechnology. By mastering the concepts and techniques in organic chemistry, students are prepared to contribute to innovative solutions in health, industry, and sustainability, addressing global challenges through scientific advancement.

Overview

This course offers an in-depth study of organic chemistry, beginning with its historical development and significance. It covers the electronic theory, structure determination, nomenclature, and functional groups of organic compounds. Students will explore reaction mechanisms, kinetics, stereochemistry, and the chemistry of hydrocarbons, alcohols, ethers, amines, and other key organic compounds. Additionally, the course introduces the chemistry of selected metals and non-metals, including comparative chemistry of group IA, IIA, and IVA elements, and transition metal chemistry, with a focus on their practical applications in various industries.

Objectives

6. understand the historical development and significance of organic chemistry.
7. describe the structure, nomenclature, and functional groups of organic compounds.
8. analyze reaction mechanisms, kinetics, and stereochemistry in organic reactions.
9. explore the chemistry of hydrocarbons, alcohols, ethers, amines, and other key organic compounds.
10. examine the chemistry of selected metals, non-metals, and transition metals, including their industrial applications.

Learning Outcomes

At the end of this course, the students should be able to:

1. state the importance and development of organic chemistry;
2. define fullerenes and its applications;
3. discuss electronic theory;
4. determine the qualitative and quantitative of structures in organic chemistry;
5. state rules guiding nomenclature and functional group classes of organic chemistry;
6. determine the rate of reaction to predict mechanisms of reaction;
7. identify classes of organic functional group with brief description of their chemistry;
8. discuss comparative chemistry of group 1A, IIA and IVA elements; and
9. describe basic properties of transition metals.

Course Contents

Historical survey of the development and importance of organic chemistry; fullerenes as fourth allotrope of carbon, uses as nanotubules, nanostructures, nanochemistry. Electronic theory in organic chemistry. Isolation and purification of organic compounds; determination of structures of organic compounds including qualitative and quantitative analysis in organic chemistry; nomenclature and functional group classes of organic compounds. Introductory reaction mechanism and kinetics. Stereochemistry. The chemistry of alkanes, alkenes, alkynes, alcohols, ethers, amines, alkyl halides, nitriles, aldehydes, ketones, carboxylic acids and derivatives. The chemistry of selected metals and non-metals. Comparative chemistry of group IA, IIA and IVA elements. Introduction to transition metal chemistry.

ABUAD- CHM 108: General Practical Chemistry II (1 Unit C: PH 45)

Senate-approved relevance

This practical course is integral to our university's focus on Quality Education (SDG 4) and Responsible Consumption and Production (SDG 12). It builds on the foundation established in CHM 107, enhancing students' proficiency in functional group analysis and quantitative analysis using volumetric methods. These skills are essential for accurate chemical analysis in research and industry, promoting sustainable practices in chemical production and environmental monitoring. By mastering these techniques, students are better prepared to contribute to scientific innovation and responsible resource management, aligning with global sustainability goals.

Overview

This course continues from CHM 107, offering advanced laboratory experiments in functional group analysis and quantitative analysis through volumetric methods. Students will refine their skills in identifying functional groups and conducting precise quantitative measurements. The course emphasizes accurate data collection, analysis, and interpretation, which are critical for success in research and industrial applications. By the end of the course, students will have a deeper understanding of practical chemical analysis and its importance in various scientific fields.

Objectives

The objectives of the course are to:

6. perform advanced functional group analysis accurately.
7. conduct quantitative analysis using volumetric methods with precision.
8. analyze and interpret experimental data effectively.
9. develop proficiency in laboratory techniques for chemical analysis.
10. apply practical skills to real-world chemical problems and research.

Learning Outcomes

At the end of this course, the students should be able to:

1. state the general laboratory rules and safety procedures;
2. collect scientific data and correctly carry out chemical experiments;
3. identify the basic glassware and equipment in the laboratory;
4. identify and carry out preliminary tests which include ignition, boiling point, melting point, test on known and unknown organic compounds;
5. carry out solubility tests on known and unknown organic compounds;
6. carry out elemental tests on known and unknown compounds; and
7. carry out functional group/confirmatory test on known and unknown compounds which could be acidic/basic/ neutral organic compounds.

Course Contents

Continuation of CHM 107. Additional laboratory experiments to include functional group analysis, quantitative analysis using volumetric methods.

200 LEVEL FIRST SEMESTER

ENT 211: Entrepreneurship and Innovation (2 Units C: LH=30)

Learning Outcomes

At the end of this course, students should be able to:

1. explain the concepts and theories of entrepreneurship, intrapreneurship, opportunity seeking, new value creation and risk-taking;
2. state the characteristics of an entrepreneur;
3. analyse the importance of micro and small businesses in wealth creation, employment generation and financial independence;
4. engage in entrepreneurial thinking;
5. identify key elements in innovation;
6. describe the stages in enterprise formation, partnership and networking, including business planning;
7. describe contemporary entrepreneurial issues in Nigeria, Africa and the rest of the world; and
8. state the basic principles of e-commerce.

The concept of entrepreneurship (entrepreneurship, intrapreneurship/corporate entrepreneurship); theories, rationale and relevance of entrepreneurship (Schumpeterian and other perspectives, risk-taking, necessity and opportunity-based entrepreneurship, and creative destruction); characteristics of entrepreneurs (opportunity seeker, risk-taker, natural and nurtured, problem solver and change agent, innovator and creative thinker); entrepreneurial thinking (critical thinking, reflective thinking and creative thinking). Innovation (The concept of innovation, dimensions of innovation, change and innovation, knowledge and innovation). Enterprise formation, partnership and networking (basics of business plan, forms of business ownership, business registration and alliance formation, and joint ventures). Contemporary entrepreneurship issues (knowledge, skills and technology, intellectual property, virtual office and networking). Entrepreneurship in Nigeria (biography of inspirational entrepreneurs, youth and women entrepreneurship, entrepreneurship support institutions, youth enterprise networks and environmental and cultural barriers to entrepreneurship). Basic principles of e-commerce.

GET 201: Applied Electricity I (3 Units C: LH=45)

Learning Outcomes

Students will be able to:
1. discuss the fundamental concepts of electricity and electrical d.c. circuits;
2. state, explain and apply the basic d.c. circuit theorems;
3. explain the basic a.c. circuit theory and
4. apply to solution of simple circuits.

Course contents

Fundamental concepts: Electric fields, charges, magnetic fields. current, B-H curves Kirchhoff's laws, superposition. Thevenin Norton theorems, Reciprocity, RL, RC, RLC circuits. DC, AC bridges, Resistance, Capacitance, Inductance measurement, Transducers, Single phase circuits, Complex j - notation, AC circuits, impedance, admittance, and susceptance.

GET 203: Engineering Graphics and Solid Modeling II (3 Units E: LH=30; PH=45)

Learning Outcomes

Students should be able to:
1. apply mastery of the use of projections to prepare detailed working drawing of objects and designs;
2. develop skills in parametric design to aid their ability to see design in the optimal specification of materials and systems to meet needs;
3. be able to analyze and optimize designs on the basis of strength and material minimization;
4. get their appetites wet in seeing the need for the theoretical perspectives that create the basis for the analysis that are possible in design and optimization, and recognize/understand the practical link to excite their creativity and ability to innovate; and
5. be able to translate their thoughts and excitements to produce shop drawings for multi physical, multidisciplinary design.

Course Contents

Projection of lines, auxiliary views and mixed projection. Preparation of detailed working production drawing; semi-detailed drawings, conventional presentation methods. Solid, surface and shell modeling. Faces, bodies and surface intersections. Component-based design. Component assembly and motion constraints. Constrained motions and animation. Introduction to electronics modeling. Electronics board layout preparation, Component libraries and Schematic design. Parametric modeling and adaptive design. Simulation for material optimization. Designing for manufacturing. Additive and subtractive manufacturing. Production for 3-D printing, Laser cutting and CNC machinery. Arrangement of engineering components to form a working plant (Assembly Drawing of a Plant).

GET 205: Fundamentals of Fluid Mechanics (3 Units C: LH=45)

Learning Outcomes

At the end of this course, the students should be able to:

1. explain the properties of fluids;
2. determine forces in static fluids and fluids in motion;
3. determine whether a floating body will be stable;
4. determine the effect of various instruments, (valves, orifices, bends and elbows) on fluid f low in pipes;
5. measure flow parameters with venturi meters, orifice meters, weirs and others;
6. perform calculations based on principles of mass, momentum and energy conservation;
7. perform dimensional analysis and simple fluid modelling problems; and
8. specify the type and capacity of pumps and turbines for engineering applications.

Course Contents

Fluid properties, hydrostatics, fluid dynamics using principles of mass, momentum and energy conservation from a control volume approach. Flow measurements in pipes, dimensional analysis, and similitude, 2-dimensional flows. Hydropower systems.

GET 207: Applied Mechanics (3 Units C: LH=45)

Learning Outcomes

Students will acquire the ability to:

1. explain the fundamental principles of applied mechanics, particularly equilibrium analysis, friction, kinematics and momentum;
2. identify, formulate, and solve complex engineering problems by applying principles of engineering, science, mathematics and applied mechanics;
3. synthesize Newtonian Physics with static analysis to determine the complete load impact (net forces, shears, torques, and bending moments) on all components (members and joints) of a given structure with a load; and
4. apply engineering design principles to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

Course Contents

Forces, moments, couples. Equilibrium of simple structures and machine parts. Friction. First and second moments of area; centroids. Kinematics of particles and rigid bodies in plane motion. Newton's laws of motion. Kinetic energy and momentum analyse

GET 209: Engineering Mathematics I (3 Units C: LH=45)

Learning Outcomes

At the end of the course, the students should be able to:

1. solve qualitative problems based on vector and matrix analyses such as linear independence and dependence of vectors, rank etc;
2. describe the concepts of limit theory and nth order differential equations and their applications to physical phenomena;
3. solve the problems of differentiation of functions of two variables and know about the maximization and minimization of functions of several variables;
4. describe the applications of double and triple integration in finding the area and volume of engineering solids, and explain the qualitative applications of Gauss, Stoke's and Green's theorem;
5. explain ordinary differential equations and applications, and develop a mathematical model of linear differential equations, as well as appreciate the necessary and sufficient conditions for total differential equations; and
6. analyse basic engineering models through partial differential equations such as wave equation, heat conduction equation, etc., as well as fourier series, initial conditions and its applications to different engineering processes.

Course Contents

Limits, continuity, differentiation, introduction to linear first order differential equations, partial and total derivatives, composite functions, matrices and determinants, vector algebra, vector calculus, directional derivatives.

GET 211: Computing and Software Engineering (3 Units C: LH=30; PH=45)

Learning Outcomes

At the end of the course, the students should be able to:

1. describe and apply computing, software engineering knowledge, best practices, and standards appropriate for complex engineering software systems;
2. develop competence in designing, evaluating, and adapting software processes and software development tools to meet the needs of an advanced development project through practical object-oriented programming exposure taught in concrete terms with a specific modern language – preferable selected from Python, Java or C++;
3. use widely available libraries to prepare them for machine learning, graphics and design simulations;
4. develop skills in eliciting user needs and designing an effective software solution;
5. recognise human, security, social, and entrepreneurial issues and responsibilities relevant to engineering software and the digitalisation of services; and

6. acquire capabilities that can further be developed to make them productively employable by means of short Internet courses in specific areas.

Course Contents

Introduction to computers and computing; computer organisation – data processing, memory, registers and addressing schemes; Boolean algebra; floating-point arithmetic; representation of non-numeric information; problem-solving and algorithm development; coding (solution design using flowcharts and pseudo codes). Data models and data structures; computer software and operating system; computer operators and operators' precedence; components of computer programs; introduction to object oriented, structured and visual programming; use of MATLAB in engineering applications. ICT fundamentals, Internet of Things (IoT). Elements of software engineering.

AFE 201: Fitness for Life (1, 0, 0) 1 Unit
AFE 203: Language Acquisition II (1, 0, 0) 1 Unit
AFE 205: Basic Skill Acquisition (1, 0, 0) 1 Unit

200 LEVEL SECOND SEMESTER

GST 212: Philosophy, Logic and Human Existence (2 Units C: LH=30)

Learning Outcomes

At the end of the course, students should be able to:

1. know the basic features of philosophy as an academic discipline;
2. identify the main branches of philosophy & the centrality of logic in philosophical discourse;
3. know the elementary rules of reasoning;
4. distinguish between valid and invalid arguments;
5. think critically and assess arguments in texts, conversations and day-to-day discussions; 6. critically asses the rationality or otherwise of human conduct under different existential conditions;
7. develop the capacity to extrapolate and deploy expertise in logic to other areas of knowledge, and 8. guide his or her actions, using the knowledge and expertise acquired in philosophy and logic.

Course Content

Scope of philosophy; notions, meanings, branches and problems of philosophy. Logic as an indispensable tool of philosophy. Elements of syllogism, symbolic logic— the first nine rules of inference. Informal fallacies, laws of thought, nature of arguments. Valid and invalid arguments, logic of form and logic of content — deduction, induction and inferences. Creative and critical thinking. Impact of philosophy on human existence. Philosophy and politics, philosophy and

human conduct, philosophy and religion, philosophy and human values, philosophy and character molding, etc.

GET 204: Students Workshop Practice (2 Units C: LH=15; PH=45)

Learning Outcomes

At the end of this course, the students should be able to:

1. identify various basic hands and machine tools, analogue and digital measurement devices and instruments, and acquire skills in their effective use and maintenance;
2. practically apply basic engineering technologies, including metrology, casting, metal forming and joining, materials removal, machine tooling (classification, cutting tool action, cutting forces, non-cutting production) and CNC machining technology;
3. master workshop and industrial safety practices, accident prevention and ergonomics;
4. physically recognize different electrical & electronic components like resistances, inductances, capacitances, diodes, transistors and their ratings;
5. connect electric circuits, understand different wiring schemes, and check ratings of common household electrical appliances and their basic maintenance; and
6. determine household and industrial energy consumption, and understand practical energy conservation measures.

Course Contents

The course comprises general, mechanical and electrical components: supervised hands-on experience in safe usage of tools and machines for selected tasks; Use of measuring instruments (calipers, micrometers, gauges, sine bar, wood planners, saws, sanders, and pattern making). Machine shop: lathe work shaping, milling, grinding, reaming, metal spinning. Hand tools, gas and arc welding, cutting, brazing and soldering. Foundry practice. Industrial safety and accident prevention, ergonomics, metrology. Casting processes. Metal forming processes: hot-working and cold-working processes (forging, press-tool work, spinning, etc.). Metal joining processes (welding, brazing and soldering). Heat treatment. Material removal processes. machine tools and classification. Simple theory of metal cutting. Tool action and cutting forces. Introduction to CNC machines. Supervised identification, use and care of various electrical and electronic components such as resistors, inductors, capacitors, diodes and transistors. Exposure to different electric circuits, wiring schemes, analogue and digital electrical and electronic measurements. Household and industrial energy consumption measurements. Practical energy conservation principles.

GET 206: Fundamentals of Engineering Thermodynamics (3 Units C: LH=45)

Learning Outcomes

At the end of this course, the students should be able to:

1. describe basic concepts of thermodynamics, quantitative relations of Zeroth, first, second and third laws;
2. define and explain system (surrounding, closed and open system), control volume and control mass, extensive and intensive properties;
3. calculate absolute and gage pressure, and absolute temperature, calculate changes in kinetic, potential, enthalpy and internal energy;
4. evaluate the properties of pure substances i.e. evaluate the state of the pure substances such as compressed liquid, saturated liquid-vapour mixture and superheated vapour using property diagrams and tables; arrange the ideal and real gas equations of state,
5. formulate the first law of thermodynamics for a closed system i.e. organize the change in energy in the closed systems via heat and work transfer;
6. distinguish heat transfer by conduction, convection and radiation, and calculate the amount of heat energy transferred;
7. calculate the changes in moving boundary work, spring work, electrical work and shaft work in closed systems;
8. apply the first law of thermodynamics for closed systems and construct conservation of mass and energy equations;
9. formulate the first law of thermodynamics to the open systems i.e. describe steady-flow open system, apply the first law of thermodynamics to the nozzles, diffusers, turbines, compressors, throttling valves, mixing chambers, heat exchangers, pipe and duct flow;
10. construct energy and mass balance for unsteady-flow processes;
11. evaluate thermodynamic applications using second law of thermodynamics;
12. calculate thermal efficiency and coefficient of performance for heat engine, refrigerators and heat pumps; and
13. restate perpetual-motion machines, reversible and irreversible processes.

Course Contents

Basic concepts, definitions and laws (quantitative relations of Zeroth, first, second and third laws of thermodynamics). Properties of pure substances: the two-property rule (P-V-T behaviour of pure substances and perfect gases); state diagrams. The principle of corresponding state; compressibility relations; reduced pressure; reduced volume; temperature; pseudo-critical constants. The ideal gas: specific heat, polytropic processes. Ideal gas cycles; Carnot; thermodynamic cycles, turbines, steam and gas, refrigeration. The first law of thermodynamics – heat and work, applications to open and closed systems. The steady f low energy equation (Bernoulli's equation) and application. Second law of thermodynamics, heat cycles and efficiencies.

GET 210: Engineering Mathematics II (3 Units C: LH=45)

Learning Outcomes

At the end of the course, the students should be able to:

1. describe physical systems using ordinary differential equations (ODEs);
2. explain the practical importance of solving ODEs, solution methods, and analytically solve a wide range of ODEs, including linear constant coefficient types;
3. numerically solve differential equations using MATLAB and other emerging applications; 4. perform calculus operations on vector-valued functions, including derivatives, integrals, curvature, displacement, velocity, acceleration, and torsion, as well as on functions of several variables, including directional derivatives and multiple integrals;
5. solve problems using the fundamental theorem of line integrals, Green's theorem, the divergence theorem, and Stokes' theorem, and perform operations with complex numbers; 6. apply the concept and consequences of analyticity and the Cauchy-Riemann equations and of results on harmonic and entire functions of complex variables, as well as the theory of conformal mapping to solve problems from various fields of engineering; and
7. evaluate complex contour integrals directly and by the fundamental theorem, apply the Cauchy integral theorem in its various versions, and the Cauchy integral formula.

Course Contents

Introduction to ordinary differential equations (ODEs); theory, applications, methods of solution; second order differential equations. Advanced topics in calculus (vectors and vector valued function, line integral, multiple integral and their applications). Elementary complex analysis including functions of complex variables, limits and continuity. Derivatives, differentiation rules and differentiation of integrals. Cauchy-Riemann equation, harmonic functions, basic theory of conformal mapping, transformation and mapping and its applications to engineering problems. Special functions.

EEE 202: Applied Electricity II (3 Units C: LH=45)

Learning outcomes

Students will be able to:

1. differentiate between various d.c. and a.c. machines;
2. explain the principles of operation of machines;
3. explain the operation of basic semiconductor devices and their basic applications; and
4. explain the principle of operation of communication systems with examples.

Course contents

Basic machines – DC, synchronous alternators, transformers, equivalent circuits. Three- phase balanced circuits, PN junction diode, BJTs, FETs, thyristors, communications fundamentals, introduction of TV, Radio, Telephone systems.

EEE 204 Electrical Engineering Materials (3 Units E: LH=45)

Learning Outcomes

Students will be able to:

1. discuss electron conduction mechanisms in semiconductors;
2. explain transport phenomena in semiconductors; and
3. describe semiconductors device fabrication techniques.

Course contents

Free electron motion in static electric and magnetic fields, electronic structure of matter, conductivity in crystalline solids. Theory of energy bands in conductors, insulators and semiconductors: electrons in metals and electron emissions; carriers and transport phenomena in semiconductors, characteristics of some electron and resistors, diodes, transistors, photo cell and light emitting diode. Elementary discrete devices fabrication techniques and IC technology.

ABUAD-GET 208: Strength of Materials (3 Units C: LH=45)

Senate-approved relevance

This course is integral to our university's commitment to Quality Education (SDG 4) and Industry, Innovation, and Infrastructure (SDG 9). It provides a deep understanding of the mechanical behavior of materials under various loading conditions, which is essential for designing safe and efficient structures. Mastery of these concepts is critical for students pursuing careers in civil, mechanical, and aerospace engineering. The course equips students with the skills needed to analyze and design structural components, contributing to innovative and sustainable infrastructure development, essential for addressing global engineering challenges.

Overview

This course explores the principles of equilibrium, stress-strain relationships, and the application of Hooke's law to various materials. It covers the effects of loading and temperature changes on stresses and strains, torsion in circular members, and shear forces in beams. The course also examines bending moments, bending stresses in beams with symmetrical and combined loadings, and stress analysis. Students will gain the knowledge necessary to analyze and design structural components under different conditions.

Objectives

The objectives of the course are to:

1. understand the principles of equilibrium in structural systems.
2. analyze stress-strain relationships in materials using generalized Hooke's law.
3. evaluate stresses and strains due to different loading conditions and temperature changes.

4. understand and analyze torsion in circular members.
5. calculate shear forces and bending moments in beams under various loadings.
6. analyze bending stresses in beams with symmetrical and combined loadings.
7. apply stress analysis techniques to structural components.
8. design safe and efficient structural components considering various mechanical factors.

Learning Outcomes

At the end of this course, the students should be able to:
1. recognize a structural system that is stable and in equilibrium;
2. determine the stress-strain relation for single and composite members based on Hooke's law;
3. estimate the stresses and strains in single and composite members due to temperature changes;
4. evaluate the distribution of shear forces and bending moments in beams with distributed and concentrated loads;
5. determine bending stresses and their use in identifying slopes and deflections in beams;
6. use Mohr's circle to evaluate the normal and shear stresses in a multi-dimensional stress system and transformation of these stresses into strains;
7. evaluate the stresses and strains due to torsion on circular members; and
8. determine the buckling loads of columns under various fixity conditions at the ends.

Course Contents

Consideration of equilibrium; composite members, stress-strain relation. Generalized Hooke's law. Stresses and strains due to loading and temperature changes. Torsion of circular members. Shear force, bending moments and bending stresses in beams with symmetrical and combined loadings. Stress

Minimum Academic Standards

As stated in the CCMAS.

GET 299: Students Industrial Work Experience I (SIWES I) (3 Units C: 9 weeks)

Learning Outcomes

SIWES I should provide opportunity for the students to:
1. acquire industrial workplace perceptions, ethics, health and safety consciousness, inter personal skills and technical capabilities needed to give them a sound engineering foundation;
2. learn and practise basic engineering techniques and processes applicable to their specialisations; 3. build machines, devices, structures or facilities relevant to their specific engineering programmes and applications; and
4. acquire competence in technical documentation (log-book) and presentation (report) of their practical experiences.

Course Contents

Practical experience in a workshop or industrial production facility, construction site or special centres in the university environment, considered suitable for relevant practical/industrial working experience but not necessarily limited to the student's major. The students are exposed to hands-on activities on workshop safety and ethics, maintenance of tools, equipment and machines, welding, fabrication and foundry equipment, production of simple devices; electrical circuits, wiring and installation, etc. (8-10 weeks during the long vacation following 200 level).

300 LEVEL FIRST SEMESTER

EEE 313: Electronics I (Physical) (2, 1, 0) 3 UNITS

Free electron motion in static electric and magnetic fields, electronic structure of matter, conductivity in crystalline solids. Theory of energy hands in conductors, insulators and semi-conductors: electrons in metals and electron emissions; carriers and transport phenomena in semi-conductors, characteristics of some electron and resistors, diodes, transistors, photo cell and light emitting diode. Elementary discrete devices fabrication techniques and IC technology.

EEE 315: Electromagnetic Field and Wave Theory I (2, 1, 0) 3 UNITS

Review of electromagnetic laws, Gauss' law, Ampere's and Faraday's laws, Electric potential, Electrostatic fields due to distribution of charges, Energy stored in Electric field, magnetic fields in and around current carrying conductors of different shapes, magnetic vector potential. Time varying magnetic and electric fields, conduction and displacement current in magnetic vector potential.

EEE 317: Electrical/Electronics Laboratory 1 (0, 0, 3) 1 UNIT

Laboratory work involves soldering techniques, measuring instrument, identification of semiconductors, application of circuit and transformer testing. A Laboratory work designed to illustrate topics covered in measurement and instrumentation, electric circuits theory transformers and DC machines.

EEE 319: Measurements and Instrumentation (2, 1, 0) 3 UNITS

Units. Theory of Errors. Direct and Indirect Measurements. General Instrumentation, Moving Coil Instruments (PMMC). Moving Iron Instruments. Basic Meter in DC measurement. Basic meter in AC measurements; rectifier voltmeter, electro-dynamometer and Wattmeter, instrument transformers; DC and AC bridges and their applications; general form of AC bridge universal impedance bridge; Electronic instruments for the measurement of voltage, current resistance and other circuit parameter, electronic voltmeters, AC voltmeters using rectifiers, electronic multimeter, digital volumeters; oscilloscope: vertical deflection system, horizontal deflection system, probes, sampling CRO, Instruments for generating and analyzing waveforms; square-wave and pulse generator, signal generators, function generators, wave analysers, Electronic counters and their applications: time base circuitry, universal counter measurement modes;

Analog and digital data acquisition systems: tape recorders, D/A and A/D conversions, sample and hold circuits.

EEE 321: Electric Circuit Theory I (2, 1, 0) 3 UNITS
Circuit elements, circuit theorems, applications. Network response to steps, ramp, impulses. Network functions, response to exponential, sinusoidal sources. Laplace transform, polezero analysis, network synthesis, resonance, two-point analysis, ladder network. Star-Delta transformation, T, P Networks.

EEE 325: Electrical Machines I (2, 1, 0) 3 UNITS
Basic principles of relays and actuators. Transformation of Electric energy. Transformer performance; equivalent circuits, efficiency, regulation, per unit values. Types of Transformers: Auto transformer, Instrument Transformer, Elements of Transformer design. Transformer in Polyphase circuits. Parallel operation of transformers. Basic principles of electromechanical energy conversion. Direct current machines: armature windings, internal torque, and methods of excitation. Armature reaction. Characteristics of D.C. generators and Motors. Basic principles of selection of motors and generators for practical application. Speed control and electric braking. Cross-field machines. Commutator machines.

EEE 341: Software Development Techniques (2, 1, 0) 3 UNITS
Software development life cycle. Top-Down design. Program, design using pseudo-code, flowchart. Flowchart ANSI symbols and usage. Extensive examples, and exercises using pseudo-code/flowchart to solve practical problems in engineering. Debugging and documentation techniques. Programming using a structural language such as C: Symbols, keywords, identifiers, data types, operators, various statements, operator precedence, type conversion, conditional and control structures, function, recursive functions. Arrays: 1-D, and multi-dimensional arrays, passing elements or whole array to a function. Simple sorting and searching on arrays, pointers, strings, dynamic memory allocation. Structures and Unions: Structure declaration and definition, accessing structures, array of structures, pointers and structures, union declaration, enumerated variables. File Handling: Concept of a file, files and streams, standard file handling functions, binary files, random access files. Advanced Topics: Command line parameters, pointers to functions, creation of header files, stacks, linked lists, bitwise manipulation. Software development in C in MS Windows, UNIX/LINUX environments, header file, preprocessor directives, make, makefile. Static and dynamic linking libraries. Extensive examples, and exercises programming in C to solve practical problems in engineering. Exercises are to be done in the Computer Laboratory.

ENG 381: Engineering Mathematics III (2, 1, 0) 3 UNITS
Linear algebra. Elements of matrices, determinants, inverses of matrices, theory of linear equations, eigen values and eigen vectors. Analytical geometry, coordinate transformation, solid geometry, polar, cylindrical and spherical coordinates. Elements of functions of several variables,

surface variables. Ordinary integrals, evaluation of double integrals, triple integrals, line integrals and surface integrals. Derivation and integrals of vectors, the gradient of scalar quantities. Flux of vectors, the curl of a vector field, gauss, greens and stoke's theorems and applications. Singular valued functions. Multivalued functions, analytical functions, cauchy riemann's equations. Singularities and zeroes, contour integration including the use of cauchy's integral theorems, bilinear transformation. Applications of microsoft excel, matlab and/or mathcad.

AFE 301 Practical Entrepreneurship Skills (1, 0, 3) 1 Unit

Profiles of business ventures in the various business sectors such as:

Soap/detergent, tooth brush and tooth paste making; photography; brick making; rope making; brewing; glassware production/ ceramic production, paper production; water treatment/conditioning/packaging; food processing/preservation/packaging; metal fabrication; tanning industry; vegetable oil extraction; farming; fisheries/aquaculture; plastic making; refrigeration/air-conditioning; carving, weaving; bakery; tailoring; printing; carpentry; interior decoration; animal husbandry etc. Case study methodology applied to the development and administration of cases that bring out key issues of business environment, start-up, pains and gains of growth of businesses, etc. With particular reference to nigerian businesses. Experience sharing by business actors in the economy with students during case presentations.

300 LEVEL SECOND SEMESTER COURSES

EEE 314: Electronics II (Analogue) (2, 1, 0) 3 UNITS

Broadband and narrow band amplifiers, power amplifiers, voltage and current stabilizing circuits. Sinusoidal RC and RLC and crystal oscillators. Switching properties of electronic devices. Switching and wave-shaping circuits. Generation of non-sinusoidal waveforms, astable, monostable and bistable multi-vibrators, comparator, Schmitt-trigger and time base generators using discrete transistor, operational amplifier fundamentals, analyses and applications, other analog integrated circuits-Time chips and their applications. Analysis and design of logic gates of various families. Diode logic, RTL, TTL, RCL, MOS and MOS of digital integrated circuits. Interfacing between various logic families. Concepts of small, medium, large and very large integration and their consequences, some digital building blocks, flip-flops, counters, register and decoders. Introduction to D/A and A/D conversion principles.

EEE 316: Electromagnetic Field and Wave Theory II (2, 1, 0) 3 UNITS

Maxwell's equations (in regular coordinates and vector-calculus notation); Derivation of Maxwell's equation, electromagnetic potential and waves, pointing vector, Boundary conditions, wave propagation in good conductors, skin effect, plane waves across boundaries of different media, propagation of electromagnetic waves in ionized media, fundamentals of transmission lines, matching, voltage reflection coefficient standing wave ratio waveguides and antennae. Applications of wave theory in communication.

EEE 318: Electrical/Electronic Laboratory II (0, 0, 3) 1 UNIT

Laboratory work involves winding of Transformer and Protective devices. Confirmation of semiconductor-Op Amp, Timing circuit. A Laboratory work designed to illustrate topics covered in electric circuits theory electronic engineering and AC machines.

EEE 322: Electric Circuit Theory II (2, 1, 0) 3 UNITS

Networks; Node, loop analysis, Realization of transfer functions, Filters: design, operation, low, high, Band pass. Butterworth, Chebychev, filters, Non-linear circuit analysis Locus diagrams. Network functions. Active network synthesis and analysis: Foster and Cauers

EEE 324: Linear Systems (2, 1, 0) 3 UNITS

Mathematical models of physical system. Analogues concepts in electrical mechanical and thermal systems. Transfer functions. Block diagrams and signal flow graphs. Feedback control system; advantages, transient and steady-state response of systems. Pole stability. The root-locus methods. Frequency response of systems; Bode and polar plots system stability. Routh and Nyquist criteria. Introduction to analogue computer simulation.

EEE 326: Electrical Machines II (2, 1, 0) 3 UNITS

Magnetic circuits. Basic principles of relays and activators; Ideal transformer. Equivalent circuits and basic analysis of practical transformers. D.C. machine contraction, characteristics of D.C. generators. Excitation of D.C. machines. Torque-speed characteristics of D.C. motors. A.C. machines; production of rotating magnetic fields. Simple theory of three-phase induction motors; torque speed characteristics, three-phase induction motors Single-phase motors-applications. Selection of motors, for practical applications. Synchronous machines.

EEE 328: Communication Principles (2, 1, 0) 3 UNITS

Introduction to communication systems; General features of point-to-point communication systems over the entire frequency, characteristics of transmitters, receivers and antennae. Radio and TV broadcasting. Telephony. Facsimile. Radar Telemetry. Analogue modulation systems. Amplitude modulation and demodulation methods, DSB, DSBSC, SSB, and VSB. Comparison of AM systems. Angle modulation and demodulation. Frequency and Phase modulation. Wide band and narrow band and FM.

Study of communication systems design and analysis. Topics include signals and spectra, error-control coding, synchronization, multiple-access techniques, spread spectrum signaling, fading channels baseband signaling and detection in noise, digital and analog modulation and demodulation techniques, and communications link budget analysis.

Transmission media; attenuation in open space, air, cable and fibre channels; construction of cables and fibres, sampling theorem, pulse amplitude modulation, pulse width modulation, multiplexing, quantization systems and pulse code modulation, delta modulation, courses and correction of errors in PCM and DM.

ENG 382 Engineering Mathematics IV (2, 1, 0) 3 UNITS

Series solution of second order linear differential equations with variable coefficients. Bessel and legendre equations. Equations with variable coefficients. Sturn-louville boundary value problems. Solutions of equations in two and three dimensions by separation of variables. Eigen

value problems. Use of operations in the solution of partial differential equations and linear integral equations. Integral transforms and their inverse including fourier, laplace, mellin and handel transforms. Convolution integrals and hilbert transforms. Calculus of finite differences. Interpolation formulae. Finite difference equations. Runge-kutta and other methods in the solutions of ode and pdes. Numerical integration and differentiation. Applications of microsoft excel, matlab and/or mathcad.

ENG 384 Engineering Economics (2, 0, 0) 2 UNITS

Economic analysis of engineering projects; value systems economic decisions on capital investments and choice of engineering alternatives; new projects, replacement and abandonment policies, risky decisions; corporate financial practices.

Objectives of valuation work/valuer's primary duty and responsibility. Valuer's obligation to his or her client, to other valuers, and to the society. Valuation methods and practices. Valuation reports. Expert witnessing. Ethics in valuation. Valuation i appraisal standards. Price, cost and value. Depreciation and obsolescence. Valuation terminology. Appraisal reporting and review. Real property valuation. Personal property valuation. Machinery and equipment valuation. Oil and gas valuation. Mines and quarries valuation. Cost and schedule management- an engineering function. Supporting skills and knowledge. Role of cost engineer during evaluation phase. Role of cost engineer during the basic design phase. Role of cost engineer in contractor selection. Role of cost engineer during detailed engineering design phase. Role of cost engineer during construction. Cost engineering function as distinct from design engineering function. Canon of ethics for cost engineers. Basic capital cost estimating. Basic operating cost estimating. Basic project planning and scheduling. Cost engineering terminology. Cost engineering standards.

400 LEVEL FIRST SEMESTER COURSES

EEE 405: Numerical Computations (2, 1, 0) 3 Units

Computer as a number crunching tool. Floating-point number representation and arithmetic error, stability, convergence theory of computational solution to problem: numerical algorithm formulation and design, numerical software systems. Iterative solutions of non-linear systems: (Newton's Method). Numerical solutions of linear systems. Numerical computation of Eigen values and Eigen vectors. Curve fitting; function approximation. Numerical differentiation and integration (Simpson's Rule etc). Explicit and implicit methods. Differential equation (Euler's method etc). The Eigen value problem and solution of ordinary differential equations-methods (ODE) of Taylor, Euler predictor-corrector and Runge-kuta. Numerical integration and differentiation. Linear algebra. Finite differences. Introduction to the use of Octave, Matlab or LabView in numerical computation and engineering applications. Emphasis is on the use of software to solve real problems.

EEE 415: Assembly Language Programming (2, 0, 0) 2 Units

Introduction: Language level of abstraction and effect on machine, characteristics of machine code, advantages, justifications of machine code programming, instruction set and dependency

on underlying processor. Intel 8086 microprocessor assembly language programming: Programming model as resources available to programmer, addressing modes, instruction format, instruction set- arithmetic, logical, string, branching, program control, machine control, input/output , etc; assembler directives, hand-assembling, additional 80x86/Pentium instructions. Modular programming. Interrupt and service routine. Interfacing of assembly language to C . Intel 80x87 floating point programming. Introduction to MMX and SSE programming. Motorola 680x0 assembly language programming. Extensive practical engineering problems solving in assembly language using MASM for Intel, and cross-assembler for Motorola.

EEE 423: Electrical/Electronic Laboratory III (0, 0, 3) 1 Unit
Laboratory work involves winding techniques. Computer hardware and software. Maintenance and repairs, communication. A laboratory work set up to illustrate topics covered in control theory, communication principles, electronic engineering, computer techniques and electric power principles.

EEE 427: Data Communication and Network (2, 1, 0) 3 Units
Introduction to Data communications: the Development of Data Communications; types and sources of data, simple communications network, transmission definitions, one way transmission, half duplex transmission, transmission codes, transmission modes, parallel transmission, serial transmission, bit synchronization, character synchronization, character synchronization, synchronous transmission, asynchronous transmission, efficiency of transmission, error detection methods and data compression. Protocols: Introduction to network protocol. Seven Layer ISO-OSI standard protocols and network architecture. Transport protocols, session services protocols, and other protocols. Institute of Electrical and Electronics Engineering 802 standards. Error control and Data Compression: Forward Error Control; error detection methods; parity checking; linear block codes, cyclic redundancy checking; feedback error control, data compression, Huffman coding and dynamic Huffman coding. Local Area Networks: medium access control techniques – Ethernet, token bus and token ring; LAN standards; fibre distributed data interface, metropolitan area network. Peer-to-peer, Client Server. Client-Server Requirements: GUI design standards, interface independence, platform independence, transaction processing, and connectivity, reliability, backup and recovery mechanisms. Information Network Software; Features and benefits of major recovery mechanisms. Information Network Software: features and benefits of major Network Operating Systems. Network OS: (e.g. Novell NetWare, UNIX/LINUX, OS/2 & Windows NT). TCP/IP and Network OS. INTERNET: Definition, architecture, services, Internet addressing. Internet protocol, IPv4, IPv6. Internet programming, Intranet. System administration, and security issues.

EEE 431: Electric Power Principles (2, 1, 0) 3 Units
Types of power station, operation, auxilliaries, economics of operation - stations, substations power supply economics, tariffs. Power factor correction. Polyphase theory. DC, AC power distribution, network calculations, Overhead line conductors. Corona effect, voltage control, circuit breakers, load forecast, siting of generating plants.

EEE 441: Control Theory and Systems (2, 1, 0) 3 Units

Review of feedback control systems. Transfer functions and System modelling (electrical, hydraulic, pneumatic and thermal systems). Review of frequency-response analysis (Polar, Bode, Nyquist, Nichols). Review of pole stability and Routh-Hurwitz's stability criterion. System classification and time response analysis. Nyquist stability criterion, M- and N- circles, inverse Nyquist Plots. Bode diagrams and determination of transfer function from asymptotic plots. Relative stability. Introduction to Feedback compensation. The tracking and regulation objectives. Compensation techniques and controller design: Lag, Lead and Lead-Lag compensation; Three-term (PID) controllers (P, PD, PI and PID); Series and parallel compensation design; controller design using Bode, Nichols and Root locus methods. Servomechanical systems. Servo-motors, tachogenerators, error detectors amplifiers, actuators, valves, etc. and their transfer functions. Servomechanism position and velocity control systems. Voltage regulators. Introduction to nonlinear systems.

EEE 471: Digital Electronics and Systems (2, 1, 0) 3 Units

Electronic instruments for the measurement of voltage, current resistance and other circuit parameter, electronic voltmeters, AC voltmeters using rectifiers, electronic multimeter, digital volumeters; oscilloscope: vertical deflection system, horizontal deflection system, probes, sampling CRO, Instruments for generating and analyzing waveforms; square-wave and pulse generator, signal generators, function generators, wave analysers, Electronic counters and their applications: time base circuitry, universal counter measurement modes; Analog and digital data acquisition systems: tape recorders, D/A and A/D conversions, sample and hold circuits. Digital Test Instruments.

Number Systems and Codes. Logic Gate Simplification of Logic expressions using Boolean Algebra. Simplification of Logic expressions using Karnaugh Method. Design combinational circuit. Flip-Flops. Application of Flip-Flops in the design of counters, registers and timers. Switching and Waves shipping circuit. Generation of non-sinusoidal signal (multi vibrators). Introduction to ADC and DAC. Design of Logic Gates (Diode, DTL, TTL, ECL, CMOS families etc). Memory devices. Latches. Decoders and encoders. Multiplexer circuits. Computer-aided design of Modular Systems. Synchronous and asynchronous operations. Analysis and synthesis of sequential switching circuits, Registers.

Design of combinatorial circuits with emphasis on application. Design of sequential circuits, synchronous and asynchronous. Design of simple CPU, peripheral devices (I/O), memory/storage devices. Stored-program concept.

Introduction to programming of microcontrollers

400 LEVEL SECOND SEMESTER

ENG 490: Students Industrial Work Experience Scheme (6 Units)

A comprehensive internship programme in which students spend a full semester in approved electrical engineering establishments (private and public) and industries. The exposure also provide opportunity for students to sharpen their technical writing skill through field reports,

keeping log-books and prepared technical documents under close supervision of competent lecturers. This will be registered alongside ENG 290 (SWEP I) and ENG 390 (SWEP II).

ENG 290: Students Work Experience Programme (SWEP) I (3 UNITS)
Practices and skills in general engineering through instruction in operation of hand and powered tools for wood and metal cutting and fabrication. Supervised hand on experience in safe usage of tools and machine for selected tasks. General practices on automobile repairs, survey, civil, electrical and other related engineering practice. It is regarded as on the job experience in industry chosen for practical working experience but not necessarily limited to the student's major (12 weeks during the long vacation following 200 level).

ENG 390: Students Work Experience Programme (SWEP) II (3 UNITS)
On the job experience in industry chosen for practical working experience but not necessarily limited to the student's major (12 weeks during the long vacation following 300 level).

Eng 490 Students Industrial Work Experience Scheme (Siwes) (6 Units)
On the job experience in industry chosen for practical working experience but not necessarily limited to the student's major (24 weeks from the end of the first semester at 400-level to the beginning of the first semester of the following session. Thus, the second semester at 400-level is spent in industry.)

500 LEVEL FIRST SEMESTER COURSES

EEE 503: Modelling and Simulation of Electrical and Electronics Systems (1, 1, 0) 2 Units
Basic introduction to modelling and simulation. Introduction to MATLAB and AUTOCAD, and their engineering applications. Simulation of modelled systems using MATLAB and others. Application programs in computer aided design (CAD) of Electrical systems. CAD/CAM systems: Introduction: Basic concepts and terminology. Introduction to computer graphic and data base design. Software systems: Interactive languages and data base design for CAD/CAM systems. Description of existing CAD/CAM systems. Application to mechanical, electronic design etc. CAD simulation tools.
Analysis and design of integrated operational amplifiers and advanced circuits such as wideband amplifiers, instrumentation amplifiers, multiplier circuits, voltage controlled oscillators, and phase locked loops, Design techniques for advanced analogue circuits containing transistors and operational amplifiers. Simulation of circuit using appropriate packages e.g. PSPICE, Electronic workbench, Visio technical etc. should be encouraged.

EEE 509: Advanced Electronics (2, 1, 0) 3 Units
Linear I.C. op-amp, linear and non-linear operations, logarithmic amplifiers, A/D and D/A converters, gyrators and negative impedance converters, the 555 timer structure and applications; four-quadrant multipliers, dynamic logic systems and RAM memory circuits, application of digital circuits to instrumentation.

EEE 511: Digital Signal processing (2, 1, 0) 3 Units
Discrete signals and Z-transform, digital Fourier Transform, Fast Fourier Transform. The approximation problem in network theory. Synthesis of low-pass filters. Spectral transforms and their application in synthesis of high-pass and band-pass filters. Digital filtering, digital transfer function aliasing, one-dimensional recursive and non-recursive filters; Computer techniques in filter synthesis, Realisation of filters in hardware and software. Basic image processing concepts.

EEE 561: Control Engineering (2, 1, 0) 3 Units
State space description of linear systems. Transfer functions and their state space realisations. Eigenvalues, eigenvectors, system response and stability. Concepts of controllability and observability. State feedback control, modal control, and observers. Combined control-observer compensator. The Linear Quadratic Regulator (LQR). Applications to include control of robotic systems, circuit synthesis, and signal processing. Computer-aided analysis and design of control systems.

ENG 583 Engineering Management and Law (2, 0, 0) 2 Units
Principles of organization; elements of organization; management by objectives. Financial management, accounting methods, financial statements, cost planning and control, budget and budgetary control. Depreciation accounting and valuation of assets. Personnel management, selection, recruitment and training, job evaluation and merit rating. Industrial psychology. Resource management; contracts, interest formulae, rate of return. Methods of economic evaluation. Planning decision making; forecasting, scheduling. Production control. Gantt Chart, CPM and PERT. Optimization, linear programming as an aid to decision making, transport and materials handling. Raw materials and equipment. Facility layout and location. Basic principles of work study. Principles of motion economy. Ergonomics in the design of equipment and process.
Common Law: Its history, definition, nature and division. Legislation codification interpretation. Equity: Definition and its main spheres. Law of contracts for Engineers: offer, acceptance, communication termination. General principles of criminal law. Law of torts: definition, classification and liabilities. Patents: requirements, application, and infringement. Registered designs: application, requirements, types and infringement. Company law. Labour law and Industrial Law.

500L FIRST SEMESTER (ELECTIVE COURSES)
EEE 513: Communication Systems Engineering (2, 1, 0) 3 Units
Microwave frequencies and uses; microwave transmission in transmission lines and wave guides, microwave circuits; impedance transformation and matching, microwave circuits; passive microwave devices, resonant and filter circuits, active microwave devices; Klystron and magnetron tubes and semiconductor devices for microwave generation. Antennae: definitions of elementary parameters related to radiation patterns; dipole and operture antennae and the related design parameters; introduction to antennae arrays. Radiowave propagation: propagation in the ionosphere, troposphere and in stratified media; principles of scatter propagation; applications in general broadcast, television and satellite communication systems.

Radar systems nature of radar and radar equations; composition of a radar system; application of different types of radars.

EEE 517: Analogue and Digital Computers (2, 1, 0) 3 Units
Analogue computation, electrical analogue of mechanical, electromechanical systems and servomechanisms. Analogue computer elements, potentiometers, operational amplifiers, multipliers, function generators, simulation of systems transfer functions. Digital computer CPU, peripherals, storage. Arithmetic processes. Hybrid computer systems, parallel, vector, RISC's workstations.

EEE 523: Industrial Electronics Design (2, 1, 0) 3 Units
Characteristics and industrial applications of thyristors and other SCR devices. Transducers and their applications in sensing light, voltage pressure, motion, current temperature, etc. Mechanical relays, solid state relays and stepping motors. Real time control and remote control concepts in instrumentation. Micro-processor and micro-computer based systems. Fire alarms, burglar alarms and general home and industrial instrumentation. Fundamentals of Solid State Electronics. Properties of semiconductors and their devices – physical and electrical properties such as band structures; Fermi functions; doping and conductivity. Charge transport properties including drift and diffusion. Presentation of the law of mass action and applications. Junction devices including p-n junction and metal semi-conduction junctions, their I-V characteristics and applications. Other major devices of interest including the bipolar junction transistors (BJTs), metal-oxide field effect transistors (MOSFETs), and C-MOS. Optoelectronic and photonic devices including LED, Lasers,

EEE 535: High Frequency & Microwave Electronics (2, 1, 0) 3 Units
A survey of microwave engineering, models in waveguides and resonators, passive components, reactive and resistive elements, directional couples and teess ferrite isolators and circulators active components. Klystrons, magnetrons, travelling wave tubes, parametric amplifiers and solid state sources, introduction to varactor. PIN, Gunn-effect diode photodiode, phototransistor.

EEE 543: Renewable Energy Systems (2, 1, 0) 3 Units
Energy Scenario: Classification of Energy Sources, Energy resources (Conventional and nonconventional). Energy efficiency and energy security. Distributed generation.
Solar Energy: Solar thermal Systems and Photo voltaic (PV) technology: characteristics of PV, equivalent circuit, efficiency, calculations, applications. sizing and economics. Peak power operation. Wind Energy: Wind speed and power relation, power extracted from wind. Wind power systems: system components, Types of Turbine, Turbine rating. Choice of generators, turbine rating, Variable speed operation, maximum power operation, control systems, system design features. Grid Integration: Stand-alone systems, Concept of Micro-Grid and its components, Hybrid systems. Hybrid system economics, Interface requirements, Stable operation. Effect on power quality, harmonic distortion, voltage transients and sags, voltage flickers, dynamic reactive power support.

EEE 553: Power System Engineering (2, 1, 0) 3 Units
Representation of power systems, one-line diagrams, per-unit quantities, power system equation and analysis, use of network analyser, load flow studies, load forecasting, state flow equations, economic operation of power systems, symmetrical three phase faults on synchronous machines. Bus Impedance matrix in fault calculations, symmetrical components, symmetrical and unsymmetrical faults, digital calculation of fault currents using the Bus impedance matrix, various types of relays used in power systems, protection systems of power transmission lines, switchgear, circuit breaker, principles of fault detection, discrimination and clearance.

EEE 555: High Voltage Engineering (2, 1, 0) 3 Units
Importance of High voltage generation and transmission. Characteristics and details of high voltage equipment with emphasis on line structure and hardware. Generation of high AC/DC and impulse voltages. Requirement of testing of internal and external insulation system. Propagation of surges in high voltage transmission lines. Protection of transmission lines and substation from direct lighting strokes. Preventive testing of insulation process in multilayer conductors and insulation coordination. Line and substation insulation. Over-head line, bus-bars, isolators and circuit breakers. Corona/Radio interferences and minimization of their effects on the lines. Brief discussion about discharge in gases, breakdown voltages in gases. Electric field calculation for different gases, breakdown voltages in gases. Electric field calculation for different electrode configuration with respect to distance, temperature etc.

EEE 565: Control Engineering (Modern) (2, 1, 0) 3 Units
Digital control: concept of sampling and quantization, z- transforms, inverse z-transforms, zero-order-hold, discrete-time transfer functions and stability analysis; discrete-time state variables of dynamic system, formulation of state vector differential equation, solution of the state equation, the state transition matrix, eigenvalues and eigenvectors; system response and stability. Digital three-term PID design. State feedback control design in discrete time. Digital control implementation and finite word length effect. Introduction to neural network and fuzzy-logic based control systems. Applications to include control of electrical machines, mechatronic and robotic systems. Computer-aided analysis and design of control systems

500 LEVEL SECOND SEMESTER COURSES

EEE 502: Electrical Services Design (2, 0, 0) 2 Units
Lighting installation, power installation, energy supply and distribution, choice of cables and conductors, wiring systems and accessories, outdoor low voltage lines and cables, protection of low voltage installation, and characteristics of low voltage equipment, earthing and testing of electrical installation, illumination.

EEE 504: Embedded Systems Design (2, 1, 0) 3 Units
Introduction to embedded system, components, characteristics, applications. Intel 8051/8031 Micro-controller: Features of the 8051/8031 family, block diagram and definitions of the pin of the 8051, I/O port structure, memory organisation: general purpose RAM, bit addressable RAM,

register bank, special function registers, external memory, memory space mapping and decoding, bus control signals timing, a typical 8051 micro-controller based system. Instruction Set and Assembly Language Programming: Addressing modes, the 8051 instruction set and typical examples, assembler operation, assembly language format, assembler directives, operation of assemblers and linkers, programming examples. On-chip Peripheral Devices: I/O ports, operations and uses of port 0, port 1, port 2, port 3, timers: their operations, programming, and applications, serial port: operations and programming, typical applications, serial port interrupt. Interfacing to external memory, keypad, seven-segment LED display, ADC and DAC chips, and input / output port expansion, description and uses of hardware development tools. MOTOROLA M6811 Micro-controller: Features of the M6811 family, block diagram and definitions of the pin of the M6811, I/O port structure, memory organisation: general purpose RAM, bit addressable RAM, register bank, special function registers, external memory, memory space mapping and decoding, bus control signals timing. Instruction Set and Assembly Language Programming. On-chip peripheral devices and I/O interfacing. Introduction to PIC microcontroller: general architecture, applications and selection of microcontroller, advantages, low-end, and high performance PIC. Specific PIC microcontrollers: Features, architecture, block diagram, pin configuration, on-chip memory, and peripheral. Instruction set and Assembly language programming. Serial I/O interfacing: I2C, and SPI interfacing and programming. Memory interfacing: external memory interfacing, EEPROM and Flash memory interfacing. Design exercises using development system.

EEE 514: Reliability Engineering (2, 0, 0) 2 Units
Introduction to reliability, maintainability, availability. Elementary reliability theory. Application to power systems and electronics systems. Test characteristics of electrical and electronic components. Types of faults. Design for higher reliability. Packaging, mounting, ventilation. Protection from humidity, dust.

EEE 562: Power Electronics (2, 1, 0) 3 Units
Switching characteristics of diodes, transistors, thyristors etc. analysis of diode circuit with reactive loads, analysis of circuits using transistors as switches, power control circuits, ACDC converters, characteristics of switching transformers, power semi-conductor device protection, examples of power electronic circuits, solar devices.

EEE 564: Seminars on Use of Engineering Software Packages in Electrical/Electronics Field (1, 0, 0) 1 Unit
This course is to address and discuss current development in the field of electrical & electronics engineering.
Students are expected to present a term paper on the use of Engineering Software packages in implementation and optimization of Electrical/Electronics system within a current trend of interest.

EEE 599: Student Project (0, 0, 18) 6 Units

This course lasts for one academic session. Each student must undertake a project under the supervision of a lecturer, submit a comprehensive project report and present a seminar at the end of the year. A project status report is to be presented at the end of the first semester. Each student must attend Engineering Seminars.

500 LEVEL SECOND SEMESTER (ELECTIVE COURSES)

EEE 506: Digital System Design with VHDL (2, 1, 0) 3 Units

Finite State Machine: definition, mealy and Moore models, state diagram, state table, transition table. Sequential circuits design using flip-flops, asynchronous, and synchronous circuit design. Algorithm State Machine. Design examples and exercises. Structured Design: Design constructs, Design Levels, Geometry-based interchange formats, Computer aided electronic system design tools, Schematic circuit capture, Hardware description languages, Design process (simulation, synthesis), Structural design decomposition. Introduction to VHDL: VHDL language abstractions, Design hierarchies, VHDL component, Lexical description, VHDL source file, Data types, Data objects, Language statements, Concurrent VHDL, Sequential VHDL, Advanced features of VHDL (library, package and subprograms). Structural level modelling, Register-Transfer level modelling, FSM with datapath level modelling, Algorithmic level modelling. Introduction of ASIC, Types of ASIC, ASIC design process, Standard cell ASIC synthesis, FPGA Design Paradigm, FPGA synthesis, FPGA/CPLD Architectures. VHDL Design: Top-down design flow, Verification, simulation alternatives, simulation speed, Formal verification, Recommendations for verification, Writing RTL VHDL code for synthesis, top-down design with FPGA. VHDL synthesis, optimization and mapping, constraints, technology library, delay calculation, synthesis tool, synthesis directives. Computer-aided design of logic circuits.

EEE 512: Digital Communication (2, 1, 0) 3 Units

Block Diagram of digital communication system sampling theorem, Shannm theorem and applications in digital communication system. Advantages of digital signals. Noise in digital system. Filtering and equalisation. Digital modulation techniques: FSK, ASK, QPSK, M-PSK, QAM, etc. Error detection and correction techniques. Encoders/Decoders. Applications of digital communication system: Satellite communication, telephoning microwave, wireless communication, optical communication, Broadband. Communication. Internet Technology.

EEE 554: Power Systems Communication and Control (2, 1, 0) 3 Units

Review of transmission line theory. High frequency communication on power lines carrier systems and power line carrier applications. Multiplexing, Telemetering, Signal processing and data transmission. Control strategy, control of power generation, voltage control, element of power system stability, automatic voltage regulators, regulating transformers. Grid integration and control of renewable energy sources, smart grid control and monitoring

EEE 556: Electric Drives and Applications (2, 1, 0) 3 Units

Historical development, types of motors shunt, series, compound wound DC, AC induction motors, repulsion, reluctance, hysteresis and synchronous motors. Speed- torque characteristics, speed-control of electric drives. Braking. Motor power rating selection. Protection, control of drive motors. Application of electric drives

EEE 558: Electrical Energy Conversion and Storage (2, 1, 0) 3 Units
Electromechanical energy conversion, sources of motive power. Waste heat recovery. Solar energy nuclear power. Other sources of energy. Wind, geothermal, primary and secondary cells, cars and heavy vehicle batteries, testing, fault diagnosis, repairs. Effect of environmental factors on battery life, small-scale power resources. Concept of distributed generation.

EEE 566: Control Engineering III (Process Automation & Non-Linear) (2, 1, 0) 3 Units
Industrial sensors and actuators. P&ID (Piping and instrumentation) diagrams. Process modelling and identification. PID control and various tuning rules. PLCs, ladder logic and sequential flow charts. Control using PLCs. Industrial control networks. SCADA and DCS systems. The model predictive control concept.
Nonlinear control, common types and effects of nonlinearities, phase-plane and describing function analysis, closed loop response and stability. Linearisation and the concept of gain-scheduling control.

EEE 576: Microcomputer Technology (2, 1, 0) 3 Units
Elements of digital computer design; control unit, micro-programming, bus organization and addressing schemes. Micro-processors, system architecture, bus control, instruction execution and addressing modes. Machine codes, assembly language and high-level language programming, Micro-processors as state machines. Microprocessor interfacing: Input/output. Technique, interrupt systems and direct memory access; interfacing to analogue systems and applications to D/A and A/D converters. System development tools: simulators, EPROM programming, assemblers and loaders, overview of an available microprocessor application.

EEE 578: Broadcasting and Internet Technology (2, 1, 0) 3 Units
Elements of broadcasting system, studio equipment, microphones, sound, disc, magnetic tape recording, and radio transmitters power amplifiers, FM systems, radiating antenna, television camera, synchronization circuits, video amplifiers. Broadcasting regulations. Frequency spectrum: allocation, assignment and licensing. Regulatory bodies. Design, configuration, and services of CATV, MATV, MMDS systems. Multipath problems. Polarization, field strength, and footprint. Transmitter power rating, beam-width, interference and minimum separation. Digital Television standards: MPEG, DVB, channel coding techniques. HDTV. Digital television/monitor set: LCD, and Plasma technology. Internet Technology: The Internet, definition and services. Internet architecture, OSI layers, TCP/IP, Internet addressing, IPv4, IPv6. Internet broadcasting: principles, components, standards, and applications.

SECTION EIGHT: STAFFING

8.1. Recruitment, Retention, Dismissal and Welfare Recruitment

Appointment to any established staff position in the University is normally through advertisements and interviews. The Registrar compiles the list of short-listed candidates from various Departments for consideration and approval of the Vice-Chancellor.

Short-listed candidates are interviewed by the Assessment and Interview Panel. Successful candidates are put before the Appointments and Promotions Committee (APC) for consideration and approval on behalf of the Governing Council.

The APC is charged with the responsibility of all matters concerning promotion and appointment of different cadres of Senior Staff (both teaching and non-teaching), matters relating to salaries and wages, general conditions of service for all staff, confirmation, or extension of appointment, and to advise on any other matters referred to it by Council or Senate. The membership of the APC consists of the Vice-Chancellor, two Council representatives (who are not members of Senate), the Registrar, the University Librarian, Bursar, Provosts of Faculty's, and Directors of Academic Units.

Retention and Welfare

The University cares for and therefore retains its staff through cordial relationship with Administration. Salaries and allowances are the same as for the Federal universities and are paid regularly. Indeed, there has not been a situation when salaries are delayed. Also, funds are made available for staff development, attendance at conferences, research grants, etc.

Dismissal

There is a Senior Staff Disciplinary Committee (SSDC), which is a joint Committee of Council and Senate, whose duties are to investigate and report on any staff disciplinary matters referred to by Council or by the Vice-Chancellor. However, any staff charged with official misconduct is given a chance of a fair hearing by the SSDC before further action is taken against such staff.

8.2. Staff Mix

The summary of the staff mixes in Electrical/Electronic Engineering is presented in Table 8.1.

Table 8.1: The summary of the teaching staff and the staff mixed in percentage.

Staff –Mix	Professorial Cadre	Senior Lecturer	Lecturer I, Lecturer II, Assistant Lecturer	Total
Core Engineering Staff	3	5	6	14
Adjunct Staff	0	0	0	0

Staff –Mix	Professorial Cadre	Senior Lecturer	Lecturer I, Lecturer II, Assistant Lecturer	Total
(½ Score)				
Total				
Percentage (%)				

The staff-mix for the Department in the ratio Professor/Senior Lecturer/Others is, therefore, 3: 5: 6 (numerically) = 21.4:35.7:42.9 (percentage). This complies closely with the recommended ratio by the National Universities Commission (NUC) and the Council for the Regulation of Engineering in Nigeria (COREN).

8.3. Staff-Student Ratio

The students involved in the learning of Electrical/Electronic Engineering and its practices are tabulated from 100 – 500 level students for 2024/2025 Academic Session and their total number are given in the same table.

Table 8.2a:Student Admissions & Enrollment into Electrical/Electronic Engineering.

Applicants		Total Number of Students Admitted		
S/N	Session	UTME	Direct Entry	Total
1	2014-2015	52	--	52
2	2015-2016	75	2	77
3	2016-2017	71	1	72
4	2017-2018	70	1	71
5	2018-2019	83	2	85
6	2019-2020	56	3	59
7	2020-2021	67	2	69
8	2021-2022	37	1	38

9	2022-2023	31	2	33
10	2023-2024	28	3	31
11	2024-2025	29	2	30

Table 8.2b: Students in 100L-500L for the 2024/2025 academic session

S/N	Academic Level	Number of Students
1	100	30
2	200	29
3	300	35
4	400	38
5	500	62

The students involved in the learning of Electrical Electronic Engineering and its practices are tabulated from 200 – 500 level students for 2023/2024 Academic Session and the total number is given in the same table.

Table 8.2c: Students in 200L-500L for the 2024/2025 academic session

	100			**200**			**300**			**400**			**500**			Program **Grand Total**		
	F	M		F	M		F	M		F	M		F	M		F	M	
Programme Grand Total	5	25	30	4	25	29	5	30	35	6	32	38	11	51	62	31	167	194
Department 200 to 500 Level Total																		

Table 8.2d: Staff-Student ratio calculation

	Number of Students	Number of Academic Staff	Staff-Student Ratio
Total	194	14	1:13.9

Table 8.2b: Students in 100L-500L for the 2025/2026 academic session

S/N	Academic Level	Number of Students
1	100	-
2	200	30
3	300	29
4	400	35
5	500	38

The students involved in the learning of Electrical Electronic Engineering and its practices are tabulated from 200 – 500 level students for 2023/2024 Academic Session and their total number are given in the same table.

Table 8.2c: Students in 200L-500L for the 2024/2025 academic session

	100			200			300			400			500			Program Grand Total		
	F	M		F	M		F	M		F	M		F	M		F	M	
Programme Grand Total				5	25	30	4	25	29	5	30	35	6	32	38	20	112	132

Department 200 to 500 Level Total			

Table 8.2d: Staff-Student ratio calculation

	Number of Students	Number of Academic Staff	Staff-Student Ratio
Total	132	14	1:9.43

8.4. Staff Qualification

Consider the table given below showing the academic and professional qualifications of the academic staff.

Table 8.3a: Academic staff educational qualifications

Academic Qualification	Full-Time	Part-Time	Total	%
Ph.D.	10	0	10	71.4%
M.Eng.	4	0	4	28.6%
Total	14	0	14	100%

Table 8.3b: Academic staff professional qualifications

COREN Professional Qualification	Full-Time	Part-Time	Total	%
Registered	13	0	13	92.9%
Non-Registered	1	0	1	7.1%
Total	14	0	14	100%

The minimum qualification for academic teaching staff is a Master's degree in the specialization area. The academic staff of the Department are well qualified. As shown in the table above, 71.4% of the academic teaching staff have a Ph.D. degree and combined years of teaching experience. The remaining 28.6% have a master's degree and most are enrolled in a Ph.D. program. Also, 92.9% of the academic teaching staff are registered with COREN as professional engineers. The credentials of each teaching staff can be further verified from the abridged CVs of all teaching staff.

8.5. Programme Academic and Teaching Staff

Table 8.4: List of Full-time Academic staff of the programme

S/N	Name of staff	Rank	Status	Qualification, Dates Obtained and Specialization, Membership of professional Associations and Number of Publications	Specialization	Other Responsibilities /Interest in curricular and extra-curricular activities
1.	Engr. Dr. OJO Adedayo Olukayode	Senior Lecturer	Permanent	Ph.D. (2024) M.Eng.(2014), B.TECH.(2008), COREN MNSE	Communications and Electronic systems	Head of Department, Level Adviser.
2.	Engr. Prof. Moses Oluwafemi ONIBONOJE	Professor	Permanent	Ph.D. (2016), M.Eng.(2012, B.Eng. (2007), HND (2005), OND (2001) COREN MNSE	Instrumentation and Control, Distributed Systems, Wireless Sensor Network	Provost

3.	Engr. Prof. Joseph Olufemi DADA	Professor	Permanent	Ph.D. (2001), M.Eng.(1991), B.Eng. (1978) COREN, MIEEE, MNSE	Power and Energy Systems	
4.	Engr. Dr. Ayodeji Olalekan SALAU	Associate Professor	Permanent	Ph.D. (2018), M.Eng.(2015), B.Eng. (2010) COREN MNSE	Communication, Electronics and Computer Vision	College PG Coordinator
5.	Engr Dr. Joseph Folorunsho ORIMOLADE	Senior Lecturer	Permanent	Ph.D. (2017), M.Sc. (2004), B.Eng. (1995)	Communication and Electronics	Laboratory Coordinator
6.	Engr. Dr. ADEJUMOBI Babatunde Segun	Senior Lecturer	Permanent		Communication Systems	Academic Secretary
7.	Engr. Dr. Oluwafunso Oluwole OSALONI	Senior Lecturer	Permanent	Ph.D. (2021), M.Sc. (2016), PGD. (2012), HND (2008), ND (2005) COREN MNSE	Power and Energy Systems	Level Adviser
8.	Engr. Dr. OGUNLADE Michael Adegoke	Senior Lecturer	Permanent	Ph.D. (2024) ,M.Eng.(2015), B.Eng. (2009), COREN MNSE	Communication Electronics and Computer Vision	SWEP/SIWES Coordinator, Level Adviser
9.	Engr. Dr. OMOGBAI Nelson Oyakhilomen	Lecturer I	Permanent	Ph.D. (2023), M.Eng.(2018), B.Eng. (2008)	Power and Machine	Examination Script officer, Level Adviser
10.	Engr. Dr. Oladapo IBITOYE	Lecturer I	Permanent	Ph.D (2024) M.Eng (2021),	Artificial Intelligence and	Final Year Project Coordinator

				B.Sc, HND(2007), PGD (2013)	Micro Electronics	
11.	Engr FABUNMI Temitayo Emmanuel	Lecturer I	Permanent	M.Eng. (2021), B.Eng. (2016), COREN MNSE	Instrumentation and Control; Electronic Systems	Level Adviser, Examination Officer, Treasurer
12.	Engr. AKINOLA Akintunde	Lecturer II	Permanent	M.Eng. (2016), B.Eng. (2010),	Communications and Electronic Systems	Level Adviser, Welfare Officer, Exam Officer
13.	Mr. ADEBAJO-AKINPELU Victor	Assistant Lecturer	Permanent	M.Eng.(2025), B.Eng. (2021)	Control and Instrumentation	Level Adviser
14.	MR. AWODOSU A. O.	Lecturer II	Permanent	B. Eng.	Power Systems	--

8.6. Shared/Visiting Staff

Table 8.5: List of Shared/Visiting Staff

S/N	Name of Academic Staff	Rank and Date of Appointment	Employment Status	Qualification	Area of Specialization	Other Responsibility Interest in Curricular and Extra-Curricular Activities
1	Nil	Nil	Nil	Nil	Nil	Nil

Table 8.6: Summary of Academic Staff for the Session

	Staff Teaching Engineering Subjects		
	M.Eng/M.Sc./M.Tech	**Ph.D.**	**TOTAL**
Programme Staff (Dedicated)	4	10	14
Shared Staff (from other programmes)	-	-	-
Adjunct Engr. Staff	-	-	-
GA / RA	-	-	-

8.7. Overall Staff Workload

This section discusses how the overall staff workload enables effective teaching (including student-teacher ratio), student-staff interaction, student advising and counselling, institutional service and research activities, professional development and interaction with industry

8.7.1. Workload and effective teaching

The Electrical/Electronic Engineering program emphasizes maintaining a balanced workload for faculty, as detailed in Tables 8.7. This balance allows lecturers to dedicate ample time to students. As mentioned in section 8.1, the university employs highly qualified instructors. Consistent positive feedback on teaching effectiveness is reflected in student evaluations and faculty monitoring.

Lectures are meticulously designed to achieve program outcomes (POs) and course learning outcomes (CLOs). Instructors employ a variety of teaching methods to enhance student engagement, including course notes, electronic smart boards, whiteboards, and multimedia facilities. This diverse approach caters to different learning styles and fosters a dynamic learning environment.

Table 8.7: Staff Workload for First Semester 2024/2025

S/No	Name of Staff	Rank	Course Code	Course Title, Units/Status	Prerequisite	No of Students Taught	Staff Contact Hours			Weekly Contact Hours
							Lecture	Tutorial	Practical	
1	Engr. Dr. A. O OJO	Senior Lecturer	EEE 315	Electromagnetic Field and Wave Theory I		35	2	0	0	
			EEE 413	Communications System I		38				
			EEE 537	Antenna Theory and Radar Systems		21				
2	Engr.Prof. M.O. Onibonoje	Professor	ENG 221	Basic Electrical Engineering I	Nil	29	2	1	0	
			EEE 319	Measurement and Instrumentation	ENG 222	35	2	0	0	
			EEE 471	Digital Electronics and Systems	EEE 319	38	2	0	0	
			EEE 561	Control Engineering		62				

3	Engr. Prof. J. O. Dada	Professor	EEE 431 EEE 553	Electronic Power Principles Power System Engineering	Nil Nil	38 35	2 2	0 0	0 0	
4	Engr. Dr. A.O. Salau	Reader	ENG 341 EEE 511 EEE 531	Software Development Techniques Digital Signal Processing Solid State Electronics	Nil Nil Nil	35 62 62	2 2	0 0	0 0	
5	Engr. Dr. O. O. Osaloni	Senior Lecturer	EEE 325 EEE 555	Electrical Machines I High Voltage Engineering		35 25	2	0	0	
6	Engr Dr. Joseph Folorunsho ORIMOLADE	Senior Lecturer	ENG 223	Computer Applications		35	2	0	0	

			EEE 313 EEE 318	Electronics I Electronics II		62 62	2	0	0	
7	Engr. Dr. M. A. Ogunlade	Senior Lecturer	EEE 313 EEE 521	Electronics Engineering I Computer Aided Design and Simulation		35 62	2	0	0	
8	Engr. Dr. ADEJUMOBI Babatunde Segun	Senior Lecturer	ENG 223 EEE 326 EEE 415	Computer Applications Communication Princples Assembly Language Programming		30 30 23	2	0	0	
9	Engr. Dr. Oladapo IBITOYE	Lecturer I	EEE 341 EEE 405	Software Development Techniques Numerical Applications		35 62	2	0	0	

			EEE 506	Digital System Design with VHDL Programming		23				
10	Engr. Dr. N.O. Omogbai	Lecturer I	EEE 565 EEE 553	ModernElectronic Power Principles Power System Engineering		30 23	2	0	0	
11	Engr. T. E. Fabunmi	Lecturer II	ENG 221 EEE 535 COE 501	Basic Electrical Engineering I Microwave and Satellite Communications Artificial Neural Network		29 17 23	2 2	0 0	0 0	
12	Engr. A. Akinola	Lecturer I	ENG 221	Basic Electrical Engineering I		29	2	0	0	

13	Mr. V. A. Akinpelu	Assistant Lecturer	ENG 221 EEE 319 EEE 471 EEE 561	Basic Electrical Engineering Measurement and Instrumentation Digital Electronics and Systems Control Engineering		29 35 38 62	2	0	0	
14	Mr.AWODOSU A.O	Lecturer II	ENG 221 EEE 341	Basic Electrical Engineering Software Development techniques		29 35	2	0	0	

Code: P = Professor AP = Associate Professor AST = Assistant Professor I = Instructor A = Adjunct O = Other

8.8. Lab/Workshop Technical Staff

The details of the Technologists in the programme are indicated in the table. The programme has three (3) Technologists – one (1) registration is presently ongoing with COREN and one (1) is registered with NATE which is a body under COREN.

Table 8.8: List of full-time technical staff of the programme

S/N	Name of Technical Staff	Discipline	Area of Specialization	Qualification	Rank and Date of Appointment	Employment Status
1	Mr. Oluwafemi Samuel AGUNBIADE.	Electrical/ Electronics	Power and Machines	M.ENG (2024), PGD (2020), HND(2006), OND (2002)	Chief Technologist	Permanent
2	Mr. Akinyele Abiodun ITAMAKINDE	Electrical/ Electronics	Communications	M.Eng (2022) PGD (2016) , HND(2000), OND (1997),	Principal Technologist	Permanent
3	Mr. Elijah Olusegun OMITOKUN	Electrical Electronics	Measurement and Instrumentaion	HND (2009), ND (2006)	Senior Technologist	Permanent
4	Mr. Adebayo Gabriel KUMUYI	Electrical /Electronic	Electronics	HND (2011), ND (2008	Senior Technologist	Permanent

5	Mr. Olayemi Emmanuel OLANIPEKUN	Electrical/ Electronics	Power and Machines	HND (2011), ND (2006)	Technologist II	Permanent
6	Mr. Kolawole Paul OLOGUNDE	Electrical/ Electronics	Computer Software Technology	HND (2019), ND (2016)	Technologist II	Permanent

8.9. Administrative Support

Table 8.9: Administrative support staffs of the faculty

Name of Staff	Qualification and Dates Obtained	Rank/Designation Salary Scale and Date of First Appointment	Post Qualification Work Experience	Remarks
Mrs M. T. Ojo	B.Sc (2018)	Admin Officer II		

8.10. Programme Workload per Academic Staff

Table 8.10: Staff Workload in the last Three Years

2023/2024 Electrical and Electronic Engineering Workload by Staff

2023/2024 First semester Workload

S/N	Name	Course Code	Course Title	Credit
1	Engr. Prof. M. O. ONIBONOJE	ENG 319	Measurement and Instrumentation	3
		EEE 441	Servomechanism and Control systems	3
		EEE 471	Electronic Instrumentation	3
		EEE 561	Process Control	3
2	Engr. Prof. J. O. DADA	EEE 325	Electrical Machines I	3
		EEE 431	Power Transmission, distribution, and Installation	3
		EEE 553	Power systems Engineering	3
4	Engr. Prof. OWOLABI	EEE 315	Electromagnetic Field and wave theory	3
5	Engr. Dr. A. O. Salau	ENG 485	Technical Communications (Report Writing)	2
		EEE 511	Digital Signal Processing	2
6	Engr. Dr. GBADAMOSI	EEE 541	Reliability of Electrical & Electronic Systems	3
		EEE 551	Power Electronics and Drives	3

7	Engr. T. EJIDOKUN	ENG 405	Numerical Computations	3
8	Engr. Dr. OGUNLADE	EEE 415	Assembly Language Programming	2
		EEE 521	Computer Aided Design and Simulation	2
		EEE 531	Solid State Electronics	3
9	Engr. Dr. O. O. OSALONI	EEE 555	High Voltage engineering	3
10	Engr IMAHE	EEE 321	Electric Circuit Theory I	3
		EEE 565	Modern Control System	3
11	Engr. Dr. A. O. OJO	EEE 315	Electromagnetic Field and Wave Theory	3
		ENG 427	Data Communications and Networks	3
		EEE 537	Antenna theory and Radar Systems	3
13	Engr. Dr. AKINWUNMI	ENG 381	Engineering Mathematics III	3
		ENG 341	Software Development Techniques	2
14	ngr. T. E. FABUNMI	EEE 471	Digital Electronics and Systems	3
		EEE 535	Microwave & Satellite Communications	3
15	Engr. AKINOLA	EEE 313	Electronics I	3

2023/2024 Second Semester Workload

S/N	Name	Course Code	Course Title	Credit
1	Engr. Prof. M. O. ONIBONOJE	ENG 324	Linear System	3
		EEE 566	Introduction to Non-Linear systems	3
2	Engr. Prof. J. O. DADA	EEE 554	Power systems Engineering II	3
3	Engr. Dr. A. O. Salau	ENG 384	Engineering Economics	2
		EEE 524	Analog IC Designs & Applications	2
4	Engr. Dr. GBADAMOSI	EEE 326	Electrical Machines II	3
5	Engr. T. EJIDOKUN	ENG 512	Digital Communications	3
6	Engr. Dr. OGUNLADE	EEE 576	Microcomputer Technology	3
7	Engr. Dr. O. O. OSALONI	EEE 326	Electrical Machines II	3
		ENG 382	Engineering mathematics IV	3
		EEE 556	Machine Application and Protection	3
8	Engr. IMAHE	EEE 322	Electric Circuit theory II	3
		EEE 578	Introduction to Biomedical Engineering	
9	Engr. Dr. A. O. OJO	EEE 316	Electromagnetic Field and Wave Theory	3
		ENG 322	Electric Circuit theory	3

		EEE 564	Current Trends in Electrical/Electronic Engineering	2
		EEE 536	Wireless Communications	3
10	Engr. AKINOLA	EEE 314	Electronics II	3
		EEE 328	Communication Principles	3
		EEE 552	Industrial Utilization of Electrical Power	2
11	Engr. Dr. EYINAGHO	EEE 522	Microprocessor Application, Organization and Embedded systems	3

2024/2025 Electrical and Electronic Engineering Workload by Staff

2024/2025 First semester Workload

S/N	Name	Course Code	Course Title	Credit
1	Engr. Prof. M. O. ONIBONOJE	ENG 319	Measurement and Instrumentation	3
		EEE 471	Digital Electronics and Systems	3
		EEE 561	Control Engineering	3
2	Engr. Prof. J. O. DADA	EEE 431	Electric Power Principles	3
		EEE 543	Renewable Energy Systems	3
		EEE 553	Power System Engineering	3
5	Engr. Dr. A. O. Salau	ENG 485	Technical Communications (Report Writing)	2
		EEE 511	Digital Signal Processing	2

6	Engr. Dr. B. S ADEJUMOBI	EEE 415	Assembly Language Programming	3
7	Engr. Dr. J. F. ORIMOLADE	ENG 313	Electronics I	3
8	Engr. Dr. OGUNLADE	EEE 313	Electronics I	3
		EEE 531	Solid state Electronics	3
		EEE 523	Industrial Electronics Designs	3
		EEE 517	Analogue and Digital Computer Control	2
9	Engr. Dr. O. O. OSALONI	EEE 555	High Voltage engineering	2
		EEE 325	Electrical Machines I	3
		EEE 541	Reliability of Electrical Systems	2
		EEE 551	Power Electronics and Drives	3
		ENG 381	Engineering Mathematics III	3
		BME 411	Digital control systems for BME	3
		EEE 565	Modern Control Engineering	3
10	Engr. Dr. A. O. OJO	EEE 315	Electromagnetic Field and Wave Theory	3
		ENG 321	Electric Circuit Theory I	3
		EEE 513	Communication systems Engineering	3
		ENG 415	Assembly Language Programming	2

11	Engr. T. E. FABUNMI	EEE 441	Control systems and servomechanisms	3
		EEE 535	High Frequency and Microwave electronics	3
12	Engr. AKINOLA	EEE 315	Electromagnetic Field and Waves	3
		EEE 427	Data Communication and Networks	3
		EEE 507	Modelling and Simulation of Electrical and Electronic Systems	2
		EEE 509	Advanced Electronics	3
13	Engr. Dr. OMOGBAI	EEE 325	Electrical Machines I	3
		EEE 431	Electric Power Principles	3
		EEE 583	Engineering Management and Law	2
		EEE 553	Power System Engineering	3
14	Mr. V. O AKINPELU	EEE 319	Measurement and Instrumentation	3
		EEE 471	Digital Electronics and Systems	3
		EEE 561	Control Engineering	3

2024/2025 Second Semester Workload

S/N	Name	Course Code	Course Title	Credit
1	Engr. Prof. M. O. ONIBONOJE	ENG 324	Linear System	3
		EEE 502	Electrical Services Design	2
		EEE 566	Introduction to Non-Linear systems	3

2	Engr. Prof. J. O. DADA	EEE 554	Power systems Engineering II	3
3	Engr. Dr. A. O. SALAU	ENG 328	Communication Principles	2
		EEE 578	Broadcasting and Internet Technology	2
4	Engr. Dr. EYINAGHO	EEE 504	Embedded System	3
5	Engr. Dr. B. S. ADEJUMOBI	EEE 328	Communication Principles	3
6	Engr. T. EJIDOKUN	ENG 512	Digital Communications	2
7	ngr. Dr. OGUNLADE	EEE 314	Electronics II(Analog)	3
		EEE 576	Microcomputer Technology	3
8	Engr. Dr. O. O. OSALONI	EEE 326	Electrical Machines II	3
		EEE 556	Machine Application and Protection	3
		EEE 322	Electric Circuit theory II	3
		EEE 326	Electrical Machines II	3
		EEE 506	Digital System Design with VHDL Programming	3
9	Engr. Dr. A. O. OJO	EEE 316	Electromagnetic Field and Wave Theory	3
		ENG 322	Electric Circuit theory II	3

		EEE 564	Seminars on Use of Engineering Software Packages in Electrical/Electronic Field	1
10	Engr. AKINOLA	EEE 314	Electronics II	3
		EEE 562	Power Electronics	3
11	Engr. Dr. J. F. ORIMOLADE	EEE 318	Electrical/Electronic Lab II	1
12	Engr. Dr. OMOGBAI	ENG 384	Engineering Economics	2
		EEE 514	Reliability Engineering	2
		EEE 558	Electrical Energy Conversion and Storage	2
		EEE 554	Power systems communication and control	3
13	Mr. V. O. AKINPELU	EEE 324	Linear Systems	3
		EEE 502	Electrical services and design	3

2025/2026 Electrical and Electronic Engineering Workload by Staff

2025/2026 First semester Workload

S/N	Name	Course Code	Course Title	Credit
1	Engr. Prof. M. O. ONIBONOJE	ENG 319	Measurement and Instrumentation	3
		EEE 471	Digital Electronics and Systems	3
		EEE 561	Control Engineering	3

2	Engr. Prof. J. O. DADA	EEE 431	Electric Power Principles	3
		EEE 543	Renewable Energy Systems	3
		EEE 553	Power System Engineering	3
3	Engr. Dr. A. O. Salau	ENG 485	Technical Communications (Report Writing)	2
		EEE 511	Digital Signal Processing	2
4	Engr. Dr. B. S ADEJUMOBI	EEE 415	Assembly Language Programming	3
5	Engr. Dr. J. F. ORIMOLADE	ENG 313	Electronics I	3
6	Engr. Dr. OGUNLADE	EEE 313	Electronics I	3
		EEE 531	Solid state Electronics	3
		EEE 523	Industrial Electronics Designs	3
		EEE 517	Analogue and Digital Computer Control	2
7	Engr. Dr. O. O. OSALONI	EEE 555	High Voltage engineering	2
		EEE 325	Electrical Machines I	3
		EEE 541	Reliability of Electrical Systems	2
		EEE 551	Power Electronics and Drives	3
8	Engr IMAHE	EEE 321	Electric Circuit Theory I	3
		ENG 381	Engineering Mathematics III	3

		BME 411	Digital control systems for BME	3
		EEE 565	Modern Control Engineering	3
9	Engr. Dr. A. O. OJO	EEE 315	Electromagnetic Field and Wave Theory	3
		ENG 321	Electric Circuit Theory I	3
		EEE 513	Communication systems Engineering	3
10	Engr. Dr. AKINWUNMI	EEE 341	Software Development Techniques	3
		ENG 415	Assembly Language Programming	2
11	Engr. T. E. FABUNMI	EEE 441	Control systems and servomechanisms	3
		EEE 535	High Frequency and Microwave electronics	3
12	Engr. AKINOLA	EEE 315	Electromagnetic Field and Waves	3
		EEE 427	Data Communication and Networks	3
		EEE 507	Modelling and Simulation of Electrical and Electronic Systems	2
		EEE 509	Advanced Electronics	3
13	Engr. Dr. OMOGBAI	EEE 325	Electrical Machines I	3
		EEE 431	Electric Power Principles	3
		EEE 583	Engineering Management and Law	2
		EEE 553	Power System Engineering	3
14	Mr. V. O	EEE 319	Measurement and Instrumentation	3

	AKINPELU	EEE 471	Digital Electronics and Systems	3
		EEE 561	Control Engineering	3

8.11. Staff Development Programme

Afe Babalola University prides herself in her staff growth and development some of the means by which this is supported by the university are detailed below.

8.10.1. Staff Development Award

All members of staff on a permanent appointment are eligible for consideration for the staff development award as appropriate. The scheme is managed by the Appointments and Promotions Committee. A member of staff who has benefited from the staff development award is bonded to the University for two years for every year of award subject to a maximum of five years.

8.10.2. Short-term Training Programme

The University encourages and sponsors short-term training programme including short courses, seminars, workshops, colloquia etc. each lasting not more than four weeks. These programme, which are for both teaching and non-teaching staff, are covered under the Staff Development Scheme and managed by the Appointments and Promotions Committee.

8.10.3. Learned Conferences

The University encourages eligible staff to participate in learned conferences and normally funds the participation of at least one local learned conference per annum for every academic staff, subject to availability of funds. A few have enjoyed part-sponsorship for overseas learned conferences.

8.10.4 Research Grants

The University has the policy to offer research grants to eligible academic staff to conduct their research, except that the award is not for studies towards higher degree awards.

8.12. List of Teaching Staff Who Have Benefited from Staff Development

Table 8.11: List of Teaching Staff Who Have Benefited from Staff Development

S/N	Name	Designation	Remark	Date

1	Engr. Dr. Adedayo Olukayode OJO	Senior Lecturer	IEEE Conference Sponsorship Recently Obtained Ph.D.at ABUAD DAAD-NIREMAS Fellowship	26th November, 2024 August 2024 October, 2023
2	Engr. Prof. Moses Oluwafemi ONIBONOJE	Professor	IEEE Conference Sponsorship	26th November, 2024
3	Engr. Prof. DADA Joseph Olufemi	Professor	IEEE Conference Sponsorship	26th November, 2024
4	Engr. Dr. Ayodeji Olalekan SALAU	Associate Professor	IEEE Conference Sponsorship Awarded best researcher by London Metropolitan University	26th November, 2024
5	Engr. Dr. Oluwafunso Oluwole OSALONI	Senior Lecturer	IEEE Conference Sponsorship	26th November, 2024
6	Engr. Dr. Michael Adegoke OGUNLADE	Senior Lecturer	IEEE Conference Sponsorship COREN annual assembly and workshop Recently Obtained Ph.D.at ABUAD	26th November, 2024 August 2025 February 2025
7	Engr. Dr. Nelson Oyakhilomen OMOGBAI	Lecturer I	IEEE Conference Sponsorship	26th November, 2024
8	Engr Temitayo Emmanuel FABUNMI	Lecturer II	IEEE Conference Sponsorship Currently Undergoing Ph.D.at ABUAD	26th November, 2024 TILL DATE

9	Engr. Akintunde AKINOLA	Lecturer II	IEEE Conference Sponsorship Currently Undergoing Ph.D.at ABUAD	26th November, 2024 Till Date
10	Mr. Victor ADEBAJO-AKINPELU	Assistant Lecturer	IEEE Conference Sponsorship Recently completed masters programme at ABUAD Currently undergoing Ph.D. at ABUAD	26th November, 2024 August 2025 Till date

Table 8.12: List of Non-Teaching Staff who benefited from Staff Development

NO	Name	Designation	Remark	Date
1.	Mr. Adebayo Gabriel KUMUYI	Senior Technologist	Currently Undergoing Masters' Degree in ABUAD	Till date
2.	Mr. Kolawole Paul OLOGUNDE	Technologist 1I	Attended COREN annual national assembly and workshop	August 2025

8.13. Staff Contribution to Engineering and Industry

Some members of staff of the department are actively contributing to engineering and industry. The contribution of our faculty to the advancement of engineering and to Industry which included but not limited to proffering solution to real-life problems in the community and in industry, making breakthrough in ground-breaking research leading to the presentation of papers in Conferences and in Engineering associations. And some are very active in professional associations such as NSE and COREN.

Table 8.13 lists some of the contributions of the Departmental Staffs to Engineering and the Industry.

Table 8.13: Staff Contribution to Engineering and Industry

S/N	Name	Designation	Contributions to Engineering and Industry
1	Engr. Dr. Adedayo Olukayode OJO	Senior Lecturer	Nigeria Innovative Research Management Structures Fellow, External Examiner, Peer reviewer, Publications and Conference paper presentation.
2	Engr. Prof. Moses Oluwafemi ONIBONOJE	Professor	Peer reviewer, Publications and Conference paper presentation
3	Engr. Prof. Joseph Olufemi DADA	Professor	External Examiner, Peer reviewer, Publications, Conference and Workshop paper presentation
4	Engr Dr. Ayodeji Olalekan SALAU	Associate Professor	External Examiner, Peer reviewer, Publications, Conference and Workshop paper presentation.
5	Engr. Dr. Oluwafunso Oluwole OSALONI	Senior Lecturer	Peer reviewer, Publications and Conference paper presentation.
6	Engr. Dr. Michael Adegoke OGUNLADE	Senior Lecturer	Peer reviewer, Publications and Conference paper presentation
7	Engr. Dr. Nelson Oyakhilomen OMOGBAI	Lecturer I	Peer reviewer, Publications and Conference paper presentation
8	Engr Temitayo Emmanuel FABUNMI	Lecturer I	Publications and Conference paper presentation
9	Engr. Akintunde AKINOLA	Lecturer II	Publications and Conference paper presentation

S/N	Name	Designation	Contributions to Engineering and Industry
10	Mr. Victor ADEBAJO-AKINPELU	Assistant Lecturer	Publications and Conference paper presentation

8.14. Recognized Staff Research Publication

Some of the journal and conference papers published by faculty staff in the past three years can be found below and others are attached in the printout of the Google Scholar and Scopus Pages of the academic Staff.

Engr. Dr. A. O. Ojo

i. **Ojo, A. O**., Omolara, O. F., Ibitoye, O. T., Gbadamosi, S. L., Onibonoje, M. O., & Hibiokpom, C. A. P. (2024, November). Performance Evaluation of an Intelligent BFGS-Based Neuroclassifier for Brain Tumour Decision Support System. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-5). IEEE.

ii. **Ojo, A. O**., Alegbeleye, O. O., & Olomowewe, R. O. (2024). A soft computing algorithmic technique for circuital analysis of a wireless mobile charger. IAES Int. J. Artif. Intell., 13(2), 1443-1449.

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Engr. Prof. J.O Dada

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Engr. Dr. A. O. Salau

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Engr. Dr. O. O. Osaloni

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ix. Rominiyi, O. L., Salau, A. O., Adaramola, B. A., **Ogunlade, M. A**., Olanibi, T. O., & Akintoye, F. A. (2023, January). Development of a precision agricultural based unmanned aerial vehicle for pest control. In 2023 International Conference On Cyber Management And Engineering (CyMaEn) (pp. 439-443). IEEE.

x. **Ogunlade, M. A**., Gbadamosi, S. L., Owolabi, I. E., & Nwulu, N. I. (2023). Noise measurement, characterization, and modeling for broadband indoor power communication system: A comprehensive survey. Energies, 16(3), 1535.

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Engr.Dr. N. O. Omogbai

i. **Omogbai, N. (2023).** Derived Functions for MATLAB Non-Linear Solvers: An optimal rotor slot and bar design stopgap. World Journal of Advanced Engineering Technology

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ii. **Omogbai, N. O. (2023).** Derived Functions for MATLAB Non-linear Solvers–an optimal Stator Slot Design Stopgap. Journal of Engineering Research and Reports, 24(4), 8-14.

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v. **Oyakhilomen, O. N**., Anazia Emmanuel, A., & Anionovo Ugochukwu, E. Exploring the Influence of Rotor Bar Taper Angle on Breakdown Torque using Machine Learning.

Engr. T. E. Fabunmi

i. **Fabunmi, T. E.**, Ojo, A. O., Onibonoje, M. O., Gbadamosi, S. L., Opara, I., & Omosor, I. (2024, November). An Adaptable Intelligent LSTM-based Technique for Fault Detection in Domestic Electrical Networks. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-5). IEEE.

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Engr. A. A. Akinola

i. **Akinola, A**., Adejumobi, B. S., Dada, J. O., Salau, A. O., Onibonoje, M. O., & Ogunlade, M. A. (2024, November). Zero Padding Schemes for Power line Communications. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-5). IEEE.

Mr. V. O. Adebajo-Akinpelu

i. Adebajo-Akinpelu, V. O, Onibonoje, M. O., & Salau, A. O. (2024, November). Artificial Intelligence in Smart Mobile Communication Systems: A Review. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-4). IEEE.

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8.15. Administration of the Programme

Electrical/Electronic Engineering is headed by the Head of Department (HOD). Level Advisors, course coordinators, Examination officers, Undergraduate Project Coordinator, SIWES coordinators, and Facility Manager, are appointed to work with the HOD for the smooth running of the programme.

8.16. Involvement of Staff in the Decision-making Process and General Administration

Decision-making process is through regular Departmental and Committee meetings. The Head of Department is the chairman at departmental meetings and in such meetings, major academic and non-academic decisions are democratically taken. At such meetings, ad hoc committees can be constituted to carry out certain activities and submit their findings to the programme for final deliberations and decisions.

The Head of Department at the Programme Board Meeting appoints the Level Advisers, Course Coordinators, and Examination Officers from among his staff to head these sub-units, within the programme. During Programme Board meetings, most issues deliberated upon are decided based on consensus or democratic ethos in line with practices that the chair considers to be in conformity with the university quality assurance and academic standards.

It is also the Head of Programme who assigns (allocates) courses to individual lecturers with

input from other lecturers at the Programme Board meetings, usually based on their areas of specialization as informed by their CVs submitted to the programme. The course allocation system is based on shared interest and or collective efforts, that is, more than one person can team up to teach a course with one of them (most likely the senior person) as coordinator. The Head of Programme also allocates students to staff for supervision of their Final year Projects.

8.17. Administrative Responsibility

Table 8.14: Administrative responsibilities

S/N	Name of Staff	Administrative Responsibilities
1.	Engr. Dr. A. O. Ojo	(a) Head of Department (b) 500 level Adviser
2.	Engr. Prof. M. O. Onibonoje	(a) Provost, College of Engineering
3.	Engr. Dr. A. O. Salau	(a) College Postgraduate Coordinator
4.	Engr. Dr. O. O. Osaloni	(a) 300 Level Adviser
5.	Engr. Dr. M. A. Ogunlade	(a) SIWES Coordinator (b) 400 Level Adviser
6.	Engr.Dr. N. O. Omogbai	(a) Examination Support Officer (b) 100 Level Adviser
7.	Engr. T. E. Fabunmi	(a) Treasurer (b) Examination Officer (c) 200 Level Adviser
8.	Engr. A. O. Akinola	(a) Welfare Officer (b) Examination Officer (c) 300 level Adviser
9.	Mr. V. O. Adebajo-Akinpelu	(a) 100 level adviser
10.	Mr. O. S. Agunbiade	(a) Facility Manager

8.18. Faculty and Programme Board

The Faculty of Engineering Board consists of the Dean as chairman, all the Heads of Programme, all Professors and Associate Professors and lecturers in the Faculty. Subject to the General Regulations of the University, the Programme Board is responsible for all academic and non-academic matters affecting the programme. The Board comprises of the Head of Programme (as the Chairman), all Lecturers cadre.

8.19. Staff Promotions

One of the University's objectives is to attract, develop and retain high quality staff. Academic promotion seeks to recognize and reward excellent staff. All academic staff are eligible and encouraged to apply for academic promotion. There are no quotas and all staff meeting the promotion Guideline can advance. Contributions to teaching, research and community service are all highly valued.

8.20. Abridged Recognized Staff Publication and Curriculum Vitae

Engr. Dr. A. O. Ojo

Education

Degree	Discipline	Institution	Year
Ph.D.	Electrical/Electronic Engineering	ABUAD	2024
M.Eng	Electrical/Electronic Engineering	University Putra Malaysia	2014
B.Eng	Electrical/Electronic Engineering	Ladoke Akintola University of Technology	2003

Academic Experience

Institution	Rank	Title	Dates Held	FT/PT
ABUAD	Senior Lecturer	Dr	9/06/2025	FT

Certifications or Current Membership in Professional Organizations

Member, Council for the Regulation of Engineering in Nigeria
Corporate Member, The Nigerian Society of Engineers

Selected Publications and Presentations from the Past Five Years

i. **Ojo, A. O**., Omolara, O. F., Ibitoye, O. T., Gbadamosi, S. L., Onibonoje, M. O., &

Hibiokpom, C. A. P. (2024, November). Performance Evaluation of an Intelligent BFGS-Based Neuroclassifier for Brain Tumour Decision Support System. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-5). IEEE.

ii. **Ojo, A. O**., Alegbeleye, O. O., & Olomowewe, R. O. (2024). A soft computing algorithmic technique for circuital analysis of a wireless mobile charger. IAES Int. J. Artif. Intell., 13(2), 1443-1449.

iii. **Ojo, A. O.**, Omolara, O. F., Ibitoye, O. T., Gbadamosi, S. L., Onibonoje, M. O., & Hibiokpom, C. A. P. (2024, November). Performance Evaluation of an Intelligent BFGS-Based Neuroclassifier for Brain Tumour Decision Support System. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-5). IEEE.

iv. Fabunmi, T. E., **Ojo, A. O**., Onibonoje, M. O., Gbadamosi, S. L., Opara, I., & Omosor, I. (2024, November). An Adaptable Intelligent LSTM-based Technique for Fault Detection in Domestic Electrical Networks. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-5). IEEE.

v. Ibitoye, O. T., **Ojo, A. O**., Bisirodipe, I. O., Ogunlade, M. A., Ogbodo, N. I., & Adetunji, O. J. (2024, November). A Deep Learning-Based Autonomous Fire Detection and Suppression Robot. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-4). IEEE.

vi. **Ojo, A. O**., Ibitoye, O. T., Adegoke, O. M., & Ibrahim-Niass, O. (2024, November). Design and Performance Analysis of Gradient-Based Optimization Techniques for Household Power Consumption Prediction. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-5). IEEE.

vii. Adegoke, O. M., **Ojo, A. O**., Adejumobi, B. S., Gbadamosi, S. L., Tolulope, I. O., & Imonion, O. N. (2024, November). Development of a Stochastic Gradient Descent-Based Neuro Model For Identifying Fraudulent Transactions. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-4). IEEE.

viii. **Ojo, A. O**., Esan, O. I., & Omitola, O. O. (2022). Deep Learning Based Software-Defined Indoor Environment for Space-time Coded Wireless Communication Using Reconfigurable Intelligent Surfaces. International Journal of Microwave & Optical Technology, 17(6).

ix. Onibonoje, M. O., & **Ojo, A. O.** (2022). A multipart distributed modelling approach for an automated protection system in open-air cloth drying. International Journal of Electrical and Computer Engineering (IJECE), 12(3), 2432-2440.

x. **Ojo, A. O**., & Oluwafemi, O. M. (2022, April). Performance Analysis of BFGS Quasi-Newton Neuro Algorithm for the Design of 30 GHz Patch Antenna for 5G Applications. In 2022 IEEE Nigeria 4th International Conference on Disruptive Technologies for Sustainable Development (NIGERCON) (pp. 1-5). IEEE.

Engr. Prof. M. O. Onibonoje

Education

Degree	Discipline	Institution	Year
Ph.D.	Electrical/Electronic Engineering	OAU	2016
M.ENG	Electrical/Electronic Engineering	OAU	2012
B.ENG	Electrical/Electronic Engineering	OAU	2007
HND	Electrical/Electronic Engineering	Federal Polytechnic, Ado Ekiti	2002
ND	Electrical/ Electronic Engineering	Federal Polytechnic, Ado-Ekiti	2001

Institution	Rank	Title	Dates Held	FT/PT
ABUAD	Professor	Professor	15/01/2024	FT

Certifications or Current Membership in Professional Organizations

Registered Member with Council for the Regulation of Engineering in Nigeria. (COREN)
Member, Nigerian Society Engineers. (NSE)
Member. Institute of Electrical and Electronics Engineers.

Selected Publications and Presentations from the Past Five Years.

i. Agunbiade, Oluwafemi Samuel, **Moses Oluwafemi Onibonoje**, and Oluwafunso Oluwole Osaloni. "Techniques for optimal sizing and placement of renewable energy generators in power distribution grid: A review." AIP Conference Proceedings. Vol. 3169. No. 1. AIP Publishing LLC, 2025.

ii. **Onibonoje, Moses Oluwafemi**, and Ayodeji Olalekan Salau. "Artificial Intelligence in Smart Mobile Communication Systems: A Review." 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON). IEEE, 2024.

iii. **Onibonoje, Moses Oluwafemi**, and Ayodeji Olalekan Salau. "Artificial Intelligence in Smart Mobile Communication Systems: A Review." 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON). IEEE, 2024.

iv. Egbuwe, K. O., **Onibonoje, M. O**., Ikpeze, O. F., Salau, A. O., Adediji, O. E., & Chima, W. E. (2024, November). Developing an Electrical Surge Protector Maintenance System. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-5). IEEE.

v. Ibitoye, O. T., **Onibonoje, M. O.**, Dada, J. O., Ikumapayi, O. M., & Laseinde, O. T. (2024). An Enhanced Edge Computing Technique for Detection of Voltage Fluctuation in Grid-tied Renewable Energy. WSEAS Transactions on Power Systems, 19, 338-349.

vi. Ojo, A. O., Owolabi, I. E., & **Onibonoje, M. O**. (2023). Improving the performance of free space optical systems: a space-time orthogonal frequency division modulation approach. Int. J. Electr. Comput. Eng., 13(6), 6435-6442.

vii. Salau, A. O., Daniel, N. I., Ayobamidele, O., Vohra, S. K., Alemran, A., & **Onibonoje, M. O.** (2024, November). Enhancing Hostel Security: Integration of Deep Learning Models for Smoking and Violence Detection. In 2024 Second International Conference

Computational and Characterization Techniques in Engineering & Sciences (IC3TES) (pp. 1-7). IEEE.

viii. Salau, A. O., Marvellous, T. J. M., Gupta, S. K., Żywiołek, J., & **Onibonoje, M. O**. (2024, November). Development of a Smart IoT-based Dustbin Level Monitoring System. In 2024 Second International Conference Computational and Characterization Techniques in Engineering & Sciences (IC3TES) (pp. 1-5). IEEE.

ix. Akinwumi, A. O., **Onibonoje, M. O.**, & Yusuff, A. S. (2024). Development of a Sustainable Multi-Objective Approach to Poultry Management System with IoT-Based Environmental Monitoring.

x. Yusuff, A. S., & **Onibonoje, M. O.** (2023). Biodiesel production from transesterified yellow grease by ZSM-5 zeolite-supported BaO catalyst: process optimization by Taguchi's experimental design approach. Materials for Renewable and Sustainable Energy, 12(3), 199-208.

xi. Yusuff, A. S., Ishola, N. B., Gbadamosi, A. O., Azeez, T. M., & **Onibonoje, M. O.** (2023). An artificial intelligence approach to model and optimize biodiesel production from used cooking oil using CaO incorporated zeolite catalyst. Energy Conversion and Management: X, 20, 100452.

Engr. Prof. J. O. Dada

Education

Degree	Discipline	Institution	Year
Ph.D.	Electrical/Electronic Engineering	University of Duisburg-Essen	2001
M.ENG	Electrical/Electronic Engineering	University of Illorin	1991
B.ENG	Electrical/Electronic Engineering	University of Illorin	1987

Present Position and Designation

Institution	Rank	Title	Dates Held	FT/PT
ABUAD	Professor	Professor	11/2016	FT

Certifications or Current Membership in Professional Organizations

Registered Member with Council for the Regulation of Engineering in Nigeria. (COREN)

Member, Nigerian Society Engineers. (NSE)

Selected Publications and Presentations from the Past Five Years.

Engr. Prof. J.O Dada

i. Adetunla, A. O., Igenewari, O. O., Salau, A. O., **Dada, J. O**., & Kit, C. C. (2025). Development of an Unmanned Aerial Vehicle for Intelligent Monitoring: An Insight in Precision Farming. Journal Européen des Systèmes Automatisés, 58(5), 931.10.1016/j.esr.2024.101337.

ii. Bolu, C. A., Musa, T. A., Domfang, M. C., Eletta, O. A. A., **Dada, J. O**., Brijmohan, Y., ... & Matemba, E. (2024, December). Collaborative Pathways: Enhancing Engineering Education through Industry-Academia Partnerships in Africa. In 2024 World Engineering Education Forum-Global Engineering Deans Council (WEEF-GEDC) (pp. 1-9). IEEE..

iii. Bolu, C. A., Musa, T. A., Domfang, M. C., Eletta, O. A. A., **Dada, J. O**., Brijmohan, Y., ... & Matemba, E. (2024, December). Exploring Artificial Intelligence Utilization for Engineering Education and Research in Africa. In 2024 World Engineering Education Forum-Global Engineering Deans Council (WEEF-GEDC) (pp. 1-8). IEEE.

iv. Ibitoye, O. T., Onibonoje, M. O., **Dada, J. O**., Ikumapayi, O. M., & Laseinde, O. T. (2024). An Enhanced Edge Computing Technique for Detection of Voltage Fluctuation in Grid-tied Renewable Energy. WSEAS Transactions on Power Systems, 19, 338-349.

v. Liman, J., Anura, F., Adamude, F. A., Ardo, F. M., Aliyu, A. S., Bello, B. A., ... & **Dada, J.** (2024). Biotechnology of biofuels: bioethanol and biodiesel. In Microbial Biotechnology for Bioenergy (pp. 219-236). Elsevier.

vi. Ayodeji, M. G., Chidera, N. C., Akang, K. N., Dyem, A. A., Daniel, A., Oluwafunso, O. O., ... & **Dada, J. O.** (2024, November). Design And Construction Of A 500-watt Iot Based Portable Solar Power Generating Station. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-5). IEEE.

vii. Ayodeji, M. G., Chidera, N. C., Akang, K. N., Dyem, A. A., Daniel, A., Oluwafunso, O. O., ... & **Dada, J. O.** (2024, November). Design And Construction Of A 500-watt Iot Based Portable Solar Power Generating Station. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-5). IEEE.

viii. Avwunuketa, A., Joseph, O., Nmoyem, D., Oyelaran, O., Quaicoe, M., & **Dada, J.** (2024, April). An Algorithm for the Visualization of Shock Waves in Supersonic Flow. In 2024 International Conference on Science, Engineering and Business for Driving

Sustainable Development Goals (SEB4SDG) (pp. 1-9). IEEE.

ix. Okokpujie, I. P., Avwunu, O., Akande, S., **Dada, O**. J., Musa, A. I., & Agbemuko, D. I. (2024). Development and Evaluation of a MQ-5 Sensor-Based Condition Monitoring System for In-Situ Pipeline Leak Detection. Instrumentation, Mesures, Métrologies, 23(1).

x. Ibitoye, O. T., Onibonoje, M. O., & **Dada, J. O.** (2023). Analysis of power quality and technical challenges in grid-tied renewable energy. Europe, 575(608), 647.

xi. Ibitoye, O. T., Onibonoje, M. O., & **Dada, J. O.** (2022, October). Machine learning based techniques for fault detection in power distribution grid: A review. In 2022 3rd International Conference on Electrical Engineering and Informatics (ICon EEI) (pp. 104-107). IEEE.

Engr. Dr. A. O. Salau

Education

Degree	Discipline	Institution	Year
Ph.D.	Electrical/Electronic Engineering	OAU	2018
M.ENG	Electrical/Electronic Engineering	OAU	2015
B.ENG	Electrical/Electronic Engineering	FUT(Minna)	2010

Present Position and Designation

Institution	Rank	Title	Dates Held	FT/PT
ABUAD	Associate Professor	Dr.	01/02/2024	FT

Certifications or Current Membership in Professional Organizations

Registered Member with Council for the Regulation of Engineering in Nigeria. (COREN)
Member, Nigerian Society Engineers. (NSE)
Member, Institute of Electrical and Electronic Engineers
Quaterly Franklin Menbership

Selected Publications and Presentations from the Past Five Years

i. Yakkala, V. S., Nusimala, K. V., Gayathri, B., Kanamarlapudi, S., Aravinth, S. S., **Salau, A. O**., & Srithar, S. (2025). Deep learning-based crop health enhancement through early disease prediction. Cogent Food & Agriculture, 11(1), 2423244.

ii. Saranya, K., Karthikeyan, U., Kumar, A. S., **Salau, A. O**., & Tin Tin, T. (2025). DenseNet-ABiLSTM: revolutionizing multiclass arrhythmia detection and classification using hybrid deep learning approach leveraging PPG signals. International Journal of Computational Intelligence Systems, 18(1), 1-19.

iii. Ali, G., Samuel, A., Mijwil, M. M., Al-Mahzoum, K., Sallam, M., **Salau, A. O**., ... & Melekoglu, E. (2025). Enhancing Cybersecurity in Smart Education with Deep Learning and Computer Vision: A Survey. Mesopotamian Journal of Computer Science, 2025, 115-158.

iv. Kassahun, E., **Salau, A. O**., Jember, K. A., Hailu, Z., & Kassahun, H. E. (2025). Efficient resource allocation scheme for dually connected non-orthogonal multiple access based LIFI-RF networks. Wireless Networks, 31(2), 1515-1528.

v. Ragupathi, C., Dhanasekaran, S., Vijayalakshmi, N., & **Salau, A. O.** (2024). Prediction of electricity consumption using an innovative deep energy predictor model for enhanced accuracy and efficiency. Energy Reports, 12, 5320-5337.

vi. **Salau, A. O.** (2025). 26 Sustainable development and multi-disciplinary research. Multi-Disciplinary Research and Sustainable Development, 184.

vii. Palanisamy, S., Thangaraju, V., Kandasamy, J., & **Salau, A. O.** (2025). Towards precision in IoT-based healthcare systems: a hybrid optimized framework for big data classification. Journal of Big Data, 12(1), 190.

viii. Itika, R. E., Belachew, H. Z., Tegegn, D. F., & **Salau, A. O.** (2025). Improving Industrial Boiler Efficiency With Quantitative Feedback Theory-Based Controllers. Advanced Control for Applications: Engineering and Industrial Systems, 7(2), e70010.

ix. Sefiani, H. M., Bouzit, F., Elmeriah, A., Bouddou, R., **Salau, A. O.**, Berrahi, K., ... & Benaziza, A. (2025). Thermal performance enhancement of elliptical

absorbers in cylindrical-parabolic solar collectors: A CFD-Based comparative study. Scientific African, 28, e02664.

x. Adetunla, A. O., Igenewari, O. O., **Salau, A. O.**, Dada, J. O., & Kit, C. C. (2025). Development of an Unmanned Aerial Vehicle for Intelligent Monitoring: An Insight in Precision Farming. Journal Européen des Systèmes Automatisés, 58(5), 931.

xi. Rout, N. K., Patro, S. G. K., Parida, N., **Salau, A. O.**, Dansana, D., & Panda, G. (2025). Estimation of COVID-19 secondary effects on cardiac anomalies using deep transfer learning approach. Sādhanā, 50(2), 91.

Engr. Dr. O. O. Osaloni

Education

Degree	Discipline	Institution	Year
Ph.D.	Electrical Engineering	University of KwaZulu-Natal	2021
M.ENG	Electrical Engineering	University of Cape Town	2016
PGD	Electrical/Electronic Engineering	FUTA	2012
HND	Electrical/Electronic Engineering	Federal Polytechnic Ado-Ekiti	2008

Present Position and Designation

Institution	Rank	Title	Dates Held	FT/PT
ABUAD	Senior Lecturer	Dr.	12/02/2024	FT

Certifications or Current Membership in Professional Organizations

Registered Member with Council for the Regulation of Engineering in Nigeria. (COREN)

Engr. Dr. M. A. Ogunlade

Education

Degree	Discipline	Institution	Year
Ph.D.	Electrical/Electronic Engineering	ABUAD	2024
M.Sc.	Electrical/Electronic Engineering	Tiajin University of technology	2015
B.Sc	Electrical/Electronic Engineering	University of Ado-Ekiti	2008
HND	Electrical/Electronic Engineering	The Federal Polytechnic, Ado Ekiti	2001

Academic Experience

Institution	Rank	Title	Dates Held	FT/PT
ABUAD	Lecturer I	Dr	2023 – 2024	FT

Certifications or Current Membership in Professional Organizations

Member, Council for the Regulation of Engineering in Nigeria
Member, The Nigerian Society of Engineers

Selected Publications and Presentations from the Past Five Years

i. Ibitoye, O. T., Ojo, A. O., Bisirodipe, I. O., **Ogunlade, M. A.**, Ogbodo, N. I., & Adetunji, O. J. (2024, November). A Deep Learning-Based Autonomous Fire Detection and Suppression Robot. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-4). IEEE.

ii. **Adegoke, O. M**., Gbadamosi, S. L., Adejumobi, B. S., Owolabi, I. E., Oke, W. A., & Nwulu, N. I. (2024). Noise modelling and mitigation for broadband in-door power line communication systems. IET Communications, 18(15), 869-881.

iii. Rominiyi, O. L., Salau, A. O., Adaramola, B. A., **Ogunlade, M. A**., Badaseraye, O. P., & Akintoye, F. A. (2023, January). Development of Pedal Driven Reciprocating Pump for Rural Usage. In 2023 International Conference On Cyber Management And Engineering (CyMaEn) (pp. 350-357). IEEE.

iv. Ibitoye, O. T., Ojo, A. O., **Ogunlade, M. A**., Adejumobi, B. S., Fabunmi, T. E., & Agunbiade, O. S. (2024, November). Remote Monitoring and Control of Solar Photovoltaic Power Generation. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-4). IEEE.

v. Ojo, A. O., Ibitoye, O. T., **Adegoke, O. M**., & Ibrahim-Niass, O. (2024, November). Design and Performance Analysis of Gradient-Based Optimization Techniques for Household Power Consumption Prediction. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-5). IEEE.

vi. Ojo, A. O., Ibitoye, O. T., **Adegoke, O. M**., Onibonoje, M. O., Udemgba, R. C., & Fawaz, Y. (2024, November). A Neuro-Intelligent Technique for Circuital Analysis of Analogue Phase Detector. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-5). IEEE.

vii. **Adegoke, O. M**., Ojo, A. O., Adejumobi, B. S., Gbadamosi, S. L., Tolulope, I. O., & Imonion, O. N. (2024, November). Development of a Stochastic Gradient Descent-Based Neuro Model For Identifying Fraudulent Transactions. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-4). IEEE.

viii. **Adegoke, O. M**., Ojo, A. O., Adejumobi, B. S., Gbadamosi, S. L., Tolulope, I. O., & Adebomehin, A. A. (2024, November). Development and Performance Evaluation of a Neuro-Based Software for Modelling of Orbital Properties of Satellites. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-4). IEEE.

ix. Rominiyi, O. L., Salau, A. O., Adaramola, B. A., **Ogunlade, M. A**., Olanibi, T. O., & Akintoye, F. A. (2023, January). Development of a precision agricultural based unmanned aerial vehicle for pest control. In 2023 International Conference On Cyber Management And Engineering (CyMaEn) (pp. 439-443). IEEE.

x. **Ogunlade, M. A**., Gbadamosi, S. L., Owolabi, I. E., & Nwulu, N. I. (2023). Noise measurement, characterization, and modeling for broadband indoor power communication system: A comprehensive survey. Energies, 16(3), 1535.

xi. **Adegoke, O. M**., Ojo, A. O., Ohai, P. C., Gbadamosi, S. L., Lawan, R. O., & Macaulay, A. J. (2006). Development of a 1KVA fuelless generator using a brushed direct current motor.

Engr. Dr. O. N. Omogbai

Education

Degree	Discipline	Institution	Year
Ph.D.	Electronic and Electrical Engineering	Nnamdi Azikwe Univeristy	2023
M.Sc.	Electronic and Electrical Engineering	UNIBEN	2018
B.Sc	Electronic and Electrical Engineering	UNIBEN	2008

Academic Experience

Institution	Rank	Title	Dates Held	FT/PT
ABUAD	Lecturer I	Engr. Dr	24/10/2023	FT

Certifications or Current Membership in Professional Organization

Registered Member, Council for the Regulation of Engineering in Nigeria (COREN)
Corporate Member Nigeria Society of Engineers (MNSE)

Selected Publications and Presentations from the Past Five Years

i. **Omogbai, N. (2023).** Derived Functions for MATLAB Non-Linear Solvers: An optimal rotor slot and bar design stopgap. World Journal of Advanced Engineering Technology and Sciences.

ii. **Omogbai, N. O. (2023).** Derived Functions for MATLAB Non-linear Solvers–an optimal Stator Slot Design Stopgap. Journal of Engineering Research and Reports, 24(4), 8-14.

iii. Orimolade, J. F., Raji, A. A., Akinyemi, L. A., Ejidokun, T. O., **Oyakhilomen, O. N**., & Olabode, O. E. (2024, November). A consensus Decision Process for Network Selection in Converged Wireless Networks. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-5). IEEE.

iv. **Oyakhilomen, O. N.**, & Aigbologa, O. C. The Peculiarity of the Load Factor Variable Assumptions in Transformer Total Owning Cost Sensitivity Analyses.

v. **Oyakhilomen, O. N**., Anazia Emmanuel, A., & Anionovo Ugochukwu, E. Exploring the Influence of Rotor Bar Taper Angle on Breakdown Torque using Machine Learning.

Engr. T. E. Fabunmi

Education

Degree	Discipline	Institution	Year
M.Sc.	Electrical/Electronic Engineering	ABUAD	2021
B.Sc	Electrical/Electronic Engineering	ABUAD	2016

Academic Experience

Institution	Rank	Title	Dates Held	FT/PT
ABUAD	Lecturer I	Engr	12/03/2024	FT

Certifications or Current Membership in Professional Organizations

Member, Institute of Electrical and Electronic

Selected Publications and Presentations from the Past Five Years

i. **Fabunmi, T. E.**, Ojo, A. O., Onibonoje, M. O., Gbadamosi, S. L., Opara, I., & Omosor, I. (2024, November). An Adaptable Intelligent LSTM-based Technique for Fault Detection in Domestic Electrical Networks. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-5). IEEE.

ii. Ambrose, O. E., Azuka, A., **Fabunmi, T. E**., Adejumobi, B. S., Onibonoje, M. O., & Aderomose, K. S. (2024, November). Development of an IoT-based Postharvest Monitoring and Control System. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-6). IEEE.

iii. Ibitoye, O. T., Ojo, A. O., Ogunlade, M. A., Adejumobi, B. S., **Fabunmi, T. E.,** & Agunbiade, O. S. (2024, November). Remote Monitoring and Control of Solar Photovoltaic Power Generation. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-4). IEEE.

iv. **Fabunmi, T. E**., Ojo, A. O., Gbadamosi, S. L., Ekweozor, C., Ashigye, J., & Chidera, C. (2024, November). Development of an Adaptive Neuro Fuzzy Inference Model for Prediction of Heating Loads in Residential Buildings. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-5). IEEE.

v. Emmanuel, F. T., Ojo, A. O., & Gbadamosi, S. L. (2022, April). A Drowsiness Detection Decision Support System using Self-Organising Map. In 2022 IEEE Nigeria 4th International Conference on Disruptive Technologies for Sustainable Development (NIGERCON) (pp. 1-4). IEEE.

Engr. A. O. Akinola

Education

Degree	Discipline	Institution	Year
M.Sc.	Electrical/Electronic Engineering	FUTA	2016
B.Sc	Electrical/Electronic Engineering	University of Ado-Ekiti	2010

Academic Experience

Institution	Rank	Title	Dates Held	FT/PT
ABUAD	Lecturer II	Engr.	12/03/2024	FT

Certifications or Current Membership in Professional Organizations

Member, Institute of Electrical and Electronic

Selected Publications and Presentations from the Past Five Years

i. **Akinola, A**., Adejumobi, B. S., Dada, J. O., Salau, A. O., Onibonoje, M. O., & Ogunlade, M. A. (2024, November). Zero Padding Schemes for Power line Communications. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-5). IEEE.

Mr. V. O. Adebajo-Akinpelu

Education

Degree	Discipline	Institution	Year
M.Sc.	Electrical/Electronic Engineering	ABUAD	2025
B.Sc	Electrical/Electronic Engineering	ABUAD	2019

Academic Experience

Institution	Rank	Title	Dates Held	FT/PT
ABUAD	Assitant Lecturer	Mr	12/03/2024	FT

Certifications or Current Membership in Professional Organizations

Member, Institute of Electrical and Electronic

Selected Publications and Presentations from the Past Five Years

i. **Adebajo-Akinpelu, V. O**, Onibonoje, M. O., & Salau, A. O. (2024, November). Artificial Intelligence in Smart Mobile Communication Systems: A Review. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-4). IEEE.

ii. **Adebajo-Akinpelu, V. O.**, Onibonoje, M. O., Gbadamosi, S. L., Femi-Onibonoje, A. O., Yusuff, A. S., & Azeez, T. M. (2024, November). Enhancing Inventory and Payment Management: A Review of Current Technologies and Future Direction. In 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON) (pp. 1-4). IEEE.

SECTION NINE: PHYSICAL FACILITIES AND INFRASTRUCTURE

9.1 Classrooms and Lecture Theatres

The classrooms and lecture theatres used for lectures delivery are adequate with good internet connectivity. Table 9.1 shows the details of the classrooms, lecture theatres and other facilities available for students used.

Table 9.1: Classrooms and Lecture Theatres

Facility	No Available	Area m^2	Students	No Shared	Quality
Lecture Room	5	83	500	Programme Owned	Excellent
Lecture Theatre	3	5,200	3,000	Shared	Excellent
Laboratories	8	80	280	Programme Owned	Excellent
Workshop	1	127.5	-	Shared	Excellent
Drawing Studio	1	85	100	Shared	Excellent
Thinking Room/Library	1	120	80	Shared	Excellent
Office Accommodation	28	16	-	Deparment Owned	Excellent
ICT	1	4,000	1,500	Shared	Excellent

9.2 Laboratories and Workshops in the Department

The programme laboratories are as follows:

1. Computer Hardware Laboratory
2. Prototyping Laboratory
3. Electrical Machine, Power and Energy Laboratory
4. Communication and Networking Laboratory
5. National Instruments Laboratory
6. Instrumentation and Process Control
7. Microprocessor and Digital System Laboratory

8. Software Laboratory
9. Applied Electricity/Electronics Laboratory

Additional List of Tools, Machines, Instruments and Equipment in the Laboratory/Workshop

1. Electrical Power and Machine/Energy Laboratory

S/N	Equipment Description	Quantity	Current State/ Condition
1.	Electrical Machine System (EM-3000)	1	Functioning
2.	Enclosed Table Top Variac 5A	2	Functioning
3.	Enclosed Variac 5A	2	Functioning
4.	Enclosed Variac 8A	2	Functioning
5.	Enclosed Variac 3A	2	Functioning
6.	Hybrid of Solar and Wind Energy trainer	1	Functioning
7.	Solar Energy trainer	2	Functioning
8.	Fuel Cell trainer	1	Functioning
9.	DC Machine lab	1	Under maintenance.
10.	AA4420 Eddy Current Brake	**2**	Under maintenance

2. Applied Electricity/Electronics Laboratory

S/N	Equipment Description	Quantity	Current State/ Condition
1.	Basic Electricity Circuit Laboratory (KL 22001)	4	3 – Functioning 1 – Under maintenance
2.	Analogue DC power supply	2	Functioning

3.	Digital DC power supply	1	Functioning
4.	Digital system laboratory	1	Functioning
5.	Modern Digital laboratory	2	Functioning
6.	Electrical circuit Lab	2	Functioning
7.	Electronics laboratory System	2	Functioning
8.	Electronics system laboratory	2	Functioning
9.	Table top Multi-meter	1	Functioning
10.	Digital Multi-meter (BIG	5	Functioning
11.	Digital Multimeter (SMALL)	1	Functioning
12.	AC / DC power supply	5	Functioning
13.	20MHZ Oscilloscope(dual trace) GOS 620	2	Functioning
14.	20MHZ Oscilloscope(dual trace) 9003bytronics	1	Functioning
15.	30MHZ Oscilloscope(dual trace) GOS 630FC	1	Functioning
16.	Digital Storage 50MHZ Oscilloscope(dual trace) GdS1052 U	4	3 – Functioning 1 – Under maintenance
17.	Analogue DC power supply	2	Functioning

3. Prototyping Laboratory

S/N	Equipment Description	Quantity	Current State/ Condition
1.	CNC probiotics machine	2	Functioning
2.	Etching machine	1	Functioning
3.	3D Printer	1	Under maintenance

4. Software Laboratory

S/N	Equipment Description	Quantity	Current State/ Condition
1.	HP Monitor	46	45 – Functioning 1 – Under maintenance
2.	LG Monitor	4	Functioning
3.	CPU	50	43 – Functioning 7 – Under maintenance
4.	VGA to HDMI adapter	1	Functioning
5.	1TB External hard drive	2	1 – Functioning 1 – Under maintenance
6.	AIR BLOWER	1	Functioning
7.	BLUE GATE 2000VA UPS (3), BLUEGATE 1530 VA UPS (1)	4	Functioning

5. Hardware Laboratory

S/N	Equipment Description	Quantity	Current State/ Condition
1.	Computer Repair Trainer Board (975)	5	4 - Functioning 1 – Under maintenance
2.	LCD Monitor	5	Functioning
3.	Bluegate UPS (720w) Tripple lite UPS (500W)	3	2 - Functioning 1 – Under maintenance

4.	Air Blower (700W)	1	Functioning
5.	Digital Multimeter	6	Functioning
6.	Power Pack Atx	2	Functioning

6. Microprocessor and Digital System Laboratory

S/N	Equipment Description	Quantity	Current State/ Condition
1.	Microprocessor Dev Kit (Intel)	10	Functioning
2.	Microprocessor Dev Kit (Zilog)	10	Functioning
3.	Digital IC Tester	6	Functioning
4.	Flip Flops Logic Boards	5	Functioning
5.	Logic Training Boards	4	Functioning
6.	Digital IC Trainer	1	Functioning
7.	8085 Microprocessor Trainer	2	Functioning
8.	Arduino Training Kits	10	Functioning
9.	PIC KIT Programmers	10	Functioning
10.	PIC Microcontroller Development Kits	10	Functioning

7. Measurement/ Instrumentation and Process Control Laboratory

S/N	Equipment Description	Quantity	Current State/ Condition
1.	Circuit Evaluator	5	3 – Functioning 2 – Under maintenance

2.	Digital Oscilloscope	1	Functioning
3.	Digital Multimeter (DDM-451)	1	Functioning
4.	Digital Multimeter (P3-7772)	1	Functioning
5.	Digital IC Tester (DICT-01)	2	Functioning

8. Communication And Network Laboratory

S/N	Equipment Description	Quantity	Current State/ Condition
1.	Wide Band Signal Generator (100KHz-150MHz)	2	Functioning
2.	Satellite Communication Trainer System	2 Sets	Functioning
3.	Local Area Network Trainer (Lant)	5 Sets	Functioning
4.	Intelligent Counter	5	Functioning
5.	Analog Functon Generator	5	Functioning
6.	Sectrum Analyzer 9KHz-3GHz	2	Functioning
7.	GSP Series Probe Set (EMI)	2	Functioning
8.	LED RGB Strip- Addressible	2	Functioning
9.	Rotary Encoder	2	Functioning
10.	Brain Wave Start Kit	5	Functioning
11.	Bluetooth Technology Trainer	2	Functioning
12.	CDMA Trainer	2	Functioning
13.	Fiber Optic Trainer Kit	5	Functioning
14.	Logic Trainer Board On Counter And Shift	1	Functioning

	Register		
15.	Passive Filter Trainer	2	Functioning
16.	CISCO Catalyst Plus 24, 10/100+2 1000RT LAN	11	functioning
17.	Console Cable RJ 45	16	Functioning
18.	CISCO Console Cable With USB	16	Functioning
19.	V.35 Cable DTE Male	14	Functioning
20.	V.35 Cable DCE Female	14	Functioning
21.	UC –IP Phone	4	Functioning
22.	ASA 5505 Adaptive Security Application	1	Functioning
23.	CISCO Catalyst E10/100-2 SFP-IPS	2	Functioning
24.	AC/Battery Cassette Recorder	2	Functioning
25.	Line Multiplexer	3	Functioning
26.	Telephone cable	10	Functioning
27.	Network Switch	3	Functioning
28.	Protocol Analyser	1	Functioning
29.	Network Tester VERSIV FINDFIBER SIMPLIFIBRE PRO. 850/1300 Visual Fault Locator FT Fiber Viewer	1 SET 1 SET	Functioning
30.	Wireless Networking		Functioning
31.	LAN Testing Equipment		Functioning
32.	Audio Analyser		Functioning

33.	Logic Analyser		Functioning
34.	GSM/GPRS Module		Functioning
35.	Voice Recognition Shield		Functioning
36.	RFID Starter Kit		Functioning
37.	Bar-Code Scanner		Functioning

9. National Instrument Laboratory

S/N	Equipment Description	Quantity	Current State/ Condition
1.	Emona Datex Telecomms Board For Elvis	10	Functioning
2.	Freescale NI Elvis Micro-Controller Prototyping Board	5	Functioning
3.	NI Digital Electronics FPGA Board	5	Functioning
4.	QNET Mechatronics Sensor Tranier Board	5	Functioning
5.	CDAQ- 9178 Compact DAQ CHASSIS (8 SLOT USB)	5	Functioning
6.	CDAQ 9188, Compact DAQ Chassis (8-SLOT Ethernet)	5	Functioning
7.	NI 9375, 16-CH Sinking DI,16-CH Sourcing DO, DI/DO C Series Module	5	Functioning
8.	NI 9403 with DSUB 32-Channel, 7 US, TTL Digital Input /Output Module.	5	Functioning

9.	NI 9234, 4 INPUT 24-BIT. 51. 2KS/S. SW Selectable IEPE AC/DC Coupling, 102 DB Dynamic Range	5	Functioning
10.	NI 9214 16-CH Isothermal TC, 24-BIT C Series Module (Includes Terminal Block)	5	Functioning
11.	NI 9237 4-Channel 50KS/S per Channel, 24-BIT Bridge Analogue Input Module.	5	Functioning
12.	NI 9206 16-Channel 10V, 250KS/S, 16-BIT Fuel Cell Analogue Input Module.	5	Functioning
13.	QNET DC Motor Control Trainer Board	5	Functioning
14.	NI ELVIS Prototyping Board	8	Functioning
15.	NI 9208 with DSUB, 16-Channel, 24 Bit Current Input Module With DSUB, 20 MA for Series Chassis	5	Functioning
16.	Weather Station Rig	1	Functioning
17.	Digital Comm Bundle: Two NI USRP-2920S, Mimo Cable and Courseware	10 (5 pairs)	Functioning
18.	Work Stations (Desktop)	25	Functioning

Instrumentation and Process Control

S/N	Equipment Description	Function	Picture
1	Circuit Evaluator IDL-800A	It functions as a training unit designed to evaluate, test, and analyze electronic circuits. In the laboratory, it is used to practice circuit building, troubleshooting, and performance evaluation,	

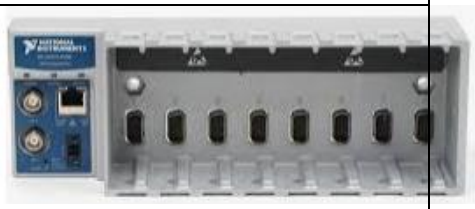
		allowing students to strengthen their understanding of electronics principles.	
2	Control System Lab	It functions as an experimental platform for studying control theory and system dynamics. In the laboratory, it is used for modeling, analyzing, and testing control systems such as feedback loops, stability analysis, and automation processes.	
3	Digital IC Tester Dict-01	It functions as a device that tests the logical operation and performance of digital integrated circuits. In the laboratory, it is used for verifying IC functionality, identifying faulty chips, and ensuring the correct operation of digital circuits.	

4	Oscilloscope GDS-1052-U	It functions as a measuring instrument that displays voltage signals as waveforms on a screen. In the laboratory, it is used for observing AC/DC signals, measuring frequency and amplitude, and troubleshooting electronic circuits.	
5	Multimeter GDM-451 & P37772	It functions as a multi-purpose measuring tool for voltage, current, resistance, and continuity. In the laboratory, it is used for diagnostics, checking circuit conditions, and verifying measurements in both electrical and electronic systems.	
6	Sensor Bundles	They function as a set of different sensors (such as temperature, light, motion, and pressure) for experimentation. In the laboratory, they are used for learning sensor principles, data collection, and interfacing sensors with microcontrollers and control systems.	

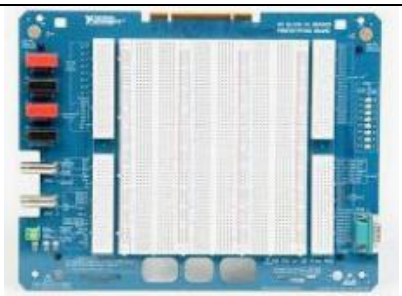
			
7	Power Supply 80W/382260	It functions as a regulated source of electrical power for circuits and devices. In the laboratory, it is used to provide stable and adjustable voltage/current for powering experiments, ensuring reliability in testing and circuit operation.	

National Instruments Laboratory

S/N	DESCRIPTION OF EQUIPMENT	MODEL	PICTURE
1.	EMONA DATEX TELECOMMS BOARD FOR ELVIS 80296-01	It functions as a telecommunications training module designed to simulate and analyze communication systems. In the laboratory, it is used for teaching modulation, demodulation, signal processing, and hands-on telecom experiments.	
2.	FREESCALE NI ELVIS MICRO-CONTROLLER PROTOTYPING BOARD 779378-01	It functions as a microcontroller prototyping platform that allows students to design, program, and test embedded systems. In the lab, it is used for microcontroller programming, interfacing, and embedded applications training.	
3.	NI DIGITAL ELECTRONICS FPGA BOARD 780125-01.	It functions as a digital design and FPGA prototyping board. In the lab, it is used to implement, simulate, and test digital logic, circuits, and FPGA-based designs.	
4.	QNET MECHATRONICS SENSOR TRANIER BOARD 747373-01	It functions as a learning platform for mechatronic sensors, allowing students to study sensing principles and signal conditioning. In the lab, it is used for sensor calibration, interfacing, and practical applications in automation.	
5.	CDAQ- 9178 COPACT DAQ CHASSIS (8 SLOT USB) 781156-01	It functions as a learning platform for mechatronic sensors, allowing students to study sensing principles and signal conditioning. In the lab, it	

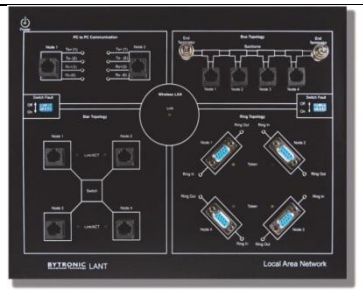
		is used for sensor calibration, interfacing, and practical applications in automation.	
6.	CDAQ 9188, COMPACT DAQ CHASSIS (8 SLOT ENET) 781424-01	It functions as an Ethernet-based data acquisition chassis. In the lab, it is used for remote data collection, distributed measurements, and networked experiments.	
7.	NI 9375, 16-CH SINKING DI,16-CH SOURCING DO, DI/DO C SERIES MODULE 779801-01	It functions as a digital input/output module for data acquisition systems. In the lab, it is used for digital signal monitoring, control, and switching applications.	
8.	NI 9403 WITH DSUB 32-CHANNEL, 7 US, TTL DIGITAL INPUT /OUTPUT MODULE 779787-01.	It functions as a high-channel digital I/O module. In the lab, it is used for digital interfacing, control of devices, and fast digital signal testing.	
9.	NI 9234, 4 INPUT 24-BIT. 51. 2KS/S. SW SELECTABLE IEPE AC/DC COUPLING, 102 DB DYNAMIC RANGE 779680-01	It functions as a vibration and acoustic signal acquisition module. In the lab, it is used for sound, vibration, and dynamic signal analysis.	

10.	NI 9214 16-CH ISOTHERMAL TC, 24-BIT C SERIES MODULE (INCLUDES TERMINAL BLOCK) 781510-01	It functions as a thermocouple input module for temperature measurement. In the lab, it is used for thermal analysis and monitoring in experiments.	
11.	NI 9237 4-CHANNEL 50KS/S PER CHANNEL, 24- BIT BRIDGE ANALOGUE INPUT MODULE 779521-01.	It functions as a strain, load, and pressure measurement module. In the lab, it is used for structural testing and sensor-based experiments.	
12.	NI 9206 16-CHANNEL 10V, 250KS/S, 16-BIT FUEL CELL ANALOGUE INPUT MODULE 779526-01.	It functions as a multi-channel analog input module. In the lab, it is used for voltage signal measurement, fuel cell monitoring, and general analog acquisition.	
13.	QNET DC MOTOR CONTROL TRAINER BOARD 746730-01	It functions as a DC motor control and analysis trainer. In the lab, it is used to study speed control, torque, and motor dynamics.	

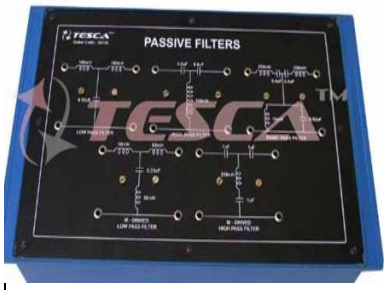
14.	NI ELVIS PROTOTYPING BOARD 780521-01	It functions as an add-on prototyping board for NI ELVIS systems. In the lab, it is used for breadboarding, circuit prototyping, and system design.	
15.	NI 9208 WITH DSUB, 16-CHANNEL ,24BIT CURRENT INPUT MODULE WITH DSUB, 20MA FOR SERIES CHASSIS 80968-01	It functions as a high-resolution current input module. In the lab, it is used for monitoring current signals and testing current-driven devices.	
16.	DIGITAL COMM BUNDLE: TWO NI USRP-2920S, MIMO CABLE AND COURSEWARE 781908-01	It functions as a software-defined radio system for wireless communication experiments. In the lab, it is used for teaching digital communication, MIMO, and RF system design	
17	WORK STATIONS(DESKSTOP) HP Pro3520	It functions as a computing workstation for experiments and analysis. In the lab, it is used for running software, simulations, and data processing.	
18	NI ELVIS II (BASE)	It functions as a complete engineering laboratory platform that integrates instruments such as oscilloscope, function generator, and digital multimeter. In the lab, it is used for hands-on learning in electronics, communications,	

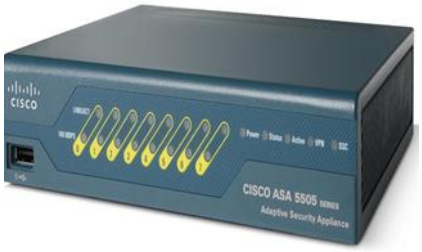
		control, and embedded systems.	

Communication and Networking Lab

S/N	DESCRIPTION OF EQUIPMENT	FUNCTION	PICTURE
1.	Wide band signal generator (100khz-150mhz) SG-4160B	This device generates continuous signals over a wide frequency range for testing, calibration, and circuit analysis, and in the lab it is used to provide test signals for amplifiers, filters, and receivers, making it essential for troubleshooting and RF experiments.	
2.	Local area network trainer (LANT)	The LAN trainer demonstrates networking concepts such as topologies, switches, and TCP/IP protocols, and in laboratory practice it is used to configure small-scale networks, monitor data flow, and troubleshoot connectivity issues.	
3.	Intelligent counter GFG-8270H	An intelligent counter measures frequency, pulse, and time intervals with high precision, and in the lab it is applied to test oscillators, digital signals, and communication systems that require accurate timing analysis.	

4.	Analog functon generator GFG-8250A	This generator produces sine, square, and triangular waveforms with adjustable settings, and in laboratory use it provides input signals for testing amplifiers, filters, and circuits to study system response and behavior	
5.	Spectrum analyzer 9khz-3ghz GSP-930	The spectrum analyzer displays signal power across frequencies for detecting harmonics, spurious signals, and interference, and in the lab it is used in wireless communication and EMI testing to verify antenna performance and transmitter outputs.	
6.	6.Brain wave start kit SEN-12805	This kit detects and processes brainwave signals for neuroscience studies, and in the lab it is used for EEG experiments, biomedical training, and learning bio-signal processing.	
7.	CDMA trainer ST2115	The CDMA trainer demonstrates multiplexing and coding principles of CDMA communication, and in labs it is used to teach spread spectrum techniques and simulate wireless network functions.	

8.	Logic trainer board on counter and shift register 38616	This board allows design and testing of digital circuits such as counters and shift registers, and in the laboratory it is used to give students hands-on practice in logic circuit design and analysis.	
9.	Rotary encoder E6B2-CWZ3E	The rotary encoder converts angular motion into digital signals, and in laboratory applications it is used for teaching motion control, robotics, and feedback systems in automation.	
10.	Bar-code scanner SEN-09166	This device reads barcodes and converts them into digital data, and in lab work it is used to demonstrate data acquisition, automation, and integration with computer systems.	
11.	CDMA trainer	Trains students on the operation of Code Division Multiple Access technology.	
12.	Passive filter trainer OC-36134	The filter trainer demonstrates low-pass, high-pass, and band-pass filter design, and in laboratory practice it is used to analyze and test filter performance in signal processing applications.	

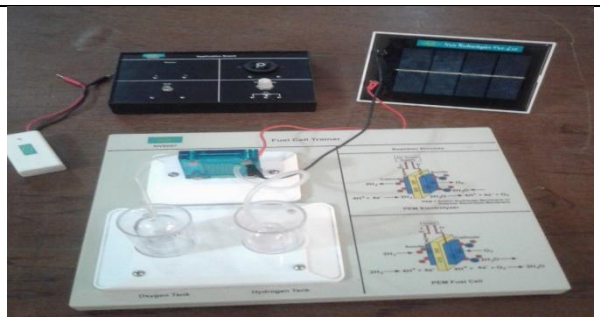
13.	Cisco catalyst plus 24, 10/100+2 1000rt lan CISCO-2960	For Local Area Networking and internetworking.	
14.	Uc –ip phone CP-7962	The IP phone supports VoIP communication systems, and in the lab it is used to configure and study voice transmission over data networks in real-world scenarios.	
15.	Asa 5505 adaptive security application 5505	This security appliance functions as a firewall and VPN gateway, and in laboratory training it is used for configuring network security and simulating enterprise cybersecurity measures.	
16.	Cisco catalyst e10/100-2 sfp-ips 3560	The Catalyst 3560 provides advanced switching and routing with Layer-3 features, and in labs it is used to explore enterprise routing, switching, and VLAN experiments.	
17.	Ac/battery cassette recorder CTR-121	This recorder captures and plays analog audio on magnetic tape, and in laboratory use it is employed for studying analog sound storage, playback, and basic communication principles.	

18.	Network switch DGS-1024D	The network switch connects multiple devices for communication within a network, and in laboratory practice it is used to build and study LAN topologies and data switching.	
19.	Protocol analyser TMA2-MO1-X	This analyzer captures and interprets network data packets, and in the lab it is used to monitor protocols, troubleshoot network errors, and study communication traffic.	
20.	Network tester • versiv • findfiber • simplifibre pro. • 850/1300 • visual fault locator • FT fiber viewer	These testers evaluate copper and fiber network integrity, and in laboratories they are used to certify cabling, detect faults, and ensure reliable communication links.	
21.	Wireless networking M5 AIRMAX	The M5 AirMax provides point-to-point and multipoint wireless communication, and in lab setups it is used to simulate Wi-Fi and broadband links for network experiments.	
22.	Lan testing equipment AT2000	The AT2000 LAN tester verifies cable continuity, wiring configuration, and signal quality, and in laboratory practice it is used to identify wiring faults, check Ethernet cable performance, and ensure reliable LAN connections	

		during networking experiments.	
23.	Audio analyser U8903A	The U8903A audio analyzer is a precision instrument used to measure audio performance parameters such as distortion, signal-to-noise ratio, frequency response, and level accuracy. In the laboratory, it is applied for testing audio circuits, amplifiers, and sound equipment to ensure high fidelity and low distortion, making it valuable in audio research, design, and quality control experiments.	

Electrical Machine, Power and Energy Laboratory

S/N	EQUIPMENT DESCRIPTION	FUNTION	PICTURE
1	Electrical Machine System EM-3000	It functions as a training system for studying electrical machines such as motors, generators, and transformers. In the laboratory, it is used for understanding machine characteristics, performance testing, and applications in power systems.	

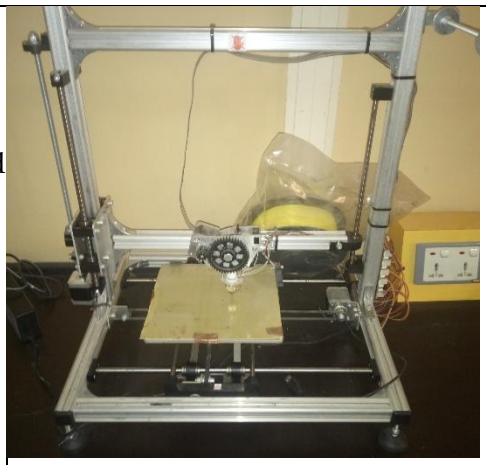
2	Fuel Cell Trainer NV6007	It functions as a trainer for demonstrating the working of fuel cells and their conversion of chemical energy into electricity. In the laboratory, it is used for renewable energy experiments, efficiency testing, and green power technology studies.	
3	Variable Transformer (Different Ratings) CMV E-1	It functions as a trainer for demonstrating the working of fuel cells and their conversion of chemical energy into electricity. In the laboratory, it is used for renewable energy experiments, efficiency testing, and green power technology studies.	
4	Hybrid of Solar and Wind Energy Trainer NVIS 6009	It functions as a combined renewable energy system that integrates solar and wind power generation. In the laboratory, it is used to study hybrid energy systems, power conversion, and efficiency analysis.	
5	Solar Energy Trainer NVIS 6005	It functions as a practical trainer for understanding solar photovoltaic technology and energy conversion. In the laboratory, it is used to measure solar panel performance, efficiency,	

		and applications in renewable energy.	
6	Solar and Energy Trainer 55502	It functions as a learning platform to demonstrate solar energy principles and applications. In the laboratory, it is used for experiments on photovoltaic systems, solar thermal concepts, and sustainable energy studies.	
7	Solar Energy System Installation Trainer	It functions as a hands-on trainer for learning solar energy system setup and installation. In the laboratory, it is used to teach practical skills in wiring, system assembly, maintenance, and solar technology deployment.	
8	DC and AC Machines EM-3330-3B EM-3330-1F EM-3330-3A EM-3330-1A EM-3330-1E EM-3330-1C EM-3330-3C	The **DC and AC Machines (EM-3330 series)** function as training units for studying the operation of motors, generators, transformers, induction, synchronous, universal, and compound machines. In the laboratory, they are used to demonstrate speed and torque control, voltage generation, power factor correction, load analysis, synchronization, and efficiency testing, giving students practical	

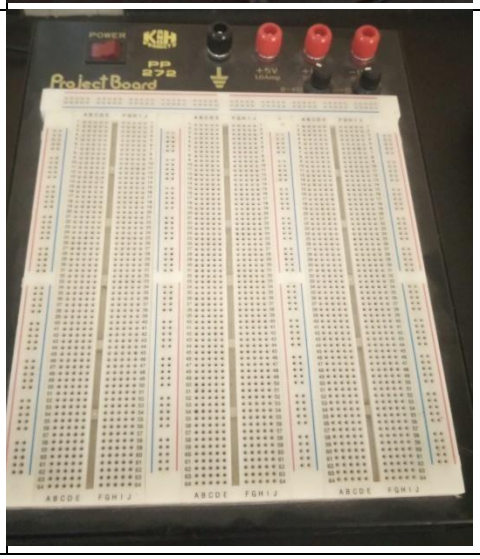
		experience in understanding and applying the principles of electrical machines.	

Prototyping Laboratory

S/N	DESCRIPTION OF EQUIPMENT	FUNCTION	PICTURE
1.	ETCHING TANK 80296-01	It functions as a chemical processing unit that uses etching solutions to remove unwanted copper from copper-clad boards, leaving behind only the desired circuit pattern. In the laboratory, it is used to fabricate printed circuit boards (PCBs), allowing students to understand the PCB manufacturing process and gain hands-on experience in creating hardware for electronic projects	
2 2.	DIGITAL OSCILLOSCOPE 779378-01	It functions as an electronic test instrument that displays voltage signals as waveforms, showing how they vary over time. In the aboratory, it is used for analyzing C and DC signals, checking noise levels, measuring frequency, and roubleshooting circuits, making it essential for both analog and digital electronics experiments.	

3.	3-D PRINTER K8200	t functions as a rapid prototyping ool that builds three-dimensional bjects layer by layer from digital esigns. In the laboratory, it is used to produce prototypes, custom enclosures for electronic circuits, and mechanical components for rojects, helping students connect design with physical implementation in engineering applications.	
4.	CNC/PCB KIT PROBOTIX MACHINE(Compact Router) M15336705	It functions as a computer-controlled milling and engraving machine capable of drilling, cutting, and routing with high precision. In the laboratory, it is used for fabricating PCBs, producing mechanical parts, and practicing computer-aided manufacturing techniques, bridging the gap between design software and physical hardware.	
5.	SOLDERING STATIONS TOL-10707 & TOL-11704	They function as tools for heating solder to join electronic components securely to PCBs. In the laboratory, they are used for assembling experimental circuits, repairing damaged boards, and performing maintenance on electronics, while also teaching students safe and effective soldering techniques.	

6.	)C VARIABLE POWER SUPPLY 17043	It functions as a regulated DC source that provides adjustable voltage and current outputs. In the laboratory, it is used to supply power to experimental circuits, test electronic devices under different voltage conditions, and ensure stable operation of projects, making it one of the most essential instruments in electronics labs.	
7.	DIGITAL IC TESTER 68027	It functions as a diagnostic tool that checks the performance and logic states of integrated circuits (ICs). In the laboratory, it is used to verify whether ICs are functioning correctly, detect faults in logic gates, and ensure that digital circuits operate as expected during experiments.	
8.	SPEAKER WITH INBUILT AMPLIFIER	It functions as an audio output device that amplifies signals before producing sound. In the laboratory, it is used in experiments involving audio electronics, communication systems, and signal processing, helping students observe the real-world application of amplified signals.	

9.	CHICAGO ELECTRIC POWER TOOL ITEM69336	It functions as a compact rotary tool that can cut, drill, grind, polish, or engrave materials. In the laboratory, it is used for mechanical work such as preparing PCB boards, shaping project enclosures, and performing small-scale fabrication tasks that support electronic experiments.	
10.	HEATERIZER XL-3000 HEAT GUN TOL-10326	It functions as a hot air tool that produces controlled heat for a variety of applications. In the laboratory, it is used for heat-shrinking cables, desoldering components, reworking PCBs, and drying coated surfaces, giving students practical exposure to repair and assembly techniques.	
11.	POWER PROJECT BOARD PP-272	It functions as a prototyping board that allows users to assemble circuits without permanent soldering. In the laboratory, it is used for quickly testing new circuit designs, experimenting with configurations, and learning circuit-building fundamentals before finalizing projects on PCBs.	

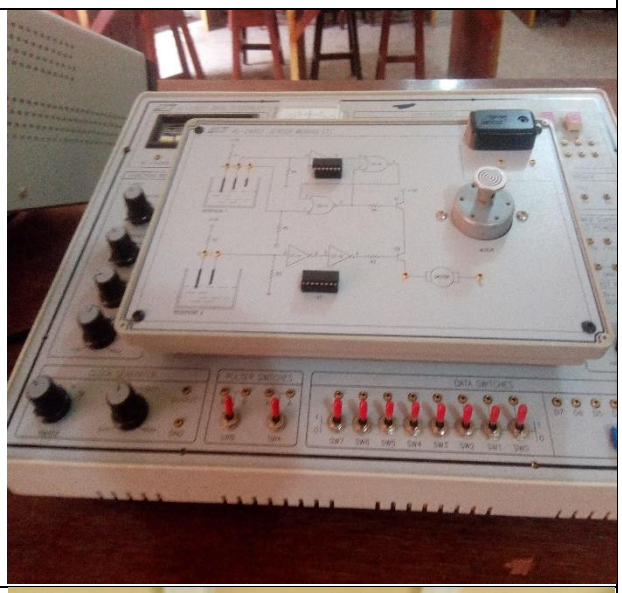
12.	DIGITAL MULTIMETER	It functions as a versatile measuring instrument capable of checking voltage, current, resistance, continuity, and sometimes additional parameters such as capacitance or frequency. In the laboratory, it is used as a primary diagnostic tool for circuit verification, fault detection, and experimental analysis, ensuring accuracy in measurements.	
13.	FLUORESCENT FILM BOX(Mega Electronics LV202E	**It functions as a light exposure unit that transfers circuit patterns onto photoresist-coated PCB boards using UV or fluorescent light. In the laboratory, it is used during the PCB fabrication process to imprint detailed circuit layouts, helping students understand the photolithography process used in real-world PCB manufacturing.**	

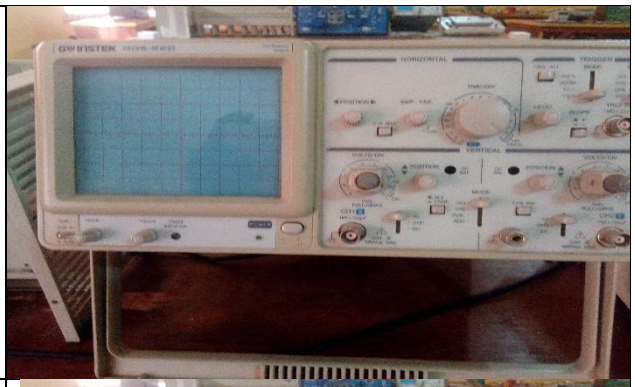
COMPUTER HARDWARE LABORATORY

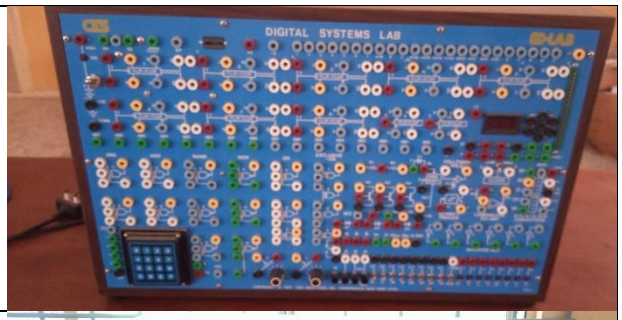
S/N	DISCRIPTION OF EQUIPMENT	FUNTION	PICTURE
1	COMPUTER REPAIR TRAINER BOARD CES 975	It functions as a training tool that simulates different faults in computer hardware, helping learners practice diagnosis and repair procedures. In the laboratory, it is used to teach computer troubleshooting, fault detection, and repair techniques.	
2	MULTIMETER	It functions as a versatile measuring device that reads voltage, current, and resistance in circuits. In the laboratory, it is used for testing, fault analysis, and verifying electrical parameters in electronic and computer systems.	

3	BLOWER	It functions as an air-supplying tool for removing dust and dirt from sensitive computer components. In the laboratory, it is used to clean hardware parts and maintain proper airflow and system performance.	
4	GSM	It functions as a communication trainer that demonstrates mobile network operations such as calls, SMS, and data transmission. In the laboratory, it is used for experiments in wireless communication and interfacing with microcontrollers or other electronic systems.	

Applied Electricity Laboratory List of Equipment

S/N	DESCRIPTION OF EQUIPMENT	FUNTION	PICTURE
1.	Basic Electricity Circuit Lab, Modules and Module Rack KL 22001	The **Basic Electricity Circuit Lab, Modules and Module Rack (KL 22001)** is designed for practical study of electrical fundamentals. It allows students to construct and analyze basic circuits using resistors, capacitors, and other components, making it useful for understanding Ohm's law, Kirchhoff's laws, and basic electrical principles in the lab.	
2.	Analogue DC Power Supply GPS 3030	The **Analogue DC Power Supply (GPS 3030)** provides a stable and adjustable DC voltage output. In the lab, it is used to power electronic circuits, test components, and ensure a reliable source of energy during experiments.	
3.	Laboratory DC Power Supply(Analogue) GPS 1850D	The **Laboratory DC Power Supply (GPS 1850D)** functions similarly, offering regulated DC outputs for circuit operations. It is commonly used in experiments where accurate and consistent voltage or current is required for testing.	

4.	20MHz Oscilloscope (Dual Trace) GOS 620	The **Laboratory DC Power Supply (GPS 1850D)** functions similarly, offering regulated DC outputs for circuit operations. It is commonly used in experiments where accurate and consistent voltage or current is required for testing.	
5.	20MHZ Oscilloscope (Dual Trace) 9003bytronics	The **20MHz Oscilloscope (Dual Trace) 9003 Bytronics** also measures signals up to 20MHz and allows side-by-side comparison of two inputs. It is applied in lab sessions to verify signal quality, phase differences, and waveform behavior.	
6.	30MHZ Oscilloscope (Dual Trace) GOS 630FC	The **20MHz Oscilloscope (Dual Trace) 9003 Bytronics** also measures signals up to 20MHz and allows side-by-side comparison of two inputs. It is applied in lab sessions to verify signal quality, phase differences, and waveform behavior.	
7.	50MHZ Digital Storage Oscilloscope GDS-1052-U	The **50MHz Digital Storage Oscilloscope (GDS-1052-U)** can capture and store signals digitally for later analysis. In laboratories, it is used to observe complex or high-speed signals, save waveforms, and provide more detailed data for advanced experiments.	

8.	Digital system Lab ED-LAB 652.A	The **Digital System Lab (ED-LAB 652.A)** is a specialized trainer for digital electronics. In lab work, it is used to design and test circuits involving logic gates, counters, flip-flops, and other fundamental digital systems.	
9.	Electronics Laboratory ED-LAB 505	The **Electronics Laboratory (ED-LAB 505)** provides facilities for studying a wide range of electronic applications. It is used in labs for experiments involving semiconductors, amplifiers, oscillators, and practical electronic device analysis.	
10.	Electrical Circuits lab ED-LAB 400	The **Electrical Circuits Lab (ED-LAB 400)** is designed for experiments on electrical principles and circuit behavior. Students use it to understand concepts such as resistance, capacitance, inductance, AC/DC analysis, and circuit theory.	
11.	Modern Digital Laboratory ED-LAB 750	The **Modern Digital Laboratory (ED-LAB 750)** supports advanced digital and embedded system experiments. It is used in labs for microprocessor studies, memory interfacing, and learning about modern computing systems.	
12.	Table top multimeter GDM 8034	The **Tabletop Multimeter (GDM 8034)** measures voltage, current, and resistance with high accuracy. In lab applications, it is used for verifying circuit values and troubleshooting electrical connections.	

13.	Table top multimeter	**Tabletop Multimeters** serve the same function of measuring electrical parameters. They are useful in lab sessions for testing circuits quickly and ensuring proper functionality.	
14.	Modern digital multimeters P37772	The **Modern Digital Multimeters (P37772)** are more precise and reliable digital instruments. In the lab, they provide accurate readings of voltage, current, resistance, and sometimes additional parameters like capacitance and frequency.	
16.	Electronics Boxex	The **Electronics Boxes** contain collections of essential components such as resistors, capacitors, ICs, and connectors. They are used in laboratories for assembling, testing, and experimenting with various circuits	
17	Bread Board	The **Breadboard** is a reusable prototyping tool that allows circuit construction without soldering. In the lab, it is widely used for quick testing and modification of experimental circuits.	
18	Soldering Iron	The **Soldering Iron** is used to permanently connect electronic components on circuit boards. In laboratories, it is applied in assembling circuits, repairing devices, and conducting hands-on practical soldering exercises.	

COMPUTER SOFTWARE LABORATORY





SOFTWARE

1. C++ PROGRAMMING LANGUAGE
2. C PROGRAMMING LANGUAGE
3. NET BEANS PROGRAMMING LANGUAGE
4. VISUAL BASIC PROGRAMMING LANGUAGE
5. MATLAB PROGRAMMING LANGUAGE
6. MATHCAD PROGRAMMING LANGUAGE
7. CHEMCAD PROGRAMMING LANGUAGE
8. AUTO CARD PROGRAMMING LANGUAGE
9. ASPEN HYSYS PROGRAMMING LANGUAGE

Table 9.4 Carpentry and Wood workshop

S/N	Description of Equipment	Model	Qty	Supplier	Location	Functionality
1	Universal machine	Kolvestratt B-8000	1	Locally sourced	Wood workshop	Functional
2	Wood lathe machine	NA	1	Locally sourced	Wood workshop	Functional
3	Planing machine	Sedgwick	1	Locally sourced	Wood workshop	Functional
4	Spindle machine	Electra Beckum	1	Locally sourced	Wood workshop	Functional
5	G-Clamp		8	Locally sourced	Wood workshop	Functional
6	Band saw	Einhell	1	Locally sourced	Wood workshop	Functional
7	Clamp		4	Locally sourced	Wood workshop	Functional
8	Bench Grinder	ower mack	1	Locally sourced	Wood workshop	Functional

Table 9.5 Festo Lab

S/N	Description of Equipment	Model	Qty	Supplier	Location	Functionality
1	Pneumatic and Hydraulic Workstation	NA	3	Locally sourced	Festo Lab	Functional
2	Factory and processing automation workstation	NA	2	Locally sourced	Festo Lab	Functional

Table 9.6 Central workshop

S/N	Description of Equipment	Model	Qty	Supplier	Location	Functionality
1	Lathe	Warco WM2550 De hol GH166	2	Locally sourced	Central workshop	Functional
2	CNC Lathe	Boxford 280	1	Locally sourced	Central workshop	Functional
3	Milling Machine	Warco 4VS Turret	2	Locally sourced	Central workshop	Functional
4	Radial Drilling Machine	WMW MEUSELWITZ BR 40	2	Locally sourced	Central workshop	Functional
5	Cylindrical Grinding Machine	Jones & shipman 1307	2	Locally sourced	Central workshop	Functional
6	Surface Grinder	MJ 7115	1	Locally sourced	entral workshop	Functional

S/N	Description of Equipment	Model	Qty	Supplier	Location	Functionality
7	Bench Grinder	NA	1	Locally sourced	entral workshop	Functional
8	Drill Press	Warco DT-25	1	Locally sourced	entral workshop	Functional

ANNEX A: KNOWLEDGE PROFILE AND DEFINITION OF COMPLEX ENGINEERING PROBLEM

Annex A-1: Knowledge Attribute Profile

The curriculum shall encompass the knowledge profile as summarised in the table below:

TableA-1: Knowledge Attribute Profile

S/No.	Attribute
K1	systematic, theory-based understanding of the **natural sciences** applicable to the discipline and awareness of relevant **social sciences**
K2	Conceptually-based **mathematics**, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline
K3	systematic, theory-based formulation of **engineering fundamentals** required in the engineering discipline
K4	Engineering **specialist knowledge** that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline
K5	nowledge, including efficient resource use, environmental impacts, whole-life ost, re-use of resources, net zero carbon, and similar concepts, that supports **engineering design and operations** in a practice area.
K6	Knowledge of **engineering practice** (technology) in the practice areas in the engineering discipline
K7	**Knowledge of** the role of engineering in society and identified issues in ngineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development*
K8	ngagement with selected knowledge in the current **research literature** of the scipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues

K9	**Ethics, inclusive behaviour and conduct**. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes

*Represented by the 17 UN Sustainable Development Goals (UN-SDG)

Annex A-2: Definition of Complex Problem Solving

The range of complex problem solving is defined as follows:

Table A-2: Range of Complex Problem Solving

Attribute	**Complex Engineering Problems** have characteristic P1 and some or all of P2 to P7:
Depth of Knowledge Required	P1: Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical proach
Range of conflicting requirements	P2: Involve wide-ranging and/or conflicting technical, non-technical issues (such as ethical, sustainability, legal, political, economic, societal) and consideration of future requirements
Depth of analysis required	P3: Have no obvious solution and require abstract thinking, creativity and originality in analysis to formulate suitable models
Familiarity of issues	P4: Involve infrequently encountered issues or novel problems
Extent of applicable codes	P5: Address problems not encompassed by standards and codes of practice for professional gineering
Extent of stakeholder involvement and conflicting requirements	P6: Involve collaboration across engineering disciplines, other fields, and/or diverse groups of stakeholders with widely varying needs

Interdependence	P7: Address high level problems with many components or sub-problems that may require a systems approach

Annex A-3: Definition of Complex Engineering Activities

The range of complex engineering activities is defined as follows:

Table A-3: Range of Complex Engineering Activities

Attribute	Complex Activities
Preamble	**Complex activities** mean (*engineering*) activities or rojects that have some or all of the following characteristics:
Range of resources	1: Involve the use of diverse resources including people, data nd information, natural, financial and physical resources and ppropriate technologies including analytical and/or design software
Level of interactions	2: Require optimal resolution of interactions between wide-ranging and/or conflicting technical, non-technical, and engineering issues
Innovation	3: Involve creative use of engineering principles, innovative solutions for a conscious purpose, and research-based knowledge
nsequences to society and the environment	A4: Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation
Familiarity	A5: Can extend beyond previous experiences by applying principles-based approaches

ANNEX B: MAPPING OF POs to PEOs

ANNEX B-1: MAPPING OF POs to PEOs

Programme Educational Outcomes	PEO1	PEO2	PEO3	PEO4	PEO5
Engineering Knowledge	x	x	x	x	x
Problem Analysis	x	x	x		x
Design/Development of Solutions	x	x	x	x	x
Investigation	x		x		
Modern Tool Usage	x	x	x	x	
The Engineer and Society:	x	x		x	x
Environment and Sustainability				x	
Ethics	x		x		x
Individual and Teamwork	x		x	x	
Communication	x			x	
Project Management			x		x
Lifelong Learning		x	x	x	x

ANNEX C: SYSTEM OF INSTRUCTIONS AND EXAMINATION

GRADE SHEETS FOR 2024/2025 SESSION

FIRST SEMESTER 500 LEVEL

				STUDENTS' GRADES				
COURSE CODE	**COURSE TITLE**	**CREDITS**	**NUMBER OF STUDENTS**	**A**	**B**	**C**	**D**	**F**
EEE 507	Modelling and Simulation of Electrical and Electronics Systems	2	59	21	27	9	2	0
EEE 509	Advanced Electronics	3	59	17	30	9	3	0
EEE 511	Digital Signal Processing	3	57	3	23	22	11	0
EEE 513	Communication Systems Engineering	3	38	12	19	7	0	0
EEE 523	Industrial Electronics Designs	3	39	12	19	8	0	0
EEE 535	High Frequency & Microwave Electronics	3	26	15	10	1	0	0
EEE 553	Power System Engineering	3	10	4	3	2	1	0
EEE 555	High Voltage Engineering	2	15	8	3	4	0	0
EEE 561	Control Engineering	3	59	15	27	14	3	0
EEE 565	Modern Control Engineering (Modern)	3	3	1	1	1	0	0
ENG 583	Engineering Management and Law	2	59	4	26	26	3	0

SECOND SEMESTER 500 LEVELS

					STUDENTS' GRADES				
S/N	**COURSE CODE**	**COURSE TITLE**	**CREDITS**	**NUMBER OF STUDENTS**	**A**	**B**	**C**	**D**	**F**
1	EEE 502	Electrical Services Design	2	59	14	33	10	2	0
2	EEE 504	Embedded System	3	59	16	16	14	13	0

3	EEE 506	Digital System Design with VHDL Programming	3	2	0	2	0	0	0
4	EEE 512	Digital Communication	2	11	2	6	3	0	0
5	EEE 514	Reliability Engineering	2	59	12	26	20	1	0
6	EEE 554	Power Systems Communication and Control	3	3	1	2	0	0	0
7	EEE 556	Electric Drives and Applications	3	27	17	6	4	0	0
8	EEE 558	Electrical Energy Conversion and Storage	2	11	5	3	2	1	0
9	EEE 562	Power Electronics	3	59	19	28	8	4	0
10	EEE 564	Seminars on Use of Engineering Software Packages in Electrical/Electronics Field	1	59	25	28	4	2	0
11	EEE 566	Control Engineering III (Non-Linear & Process Automation)	3	19	11	5	3	0	0
12	EEE 576	Microcomputer Technology	3	54	15	23	16	0	0
13	EEE 599	Student Project	6	59	35	24	0	0	0

FIRST SEMESTER 400 LEVEL

					STUDENTS' GRADES				
S/N	**COURSE CODE**	**COURSE TITLE**	**CREDITS**	**NUMBER OF STUDENTS**	**A**	**B**	**C**	**D**	**F**
1	EEE 405	Numerical Computations	3	39	12	13	10	3	1

2	EEE 415	Assembly Language Programming	2	37	10	19	7	1	0
3	EEE 423	Electrical/Electronic Laboratory III	1	36	27	7	7	1	0
4	EEE 427	Data Communication and Networks	3	36	9	14	7	5	1
5	EEE 431	Electric Power Principles	3	39	10	9	9	8	3
6	EEE 441	Control Theory and Systems	3	40	11	18	10	1	0
7	EEE 471	Digital Electronics and Systems	3	42	9	17	14	2	0
8	ENG 485	Technical Communications (Report Writing)	2	37	14	17	4	2	0
9	ENG 390	Student Work Experience Programme II (SWEP II)	3	36	20	15	1	0	0

FIRST SEMESTER 300 LEVEL

					STUDENTS' GRADES				
S/N	**COURSE CODE**	**COURSE TITLE**	**CREDITS**	**NUMBER OF STUDENTS**	**A**	**B**	**C**	**D**	**F**
1	AFE 301	Practical Entrepreneurial Studies	1	34	16	13	5	0	0
2	EEE 313	Electronics I (Physical)	3	35	6	13	10	6	0
3	EEE 315	Electromagnetic Field and Wave Theory I	3	35	7	8	6	8	6

4	EEE 317	Electrical/Electronic Laboratory I	1	33	12	8	8	5	0
5	EEE 319	Measurements and Instrumentation	3	31	11	6	11	3	0
6	EEE 321	Electric Circuit Theory I	3	35	5	11	10	7	2
7	EEE 325	Electrical Machines I	3	37	12	13	5	4	3
8	EEE 341	Software Development Techniques	3	34	4	14	9	4	3
9	ENG 290	Student Work Experience Programme I (SWEP I)	3	35	18	16	0	0	1
10	ENG 381	Engineering Mathematics III	3	35	3	10	12	6	4

SECOND SEMESTER 300 LEVEL

					STUDENTS' GRADES				
S/N	**COURSE CODE**	**COURSE TITLE**	**CREDITS**	**NUMBER OF STUDENTS**	**A**	**B**	**C**	**D**	**F**
1	EEE 314	Electronics II (Analogue)	3	35	10	11	10	2	2
2	EEE 316	Electromagnetic Field and Wave Theory II	3	36	6	8	11	6	5
3	EEE 318	Electrical/Electronic Laboratory II	1	36	21	8	6	1	0
4	EEE 322	Electric Circuit Theory II	3	37	4	9	13	7	4
5	EEE 324	Linear Systems	3	40	12	12	7	7	2
6	EEE 326	Electrical Machines II	3	37	9	15	6	5	2
7	EEE 328	Communication Principles	3	34	9	12	12	1	0

8	ENG 382	Engineering Mathematics IV	3	33	8	16	5	3	1
9	ENG 384	Engineering Economics	2	34	4	7	15	8	0

FIRST SEMESTER 200 LEVEL

					STUDENTS' GRADES				
S/N	**COURSE CODE**	**COURSE TITLE**	**CREDITS**	**NUMBER OF STUDENTS**	**A**	**B**	**C**	**D**	**F**
1	AFE 203	French Language Acquisition II	1	25	10	9	2	4	0
2	AFE 205	Chinese Language II	1	8	1	2	1	3	1
3	AFE 207	Igbo Language II	1	19	9	3	5	2	0
4	AFE 209	Yoruba Language II	1	8	8	0	0	0	0
5	AFE 211	Hausa Language II	1	3	1	1	1	0	0
6	ENG 201	Applied Electricity I	2	33	6	10	10	5	2
7	ENG 203	Engineering Graphics and Solid Modelling II	3	32	11	12	5	3	1
8	ENG 221	Basic Electrical Engineering I	3	32	9	8	5	8	2
9	ENG 223	Computer Applications	2	32	1	14	12	3	2
10	ENG 231	Engineering Drawing I	2	34	8	8	10	6	2
11	ENG 233	Engineering Nechanics I	2	35	6	11	11	6	1
12	ENG 235	Engineering Thermodynamics I	2	36	3	8	16	7	2
13	ENG 261	Workshop Technology I	2	30	3	13	11	2	1
14	ENG 281	Engineering Mathematics I	3	30	8	11	10	11	0

15	GST 201	Peace and Conflict Studies	2	30	9	9	9	2	1

SECOND SEMESTER 200 LEVEL

					STUDENTS' GRADES				
S/N	**COURSE CODE**	**COURSE TITLE**	**CREDITS**	**NUMBER OF STUDENTS**	**A**	**B**	**C**	**D**	**F**
1	**AFE 202**	Food Production and Health Awareness	1	29	21	7	1	0	0
2	**ENG 204**	Introduction to Mechatronics Engineering	2	29	7	10	5	6	1
3	**ENG 214**	Fluid Mechanics	3	29	5	10	7	4	3
4	**ENG 222**	Basic Electrical Engineering II	3	29	10	1	7	10	1
5	**ENG 224**	Computer Programming and Applications	2	29	4	7	7	11	0
6	**ENG 232**	Engineering Drawing II	2	29	7	8	8	4	2
7	**ENG 234**	Engineering Mechanics II	2	29	5	9	7	5	3
8	**ENG 262**	Workshop Technology II	2	29	8	9	7	5	0
9	**ENG 282**	Engineering Mathematics II	3	29	4	6	12	7	0
10	**ENG 284**	Engineer in Society	2	29	10	10	6	3	0
11	**GST 202**	Philosophy and Sustainable Development	2	29	5	17	6	1	0
12	**GST 212**	Introduction to Entrepreneurship	2	29	25	3	0	0	1

FIRST SEMESTER 100 LEVEL

					STUDENTS' GRADES

S/N	COURSE CODE	COURSE TITLE	CREDITS	NUMBER OF STUDENTS	A	B	C	D	F
1	AFE 101	Health Awareness Education	1	34	27	6	1	0	0
2	AFE 103	Introduction to Leadership	1	35	9	15	10	0	1
3	CHM 101	General Chemistry I	2	32	19	9	4	0	0
4	CHM 107	General Practical Chemistry I	1	30	28	2	0	0	0
5	EEE 102	Introduction to Electrical and Electronics Engineering	2	30	9	14	7	0	0
6	GET 101	Engineer in Society	1	30	13	13	2	2	0
7	GST 111	Communication in English	2	30	7	12	9	1	1
8	MTH 101	Elementary Mathematics I	2	30	21	8	1	0	0
9	MTH 103	Elementary Mathematics III	2	30	21	4	5	0	0
10	PHY 101	General Physics I	2	31	21	7	2	0	1
11	PHY 103	General Physics III	2	30	16	8	5	1	0
12	PHY 107	General Practical Physics I	1	32	11	12	6	1	2

SECOND SEMESTER 100 LEVEL

					STUDENTS' GRADES				
S/N	COURSE CODE	COURSE TITLE	CREDITS	NUMBER OF STUDENTS	A	B	C	D	F
1	**AFE 122**	Use of English	1	30	11	14	5	0	0
2	**AFE 124**	French Language I	1	21	10	4	6	1	0

3	**AFE 126**	Chinese Language	1	9	7	1	0	1	0
4	**AFE 128**	Igbo Language	1	12	10	0	2	0	0
5	**AFE 130**	Yoruba Language	1	17	16	0	1	0	0
6	**AFE 132**	Hausa Language	1	1	1	0	0	0	0
7	**CHM 102**	General Chemistry II	2	30	9	11	7	3	0
8	**CHM 108**	General Practical Chemistry II	1	30	16	7	6	1	0
9	**GET 102**	Engineering Graphics and Solid Modelling I	2	30	12	12	6	0	0
10	**GST 112**	Nigerian Peoples and Culture	2	30	9	8	9	4	0
11	**MTH 102**	Elementary Mathematics II: Calculus	2	30	23	3	3	1	0
12	**PHY 102**	General Physics II: Behaviour of Matter	2	30	22	3	4	1	0
13	**PHY 108**	General Practical Physics II	1	31	25	6	0	0	0

ANNEX D: MAPPING OF COURSES TO POs

ANNEX D: MAPPING OF COURSES TO POs

100 Level

Semester No.	Course Code	Course Title	Level of Emphasis of PO (3 = High; 2 = Medium; 1 = Low)											
			1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
			POs as defined by COREN (more may be added by the Programme)											
1	AFE 101	Health Awareness Education						1	1	2	2	3	1	2
	AFE 103	Introduction to Leadership						2	2	2	2	2	3	3
	MTH 101	Elementary Mathematics I	3	2	2	1				1				
	GST 111	Communication in English				1			3	2	2	2	1	2
	GET 101	Engineer in Society	1	1	1	1		1	2	2	3	3	3	2
	PHY 101	General Physics I: Mechanics	3	2	2	1	2			1				
	PHY 107	General Practical Physics I	1	1	1	1	3	1		1				
	CHM 101	General Chemistry I	2	1	1	1	2			1				
	CHM 107	General Practical Chemistry I	1	1	1	1	3	1		1				
2	GST 112	Nigerian Peoples and Culture							2	1	2	3	2	2
	GET 102	Engineering Graphics and Solid	2	1	3	2	1	1		1		1	1	

		Modelling I												
	MTH 102	Elementary Mathematics II: Calculus	3	2	2	1				1				
	PHY 102	General Physics II: Behaviour of Matter	3	2	2	1	2			1				
	PHY 108	General Practical Physics II	2	1	1	1	3	1		1				
	ABUAD-CHM 102	General Chemistry II	2	1	1	1	2			1				
	ABUAD-CHM 108	General Practical Chemistry II	1	1	1	1	3	1		1				
	AFE 124	French Language I							3	2			1	1
	AFE 126	Chinese Language							3	2			1	1
	AFE 128	Igbo Language							3	2			1	1
	AFE 130	Yoruba Language							3	2			1	1
	AFE 132	Hausa Language							3	2			1	1

200 LEVEL

Semester	**Course**	**Course Title**	**Level of Emphasis of PO (3 = High; 2 = Medium; 1= Low)**

No.	Code		1	2	3	4	5	6	7	8	9	10	11	12
			POs as defined by COREN (more may be added by the Programme)											
1	ENT 211	Enterpreneurship and Innovation		1	2	1		2	2	2	2	2	3	3
	GET 201	Applied Electricity I	3	3	3	2	2	1		1				
	GET 203	Engineering Graphics and Solid Modelling II	2	2	3	2	1	2		1		1	1	1
	ABUAD-GET 205	Fundamentals of Fluid Mechanics	3	2	2	1	2			1		1		
	GET 207	Applied Mechanics	3	2	2	1	2	1		1		1		
	GET 209	Engineering Mathematics I	3	2	2	1				1				
	GET 211	Computing and Software Engineering	2	2	3	3	1	2		2		1	2	1
	AFE 203	French Language II							3	2			1	1
	AFE 205	Chinese Language II							3	2			1	1
	AFE 207	Igbo Language II							3	2			1	1
	AFE 209	Yoruba Language II							3	2			1	1

	AFE 211	Hausa Language II							3	2			1	1
	ABUAD-CPE 201	MATLAB Programming												
2	GST 212	Philosophy, Logic and Human Existence	1	1	1				2	2	2	2	1	1
	GET 204	Students Workshop Practice	1	1	2	2	3	3		1	2	2	2	2
	GET 206	Fundamentals of Thermodynamics	3	2	2	1	2			1		1		
	GET 210	Engineering Mathematics II	3	2	2	1				1				
	EEE 202	Applied Electricity II	3	3	3	2	2	1		1				
	EEE 204	Electrical Engineering Materials	2	2	2	1	1			1		1	1	
	ABUAD-GET 208	Strength of Materials	3	2	2	1	2			1		1		
	AFE 202	Food Production and Health Awareness						1	1	2	2	3	2	2
	*GET 299	SIWES I	1	2	2	2	3	3	1	2	2	3	2	3

300 LEVEL

Semester No.	Course Code	Course Title	Level of Emphasis of PO (3 = High; 2 = Medium; 1 = Low)											
			1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
			POs as defined by COREN (more may be added by the Programme)											
1	ENG 290	Students Work Experience Programme(SWEP) I												
	EEE 313	Electronics I	3	2	2	2	2			1				
	EEE 315	Electromagnetic Field and Wave Theory	3	2	2	1	1			1				
	EEE 317	Electrical/Electronic Laboratory I	2	2	2	2	3	2		1				1
	EEE 319	Measurement and Instrumentation	3	2	2	2	3	1		1				
	EEE 321	Electric Circuit Theory I	3	3	3	2	2			1				
	EEE 325	Electrical Machines I	3	2	2	1	2	1		1				
	EEE 341	Software Development Techniques	2	2	3	3	1	2		2		1	2	1
	ENG 381	Engineering Mathematics III	3	2	2	1				1				
	AFE 301	Practical Entrepreneurship		1	2	1		2	2	2	2	2	3	3

2	EEE 314	Electronics II (Analogue)	3	2	2	2	2			1				
	EEE 316	Electromagnetic Field and Wave Theory II	3	2	2	1	1			1				
	EEE 318	Electrical/Electronic Laboratory II	2	2	2	2	3			1				1
	EEE 322	Electric Circuit Theory II	3	3	3	2	2			1				
	EEE 324	Linear Systems	3	3	3	2	2			1				
	EEE 326	Electrical Machines II	3	2	2	1	2			1				
	EEE 328	Communication Principles	3	2	3	2	1			1		1		
	ENG 382	Engineering Mathematics IV	3	2	2	1				1				
	ENG 384	Engineering Economics	1	1	1	1			1	2	1	2	3	2

400 Level

Semester No.	Course Code	Course Title	Level of Emphasis of PO (3 = High; 2 = Medium; 1 = Low)											
			1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
			POs as defined by COREN (more may be added by the Programme)											
1	ENG 390	Students Work Experience Programme (SWEP) II	2	2	2	2	3	3	2	2	2	3	2	3
	ENG 485	Technical Writing and Presentation			1	1	1		3	2	1	1	1	2
	EEE 405	Numerical Computations	3	2	2	2	1			1				
	EEE 415	Assembly Language Programming	2	2	3	3	1	1		1				
	EEE 423	Electrical/Electronic Laboratory III	2	2	2	2	3	2		1				1
	EEE 427	Data Communication and Networks	3	2	3	2	1	1		1		1	1	
	EEE 431	Electric Power Principles	3	3	3	2	2	1		1		1	1	
	EEE 441	Control Theory and Servomechanism	3	3	3	2	2	1		1		1	1	
	EEE 471	Digital Electronics and Systems	3	2	3	2	2	1		1				

2	ENG 290	Student Work Experience Programme I (SWEP I)	2	2	2	2	3	3	2	2	2	3	2	3
	ENG 390	Student Work Experience Programme II (SWEP II)	2	2	2	2	3	3	2	2	2	3	2	3
	ENG 490	Student Industrial Work Experience Scheme (SIWES)	2	2	2	2	3	3	2	2	2	3	2	3

500 Level

Semester No.	**Course Code**	**Course Title**	**Level of Emphasis of PO (3 = High; 2 = Medium; 1 = Low)**											
			1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
			POs as defined by COREN (more may be added by the Programme)											
1	ENG 583	Engineering Management and Law	1	1	2	1		1	2	2	2	3	3	3
	EEE 507	Modelling and Simulation of Electrical and Electronics Systems	3	3	3	2	2			1				
	EEE 509	Advanced Electronics	3	3	3	2	2			1				

	EEE 511	Digital Signal Processing	3	3	3	2	2			1				
	EEE 561	Control Engineering	3	3	3	2	2			1		1		
	EEE 513	Communication Systems Engineering	3	2	3	2	1			1		1	1	
	EEE 535	High Frequency & Microwave Electronics	3	2	3	2	1			1				
	EEE 553	Power System Engineering	3	3	3	1	2			1		1	1	
	EEE 555	High Voltage Engineering	3	2	3	1	1			1		1	1	
	EEE 565	Modern Control Engineering	3	3	3	2	2			1		1	1	
	EEE 523	Industrial Electronics Designs	3	2	3	2	2	1		1			1	
	EEE 543	Renewable Energy System	3	2	2	1	1			1		2	2	1
	EEE 517	Analogue and Digital Computer Control	3	2	2	2	1			1				
2	EEE 502	Electrical Services Design	2	2	3	2	1			1		1	1	
	EEE 504	Embedded System	3	2	3	3	1			1				
	EEE 514	Reliability Engineering	2	2	2	1	1			1		1	1	
	EEE 562	Power Electronics	3	2	3	2	2			1		1		

	EEE 564	Seminars on Use of Engineering Software Packages in Electrical/Electronics Field	1	1	2	3	1	1	2	3	1	1	1	2
	EEE 599	Student Project	3	3	3	3	3	3	2	2	2	3	3	3
	EEE 512	Digital Communication	3	2	3	2	1			1				
	EEE 506	Digital System Design with VHDL Programming	3	2	3	2	1			1				
	EEE 578	Broadcasting and Internet Technology	2	1	2	2	1			2		1	1	
	EEE 576	Microcomputer Technology	3	2	3	2	1			1				
	EEE 554	Power Systems Communication and Control	3	3	3	2	2			1		1	1	
	EEE 556	Electric Drives and Applications	3	3	3	2	2			1		1		
	EEE 558	Electrical Energy Conversion and Storage	3	2	2	1	1			1		1	1	
	EEE 566	Control Engineering III (Non-Linear & Process Automation)	3	3	3	2	2			1		1	1	

ANNEX E: DESIGN OF ENGINEERING CURRICULUM

Domain	Knowledge Area	COREN/NUC Recommended		Institute's Program Breakup	
			Overall	Total	Overall
		Total Credit	%	Credits	%
Non-Engineering	Humanity	As per discipline specific COREN BMAS guidelines	25% - 40%	21	29%
	Management Sciences			8	
	Natural Sciences			28	
Engineering	Computing	As per discipline specific COREN BMAS guidelines	60% - 75%	13	71%
	Engineering Foundation			36	
	Major Based Core (Breadth)			20	
	Major Based Core (Depth)			40	
	Inter-Disciplinary Engineering Breadth (Electives)			12	

	Final Year Design Project	6		6	
	Industrial Training SIWES	13		12	
Total		**130-138**		**197**	

ANNEX F: CURRICULUM

ANNEX F-1: COURSE OFFERINGS

Semester	No.	Course Code	Course Title	Credit Hours	Knowledge Area	Pre-requisite Courses
1	1	GST 111	Communication in English	2	General Studies	-
	2	GET 101	Engineer in Society	1	General Engineering Course	-
	3	MTH 101	Elementary Mathematics I	2	Basic Science	-
	4	PHY 101	General Physics I	2	Basic Science	-
	5	PHY 107	General Practical Physics I	1	Basic Science	-
	6	EEE 102	Intro. to Electrical/Electronics Engineering	2	Core discipline course	-
	7	CHM 101	General Chemistry I	2	Basic Science	-
	8	CHM 107	General Practical Chemistry I	1	Basic Science	-
	9	AFE 101	Health Awareness Education	1	General Studies	-
	10	AFE 103	Leadership	1	General Studies	-
Total				**15**		
2	1	GST 112	Nigerian Peoples and Culture	2	General Studies	-
	2	GET 102	Engineering Graphics & Solid Modelling I	2	General Engineering Course	-
	3	MTH 102	Elementary Mathematics II	2	Basic Science	-
	4	PHY 102	General Physics II	2	Basic Science	-
	5	PHY 108	General Practical Physics II	1	Basic Science	-
	6	CHM 102	General Chemistry II	2	Basic Science	-
	7	CHM 108	General Practical Chemistry II	1	Basic Science	-
	8	AFE ---	Language Elective I (Foreign/Nigerian)	1	General Studies	-
Total				**13**		
	1	ENT 211	Entrepreneurship and Innovation	2	General Studies	-
	2	GET 201	Applied Electricity I	3	General Engineering Course	-

3	3	GET 203	Engineering Graphics & Solid Modeling II	2	General Engineering Course	-
	4	GET 205	Fundamentals of Fluid Mechanics	3	General Engineering Course	-
	5	GET 207	Applied Mechanics	3	General Engineering Course	-
	6	GET 209	Engineering Mathematics I	3	General Engineering Course	-
	7	GET 211	Computing and Software Engineering	3	General Engineering Course	-
	8	AFE ---	Language Elective II	1	General Studies	-
Total				**20**		
4	1	GST 212	Philosophy, Logic and Human Existence	2	General Studies	-
	2	GET 204	Students Workshop Practice	2	General Engineering Course	-
	3	GET 206	Fundamentals of Thermodynamics	3	General Engineering Course	-
	4	GET 210	Engineering Mathematics II	3	General Engineering Course	-
	5	EEE 202	Applied Electricity II	3	Core discipline course	-
	6	EEE 204	Electrical Engineering Materials	3	Core discipline course	-
	7	GET 208	Strength of Materials	3	General Engineering Course	-
	8	AFE 202	Food Production and Health Awareness	1	General Studies	-
	9	GET 299	SIWES I (Off-campus)	3	SIWES	-
Total				**23**		
5	1	EEE 313	Electronics I (Physical)	3	Core discipline course	-
	2	EEE 315	Electromagnetic Field and Wave Theory I	3	Core discipline course	-
	3	EEE 317	Electrical/Electronic Laboratory I	1	Core discipline course	-
	4	EEE 319	Measurements and Instrumentation	3	Core discipline course	-
	5	EEE 321	Electric Circuit Theory I	3	Core discipline course	-
	6	EEE 325	Electrical Machines I	3	Core discipline course	-
	7	EEE 341	Software Development Techniques	3	Core discipline course	-

	8	ENG 381	Engineering Mathematics III	3	General Engineering Course	-
	9	AFE 301	Practical Entrepreneurship	1	General Studies	-
Total				**23**		
6	1	EEE 314	Electronics II (Analogue)	3	Core discipline course	EEE 313
	2	EEE 316	Electromagnetic Field & Wave Theory II	3	Core discipline course	EEE 315
	3	EEE 318	Electrical/Electronic Laboratory II	1	Core discipline course	-
	4	EEE 322	Electric Circuit Theory II	3	Core discipline course	EEE 321
	5	EEE 324	Linear Systems	3	Core discipline course	-
	6	EEE 326	Electrical Machines II	3	Core discipline course	EEE 325
	7	EEE 328	Communication Principles	3	Core discipline course	-
	8	ENG 382	Engineering Mathematics IV	3	General Engineering Course	-
	9	ENG 384	Engineering Economics	2	General Engineering Course	-
Total				**24**		
7	1	EEE 405	Numerical Computations	3	Core discipline course	-
	2	EEE 415	Assembly Language Programming	2	Core discipline course	-
	3	EEE 423	Electrical/Electronic Laboratory III	1	Core discipline course	EEE 318
	4	EEE 427	Data Communication and Networks	3	Core discipline course	-
	5	EEE 431	Electric Power Principles	3	Core discipline course	-
	6	EEE 441	Control Theory and Servomechanism	3	Core discipline course	-
	7	EEE 471	Digital Electronics and Systems	3	Core discipline course	EEE 314
	8	ENG 485	Technical Communications (Report Writing)	2	General Engineering Course	-
Total				**20**		
8	1	ENG 290	Student Work Experience (SWEP I)	3	SIWES	-
	2	ENG 390	Student Work Experience (SWEP II)	3	SIWES	-
	3	ENG 490	Industrial Work Experience (SIWES)	6	SIWES	-
	Total			**12**		

9	1	ENG 583	Engineering Management and Law	2	General Engineering Course	-
	2	EEE 507	Modelling & Simulation of EEE Systems	2	Core discipline course	-
	3	EEE 509	Advanced Electronics	3	Core discipline course	-
	4	EEE 511	Digital Signal Processing	3	Core discipline course	-
	5	EEE 561	Control Engineering	3	Core discipline course	EEE 441
	6	EEE ---	*Elective Course I*	3	Interdisciplinary Engineering course	-
	7	EEE ---	*Elective Course II*	3	Interdisciplinary Engineering course	-
Total				**19**		
10	1	EEE 502	Electrical Services Design	2	Core discipline course	-
	2	EEE 504	Embedded System Design	3	Core discipline course	-
	3	EEE 514	Reliability Engineering	2	Core discipline course	-
	4	EEE 562	Power Electronics	3	Core discipline course	EEE 326
	5	EEE 564	Seminars on Engineering Software	1	Core discipline course	-
	6	EEE 599	Student Project	6	Core discipline course	-
	7	EEE ---	*Elective Course III*	2	Interdisciplinary Engineering course	-
Total				**19**		

ANNEX F-2: LIST OF ELECTIVES

Area of Specialization/ Options	Sr. No.	Course Code	Course Title	Credit Hours	Knowledge Area	Pre-requisite Courses
Power Systems	1	EEE 553	Power System Engineering	3	Core discipline course	-
	2	EEE 555	High Voltage Engineering	2	Core discipline course	-
	3	EEE 543	Renewable Energy System	2	Core discipline course	-
	4	EEE 554	Power Systems Comm. & Control	3	Core discipline course	EEE 553
	5	EEE 556	Electric Drives and Applications	3	Core discipline course	EEE 326
	6	EEE 558	Electrical Energy Conv. & Storage	2	Core discipline course	-
Communication Engineering	1	EEE 513	Communication Systems Engineering	3	Core discipline course	-
	2	EEE 535	High Frequency & Microwave Elec.	3	Core discipline course	-
	3	EEE 512	Digital Communication	2	Core discipline course	EEE 328
	4	EEE 578	Broadcasting and Internet Tech.	2	Core discipline course	-
	5	EEE 576	Microcomputer Technology	3	Core discipline course	-
	1	EEE 565	Modern Control Engineering	3	Core discipline course	-

Instrumentation and Control	2	EEE 523	Industrial Electronics Designs	3	Core discipline course	-
	3	EEE 517	Analogue and Digital Computers	2	Core discipline course	-
	4	EEE 566	Control Engineering III (Non-Linear)	3	Core discipline course	EEE 324
	5	EEE 506	Digital System Design with VHDL	3	Core discipline course	-

ANNEX G: LABORATORIES & LABORATORY WORKS

Electrical/Electronics Engineering Laboratory Technologists and Workstations

S/N	Name of Laboratory (Staff Name & Qualifications)	Titles of Laboratory Course(s)	Type(s) of Workstations (No.)	Nature of Experiments	No. of Students per Workstation
1.	**Communication Systems Lab** Engr. UWAEZUOKE Emmanuel (HND, B.Eng., M.Eng.)	1. Communication Theory 2. Wave Propagation & Antennas 3. Microwave Engineering	1. Local Area Network (LANT) Trainers (5) 2. CDMA trainer (2) 3. Spectrum Analyzer (2) 4. Satellite communication trainer (4) 5. Fiber Optics Trainers (5) 6. CISCO Router (9) / Switches (11)	Demonstration Demonstration Demonstration Demonstration Demonstration Hands-on	3 to 5 2 to 4 3 to 5 3 to 4 4 to 5 2 to 5
2.	**Electrical Power & Machine Lab** Engr. AGUNBIADE Oluwafemi S. (M.Eng, PGD, HND, OND)	1. Electrical Machines I (EEE 325) 2. Electrical Machines II (EEE 326)	1. Electrical Machine System EM-3000 (1) 2. Enclosed Variacs (8)	Experimentation, demonstration of speed/torque control, and efficiency testing	4 to 5 (General) 8 to 10 (Trainers)

		3. Electro-Mechanical Machine Lab	3. Solar Energy trainers (2) 4. Hybrid Solar/Wind Energy trainer (1) 5. Fuel Cell trainer (1)		
3.	**Measurement & Control Lab** Mr. OMITOKUN Elijah Olusegun (HND, ND, B.Eng.)	1. Measurement and Instrumentation 2. Applied Electricity 3. Electronics Engineering 4. Robotics	1. Power Supply and Breadboards (5) 2. Circuit Evaluators and Multimeters (5) 3. Step down Transformers (5) 4. Robotics Kits and chassis Tank	Hands-on practicals involving circuit building, troubleshooting, and stability analysis	5 to 8
4.	**Basic and Applied Electricity Lab** Mr. OLANIPEKUN Olayemi E. (HND)	1. Basic Electrical Engineering Practical (ENG 221/222)	1. Wave rectifier unit 2. Transistor characteristics unit 3. Diode characteristics unit	Experimentation involving electrical fundamentals and device characteristics	5 to 7

			4. Ohms Law verification kit		
5.	**National Instruments (NI) / Data Acquisition Lab** Engr. ITAMAKINDE Akinyele A. (M.Eng, PGD, HND, OND)	1. Communication Theory 2. Wave Propagation & Antennas 3. Prototyping	1. NI Elvis Prototyping Boards (10) 2. NI USRP-2920S SDR (5 pairs) 3. Emona Datex Telecomms Boards (10)	Data acquisition, virtual instrumentation, and Software-Defined Radio (SDR) platforms	3 to 4
6.	**Prototyping Workshop & Demonstration Lab** Mr. KUMUYI Adebayo Gabriel (HND, B.Eng.)	1. Applied Electricity (GET 201) 2. Workshop practice (GET 204) 3. Prototyping Techniques (COE 409)	1. Project circuit boards (3) 2. ED Lab Trainer Boards (2) 3. Soldering stations (9) 4. Etching Tank (1) 5. PC System (Arduino/IoT)	Circuit design, soldering/desoldering techniques, etching, and IoT projects	5 to 8 (Wiring) 3 to 5 (Design) 5 (IoT)
7.	**Software Laboratory** Mr. OLOGUNDE Kolawole Paul (HND)	1. Computer Engineering Lab I 2. Computer Engineering Lab II	1. HP Monitor & CPU (44) 2. BLUE GATE UPS (4)	Live coding demonstration and Guided software practice	1

			3. Live Projector		

ANNEX H: STUDENTS

Annex H-1: Student Admissions & Enrolments

Session	Applicants			Number of Students Admitted		
	UTME	Direct Entry	Total	UTME	Direct Entry	Total
2020/2021	75	2	75	67	2	69
2021/2022	42	1	43	37	1	38
2022/2023	37	3	40	31	2	33
2023/2024	33	3	36	28	3	31
2024/2025	32	2	34	29	1	30

Annex H-2: List of Registered Students in the Programme for the past 5 Sessions

Level	Number of Registered Students				
	2020/2021	2021/2022	2022/2023	2023/2024	2024/2025
100	67	37	31	28	**30**
200	56	67	37	31	**29**
300	83	56	67	37	**35**
400	70	83	56	67	**38**
500	71	70	83	56	**62**
Total	**347**	**313**	**274**	**219**	**198**

ANNEX I: STAFFING

Annex I-1: List of Full-Time Departmental Academic Staff

S/N	Name	COREN #	Rank	Date of first appointment	Details of Qualifications			Specialization	Experience Teaching (Total) Years	Dedicated /Shared	Credit Hours taught in the Current & Last *Semesters*	
					Degree	Year	Institution				First	Second
1.	Engr. Dr. OJO Adedayo Olukayode	R33712	Senior Lecturer	14/10/2014	Ph.D.	2024	ABUAD	Communication and Intelligent Systems	11	Dedicated	12	6
					M.Eng	2014	UPM					
					B.Tech	2008	LAUTECH					
2.	Engr. Prof. Moses Oluwafemi ONIBONOJE	R30783	Professor	19/10/2015	Ph.D.	2016	OAU	Instrumentation and Control, Distributed Systems, Wireless Sensor Network	12	Dedicated	12	6
					M.Eng	2012	OAU					
					B.Eng	2007	OAU					
					HND/ND	2002	Federal Polytechnic Ado Ekiti					
3.	Engr. Prof. Joseph Olufemi DADA	R5561	Professor	1/10/2016	Ph.D.	2001	Uni. Of Duisburg Essen	Power and Energy Systems	31	Dedicated	6	4
					MEng	2014	Unilorin					
					BEng	1987	Unilorin					

4.	Engr. Dr. Ayodeji Olalekan SALAU	R35396	Associate Professor	15/02/2013	Ph.D. M.Eng B.Eng	2018 2015 2010	OAU OAU FUT(minna)	Communication, Electronics and Computer Vision	12	Dedicated	11	6
5.	Engr. Dr. Joseph Folorunso ORIMOLADE	R12,674	Senior Lecturer	19/04/2024	Ph.D. M.Sc. B.Eng.	(2017) (2004) (1995)	Capetown ABU ABU	Cybersecurity and Communication Systems	15			
6.	Engr. Dr. ADEJUMOBI Babatunde Segun	R56,341	Senior Lecturer	19/10/2020	Ph.D. M. Sc. B. Sc	2021 2016 2012	South Africa LASU Ibadan	Communication Systems				
7.	Engr. Dr. Oluwafunso Oluwole OSALONI	R54620	Senior Lecturer	21/10/2023	Ph.D. M.Sc. B.Sc	2021 2016 2012	Uni. Kwazulu-natal Uni. Of Cape town FUTA	Power and Energy Systems	9	Dedicated	8	6
8.	Engr. Dr. OGUNLADE Michael Adegoke	R35098	Senior Lecturer	19/10/2015	Ph.D. M.Eng B.Sc	2024 2015 2008	ABUAD Tiajun University of technology Uni. Of Ado-Ekiti	Communication Electronics and Computer Vision	10	Dedicated	12	4

9.	Engr. Dr. OMOGBAI Nelson Oyakhilomen	R59849	Lecturer I	19/04/2024	Ph.d M. Sc B.sc	2023 2018 2008	Nnamdi Azikwe University UNIBEN UNIBEN	Power and Machine	2	Dedicated	11	6
10.	Engr. Dr. Oladapo IBITOYE	R57,893	Lecturer I	11/2016	Ph.D M.Eng.	2024 2021	ABUAD ABUAD	Artificial Intelligence and Embedded Systems				
11.	Engr. FABUNMI Temitayo Emmanuel	R60980	Lecturer I	19/12/2019	M.Sc. B.Sc	2021 2016	ABUAD ABUAD	Instrumentation and Control; Electronic Systems	6	Dedicated	10	4
12.	Engr. AKINOLA Akintunde	R78947	Lecturer II	11/05/2023	M.Sc. B.Sc	2016 2010	FUTA University of Ado-Ekiti	Communications and Electronic Systems	8	Dedicated	11	4
13.	Mr. ADEBAJO-AKINPELU Victor	NYR	Assistant Lecturer	19/04/2024	M.eng B.eng	2025 2019	ABUAD	Control and Instrumentation	2	Dedicated	8	6

14.	Engr. AWODOSU A. O.	Registered	Lecturer I							Dedicated		

NB: Sort by Rank ** Taught to other Programme/Degrees*

Annex I-2: List of Shared/Visiting Staff from other Departments/Organizations

S/N	Name	COREN No	Rank	Details of Qualifications			Specialization	Programme / Organization	Credit Hours taught in the Current & Last Semesters	
				Degree	Year	Institution			Current Semester	Last Semester
1	NIL	NIL	NIL				NIL	NIL	NIL	NIL

NB: Names to be sorted by Rank

Annex I-3: List of Full-Time Laboratory Technologists

S/N	Name	COREN #	Rank	Details of Qualifications			Specialization	Date of First Appointment	Laboratory Works Conducted (Contact Hours)	
				Degree	Year	Institution			Current Semester	Last Semester
1.	Mr. Oluwafemi Samuel AGUNBIADE.	R63840	Chief Technologist	M.Eng PGD HND OND	2024 2020 2006 2002	Unical	Power and Machines	3/01/2010	EEE 325	EEE 325
2.	Mr. Akinyele Abiodun ITAMAKINDE	NYR	Principal Technologist	M.eng PGD HND OND	2024 2020 2006 2002	ABUAD National Open Univerisity Federal Polytechnic	Communications	2/11/2011	EEE 314	EEE 311

						Federal Polytechnic Ado Ekiti				
3.	Mr. Elijah Olusegun OMITOKUN	R2876	Senior Technologist	HND ND B.eng	2009 2006 2023	Federal Polytechnic Federal Polytechnic Ado Ekiti Federal University, oye-Ekiti	Measurement and Instrumentaion	23/10/2013	ENG 221	ENG 222
4.	Mr. Adebayo Gabriel KUMUYI	R3257 ET	Senior Technologist	HND ND	2011 2008	LAUTECH	Electronics	1/12013		
5.	Mr. Olayemi Emmanuel OLANIPEKUN	NYR	Technologist II	HND ND	2011 2006	Federal Polytechnic Federal	Power and Machines	21/11/2023	ENG 221	ENG 222

						Polytechnic Ado Ekiti				
6.	Mr. Kolawole Paul OLOGUNDE	NYR	Technologist II	HND ND	2019 2016	Federal Polytechnic Federal Polytechnic Ado Ekiti	Computer Software Technology	17/11/2023		

Annex I-4: Academic Staff Summary

	Staff Teaching Engineering Subjects			
	B.Eng	M.Eng	Ph.D.	TOTAL
Programme Staff (Dedicated)	1	3	10	14
Programme Staff (shared with other programs)	-	-	-	-
Shared Staff (from other programmes)	-	-	-	-
Adjunct Eng. Staff	-	-	-	-
GA / RA	-	-	-	-

Annex I-5: Scenario at the time of Last COREN Visit

	Staff Teaching Engineering Subjects			
	B.Eng	M.Eng	Ph.D.	TOTAL
Program Staff (Dedicated)	-	6	2	10
Shared Staff (from other programs)	-	-	-	-
Adjunct Engineering Staff	-	-	2	2
GA/RA	-	-	-	-

ANNEX J: TEMPLATE FOR SELF-STUDY REPORT (SSR)

ANNEX J: SELF-STUDY REPORT (SSR)

Council for the Regulation of
Engineering in Nigeria (COREN)

Self-Study Report

For the

Electrical/Electronics Engineering

College of Engineering

AFE BABALOLA UNIVERSITY
ADO-EKITI (ABUAD)
KM. 8.5, AFE BABALOLA WAY,
ADO-EKITI, EKITI STATE, NIGERIA.
P.M.B 5454 ADO-EKITI.

Afe Babalola University Ado-Ekiti
Ekiti State

May, 2026.

ANNEXE K: STAFF WORKLOAD FOR FIRST SEMESTER

Staff Workload for First Semester 2024/2025

S/No	Name of Staff	Rank	Course Code	Course Title, Units/Status	Prerequisite	No of Students Taught	Staff Contact Hours			Weekly Contact Hours
							Lecture	Tutorial	Practical	
1	Engr. Dr. A. O OJO	Senior Lecturer	EEE 315 EEE 413 EEE 537	Electromagnetic Field and Wave Theory I Communications System I Antenna Theory and Radar Systems		35 38 21	2	0	0	
2			ENG 221	Basic Electrical Engineering I	Nil	29	2	1	0	

	Engr.Prof. M.O. Onibonoje	Professor	EEE 319 EEE 471 EEE 561	Measurement and Instrumentation Digital Electronics and Systems Control Engineering	ENG 222 EEE 319	35 38 62	2 2	0 0	0 0	
3	Engr. Prof. J. O. Dada	Professor	EEE 431 EEE 553	Electronic Power Principles Power System Engineering	Nil Nil	38 35	2 2	0 0	0 0	
4	Engr. Dr. A.O. Salau	Reader	ENG 341 EEE 511 EEE 531	Software Development Techniques Digital Signal Processing Solid State Electronics	Nil Nil Nil	35 62 62	2 2	0 0	0 0	

5	Engr. Dr. O. O. Osaloni	Senior Lecturer	EEE 325 EEE 555	Electrical Machines I High Voltage Engineering		35 25	2	0	0	
6	Engr. Dr. M. A. Ogunlade	Senior Lecturer	EEE 313 EEE 521	Electronics Engineering I Computer Aided Design and Simulation		35 62	2	0	0	
7	Engr. Dr. N.O. Omogbai	Lecturer I	EEE 565 EEE 553	ModernElectronic Power Principles Power System Engineering		30 23	2	0	0	
8	Engr. T. E.	Lecturer II	ENG 221	Basic Electrical Engineering I		29	2	0	0	

	Fabunmi		EEE 535 COE 501	Microwave and Satellite Communications Artificial Neural Network		17 23	2	0	0	
9	Engr. A. Akinola	Lecturer I	ENG 221	Basic Electrical Engineering I		29	2	0	0	
10	Mr. V. A. Akinpelu	Assistant Lecturer	ENG 221 EEE 319 EEE 471 EEE 561	Basic Electrical Engineering Measurement and Instrumentation Digital Electronics and Systems Control Engineering		29 35 38 62	2	0	0	

Code: P = Professor AP = Associate Professor AST = Assistant Professor I = Instructor A = Adjunct O = Other

ANNEX L: INSTITUTIONAL SUPPORT & FUNDING

ANNEX M: QUALIFYING REQUIREMENTS FOR RESOURCE VERIFICATION, PRE-ACCREDITATION & ACCREDITATION VISITATION

AFE BABALOLA UNIVERSITY
ADO-EKITI (ABUAD)
KM. 8.5, AFE BABALOLA WAY,
ADO-EKITI, EKITI STATE, NIGERIA.
P.M.B 5454 ADO-EKITI.

B. Eng Degree Programme in
Electrical/Electronics Engineering

SELF ASSESSMENT REPORT

ELECTRICAL/ELECTRONICS ENGINEERING

COLLEGE OF ENGINEERING

Submitted to

EDUCATION & TRAINING DEPARTMENT
COUNCIL FOR THE REGULATION OF ENGINEERING IN NIGERIA

November, 2025

AFE BABALOLA UNIVERSITY
ADO-EKITI (ABUAD)
KM. 8.5, AFE BABALOLA WAY,
ADO-EKITI, EKITI STATE, NIGERIA.
P.M.B 5454 ADO-EKITI.

B. Eng. Degree Programme in
Electrical/Electronics Engineering

Please tick: Accreditation √
Approval of New Programme

Subject: SAR for the Programme of Bachelor of Engineering in Electrical/Electronics Engineering

1. The requirements as per the Check List below to qualify for the process of accreditation under the COREN OBE Manual of Accreditation-2019 have been addressed / Verified:

S/N	Qualifying Requirement	Institution Check / Remarks	COREN Check Remarks
1	A minimum of 160 credit units of which 85 credit units must be core engineering courses offered over a	√	
2	Final year project (minimum 6 credit hours)	√	
3	Industrial training /SWEP	√	
4	Full-time engineering faculty (minimum	√	
5	Teaching Staff: student ratio of 1:15 minimum	√	
6	External examiner's report	√	
7	Programme Educational Objectives	√	
8	Programme Outcomes	√	

2. The Self-Assessment Report (SAR) is hereby submitted for consideration of E & T Department, COREN to process for accreditation of the programme of Electrical/Electronic Engineering.

Signature:_____ Signature:_____

(Head of Department) (Dean/Head of the Institution)

Date: Date

ANNEX N: CHECKLIST OF DOCUMENT FOR RESOURCE VERIFICATION, PRE-ACCREDITATION & ACCREDITATION VISITATION

ANNEX N: CHECKLIST OF DOCUMENTS FOR RESOURCE VERIFICATION, PRE- ACCREDITATION AND ACCREDITATION VISITATION

Annex N-1: General Information

S/N	Refer to Section 4.2	To be filled out by the institution	Checked by
1	Name of the institution	Afe Babalola University Ado-Ekiti	
2	Address of institution	KM 8.5, Afe Babalola Way, P.M.B 5454, Ado-Ekiti. Ado-Ekiti, Nigeria.	
3	Name of Faculty/School/Department	College of Engineering	
4	Name and phone number of Staff to be contacted	Engr Dr. A. O. Ojo 08109976284	
5	Programme for Accreditation	Electrical/Electronics Engineering	
6	COREN Reference Number	R33,712	
7	Degree to be Awarded and Abbreviation	Bachelor of Engineering (B. Eng.)	
8	Duration of Programme (in years)	5 years	
9	Institution Academic Session	2025/2026	
10	URL Address; institution website	www.abuad.edu.ng	

Annex N-2: History of Programme Accreditation

S/N	Refer to Section 4.2	To be filled out by the institution	Checked by Evaluator
1	Introduction Year of Programme	2010	
2	Year of Last Accreditation for this Programme	2021	

3	Decision (if any) from Previous Accreditation	Full accreditation (88%)	
4	Action taken on the decision above	N/A	
5	Major Changes (Self-Initiated) Reasons and Year of Changes	Staff development has been improved significantly, and the laboratories are now under the supervision of Lecturers.	

Annex N-3: Criterion 1 - Programme Educational Objectives (PEOs)

S/N	Refer to Sections 2.1	Indicate the location of these items in the submitted SSR	Checked by Evaluator
1	State the Vision and mission of the institution and/or faculty/Programme	Page 9	
2	List the PEOs and state where they are published and publicized	Page 24	
3	Describe how the PEOs are consistent with the Vision and mission of the institution and or faculty and stakeholders'	Page 25	
4	Describe the processes used to evaluate the level of achievement of the PEOs This includes describing graduates/alumni database, tools (survey, meetings, interviews, etc.) and frequency of activities and timelines.	Page 21	
5	Discuss the PEOs achievement results by the graduates/alumni.	Page 22	
6	Describe how the feedback and results obtained from the above processes are being used for the CQI of the programme.	Page 23	
7	Describe the extent to which the programme's various stakeholders are involved in these processes	Page 19	
8	Describe CQI strategies to be implemented in relation to PEOs	Page 23	

9	Self-assessment on programme performance related to PEOs based on the following scale (with justifications): *Poor/Satisfactory/Good	Good	

Annex N-4: Criterion 2 - Programme Outcomes (POs)

S/N	Refer to Sections 3.1	Indicate the location of these items in the submitted SSR	Checked by Evaluator
1	List the POs and state where they are published	Page 35	
2	Describe how the POs relate to the PEOs	Page 37	
3	Describe how the POs listed encompass and consistent with the 12 COREN POs	Page 35	
4	Describe the PO definition or elements/performance indicators	Page 38	
5	Describe the processes used to establish and review the POs, and the extent to which the programmes Various stakeholders are involved in these processes (where applicable). This includes describing the tools used in the processes (survey, meetings, interviews, etc.) and frequency of activities and timelines.	Page 39	
6	Describe the mapping of courses with POs	Page 39	
7	Explain how the assessment results are applied to further develop and improve the POs	Page 40	
8	Describe the materials, including student work and other evidence, that demonstrate achievement of the POs	Page 40	
9	Describe the extent to which the programme's Various stakeholders are involved in the processes	Page 40-41	
10	Describe CQI strategies to be implemented in relation to POs	Page 41	
11	Self-assessment on programme performance related to POs based on the following scale (with justifications): *Poor/Satisfactory/Good	Good	

Annex N-5: Criterion 3 - Course Learning Outcomes (CLOs)

S/N	Refer to Sections 4.1	Indicate the location of these items in the submitted SSR	Checked by Evaluator
1	List the CLOs and state where they are published	Page 47	
2	Mapping of CLOs to related POs (as per template given in **Annex B-2)**	Pages 48	
3	Describe the relationship between the CLOs and the POs	Pages 39-40	
4	Self-assess on programme performance related to CLOs based on the following scale (with justifications): *Poor/Satisfactory/Good	Good	

Annex N-6: Criterion 4 - Curriculum and Learning Process

S/N	Refer to Sections s.1	Indicate the location of these items in the	Checked by
1	Discuss the programme structure and course contents to show how they are appropriate to, consistent with, and support the development of the range of intellectual and practical skills and attainment or achievement of the POs	Page 152	
2	Discuss the programme delivery and assessment methods and how these are appropriate to, consistent with, and support the development of the range of intellectual and practical skills and attainment or achievement of the POs	Page 152	
3	Provide evidence of the use of tutorials and non- conventional delivery methods such as Problem Based Learning (PBL) techniques alongside traditional lectures	52	

4	Provide evidence that students performed at least 10 laboratory practicals per semester (for at least 5 semesters from 300 level)	Page 524	
5	Provide evidence that students performed at least 10 engineering drawing assignments per semester in 200 level	Page 525	
6	Describe how the requirements of Complex Problem Solving (CPS) and Complex Engineering Activities (CEA) have been addressed	Page 292	
7	A matrix linking courses to POs to identify and track the contribution of each course to the POs (as per template given in **Annex-D**)	Page 471	
8	Distribution of the engineering courses according to areas specific to each programme (as per template given in **Annex-E**)	Page 247, 482	
9	Distribution of the related non-engineering (general education) courses	Page 247, 482	
10	Distribution of the courses offered according to semester (as per template given in **Annex-F**)	Page 485	
11	Details of Laboratory equipment / workstations and experiments conducted (as per template given in **Annex-G**)	Page 422, 492	
12	Self-assess on programme performance related to Curriculum Learning Process based on the following scale (with justifications): *Poor/Satisfactory/Good	Good	

Annex N-7: Criterion 5 - Students

S/N	Refer to Sections 3.2.s and 4.7	Indicate the location of these items in the submitted SSR	Checked by Evaluator
1	Discuss the requirement and process for	Page 17	

	admission of students to the programme, response and annual intake (in addition to template given in **Annex-H**)		
2	Discuss the policies and processes for students transfer and credit transfer/exemption	Page 17	
3	Discuss mechanism for providing guidance to students on academic, career and aspects pertaining to wellness.	Page 20	
4	Discuss students' workload, class sizes for theory as well as laboratory sessions and completion of courses	Page 20	
5	Describe formal or informal feedback platform/channel to obtain students feedback and suggestions for further programme improvement, and how have the feedback resulted in programme improvement	Page 19	
6	Summarize the graduation requirements for the programme, the process for ensuring and documenting that each graduate completes all graduation requirements for the programme (as per template given in **Annex M**).	Page 18, 300	
7	Describe CQI strategies to be implemented in relation to Students	Page 18, 300	
8	Self-assess on programme performance related to Students based on the following scale (with justifications): *Poor/Satisfactory/Good	Good	

Annex N-8: Criterion 6 - Continuous Quality Improvement (CQI)

	Refer to Sections 3.2.6 and 4.8	dicate the location of these items in the submitted SSR	Checked by Evaluator
1	Discuss the mechanism for: programme planning; curriculum development; curriculum and content review; responding to feedback and inputs from stakeholders including industrial advisors, students and alumni; tracking the contribution of individual courses to POs; tracking outcomes of performance through assessment, including rubrics; reviewing of PEOs and POs; and continuous quality improvement	Page 32, 39, 307	
2	Discuss the implementation plan based on the observations of the last accreditation Visit and the remedial actions taken	Page 21, 42	
3	Evidence on the participation of faculty members and support staff as well as students in the continuous quality improvement process	Page 33, 305, 307	
4	Evidence on the development of academic staff through opportunities in further education, industrial exposure, as well as research and development	Page 382	
5	Policies, internal processes and practices that are in place at all levels within the institution relating to the accreditation criteria as stated in Chapter 3 of this Manual	Page 42	
6	Summarize responses to the external examiner's report	Page 313, 533-556	

7	Discuss how the quality management system of the institution provides quality assurance and benchmarking with renowned national/international universities offering similar programme	Page 42, 148	
8	Evidence of the on-going participation of industrial advisors in discussions and forums, professional practice exposure, and collaborative projects	Page 27, 44	
9	Provide at least ten (10) employers' feedback report on the performance of students who graduated within the last five (5) years	Page 308-313	
10	Self-assessment on programme performance related to CQI based on the following scale (with justifications): *Poor/Satisfactory/Good	Good	

Annex N-9: Criterion 7 - Staffing

S/N		Indicate the location of these items in the submitted SSR	Checked by Evaluator
1	Discuss the strength and competencies of the academic staff in covering all areas of the programme, and in implementing the outcome- based approach to education (as per template given in **Annexes-I - K**)	Page 43, 361, 500-503	

2	Discuss how the overall staff workload enables effective teaching (including student-teacher ratio), student-staff interaction, student advising and counselling, institutional service and research activities, professional development and interaction with industry	Page 363	
3	Discuss processes for faculty development, training and retention. Produce retention index of staff of the programme	Page 356, 382	
4	Describe the role played by the faculty with respect to course creation, modification, and evaluation, their role in the definition and revision of Programme Educational Objectives and Programme Outcomes, and their role in the attainment of the Programme Outcomes. Describe the roles of others on campus, e.g., dean or provost, with respect to these areas	Page 14	
5	Discuss the sufficiency and competency of technical and administrative staff in providing adequate support to the educational programme	Page 369	
6	Produce evidence of how many staff has at least 3 publications in ISI-indexed journals in 5 years	Page 386	
7	Self-assess on programme performance related to Staffing based on the following scale (with justifications): *Poor/Satisfactory/Good	Good	

Annex N-10: Criterion 8 - Physical Facilities and Infrastructures

S/N	Refer to Sections 3.2.8 and 4.10	Indicate the location of these items in the submitted SSR	Checked by Evaluator
1	Discuss the adequacy of teaching and learning facilities such as classrooms, learning-support facilities, study areas, information resources (library), computing and information-technology systems, laboratories and workshops, and associated equipment to cater for multi-delivery modes	Page 43, 154. 414	
2	Describe the adequacy of support facilities such as hostels, sport and recreational centers, health centers, student centers, power and internet Services, fire/security services, and transport in facilitating students' life on campus and enhancing character building	Page 305, 414	
3	A summary, in tabulated form, of the lecture facilities (give number, capacity, and audio/video facilities available)	Page 414	
4	A summary, in tabulated form, of the laboratories (list down the details of workstation available in each laboratory).	Page 493-496	
5	A summary, in tabulated form, of the workshops/drawing studio (list down the equipment/machinery available in each workshop / drawing studio)		

6	A summary, in tabulated form, of the computer laboratories (list down the hardware and software available)	Page 417	
7	A summary, in tabulated form, of recreational facilities	Page 502	
8	A summary, in tabulated form, of information on recent improvements and planned improvements in facilities	Page 21	
9	Self-assess on programme performance related to Physical Facilities and Infrastructures based on the following scale (with justifications): *Poor/Satisfactory/Good	Good	

Annex N-11: Criterion 9 - Industrial Linkage and Community Services

S/N	Refer to Sections 3.2.9 and 4.11	Indicate the location of these items in the submitted SSR	Checked by Evaluator
1	Discuss the involvement of industry in discussions and fora, professional practice exposure, and collaborative projects/research for the solutions to engineering problems	Page 44, 156	
2	Discuss students' activities and involvement in student organizations that provide experience in management and governance, representation in education and related matters and social activities	Page 304	
3	Self-assess on programme performance related to Industrial Linkages and Community Services based on the following scale (with justifications): *Poor/Satisfactory/Good	Good	

Annex N-12: Criterion 10 - Institutional Support and Funding

S/N	Refer to Sections 3.2.10 and 4.12	Indicate the location of these items in the submitted SSR	Checked by Evaluator
1	Discuss institution's financial commitment and support to sustain and enhance the quality of programme. Also summarize the salient features in a tabular form (as per the template given in **Annex-L**)		
2	Self-assess on programme performance related to Institutional Support and Funding based on the following scale (with justifications): *Poor/Satisfactory/Good		

ANNEX O: EXTERNAL EXAMINER'S REPORT

Engr. Prof. POPOOLA, Jide Julius PhD. (MNSE, MIEEE)
Department of Electrical and Electronics Engineering
School of Engineering and Engineering Technology
The Federal University of Technology, PMB 704, Akure

7th July, 2025.

The Vice Chancellor

Afe Babalola University,

Ado-Ekiti,

Ekiti State.

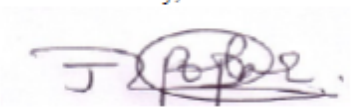
Dear Sir/Ma,

REPORT ON MODERATIONS OF BOTH 2024/2025 FIRST AND SECOND SEMESTERS EXAMINATIONS FOR BACHELOR OF ENGINEERING (B.ENG.) IN ELECTRICAL/ ELECTRONICS ENGINEERING PROGRAMME

I hereby submit reports of the moderations exercises for 2024/2025 First and Second Semesters Examinations for Electrical and Electronics Engineering Programme (B. Eng. Electrical and Electronics Engineering) in the Department of Electrical/Electronics and Computer Engineering of your University, Afe Babalola University. The report also covers the physical oral examination or project defence conducted for the 500 Level Electrical and Electronics Engineering students on Thursday June 26, 2028.

Please accept my sincere appreciation for the opportunity to serve the University in this capacity.

Yours faithfully,



7th July, 2025.

Engr. Prof. Jide Julius Popoola

Email: jjpopoola@futa.edu.ng; jidejulius2001@gmail.com Phone: +234(0)-803-413-1860

REPORT ON MODERATION OF 2024/2025 FIRST SEMESTER FINAL YEAR (500 LEVEL) EXAMINATION QUESTIONS FOR BACHELOR OF ENGINEERING (B. ENG.) ELECTRICAL AND ELECTRONICS ENGINEERING PROGRAMME

1.0 Introduction

With respect to my appointment as External Examiner to the Bachelor of Engineering (B. Eng) Electrical and Electronics Engineering programme of Afe Babalola University, Ado-Ekiti for the First Semester of 2024/2025 Academic Session, I received for moderation, the examination questions and marking schemes/guides for the following courses from the Head of Department of Electrical/Electronics and Computer Engineering:

EEE 507: Modelling and Simulations of Electrical/Electronics

EEE 509: Advance Electronics

EEE 511: Digital Signal Processing

EEE 513: Communication Systems Engineering

EEE 523: Industrial Electronics Design

EEE 535: High Frequency and Microwave Electronics

EEE 553: Power Systems Engineering I

EEE 555: High Voltage Engineering

EEE 561: Control Engineering

EEE 565: Modern Control Engineering

After I have carefully gone through each of the examination questions and marking schemes, some amendments, suggestions and modifications were made. The moderated questions have been duly forwarded to the Head of Department. Presented herein is a report on the moderation process and observations made.

2.0 Number of Questions

The number of examination questions for all the courses were adequate and in accordance to both National Universities Commission (NUC) and Council for Regulation of Engineering in Nigeria (COREN) standards.

3.0 Time Allowed for Examination

The time allowed for each of the examination was adequate. Most of the questions followed the standardized model of either three hours for three (3) credit unit courses where students are to answer five (5) questions out of seven (7) questions or two and half (2½) hours for examination with six (6) questions when students are to answer four (4) questions. Similarly, a standardized model of two hours for two (2) credit unit courses where students are to answer four (4) questions out of six (6) questions was followed.

4.0 Clarity of Instructions

Instructions for the examinations were clearly defined for all the courses.

5.0 Clarity of Questions

It is commendable that all the examination questions were very clear.

6.0 Grammatical Expressions

The grammatical constructions in all the examination questions are good and standard. Only a few typographical errors were observed and corrected.

7.0 Coverage of Course Syllabus

The course syllabi were adequately covered but there is still room for improvement in this area.

8.0 Level of Cognitive Learning Objective

As engineering students, it is important that the minds of the students are tasked for them to achieve a very high level of cognitive capacity, especially in application of knowledge. It was suggested that more applied or application questions are included at this level to adequately assess the level of cognitive objectives attained by the students. Outcome Based Education, the new Engineering Education methodology approved by the COREN should be followed by the Department.

9.0 Allocation of Marks

The allocation of marks for the questions was judicious, though there were few lumped marks.

10.0 Marking Scheme or Guide Clarity

The marking schemes or guides prepared and submitted with the examination questions were good and all reflected what were expected from the students. Marks were well distributed on most of the prepared marking guides except in one or two where the mark were lump sum.

11.0 Ease of Assessment

The Department is advised to further intimate examiners of the importance of writing marking guides that any other qualified academic will be able to use.

12.0 Summary of Assessment of Moderated Examination Questions

Table 1 presents summary of my objective assessment of the examination questions and marking guides submitted for moderation.

13.0 Conclusion

I want to thank the Vice-Chancellor and the Management of Afe Babalola University, Ado-Ekiti for the opportunity giving me to render my services to the University as an External Examiner to Electrical and Electronics Engineering programme in the Department of Electrical/Electronics and Computer Engineering for 2024/2025 Academic Session.

Table 1: Courses moderation and Comments

S/N	Course Code and Course Title	Comment
1.	EEE 507: Modelling and Simulations of Electrical /Electronics	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • The questions were well-distributed and covered over 85% of the course synopsis. • Marks distributions on the question paper were properly done. • The marking guide indeed reflects what is expected from the students. • Some of the questions are creative questions, which is highly commendable at 500 level
2.	EEE 509: Advance Electronics	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • The questions were well-structured and well-distributed. • The questions cover more than 85% of the course synopsis • The marking guide/scheme was well prepared and reflects what is expected from the students.
3.	EEE 511: Digital Signal Processing	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • The questions were well-distributed with the total marks written on each question. • The marking scheme/guide with the question is well-prepared and reflected what is expected from the students. • The questions are creative questions, which is highly commendable at 500 level. In actual fact, the question is professionally sound and well-prepared.
4.	EEE 513: Communication Systems Engineering	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • Some of the questions were creative questions, which are commendable at 500 Level.

		• The questions were well distributed. Also, the mark distributions on the question paper were properly done. • The marking scheme that followed the question paper indeed was well-prepared and reflected what is expected from the students.
5.	EEE 523: Industrial Electronics Design	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • The questions were well-distributed. Also, mark distribution on the question paper was properly done • Some of the questions were practical and creative questions, which are commendable at 500 Level. • The questions were appropriate, well-structured and well distributed across the course synopsis or contents. • The marking scheme/guide was well prepared and reflected what is expected from the students.
6.	EEE 535: High Frequency and Microwave Electronics	• Appropriate number of questions were set in accordance to both NUC and COREN standards • Some of the questions were creative questions, which is highly commendable at 500 Level • The questions were well-distributed. Also, mark distribution on the question paper was properly done. • The marking guide or scheme actually reflects what is expected from the student and well-prepared.
7.	EEE 553: Power Systems Engineering I	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • The questions creative questions, which are commendable at 500 Level. • The questions were appropriate, well-structured, properly-done and professionally sound. • The questions were well distributed and cover the entire course synopsis. • The marking guide was well prepared with mark distributions on the entire questions. • The marking scheme or guide reflected what is reflected from the students and perfectly prepared.
8.	EEE 555: High Voltage Engineering	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • The questions were well distributed. Also, mark distribution on the question paper was properly done. • The questions were appropriate and well-structured. • Some of the questions are creative questions, which are commendable at 500 Level. • The questions were well distributed and cover more than 85% of the course synopsis..
9.	EEE 561: Control Engineering	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • Some of the questions are creative questions, which

		is commendable at 500 Level. • In actual fact, the question paper was well prepared and well-structured. • The marking scheme/guide was well-prepared and reflected what is expected from the students. • The questions were well-distributed and cover more than 85% of the course synopsis.
10.	EEE 565: Modern Control Engineering	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • Some of the questions are practical and creative questions, which is commendable at 500 Level. • The marking scheme/guide was well-prepared and reflected what is expected from the students.

REPORT ON MODERATION OF 2024/2025 SECOND SEMESTER FINAL YEAR (500 LEVEL) EXAMINATION QUESTIONS AND ORAL/PROJECT DEFENCE EXAMINATION FOR BACHELOR OF ENGINEERING (B. ENG.) ELECTRICAL AND ELECTRONICS ENGINEERING PROGRAMME

1.0 Introduction

With respect to my appointment as External Examiner to the Bachelor of Engineering (B. Eng) Electrical and Electronics Engineering programme of Afe Babalola University, Ado-Ekiti for the Second Semester of 2024/2025 Academic Session, I received for moderation, the examination questions for the following courses from the Head of Department of Electrical/Electronics and Computer Engineering:

EEE 502: Electrical Service and Design

EEE 512: Digital Communications

EEE 514: Reliability Engineering

EEE 524: Analog IC Design and Applications

EEE 554: Power Systems Engineering II

EEE 556: Electrical Drives and Applications

EEE 558: Energy Conversion and Storage

EEE 562: Power Electronics

After I have carefully gone through each of the examination questions and marking schemes, some amendments, suggestions and modifications were made. The moderated questions have been duly forwarded to the Head of Department. Presented herein is a report on the moderation process and observations made.

2.0 Number of Questions

The number of examination questions for all the courses were adequate and in accordance to both National Universities Commission (NUC) and Council for Regulation of Engineering in Nigeria (COREN) standards.

3.0 Time Allowed for Examination

The time allowed for each of the examination was adequate. Most of the questions followed the standardized model of either three hours for three (3) credit unit courses where students are to answer five (5) questions out of seven (7) questions or two and half (2½) hours for examination with six (6) questions and students are to answer four (4) questions. Similarly, a standardized model of two hours for two (2) credit unit courses where students are to answer four (4) questions out of six (6) questions was followed.

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Instructions for the examinations were clearly defined for all the courses.

5.0 Clarity of Questions

It is commendable that all the examination questions were very clear.

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The course syllabi were adequately covered but there is still room for improvement in this area.

8.0 Level of Cognitive Learning Objective

As engineering students, it is important that the minds of the students are tasked for them to achieve a very high level of cognitive capacity, especially in application of knowledge. It was suggested that more applied questions are included at this level to adequately assess the level of cognitive objectives attained by the students. Outcome Based Education, the new Engineering Education methodology approved by the COREN should be followed by the Department.

9.0 Allocation of Marks

The allocation of marks for the questions was judicious, though there were few lumped marks.

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The marking schemes or guides prepared and submitted with the question papers were good and all reflected what were expected from the students. Marks were well distributed on the prepared marking guides except in one or two where the mark distribution was lump sum.

11.0 Ease of Assessment

The Department is advised to further intimate examiners of the importance of writing marking guides that any other qualified academic will be able to use.

12.0 Summary of Assessment of Moderated Examination Questions

Table 1 presents summary of my objective assessment of the examination questions and marking guides submitted for moderation.

13.0 Projects and Project Reports

An oral examination was carried out on final year project s and project reports of sixty-two (62) students. The diversity of the project titles or topics carried out by the students was impressive, and it covers various bodies of knowledge areas in Electrical and Electronics Engineering. Each student presented

his/her project confidently with the implementation skilfully demonstrated. This showed good understanding of the discipline and its application to addressing real life problems. The project reports of the students were also very good showing the painstaking efforts of the project supervisors to improve the quality of the work. In the project reports, minor corrections such as poor citation and referencing style as well as lexical and typographical errors were observed, which were pointed out for correction. The students' composure and response during questions section are worthy of note. Indeed, all the students demonstrated good understanding of their projects and presented very well. Majority of them responded very well to most of the questions posted to them. This indicates that they were both academically and practically sound. Without any doubt, their performances is really encouraging and commendable.

14.0 Conclusion

I want to thank the Vice-Chancellor and the Management of Afe Babalola University, Ado-Ekiti for the opportunity giving me to render my services to the University as an External Examiner to Electrical and Electronics Engineering programme in the Department of Electrical/Electronics and Computer Engineering for 2024/2025 Academic Session.

Table 1: Courses moderation and Comments

S/N	Course Code and Course Title	Comment
1.	EEE 502: Electrical Services and Design	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • The questions are practical and creative questions, which are indeed commendable at 500 Levels. • Mark distribution on the question paper was properly done. • The mark distribution on the marking scheme was ideal and properly done. • The marking scheme reflects what is expected from the students. Also, the question and the marking scheme are well prepared.
2.	EEE 512: Digital Communications	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • The questions were creative questions, which are commendable at 500 Level • The questions were well distributed. • The marking scheme reflects what is expected from the students.
3.	EEE 514: Reliability Engineering	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • Some of questions were practical and creative questions, which are indeed commendable at 500 Levels . • The question and marking scheme were well-prepared. • Mark distribution on the question paper was properly

		done. • The marking scheme reflects what is expected from the students
4.	EEE 524: Analog IC Design and Applications	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • The questions were well distributed. • Some of the questions were creative questions, which indeed are commendable at 500 level • Mark distribution on both the question paper and the marking scheme were properly done. • The marking scheme reflected what is expected from the students.
5.	EEE 554: Power Systems Engineering II	• Appropriate number of questions were set in accordance to both NUC and COREN standards • The questions were creative questions, which are commendable at 500 Level. • The questions were properly done and professionally sound. • The questions were well distributed. Also, mark distribution on both the question and marking scheme were properly done. • The marking scheme reflected what is expected from the students.
6.	EEE 556: Electrical Drives and Applications	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • The questions were well distributed. • Marks distributions on the question were perfectly done. However, the marks distributions on the marking scheme were lump. • The marking scheme reflected what is expected from the students.
7.	EEE 558: Energy Conversion and Storage	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • Marks distributions on the question were perfectly done. However, the marks distributions on the marking scheme were lump. • The marking scheme reflected what is expected from the students. • Some of the questions were creative questions, which are commendable at 500 Level. • The questions were well structured.
8.	EEE 562: Power	• Appropriate numbers of questions were set in

	Electronics	accordance to both NUC and COREN standards. • Marks distributions on the question were perfectly done. Likewise, marks distributions on the marking scheme were perfectly done. • The marking scheme reflected what is expected from the students. • Some of the questions were creative questions, which are indeed commendable at 500 Level. • The questions were well structured.
9.	EEE 566: Control Engineering III	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • The questions were well distributed. • Mark distributions on the question paper were lump sum. However, the marks distributions on the marking scheme were perfectly done. • The marking scheme reflected what is expected from the students. • The questions were well structured.

Engr. Prof. POPOOLA, Jide Julius PhD. (MNSE, MIEEE)
Department of Electrical and Electronics Engineering
School of Engineering and Engineering Technology
The Federal University of Technology, PMB 704, Akure

7th June, 2024.

The Vice Chancellor

Afe Babalola University,

Ado-Ekiti,

Ekiti State.

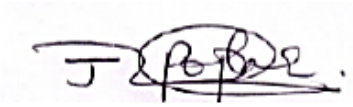
Dear Sir/Ma,

REPORTS ON MODERATIONS OF BOTH 2023/2024 FIRST AND SECOND SEMESTERS EXAMINATIONS FOR BACHELOR OF ENGINEERING (B.ENG.) IN ELECTRICAL/ ELECTRONICS ENGINEERING PROGRAMME

I hereby submit reports of the moderations exercises for 2023/2024 First and Second Semesters examinations for Electrical and Electronics Engineering Programme (B. Eng. Electrical and Electronics Engineering) in the Department of Electrical/Electronics and Computer Engineering of your University, Afe Babalola University.

Please accept my sincere appreciation for the opportunity to serve the University in this regard.

Yours faithfully,


7th June, 2024.

Engr. Prof. Jide Julius Popoola

Email: jjpopoola@futa.edu.ng; jidejulius2001@gmail.com Phone: +234(0)-803-413-1860

REPORT ON MODERATION OF 2023/2024 FIRST SEMESTER FINAL YEAR (500 LEVEL) EXAMINATION QUESTIONS FOR BACHELOR OF ENGINEERING (B. ENG.) ELECTRICAL AND ELECTRONICS ENGINEERING PROGRAMME

1.0 Introduction

With respect to my appointment as External Examiner to the Bachelor of Engineering (B. Eng) Electrical and Electronics Engineering programme of Afe Babalola University, Ado-Ekiti for the First Semester of 2023/2024 Academic Session, I received for moderation, the examination questions and marking schemes/guides for the following courses from the Head of Department of Electrical/Electronics and Computer Engineering:

EEE 511: Digital Signal Processing

EEE 521: Computer Aided Design and Simulation

EEE 531: Solid-State Electronics

EEE 535: Microwave and Satellite Communication

EEE 537: Antenna and Radar Systems

EEE 541: Reliability Engineering

EEE 551: Power Electronics and Drives

EEE 553: Power Systems Engineering I

EEE 555: High Voltage Engineering

EEE 561: Process Control and Automation

After I have carefully gone through each of the examination questions and marking schemes, some amendments, suggestions and modifications were made. The moderated questions have been duly forwarded to the Head of Department. Presented herein is a report on the moderation process and observations made.

2.0 Number of Questions

The number of examination questions for all the courses were adequate and in accordance to both National Universities Commission (NUC) and Council for Regulation of Engineering in Nigeria (COREN) standards.

3.0 Time Allowed for Examination

The time allowed for each of the examination was adequate. Most of the questions followed the standardized model of either three hours for three (3) credit unit courses where students are to answer five (5) questions out of seven (7) questions or two and half (2½) hours for examination with six (6) questions and students are to answer four (4) questions. Similarly, a standardized model of two hours for two (2) credit unit courses where students are to answer four (4) questions out of six (6) questions was followed.

4.0 Clarity of Instructions

Instructions for the examinations were clearly defined for all the courses.

5.0 Clarity of Questions

It is commendable that all the examination questions were very clear.

6.0 Grammatical Expressions

The grammatical constructions in all the examination questions are good and standard. Only a few typographical errors were observed and corrected.

7.0 Coverage of Course Syllabus

The course syllabi were adequately covered but there is still room for improvement in this area.

8.0 Level of Cognitive Learning Objective

As engineering students, it is important that the minds of the students are tasked for them to achieve a very high level of cognitive capacity, especially in application of knowledge. It was suggested that more applied or application questions are included at this level to adequately assess the level of cognitive objectives attained by the students. Outcome Based Education, the new Engineering Education methodology approved by the COREN should be adopted by the Department.

9.0 Allocation of Marks

The allocation of marks for the questions was judicious, though there were few lumped marks.

10.0 Marking Scheme or Guide Clarity

The marking schemes or guides prepared and submitted with the examination questions were good and largely reflected what were expected from the students. Marks were well distributed on most of the prepared marking guides except in one or two where there were no mark distributions.

11.0 Ease of Assessment

The Department is advised to further intimate examiners of the importance of writing marking guides that any other qualified academic will be able to use.

12.0 Summary of Assessment of Moderated Examination Questions

Table 1 presents summary of my objective assessment of the examination questions and marking guides submitted for moderation.

13.0 Conclusion

I want to thank the Vice-Chancellor and the Management of Afe Babalola University, Ado-Ekiti for the opportunity giving me to render my services to the University as an External Examiner to Electrical and Electronics Engineering programme in the Department of Electrical/Electronics and Computer Engineering for 2023/2024 Academic Session.

Table 1: Courses moderation and Comments

S/N	Course Code and Course Title	Comment
1.	EEE 511: Digital Signal Processing	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • The questions were well-distributed and covered substantive parts of the course synopsis. • Marks on the question paper were well-distributed but the marks on the marking scheme or guide were not well distributed on few of the questions. • The marking guide indeed reflects what is expected from the students.
2.	EEE 521: Computer Aided Design and Simulation	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • The questions were practical and creative questions, which are commendable at 500 Level. • The questions were appropriate and well-structured. • The questions were well distributed and cover more than 85% of the course synopsis • The marking guide was well prepared. However, while mark distributions were indicated in section A of the marking scheme, mark distribution were not indicated in section B
3.	EEE 531: Solid-State Electronics	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • The questions were creative questions, which are commendable at 500 Level • The questions were appropriate and well-structured. The questions cover more than 75% of the course synopsis • The marks allocated were well distributed on all the questions except only on Question 6(a) where the mark distribution was omitted.
4.	EEE 535: Microwave and Satellite Communication	• Some of the questions were practical and creative questions, which are commendable at 500 Level. • The questions and the marking guide or scheme were

		well prepared. • The questions cover more than 75% of the synopsis.
5.	EEE 537: Antenna and Radar Systems	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • Some of the questions were practical and creative questions, which are commendable at 500 Level. • The questions were appropriate, well-structured and well distributed across the course synopsis or contents. • The marking scheme/guide was well prepared except that marks allocation in some questions were lumped and bulk
6.	EEE 541: Reliability Engineering	• Appropriate number of questions were set in accordance to both NUC and COREN standards • The questions were practical and creative questions, which are commendable at 500 Level • The questions were well-structured and professionally sound. • The question and the marking guide were well-prepared. • Marks on both the question paper and marking guide were well-distributed. • The questions covered over 85% of the course synopsis
7.	EEE 551: Power Electronics and Drives	• Appropriate number of questions were set in accordance to both NUC and COREN standards • The questions were well-distributed and cover more than 85% of the course synopsis. • The question and the marking guide were well prepared. • Some of the questions were applied and creative questions, which are commendable at 500 Level.
8.	EEE 553: Power Systems Engineering I	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • The questions were practical and creative questions, which are commendable at 500 Level. • The questions were appropriate, well-structured and professionally sound/prepared. • The questions were well distributed and cover the entire course synopsis. • The marking guide was well prepared with mark distributions on the entire questions.
9.	EEE 555: High Voltage Engineering	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • The questions were practical and creative questions, which are commendable at 500 Level. • The questions were appropriate and well-structured.

		• The questions were well distributed and cover more than 85% of the course synopsis. • The marking guide was well prepared but the mark distributions were lumped and in bulk.
10.	EEE 561: Process Control and Automation	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • Some of the questions were practical and creative questions, which are commendable at 500 Level. • The marking scheme/guide was well written except that marks allocations in some questions were lumped and bulk. • The questions were well-distributed and cover more than 85% of the course synopsis.

REPORT ON MODERATION OF 2023/2024 SECOND SEMESTER FINAL YEAR (500 LEVEL) EXAMINATION QUESTIONS AND ORAL/PROJECT DEFENCE EXAMINATION FOR BACHELOR OF ENGINEERING (B. ENG.) ELECTRICAL AND ELECTRONICS ENGINEERING PROGRAMME

1.0 Introduction

With respect to my appointment as External Examiner to the Bachelor of Engineering (B. Eng) Electrical and Electronics Engineering programme of Afe Babalola University, Ado-Ekiti for the Second Semester of 2023/2024 Academic Session, I received for moderation, the examination questions for the following courses from the Head of Department of Electrical/Electronics and Computer Engineering:

EEE 512: Digital Communications

EEE 522: Microprocessor Application, Organization and Embedded Systems

EEE 524: Analog IC Design and Application

EEE 536: Wireless Communications

EEE 552: Industrial Utilization of Electric Power

EEE 554: Power Systems Engineering II

EEE 556: Machine Applications

EEE 564: Current Trends in Electrical/Electronic Engineering

EEE 566: Introduction to Non-Linear Systems

EEE 576: Microcomputer Technology

After I have carefully gone through each of the examination questions and marking schemes, some amendments, suggestions and modifications were made. The moderated questions have been duly forwarded to the Head of Department. Presented herein is a report on the moderation process and observations made.

2.0 Number of Questions

The number of examination questions for all the courses were adequate and in accordance to both National Universities Commission (NUC) and Council for Regulation of Engineering in Nigeria (COREN) standards.

3.0 Time Allowed for Examination

The time allowed for each of the examination was adequate. Most of the questions followed the standardized model of either three hours for three (3) credit unit courses where students are to answer five (5) questions out of seven (7) questions or two and half (2½) hours for examination with six (6) questions and students are to answer four (4) questions. Similarly, a standardized model of two hours for two (2) credit unit courses where students are to answer four (4) questions out of six (6) questions was followed.

4.0 Clarity of Instructions

Instructions for the examinations were clearly defined for all the courses.

5.0 Clarity of Questions

It is commendable that all the examination questions were very clear.

6.0 Grammatical Expressions

The grammatical constructions in all the examination questions are good and standard. Only a few typographical errors were observed and corrected.

7.0 Coverage of Course Syllabus

The course syllabi were adequately covered but there is still room for improvement in this area.

8.0 Level of Cognitive Learning Objective

As engineering students, it is important that the minds of the students are tasked for them to achieve a very high level of cognitive capacity, especially in application of knowledge. It was suggested that more applied or application questions are included at this level to adequately assess the level of cognitive objectives attained by the students. Outcome Based Education, the new Engineering Education methodology approved by the COREN should be adopted by the Department.

9.0 Allocation of Marks

The allocation of marks for the questions was judicious, though there were few lumped marks.

10.0 Marking Scheme or Guide Clarity

The marking schemes or guides prepared and submitted with the examinations were good and largely reflected what were expected from the students. Marks were well distributed on most of the prepared marking guides except in one or two where there were no mark distributions.

11.0 Ease of Assessment

The Department is advised to further intimate examiners of the importance of writing marking guides that any other qualified academic will be able to use.

12.0 Summary of Assessment of Moderated Examination Questions

Table 1 presents summary of my objective assessment of the examination questions and marking guides submitted for moderation.

13.0 Projects and Project Reports

An oral examination was carried out on final year project s and project reports of fifty-four (54) students. The diversity of the projects was impressive, and it covers various bodies of knowledge areas in Electrical and Electronics Engineering. Each student presented his/her project confidently with the implementation skilfully demonstrated. This showed good understanding of the discipline and its application to addressing real life problem. The project reports of the students were also very good showing the painstaking efforts of the project supervisors to improve the quality of the work. In the project reports, minor corrections such as poor citation and referencing style as well as lexical and typographical errors were observed, which were pointed out to be corrected. The students' composure and response during questions section are worthy of note. Indeed, all the students demonstrated good understanding of their projects and presented very well

14.0 Conclusion

I want to thank the Vice-Chancellor and the Management of Afe Babalola University, Ado-Ekiti for the opportunity giving me to render my services to the University as an External Examiner to Electrical and Electronics Engineering programme in the Department of Electrical/Electronics and Computer Engineering for 2023/2024 Academic Session.

Table 1: Courses moderation and Comments

S/N	Course Code and Course Title	Comment
1.	EEE 512: Digital Communications	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • Some of the questions were practical and creative questions, which are commendable at 500 Level • The questions were well distributed. • Mark distribution on both the question and marking scheme were properly done.
2.	EEE 522: Microprocessor Application, Organization and Embedded Systems	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • The questions were practical and creative questions, which are commendable at 500 Level • The questions were well distributed. • Mark distribution on both the question and marking scheme were properly done.
3.	EEE 524: Analog IC Design and Application	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • The questions were well distributed. • Mark distribution on both the question paper and the marking scheme were properly done except on question 6 where there is no mark distribution on the question. However, the mark distribution s were indicated on the marking scheme for question 6.
4.	EEE 536: Wireless	• Appropriate number of questions were set in accordance

	Communications	to both NUC and COREN standards • The questions were well distributed • Mark distributions on the question paper were properly done. However, the mark distributions on the marking scheme were lump.
5.	EEE 552: Industrial Utilization of Electric Power	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • Some of the questions were practical and creative questions, which are commendable at 500 Level. • The questions were appropriate, well-structured and well distributed. • Mark distributions on both the question and marking scheme were properly done.
6.	EEE 554: Power Systems Engineering II	• Appropriate number of questions were set in accordance to both NUC and COREN standards • Most of the questions were practical and creative questions, which are commendable at 500 Level. • The questions were well distributed. • Mark distribution on both the question and marking scheme were properly done.
7.	EEE 556: Machine Applications	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • The questions were well distributed. • Marks distribution on the question was ideal. However, the marks distributions on the marking scheme were lump.
8.	EEE 564: Current Trends in Electrical/Electronic Engineering	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • The questions were well distributed. • Mark distribution on the question was ideal. However, the marks distributions on the marking scheme were lump.
9.	EEE 566: Introduction to Non-Linear Systems	• Appropriate numbers of questions were set in accordance to both NUC and COREN standards. • The questions were well distributed. • Marks distributions on both the question paper and the marking scheme were properly distributed.
10.	EEE 576: Microcomputer Technology	• The questions were well distributed. • Marks distribution on the question paper was indicated. However, the marks distributions on the marking scheme were not done.